

NEET **CHEMISTRY** MODULE 6

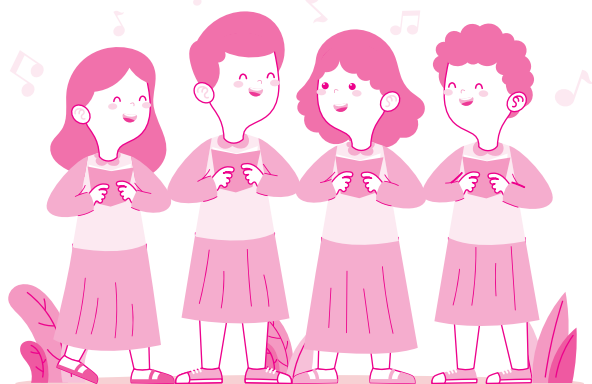
- d & f-block Elements
- Co-ordination Compounds
- The p-block Elements
- Salt Analysis

सर्वम प्रार्थना

हमको मन की शक्ति, देना मन विजय करें
दूसरों की जय से पहले खुद की जय करें
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भेद—भाव अपने दिल से साफ कर सकें
दूसरों से भूल हो तो माफ कर सकें
झूठ से बचे रहें, सच का दम भरें
दूसरों की जय से पहले खुद की जय करें
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मुश्किलें पड़े तो हम पे, इतना कर्म करें
साथ दे तो धर्म का, चलें तो धर्म पर
खुदपे होसला रहे, बदी से ना डरें
दूसरों की जय से पहले, खुद को जय करें



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Chapter 1

The d & f-BLOCK ELEMENTS

Chapter Summary

- Introduction
- General Properties of d-Block Elements
- Preparation of KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$
 - Potassium Permanganate (KMnO_4)
 - Potassium Dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$)
- Physical & Chemical Properties of KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$
- f-Block Elements (Inner Transition Elements)
- Lanthanoids
 - Electronic Configuration
- Chemical Properties of Lanthanoids
- Actinoids
- General Characteristics and Comparison with Lanthanoids

Introduction

The d-block of the periodic table contains the elements of the groups 3-12 in which the d orbitals are progressively filled in each of the four long periods. The f-block consists of elements in which 4f and 5f orbitals are progressively filled. They are placed in a separate panel at the bottom of the periodic table. The names transition metals and inner transition metals are often used to refer to the elements of d & f-blocks respectively.

Position in the Periodic Table :

The d-block occupies the large middle section of the periodic table flanked between s- and p-blocks in the periodic table. The d-orbitals of the penultimate energy level of atoms receive electrons giving rise to four rows of the transition metals, i.e., 3d, 4d, 5d and 6d.

Electronic Configurations of the d-Block Elements :

In general, the electronic configuration of outer orbitals of these elements is $(n-1) d^{1-10} ns^{1-2}$ except for Pd where its electronic configuration is $4d^{10}5s^0$. The $(n-1)$ stands for the inner d orbitals which may have one to ten electrons and the outermost ns orbital may have one or two electrons.

Classification of Transition Series:

(a) 1st Transition Series:

They are classified in fourth period and are called as '3d' series of elements. Their atomic numbers are 21(Sc) to 30 (Zn).

(b) 2nd Transition Series:

They are classified in fifth period and are called as '4d' series of elements. Their atomic numbers are 39(Y) to 48(Cd).

(c) 3rd Transition Series:

They are classified in sixth period and are called as '5d' series of elements. Their atomic numbers are 57(La), 72(Hf) to 80(Hg)

(d) 4th Transition Series:

They are classified in the seventh period and are called as '6d' series of elements. Their atomic numbers are 89(Ac) 104(Rf) to 112 (Cn).

Electronic Configuration :**Ist Transition Series**

Symbol	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Atomic No.	21	22	23	24	25	26	27	28	29	30
3d electrons	1	2	3	5	5	6	7	8	10	10
4s electrons	2	2	2	1	2	2	2	2	1	2

IInd Transition Series

Symbol	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
Atomic No.	39	40	41	42	43	44	45	46	47	48
4d electrons	1	2	4	5	6	7	8	10	10	10
5s electrons	2	2	1	1	1	1	1	0	1	2

IIIrd Transition Series

Symbol	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
Atomic No.	57	72	73	74	75	76	77	78	79	80
5d electrons	1	2	3	4	5	6	7	9	10	10
6s electrons	2	2	2	2	2	2	2	1	1	2

IVth Transition Series

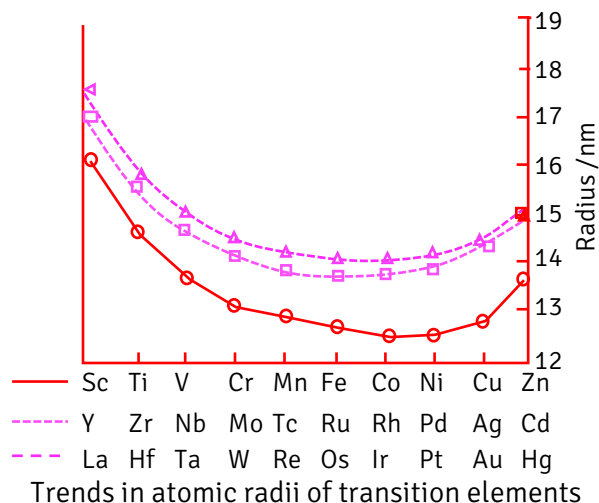
Symbol	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn
Atomic No.	89	104	105	106	107	108	109	110	111	112
6d electrons	1	2	3	4	5	6	7	8	10	10
7s electrons	2	2	2	2	2	2	2	2	1	2

General Properties of d-Block Elements**Physical Properties:**

Nearly all the transition elements display typical metallic properties such as high tensile strength, ductility, malleability, high thermal and electrical conductivity and metallic lustre. With the exceptions of Zn, Cd, Hg and Mn, they have one or more typical metallic structures at normal temperatures.

(A) Atomic Size:

In general, ions of the same charge in a given series show progressive decrease in radius with increasing atomic number. This is because the new electron enters a d orbital each time the nuclear charge increases by unity. It may be recalled that the shielding effect of a d electron is not that effective, hence the net electrostatic attraction between the nuclear charge and the outermost electron increases and the ionic radius decreases. The same trend is observed in the atomic radii of a given series. However, the variation within a series is quite small. An interesting point emerges when atomic sizes of one series are compared with those of the



corresponding elements in the other series. An increase from the first (3d) to the second (4d) series of the elements but the radii of the third (5d) series are virtually the same as those of the corresponding members of the second series. This phenomenon is associated with the intervention of the 4f orbitals which must be filled before the 5d series of elements begin. The filling of 4f before 5d orbital results in a regular decrease in atomic radii called Lanthanoid contraction which essentially compensates for the expected increase in atomic size with increasing atomic number. The net result of the lanthanoid contraction is that the second and the third d series exhibit similar radii (e.g., Zr 160 pm, Hf 159 pm) and have very similar physical and chemical properties much more than that expected on the basis of usual family relationship.

The factor responsible for the lanthanoid contraction is somewhat similar to that observed in an ordinary transition series and is attributed to similar cause, i.e., the imperfect shielding of one electron by another in the same set of orbitals. However, the shielding of one 4f electron by another is less than that of one d electron by another, and as the nuclear charge increases along the series, there is fairly regular decrease in the size of the entire $4f^n$ orbitals.

(B) Melting and Boiling Points:

- (i) Melting and boiling point of d-block > s-block

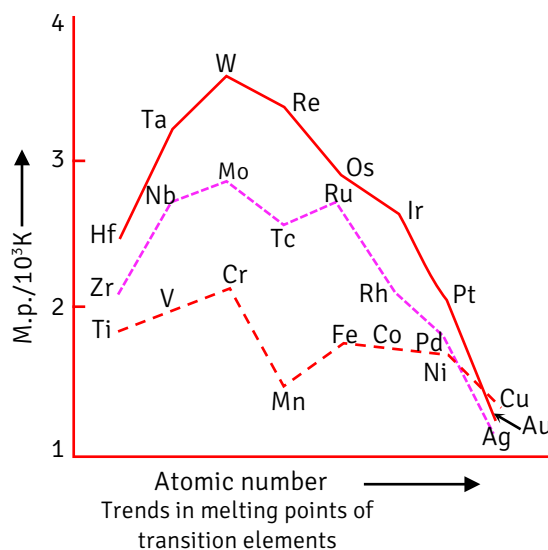
Reason: Stronger metallic bond and presence of covalent bond formed by unpaired d-electrons.)

- (ii) In Zn, Cd, and Hg there is no unpaired electron present in d-orbital, hence due to absence of covalent bond melting and boiling point are very low in series. (Volatile metals Zn, Cd, Hg)

- (iii) In 3d series Sc to Cr melting and boiling point increases then Mn to Zn melting and boiling point decreases

- (iv) With the increase in the number of unpaired d-electrons, the number of covalent bonds between atoms and the bond energy increase up to the Cr-Mo-W family and hence the melting point increases.

- (v) Mn and Tc have comparatively low melting point, due to weak metallic bond because of stable Half filled (d^5) configuration and high IP.



(C) Metallic Character:

- (i) They are hard, malleable and ductile (except Hg). IB group elements Cu, Ag and Au are most ductile and soft.
- (ii) These are good conductor of heat and electricity (due to free e^-) Elements of IB group are most conductive in nature.
- (iii) Covalent and metallic bonding both exist in the atom of transition metals.
- (iv) The presence of unfilled of d-subshell favour covalent bonding and metallic bonding is due to possession of one or two electron in outermost energy shell.

(D) Density :

The decrease in metallic radius coupled with increase in atomic mass results in a general increase in the density of these elements. Thus, from titanium ($Z = 22$) to copper ($Z = 29$) the significant increase in the density may be noted

Element		Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Atomic number		21	22	23	24	25	26	27	28	29	30
Electronic configuration											
	M	3d ¹ 4s ²	3d ² 4s ²	3d ³ 4s ²	3d ⁵ 4s ¹	3d ⁵ 4s ²	3d ⁶ 4s ²	3d ⁷ 4s ²	3d ⁸ 4s ²	3d ¹⁰ 4s ¹	3d ¹⁰ 4s ²
	M ⁺	3d ¹ 4s ¹	3d ² 4s ¹	3d ³ 4s ¹	3d ⁵	3d ⁵ 4s ¹	3d ⁶ 4s ¹	3d ⁷ 4s ¹	3d ⁸ 4s ¹	3d ¹⁰	3d ¹⁰ 4s ¹
	M ²⁺	3d ¹	3d ²	3d ³	3d ⁴	3d ⁵	3d ⁶	3d ⁷	3d ⁸	3d ⁹	3d ¹⁰
	M ³⁺	[Ar]	3d ¹	3d ²	3d ³	3d ⁴	3d ⁵	3d ⁶	3d ⁷	–	–
Enthalpy of atomisation, $\Delta_a H^\circ / \text{kJ mol}^{-1}$											
		326	473	515	397	281	416	425	430	339	126
Ionisation enthalpy/ $\Delta_i H^\circ / \text{kJ mol}^{-1}$											
$\Delta_i H^\circ$	I	631	656	650	653	717	762	758	736	736	906
$\Delta_i H^\circ$	II	1235	1309	1414	1592	1509	1561	1644	1752	1958	1734
$\Delta_i H^\circ$	III	2393	2657	2833	2990	3260	2962	3243	3402	3556	3837
Metallic/ionic	M	164	147	135	129	137	126	125	125	128	137
radii/pm	M ²⁺	–	–	79	82	82	77	74	70	73	75
	M ³⁺	73	67	64	62	65	65	61	60	–	–
Standard											
electrode	M ²⁺ /M	–	–1.63	–1.18	–0.90	–1.18	–0.44	–0.28	–0.25	+0.34	–0.76
potential E°/V	M ³⁺ /M ²⁺	–	–0.37	–0.26	–0.41	+1.57	+0.77	+1.97	–	–	–
Density/g cm^{–3}		3.43	4.1	6.07	7.19	7.21	7.8	8.7	8.9	8.9	7.1

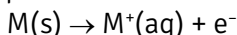
(E) Ionisation Enthalpies :

There is an increase in ionisation enthalpy along each series of the transition elements from left to right due to an increase in nuclear charge which accompanies the filling of the inner d orbitals. The variation in ionisation enthalpy along a series of transition elements is much less in comparison to the variation along a period of non-transition elements. The first ionisation enthalpy, in general, increases, but the magnitude of the increase in the second and third ionisation enthalpies for the successive elements, is much higher along a series. The irregular trend in the first ionisation enthalpy of the metals of 3d series, though of little chemical significance, can be accounted for by considering that the removal of one electron alters the relative energies of 4s and 3d orbitals. When d-block elements form ions, ns electrons are lost before (n – 1) d electrons. As we move along the period in 3d series, nuclear charge increases from scandium to zinc but electrons are added to the orbital of inner subshell, i.e., 3d orbitals. These 3d electrons shield the 4s electrons from the increasing nuclear charge somewhat more effectively than the outer shell electrons can shield one another. Therefore, the atomic radii decrease less rapidly. Thus, ionization energies increase only slightly along the 3d series. The doubly or more highly charged ions have d n configurations with no 4s electrons. A general trend of increasing values of second ionisation enthalpy is expected as the effective nuclear charge increases because one d electron does not shield another electron from the influence of nuclear charge because d-orbitals differ in direction. However, the trend of steady increase in second and third ionisation enthalpy breaks for the formation of Mn²⁺ and Fe³⁺ respectively. In both the cases, ions have d⁵ configuration. Similar breaks occur at corresponding elements in the later transition series.

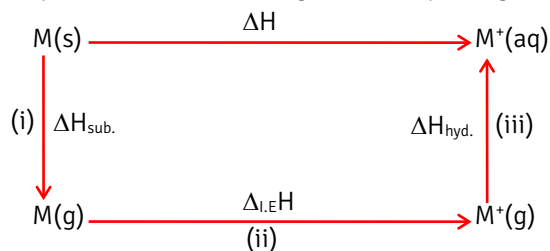
The three terms responsible for the value of ionisation enthalpy are attraction of each electron towards nucleus, repulsion between the electrons and the exchange energy. Exchange energy is responsible for the stabilisation of energy state. Exchange energy is approximately proportional to the total number of possible pairs of parallel spins in the degenerate orbitals. When several electrons occupy a set of degenerate orbitals, the lowest energy state corresponds to the maximum possible extent of single occupation of orbital and parallel spins (Hunds rule). The loss of exchange energy increases the stability. As the stability increases, the ionisation becomes more difficult. There is no loss of exchange energy at d^6 configuration. Mn^{+} has $3d^5 4s^1$ configuration and configuration of Cr^{+} is d^5 , therefore, ionisation enthalpy of Mn^{+} is lower than Cr^{+} . In the same way, Fe^{2+} has d^6 configuration and Mn^{2+} has $3d^5$ configuration. Hence, ionisation enthalpy of Fe^{2+} is lower than the Mn^{2+} . In other words, we can say that the third ionisation enthalpy of Fe is lower than that of Mn. The lowest common oxidation state of these metals is +2. To form the M^{2+} ions from the gaseous atoms, the sum of the first and second ionisation enthalpy is required in addition to the enthalpy of atomisation. The dominant term is the second ionisation enthalpy which shows unusually high values for Cr and Cu where M^{+} ions have the d^5 and d^{10} configurations respectively. The value for Zn is correspondingly low as the ionisation causes the removal of one 4s electron which results in the formation of stable d^{10} configuration. The trend in the third ionisation enthalpies is not complicated by the 4s orbital factor and shows the greater difficulty of removing an electron from the d^5 (Mn^{2+}) and d^{10} (Zn^{2+}) ions. In general, the third ionisation enthalpies are quite high. Also the high values for third ionisation enthalpies of copper, nickel and zinc indicate why it is difficult to obtain oxidation state greater than two for these elements. Although ionisation enthalpies give some guidance concerning the relative stabilities of oxidation states, this problem is very complex and not amenable to ready generalisation.

(F) Electrode Potentials:

In addition to ionisation enthalpy, the other factors such as enthalpy of sublimation, hydration enthalpy etc. Determine the stability of a particular oxidation state in solution. This can be explained in terms of their electrode potential values. The oxidation potential of a metal involves the following process:



This process actually takes place in the following three steps as given in following flowchart:



The oxidation potential which gives the tendency of the overall change to occur, depends upon the net effect of these steps. The overall energy change is $\Delta H = \Delta_{\text{sub}} H + \Delta_{\text{i.EH}} + \Delta_{\text{hyd}} H$

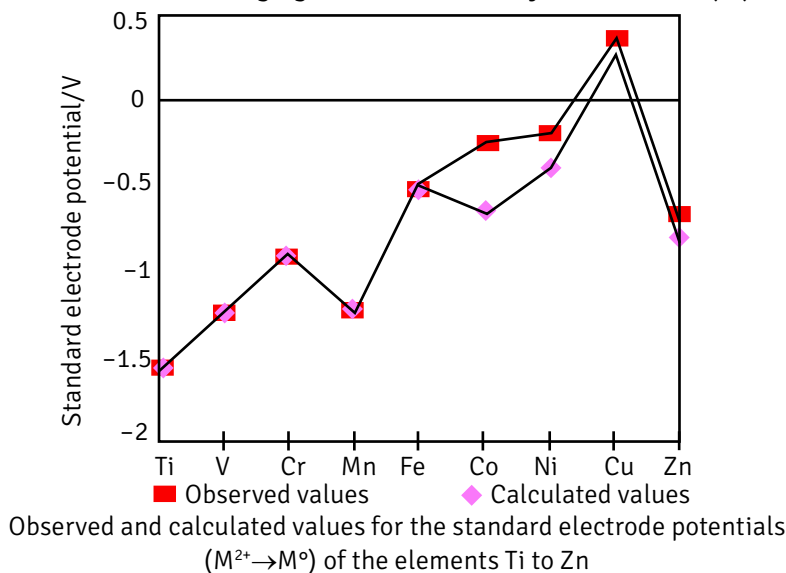
If SOP is +ve that means oxidation will be easy

If SRP is +ve that means reduction will be easy.

Some Important Examples:

1. $E^0_{M^{+2}/M} \Rightarrow +ve$ only for Cu among 3d elements because hydration energy of Cu^{+2} is not enough to compensate for sublimation energy, IE_1 and IE_2 for Cu.
2. $E^0_{M^{+3}/M^{+2}}$
 - (a) $E^0_{Cr^{+3}/Cr^{+2}} = -ve$
That means Cr^{+2} acts as a reducing agent because Cr^{+3} has t_{2g} half filled stable configuration.
 - (b) $E^0_{Mn^{+3}/Mn^{+2}} = +ve$

That means Mn^{+3} acts as an oxidizing agent due to stability of half filled (d^5) configuration.



(G) Oxidation States:

- (i) They exhibit variable oxidation state due to involvement of (ns) and ($(n-1)d$) electrons. Due to less energy difference between these electrons.

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
	+2	+2	+2	+2	+2	+2	+2	+1	+2
+3	+3	+3	+3	+3	+3	+3	+3	+2	
	+4	+4	+4	+4	+4	+4	+4		
		+5	+5	+5					
			+6	+6	+6				
				+7					

- (ii) The transition metal ions having stable configuration like d^0 , d^5 or d^{10} are more stable.

Example:

Sc^{+3} , Ti^{+4} , V^{+5} , Fe^{+3} , Mn^{+2} , Zn^{+2} etc.

- (iii) In aqueous medium Cr^{+3} is stable due to half-filled t_{2g} set.
 (iv) Co^{+3} and Ni^{+2} are stable in complexes.
 (v) In aqueous medium due to disproportionation Cu^{+1} is less stable than Cu^{+2} while its configuration is $3d^{10}$.
 (vi) Most common oxidation state among the transition elements is +2.
 (vii) Highest oxidation state shown by transition elements of '4d' and '5d' series is +8 by Ru and Os.
 (viii) The common oxidation state shown by elements of IIIB i.e., Sc, Y, La and Ac is +3.
 (ix) In lower oxidation state transition elements form ionic compounds and in higher oxidation state their compounds are covalent.
 (x) They also show zero oxidation state in their carbonyl compounds like $\text{Ni}(\text{CO})_4$.
 (xi) Usually transition metal ions in their lower oxidation state act as reducing agents and in higher oxidation state they are oxidising agents.

Example:

Sc^{+2} , Ti^{+2} , V^{+2} , Fe^{+2} , Co^{+2} etc are reducing agents Cr^{+6} , Mn^{+7} , Mn^{+6} , Mn^{+5} , Mn^{+4} etc are oxidising agents.

The Relative Stability of Various Oxidation States:

- (i) The relative stabilities of various oxidation states of 3d-series element can be correlated with the extra stability of $3d^0$, $3d^5$ & $3d^{10}$ configuration to some extent.

Example:

Stability of $Ti^{4+} > Ti^{3+}$, $Mn^{2+} > Mn^{3+}$

- (ii) The higher oxidation state of 4d and 5d series element are generally more stable than the elements of 3d series.

Example:

- (a) MoO_4^{2-} , TcO_4^- , (4d series) & WO_4^{2-} , ReO_4^- (5d series) are more stable due to their maximum oxidation state.
- (b) CrO_4^{2-} & MnO_4^- (3d-series) are strong oxidizing agents. Down the group stability of higher oxidation State increases hence oxidizing character decreases.
- (iii) Cu^{2+} is highly unstable with I^-
 $2Cu^{2+} + 4I^- \rightarrow Cu_2I_2 + I_2$

Trends in Stability of Higher Oxidation States:

The highest oxidation numbers are achieved in TiX_4 (tetrahalides), VF_5 and CrF_6 . The +7 state for Mn is not represented in simple halides but MnO_3F is known, and beyond Mn no metal has a trihalide except FeX_3 and CoF_3 . The ability of fluorine to stabilise the highest oxidation state is due to either higher lattice energy as in the case of CoF_3 , or higher bond enthalpy terms for the higher covalent compounds, e.g., VF_5 and CrF_6 . Although V^{+5} is represented only by VF_5 , the other halides, however, undergo hydrolysis to give oxohalides, VOX_3 . Another feature of fluorides is their instability in the low oxidation states

Example :

VX_2 (X = Cl, Br or I)

The stability of Cu^{2+} (aq) rather than Cu^+ (aq) is due to the much more negative $\Delta_{hyd}H^\ominus$ of Cu^{2+} (aq) than Cu^+ . Which more than compensates for the second ionisation enthalpy of Cu. Coincides with the group number and is attained in Sc_2O_3 to Mn_2O_7 . Beyond group 7, no higher oxides of Fe above Fe_2O_3 , are known. Although ferrates (VI) (FeO_4) $^{2-}$ are formed in alkaline media but they readily decompose to Fe_2O_3 and O_2 . Besides the oxides, oxocations stabilize V^V as VO_2 , V^{IV} as VO^{2+} and TiO^{2+} . The ability of oxygen the highest Mn fluoride is MnF_4 whereas the highest oxide is Mn_2O_7 . The ability of oxygen to form multiple bonds to metals explains its superiority. In the covalent oxide Mn_2O_2 , each Mn is tetrahedrally surrounded by o's including a Mn–O–Mn bridge. The tetrahedral $[MO_4]^{n-}$ ions are known for V^V , Cr^{VI} , Mn^V , Mn^{VI} and Mn^{VII}

Oxides of 3d Metals

Oxidation Number	Groups									
	3	4	5	6	7	8	9	10	11	12
+ 7					Mn_2O_7					
+ 6				CrO_3						
+ 5			V_2O_5							
+ 4		TiO_2	V_2O_4	CrO_2	MnO_2					
+ 3	Sc_2O_3	Ti_2O_3	V_2O_3	Cr_2O_3	Mn_2O_3 $Mn_3O_4^*$	Fe_2O_3 $Fe_3O_4^*$	$Co_3O_4^*$			
+2		TiO	VO	(CrO)	MnO	FeO	CoO	NiO	CuO	ZnO
+1									Cu_2O	

*mixed oxides

Formulas of Halides of 3d Metals :

Oxidation Number									
+ 6			CrF ₆						
+ 5		VF ₅	CrF ₅						
+ 4	TiX ₄	VX ₄ ^I	CrX ₄	MnF ₄					
+ 3	TiX ₃	VX ₃	CrX ₃	MnF ₃	FeX ₃ ^I	CoF ₃			
+ 2	TiX ₂ ^{III}	VX ₂	CrX ₂	MnX ₂	FeX ₂	CoX ₂	NiX ₂	CuX ₂ ^{II}	ZnX ₂
+ 1								CuX ^{III}	

NCERT QUESTIONS (Solve Yourself)

- Which of the 3d series of the transition metals exhibits the largest number of oxidation states and why?
- Why is Cr²⁺ reducing and Mn³⁺ oxidising when both have d⁴ configuration?
- The E⁰ (M²⁺ / M) value for copper is positive (+0.34V). What is possible reason for this?
(Hint: consider its high $\Delta_a H^0$ and low $\Delta_{hyd} H^0$)
- For the first row transition metals the E⁰ values are:

E ⁰	V	Cr	Mn	Fe	Co	Ni	Cu
(M ²⁺ / M)	-1.18	-0.91	-1.18	-0.44	-0.28	-0.25	+0.34

Explain the irregularity in the above values.
- Why is the E⁰ value for the Mn³⁺ / Mn²⁺ couple much more positive than that for Cr³⁺ / Cr²⁺ or Fe³⁺ / Fe²⁺? Explain.
- Why is the highest oxidation state of a metal exhibited in its oxide or fluoride only?
- Which is a stronger reducing agent Cr²⁺ or Fe²⁺ and why?
- Why are Mn²⁺ compounds more stable than Fe²⁺ towards oxidation to their +3 state?
- For M²⁺/M and M³⁺/M²⁺ systems the E⁰ values for some metals are as follows:

Cr ²⁺ /Cr	-0.9 V	Cr ³⁺ /Cr ²⁺	-0.4 V
Mn ²⁺ /Mn	-1.2 V	Mn ³⁺ /Mn ²⁺	+1.5 V
Fe ²⁺ /Fe	-0.4 V	Fe ³⁺ /Fe ²⁺	+0.8 V

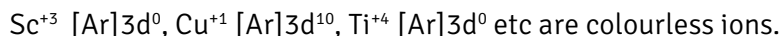
Use this data to comment upon:
 (i) the stability of Fe³⁺ in acid solution as compared to that of Cr³⁺ or Mn³⁺ and
 (ii) the ease with which iron can be oxidised as compared to a similar process for either chromium or manganese metal.
- Compare the stability of +2 oxidation state for the elements of the first transition series.
- Write down the number of 3d electrons in each of the following ions:
 Ti²⁺, V²⁺, Cr³⁺, Mn²⁺, Fe²⁺, Fe³⁺, Co²⁺, Ni²⁺ and Cu²⁺.
 Indicate how would you expect the five 3d orbitals to be occupied for these hydrated ions (octahedral).

(H) Colour Property:

- (i) Most of the transition metal ions exhibit colour property.
- (ii) This is due to d-d transition of unpaired electrons in their t_{2g} and e_g sets of 'd' orbitals.
- (iii) They require less amount of energy to undergo excitation of electrons. Hence, they absorb visible region of light exhibiting colour.

Example:

- (iv) Transition metal ions which do not have any unpaired electrons in their 'd' orbitals like $3d^0$ and $3d^{10}$ configurations, do not exhibit any colour property.

Example:

- (v) A transition metal ion absorbs a part of visible region of light and emits rest of the six colours, the combination of which, is the colour of emitted light. The colour of metal ion is the colour of the emitted light.
- (vi) In transition metal ion the 'd' orbitals split into lower energy set t_{2g} orbitals and higher energy set e_g orbitals. The electrons from t_{2g} set get excited to higher energy set i.e., e_g set. This excitation of electrons is called as 'd-d' transition. Due to this 'd-d' transition the transition metal ions exhibit colour property.

Factors Affecting the Colour of Complex:

The colour of a transition metal complex depends on-

- (i) The magnitude of energy difference Δ_0 between the two d-levels t_{2g} and e_g .
- (ii) An increase in the magnitude of Δ_0 decreases the wave length (λ) of the light absorbed by the complexes.

$$\Delta_0 \propto \frac{1}{\lambda(\text{Wav length of light absorb})}$$

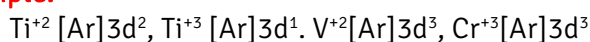
- (iii) Thus with a decrease in the λ the colour of complex changes from Red to Violet.
- (iv) KMnO_4 (dark pink), $\text{K}_2\text{Cr}_2\text{O}_7$ (orange) having d^0 configuration but they are coloured due to charge Transfer spectrum and charge is transferred from anion to cation.
(LMCT : Ligand to metal charge transfer spectrum)

Example of Some Coloured Metal Ions:

Ti^{+3}	Purple	Cr^{+3}	Green	Mn^{+2}	Light pink	Fe^{+2}	Green
Fe^{+3}	Yellow	Co^{+3}	Pink	Ni^{+2}	Green	Cu^{+2}	Blue

(I) Magnetic Properties:

- (i) Generally transition elements exhibits the magnetic property. A **paramagnetic** substance is one which is attracted into a magnetic field. Paramagnetism is mainly due to the presence of unpaired electrons in atoms or ions or molecules. It varies inversely with temperatures.
- (ii) Diamagnetic substance is one which is slightly repelled by a magnetic field. It's independent of temperature.
- (iii) As is evident most of the transition metal ions have unpaired electrons in their 'd' orbitals. Hence most of the transition metal ions are paramagnetic in nature.

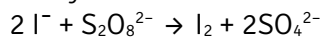
Example:

- (iv) Transition metal ions having $3d^0$ and $3d^{10}$ configuration exhibit diamagnetic nature.
- (v) The total magnetic moment of a substance is the resultant of the magnetic moments of all the individual electrons.
- (vi) The magnetic moment (μ) created due to spinning of unpaired electrons can be calculated by using $\mu = \sqrt{n(n+2)}$ Where - 'n' is the number of unpaired electrons in d-orbitals of metal ion.
 μ = Magnetic moment in Bohr Magnetons (B.M.)
- (vii) The magnetic moment of diamagnetic substances will be zero.
- (viii) Transition metal ions having d^5 configuration will have maximum number of unpaired electrons therefore, they will be maximum paramagnetic in nature.

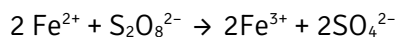
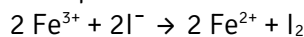
Metal Ions Configuration	Outer Most	Number of unpaired electron	Magnetic Moment μ B.M.
Sc^{3+}	$3d^0$	0	0
$\text{Ti}^{3+}, \text{V}^{4+}$	$3d^1$	1	1.73
$\text{Ti}^{2+}, \text{V}^{3+}$	$3d^2$	2	2.82
$\text{V}^{2+}, \text{Cr}^{3+}$	$3d^3$	3	3.87
$\text{Cr}^{2+}, \text{Mn}^{3+}$	$3d^4$	4	4.89
$\text{Mn}^{2+}, \text{Fe}^{3+}$	$3d^5$	5	5.91
$\text{Fe}^{2+}, \text{Co}^{3+}$	$3d^6$	4	4.89
Co^{2+}	$3d^7$	3	3.87
Ni^{2+}	$3d^8$	2	2.82
Cu^{2+}	$3d^9$	1	1.73
Zn^{2+}	$3d^{10}$	0	0

(J) Catalytic Property:

The transition metals and their compounds are known for their catalytic activity. This activity is ascribed to their ability to adopt multiple oxidation states and to form complexes. Vanadium(V) oxide (in Contact Process), finely divided iron (in Haber's Process), and nickel (in Catalytic Hydrogenation) are some of the examples. Catalysts at a solid surface involve the formation of bonds between reactant molecules and atoms of the surface of the catalyst (first row transition metals utilise 3d and 4s electrons for bonding). This has the effect of increasing the concentration of the reactants at the catalyst surface and also weakening of the bonds in the reacting molecules (the activation energy is lowering). Also because the transition metal ions can change their oxidation states, they become more effective as catalysts. For example, iron(III) catalyses the reaction between iodide and persulphate ions.



An explanation of this catalytic action can be given as:



(K) Formation of Alloy:

An alloy is a blend of metals prepared by mixing the components. Alloys may be homogeneous solid solutions in which the atoms of one metal are distributed randomly among the atoms of the other. Such alloys are formed by atoms with metallic radii that are within about 15 percent of each other. Because of similar radii and other characteristics of transition metals, alloys are readily formed by these metals. The alloys so formed are hard and have often high melting points. The best known are ferrous alloys: chromium, vanadium, tungsten, molybdenum and manganese are used for the production of a variety of steels and stainless steel. Alloys of transition metals with non transition metals such as brass (copper-zinc) and bronze (copper-tin), are also of considerable industrial importance.

Some Important Alloy:

(a)	Bronze	Cu (75 - 90 %) + Sn (10 - 25 %)
(b)	Brass	Cu (60 - 80 %) + Zn (20 - 40 %)
(c)	German Silver	Cu + Zn + Ni (2: 1: 1)
(d)	Stainless steel	Cr (12 - 14 %) & Ni (2 - 4 %)

(L) Formation of Interstitial Compounds:

Interstitial compounds are those which are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals. They are usually non stoichiometric and are neither typically ionic nor covalent, for example, TiC, Mn₄N, Fe₃H, VH_{0.56} and TiH_{1.7}, etc. The formulas quoted do not, of course, correspond to any normal oxidation state of the metal. Because of the nature of their composition, these compounds are referred to as interstitial compounds. The principal **physical and chemical Characteristics of these Compounds are as follows:**

- (1) They have high melting points, higher than those of pure metals.
- (2) They are very hard, some borides approach diamond in hardness.
- (3) They retain metallic conductivity.
- (4) They are chemically inert.

NCERT QUESTIONS (Solve Yourself)

1. What are interstitial compounds? Why are such compounds well known for transition metals?

Some Important Compounds of Transition Elements :**Oxides and Oxoanions of Metals :**

These oxides are generally formed by the reaction of metals with oxygen at high temperatures. All the metals except scandium form MO oxides which are ionic. The highest oxidation number in the oxides, coincides with the group number and is attained in Sc₂O₃ to Mn₂O₇. Beyond group 7, no higher oxides of iron above Fe₂O₃ are known. Besides the oxides, the oxocations stabilise V^V as VO₂⁺, V^{IV} as VO²⁺ and Ti^{IV} as TiO²⁺. As the oxidation number of a metal increases, ionic character decreases. In the case of Mn, Mn₂O₇ is a covalent green oil. Even CrO₃ and V₂O₅ have low melting points. In these higher oxides, the acidic character is predominant. Thus, Mn₂O₇ gives HMnO₄ and CrO₃ gives H₂CrO₄ and H₂Cr₂O₇. V₂O₅ is, however, amphoteric though mainly acidic and it gives VO₄³⁻ as well as VO₂⁺ salts. In vanadium there is gradual change from the basic V₂O₃ to less basic V₂O₄ and to amphoteric V₂O₅. V₂O₄ dissolves in acids to give VO²⁺ salts. Similarly, V₂O₅ reacts with alkalis as well as acids to give VO₄³⁻ and VO₂⁺ respectively. The well characterised CrO is basic but Cr₂O₃ is amphoteric.

NCERT QUESTIONS (Solve Yourself)

1. On what ground can you say that scandium (Z = 21) is a transition element but zinc (Z = 30) is not?
2. Silver atom has completely filled *d* orbitals (4d¹⁰) in its ground state. How can you say that it is a transition element?
3. Why do the transition elements exhibit higher enthalpies of atomisation?
4. In the series Sc (Z = 21) to Zn (Z = 30), the enthalpy of atomisation of zinc is the lowest, i.e., 126 kJ mol⁻¹. Why?
5. Name a transition element which does not exhibit variable oxidation states.

6. How would you account for the increasing oxidising power in the Example 4.5 series $\text{VO}_2^+ < \text{Cr}_2\text{O}_7^{2-} < \text{MnO}_4^-$?
7. Calculate the magnetic moment of a divalent ion in aqueous solution if its atomic number is 25.
8. Calculate the 'spin only' magnetic moment of $\text{M}^{2+}_{(\text{aq})}$ ion ($Z = 27$).
9. What is meant by 'disproportionation' of an oxidation state? Give an example.
10. Explain why Cu^+ ion is not stable in aqueous solutions?
11. Write down the electronic configuration of:
 - (i) Cr^{3+} (ii) Pm^{3+} (iii) Cu^+ (iv) Ce^{4+} (v) Co^{2+} (vi) Lu^{2+}
 - (vii) Mn^{2+} (viii) Th^{4+}
12. Explain briefly how +2 state becomes more and more stable in the first half of the first row transition elements with increasing atomic number?
13. To what extent do the electronic configurations decide the stability of oxidation states in the first series of the transition elements? Illustrate your answer with examples.
14. What may be the stable oxidation state of the transition element with the following *d* electron configurations in the ground state of their atoms: $3d^3$, $3d^5$, $3d^8$ and $3d^4$?
15. Name the oxometal anions of the first series of the transition metals in which the metal exhibits the oxidation state equal to its group number.
16. What are the characteristics of the transition elements and why are they called transition elements? Which of the d-block elements may not be regarded as the transition elements?
17. Explain giving reasons:
 - (i) Transition metals and many of their compounds show paramagnetic behaviour.
 - (ii) The enthalpies of atomisation of the transition metals are high.
 - (iii) The transition metals generally form coloured compounds.
 - (iv) Transition metals and their many compounds act as good catalyst.
18. How is the variability in oxidation states of transition metals different from that of the non transition metals? Illustrate with examples.
19. Predict which of the following will be coloured in aqueous solution? Ti^{3+} , V^{3+} , Cu^+ , Sc^{3+} , Mn^{2+} , Fe^{3+} and Co^{2+} . Give reasons for each.
20. How would you account for the following:
 - (i) Of the d^4 species, Cr^{2+} is strongly reducing while manganese(III) is strongly oxidising.
 - (ii) Cobalt(II) is stable in aqueous solution but in the presence of complexing reagents it is easily oxidised.
 - (iii) The d^1 configuration is very unstable in ions.
21. Which metal in the first series of transition metals exhibits +1 oxidation state most frequently and why?
22. Calculate the number of unpaired electrons in the following gaseous ions: Mn^{3+} , Cr^{3+} , V^{3+} and Ti^{3+} . Which one of these is the most stable in aqueous solution?
23. Give examples and suggest reasons for the following features of the transition metal chemistry:
 - (i) The lowest oxide of transition metal is basic, the highest is amphoteric/acidic.
 - (ii) A transition metal exhibits highest oxidation state in oxides and fluorides.
 - (iii) The highest oxidation state is exhibited in oxoanions of a metal.

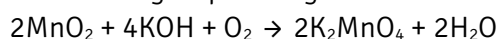
24.	Use Hund's rule to derive the electronic configuration of Ce^{3+} ion, and calculate its magnetic moment on the basis of 'spin-only' formula.								
25.	Compare the general characteristics of the first series of the transition metals with those of the second and third series metals in the respective vertical columns. Give special emphasis on the following points: (i) electronic configurations (ii) oxidation states (iii) ionisation enthalpies and (iv) atomic sizes.								
26.	Comment on the statement that elements of the first transition series possess many properties different from those of heavier transition elements.								
27.	What can be inferred from the magnetic moment values of the following complex species ?								
<table> <thead> <tr> <th>Example</th><th>Magnetic Moment (BM)</th></tr> </thead> <tbody> <tr> <td>$\text{K}_4[\text{Mn}(\text{CN})_6]$</td><td>2.2</td></tr> <tr> <td>$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$</td><td>5.3</td></tr> <tr> <td>$\text{K}_2[\text{MnCl}_4]$</td><td>5.9</td></tr> </tbody> </table>		Example	Magnetic Moment (BM)	$\text{K}_4[\text{Mn}(\text{CN})_6]$	2.2	$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	5.3	$\text{K}_2[\text{MnCl}_4]$	5.9
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Preparation of KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$

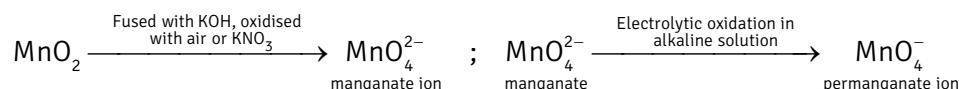
Potassium Permanganate (KMnO_4)

Preparation:

Potassium permanganate is prepared by fusion of MnO_2 with an alkali metal hydroxide and an oxidising agent like KNO_3 . This produces the dark green K_2MnO_4 which disproportionates in a neutral or acidic solution to give permanganate.



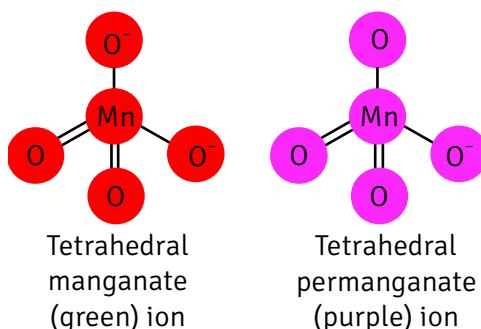
Commercially it is prepared by the alkaline oxidative fusion of MnO_2 followed by the electrolytic oxidation of manganate (VI).



Physical Properties :

Purple coloured crystalline compound.

Moderately soluble in water at room temperature.



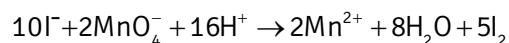
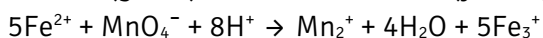
It has two physical properties of considerable interest: its intense colour and its diamagnetism along with temperature-dependent weak paramagnetism. These can be explained by the use of molecular orbital theory which is beyond the present scope. The manganate and permanganate ions are tetrahedral; the π bonding takes place by overlap of p orbitals of oxygen with d orbitals of manganese. The green manganate is paramagnetic because of one unpaired electron but the permanganate is diamagnetic due to the absence of unpaired electron. Structure of manganate and permanganate ion.

Chemical Properties :

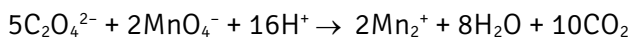
(i) Heating effect

**Oxidising Agent :****I.** In acid solutions:

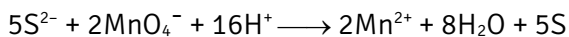
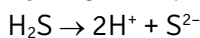
(a) Iodine is liberated from potassium iodide :

(b) Fe^{2+} ion (green) is converted to Fe^{3+} (yellow):

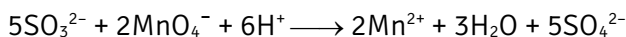
(c) Oxalate ion or oxalic acid is oxidised at 333 K:



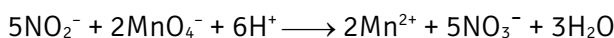
(d) Hydrogen sulphide is oxidised, sulphur being precipitated:



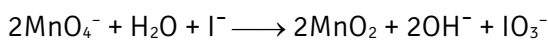
(e) Sulphurous acid or sulphite is oxidised to a sulphate or sulphuric acid:



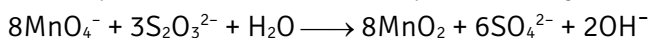
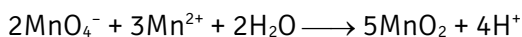
(f) Nitrite is oxidised to nitrate:

**II.** In neutral or faintly alkaline solutions:

(a) A notable reaction is the oxidation of iodide to iodate:



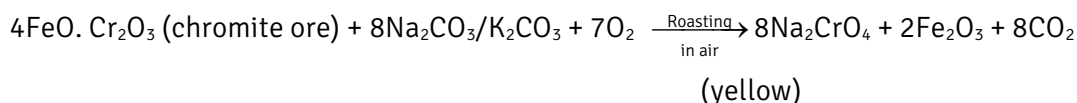
(b) Thiosulphate is oxidised almost quantitatively to sulphate:

(c) Manganous salt is oxidised to MnO_2 ; the presence of zinc sulphate or zinc oxide catalyses the oxidation:**Note:** Permanganate titrations in presence of hydrochloric acid are unsatisfactory since hydrochloric acid is oxidised to chlorine.**III. Uses:**

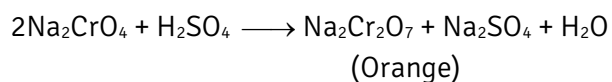
Besides its use in analytical chemistry, potassium permanganate is used as a favourite oxidant in preparative organic chemistry. Its uses for the bleaching of wool, cotton, silk and other textile fibres and for the decolourisation of oils are also dependent on its strong oxidising power.

Potassium Dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$)**Preparation :**

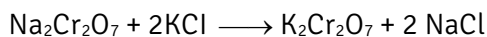
The chromite ore is roasted with sodium carbonate in presence of air in a reverberatory furnace.



The roasted mass is extracted with water when Na_2CrO_4 goes into the solution leaving behind insoluble Fe_2O_3 . The solution is then treated with calculated amount of H_2SO_4 .



The solution is concentrated when less soluble Na_2SO_4 crystallises out. The solution is further concentrated when crystals of $\text{Na}_2\text{Cr}_2\text{O}_7$ are obtained. Hot saturated solution of $\text{Na}_2\text{Cr}_2\text{O}_7$ is then treated with KCl when orange red crystals of $\text{K}_2\text{Cr}_2\text{O}_7$ are obtained on crystallisation. Sodium dichromate is more soluble than potassium dichromate.



Orange Red

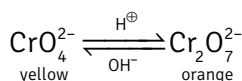
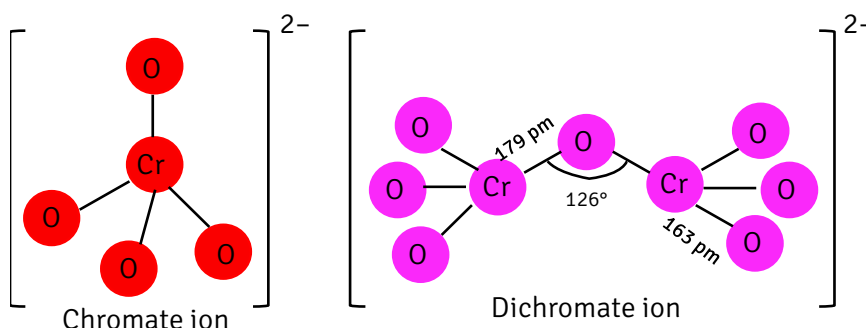
$\text{K}_2\text{Cr}_2\text{O}_7$ is preferred over $\text{Na}_2\text{Cr}_2\text{O}_7$ as a primary standard in volumetric estimation because $\text{Na}_2\text{Cr}_2\text{O}_7$ is hygroscopic in nature but $\text{K}_2\text{Cr}_2\text{O}_7$ is not.

Physical & Chemical Properties of KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$

(a) Physical Properties :

It is orange-red coloured crystalline compound. It is moderately soluble in cold water but freely soluble in hot water.

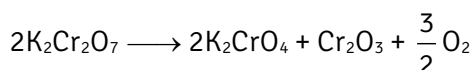
Structure of Chromate and Dichromate ion



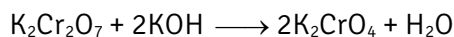
(b) Chemical Properties :

(i.) Effect of heating :

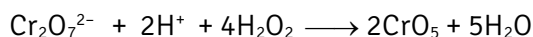
On heating strongly, it decomposes liberating oxygen.



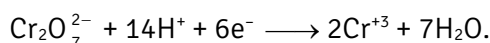
(ii.) On heating with alkalis, it is converted to chromate, i.e., the colour changes from orange to yellow. On acidifying, yellow colour again changes to orange.



(iii.) Acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution reacts with H_2O_2 to give a deep blue solution due to the formation of CrO_5 .



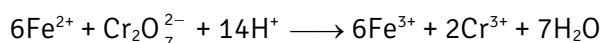
(iv.) It acts as a powerful oxidising agent in acidic medium (dilute H_2SO_4)



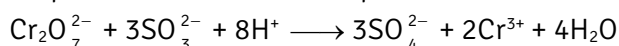
(a.) Iodine is liberated from potassium iodide :



(b.) Ferrous salts are oxidised to ferric salts :



(c.) Sulphites are oxidised to sulphates :



- (d.) H_2S is oxidised to sulphur :

$$\text{Cr}_2\text{O}_7^{2-} + 3\text{H}_2\text{S} + 8\text{H}^+ \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O} + 3\text{S}$$
- (e.) SO_2 is oxidised to H_2SO_4 :

$$\text{Cr}_2\text{O}_7^{2-} + 3\text{SO}_2 + 2\text{H}^+ \longrightarrow 2\text{Cr}^{3+} + 3\text{SO}_4^{2-} + \text{H}_2\text{O}$$

 Chrome alum is obtained when acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution is saturated with SO_2 .

$$\text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 + 3\text{SO}_2 + 23\text{H}_2\text{O} \xrightarrow{T < 70^\circ\text{C}} \text{K}_2\text{SO}_4 \cdot \text{Cr}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$$
- (v.) Chromyl chloride test : $4\text{Cl}^- + \text{Cr}_2\text{O}_7^{2-} + 6\text{H}^+ \longrightarrow 2\text{CrO}_2\text{Cl}_2 \uparrow (\text{deep red}) + 3\text{H}_2\text{O}$

(c) Uses :

- volumetric reagent in the estimation of reducing agents such as oxalic acid, ferrous ions, iodide ions, etc. It is used as a primary standard.
- for the preparation of several chromium compounds such as chrome alum, chrome yellow, chrome red, zinc yellow, etc.
- in dyeing, chrome tanning, calico printing, photography etc.
- as a cleansing agent for glass ware in the form of chromic acid.
- in leather industry and as an oxidant for preparation of azo compounds.

NCERT QUESTIONS (Solve Yourself)

- Describe the preparation of potassium dichromate from iron chromite ore. What is the effect of increasing pH on a solution of potassium dichromate?
- Describe the oxidising action of potassium dichromate and write the ionic equations for its reaction with: (i) iodide (ii) iron(II) solution and (iii) H_2S
- Describe the preparation of potassium permanganate. How does the acidified permanganate solution react with (i) iron(II) ions (ii) SO_2 and (iii) oxalic acid? Write the ionic equations for the reactions.
- Indicate the steps in the preparation of: (i) $\text{K}_2\text{Cr}_2\text{O}_7$ from chromite ore. (ii) KMnO_4 from pyrolusite ore.

Exercise 1.1

- Correct order of oxidising strength of the oxoanions will follow the order
 (1) $\text{CrO}_4^{2-} > \text{MoO}_4^{2-} > \text{WO}_4^{2-}$
 (2) $\text{CrO}_4^{2-} < \text{MoO}_4^{2-} > \text{WO}_4^{2-}$
 (3) $\text{WO}_4^{2-} > \text{CrO}_4^{2-} > \text{MoO}_4^{2-}$
 (4) $\text{MoO}_4^{2-} > \text{WO}_4^{2-} > \text{CrO}_4^{2-}$
- The colour of KMnO_4 is purple due to
 (1) d-d electronic transition
 (2) ligand to metal charge transfer spectra
 (3) metal to ligand charge transfer spectra
 (4) transition of 4s electrons in Mn
- Which one has maximum second ionisation enthalpy?
 (1) Fe (2) Cr (3) Mn (4) Ti
- Amongst TiF_6^{2-} , CoF_6^{3-} , Cu_2Cl_2 and NiCl_4^{2-} the colourless species are
 (1) CoF_6^{3-} and NiCl_4^{2-} (2) TiF_6^{2-} and CoF_6^{3-}
 (3) Cu_2Cl_2 and NiCl_4^{2-} (4) TiF_6^{2-} and Cu_2Cl_2
- Which one pair of ions having similar magnetic moment?
 (1) Ti^{3+} , V^{3+} (2) Cr^{3+} , Mn^{2+}
 (3) Fe^{2+} , Mn^{2+} (4) Mn^{2+} , Fe^{+3}
- Find the most basic oxide from the following
 (1) ZnO (2) Cr_2O_3
 (3) Mn_2O_7 (4) FeO
- Which of the following set has all the coloured ions due to d-d transition?
 (1) Cu^+ , Co^{3+} , Ni^{2+} (2) Cu^{2+} , Sc^{3+} , Co^{2+}
 (3) Cu^{2+} , Fe^{2+} , Ti^{4+} (4) Cr^{3+} , Mn^{2+} , Ni^{2+}
- The lowest S.R.P. value for the M^{2+}/M couple is for
 (1) Mn (2) Cu (3) Zn (4) Fe
- Which one is most acidic in nature?
 (1) MnO (2) Mn_2O_3
 (3) MnO_2 (4) MnO_3

10. In 3d-series which one of the following has positive standard reduction potential
(1) Chromium (2) Cobalt
(3) Copper (4) Gold
11. $\text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{O}_2 + \text{H}^+ \longrightarrow$ products Product of the above reaction is
(1) Cr_2O_3 (2) K_2CrO_4
(3) CrO_5 (4) CrO_3
12. Which of the following element is most unlikely to show +2 oxidation state?
(1) Cu (2) Mn (3) Cr (4) Sc
13. The transition element having highest melting point is:
(1) Ru (2) Os (3) Rh (4) W
14. Metallic radii of some transition elements are given below. Which of these elements will have highest density?

Element	Fe	Co	Ni	Cu
Metallic radii/ pm	126	125	125	128

(1) Fe (2) Ni (3) Co (4) Cu
15. When KMnO_4 solution is added to oxalic acid solution, the decolourisation is slow in the beginning but becomes instantaneous after some time because
(1) CO_2 is formed as the product.
(2) Reaction is exothermic.
(3) MnO_4^- catalyses the reaction.
(4) Mn^{2+} acts as autocatalyst.
16. Formula of Rust is:
(1) Fe_2O_3 (2) $\text{FeO} \cdot x\text{H}_2\text{O}$
(3) $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ (4) $\text{Fe}_3\text{O}_4 \cdot x\text{H}_2\text{O}$
17. White vitriol is:
(1) ZnS (2) ZnSO_3
(3) $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ (4) ZnCO_3
18. In a transition series, as the atomic number increases, paramagnetism:
(1) Increases gradually
(2) Decreases gradually
(3) First increases to a maximum and then decreases
(4) First decreases to a minimum and then increases
19. Oxide of metal cation which is not amphoteric?
(1) Al^{3+} (2) Cr^{3+} (3) Fe^{3+} (4) Zn^{2+}
20. Iron is rendered passive by treatment with:
(1) H_2SO_4 (dil.) (2) H_3PO_4
(3) conc. HNO_3 (4) HCl
21. The solubility of silver bromide in hypo solution is due to the formation of:
(1) Ag_2SO_3 (2) $\text{Ag}_2\text{S}_2\text{O}_3$
(3) $[\text{Ag}(\text{S}_2\text{O}_3)]^-$ (4) $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$
22. Which of the following is called Wilkinson's catalyst?
(1) $[\text{Ph}_3\text{P}]_3\text{RhCl}$ (2) $\text{TiCl}_4 + (\text{C}_2\text{H}_5)_3\text{Al}$
(3) $(\text{C}_2\text{H}_5)_4\text{Pb}$ (4) $[\text{PtCl}_2(\text{NH}_3)_2]$
23. Magnetic moment of Cr^{2+} is nearest to:-
(1) Fe^{2+} (2) Mn^{2+} (3) Co^{2+} (4) Ni^{2+}
24. Which of the following element has lowest melting point:-
(1) Cr (2) Fe (3) Ni (4) Cu
25. Which pair of ions is colourless:
(1) Mn^{3+} , Co^{3+} (2) Fe^{3+} , Cr^{3+}
(3) Zn^{2+} , Sc^{3+} (4) Ti^{2+} , Cu^{2+}
26. The higher oxidation states of transition elements are found to be the combination with A and B, which are:
(1) F, O (2) O, N (3) O, Cl (4) F, Cl
27. Compared to Cu^{2+} having $3d^9$ configuration, Cu^+ having $3d^{10}$ configuration (aq. solution):-
(1) Is more stable
(2) Is equally stable
(3) Is less stable
(4) Stability depends upon nature of copper salt
28. Which of the following transition metal do not form amalgam with Hg:-
(1) Fe (2) Co
(3) Ni (4) All of above
29. Which halide of copper does not exist?
(1) CuF_2 (2) CuCl_2
(3) CuI_2 (4) CuI

30. Which order is correct for oxidizing power?

- (1) $\text{Cr}_2\text{O}_7^{-2} < \text{VO}_2^+$ (2) $\text{VO}_2^+ > \text{MnO}_4^-$
 (3) $\text{VO}_2^+ < \text{Cr}_2\text{O}_7^{-2}$ (4) $\text{Cr}_2\text{O}_7^{-2} > \text{MnO}_4^-$

31. Which is correct match

- (1) Brass; Cu + Zn
 (2) Bronze; Cu + Sn
 (3) Stainless steel; Cr + Ni
 (4) All of the above

32. Which of the following 3d series metal has positive value of $E^\circ(\text{M}^{+2}/\text{M})$?

- (1) Zn (2) Cu (3) Mn (4) Cr

33. In which of the following pairs are both the ions coloured in aqueous solution?

(At. no.: Sc = 21, Ti = 22, Ni = 28, Cu = 29, Co = 27)

- (1) Ni^{2+} , Cu^+ (2) Ni^{2+} , Ti^{3+}
 (3) Sc^{3+} , Ti^{3+} (4) Sc^{3+} , Co^{2+}

34. Which one of the following ions is the most stable in aqueous solution

- (1) V^{3+} (2) Ti^{3+} (3) Mn^{3+} (4) Cr^{3+}

35. Which of the following are amphoteric oxides?

Mn_2O_7 , CrO_3 , Cr_2O_3 , CrO , V_2O_5 , V_2O_4

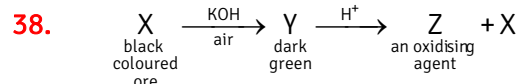
- (1) V_2O_5 , Cr_2O_3 (2) Mn_2O_7 , CrO_3
 (3) CrO , V_2O_5 (4) V_2O_5 , V_2O_4

36. Interstitial compounds are formed when small atoms are trapped inside the crystal lattice of metals. Which of the following is not the characteristic property of interstitial compounds?

- (1) They have high melting points in comparison to pure metals
 (2) They are very hard
 (3) They retain metallic conductivity
 (4) They are chemically very reactive

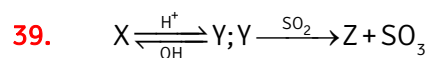
37. The magnetic moment is associated with its spin angular momentum and orbital angular momentum. Spin only magnetic moment value of Cr^{3+} ion is.....

- (1) 2.87 B.M.
 (2) 3.87 B.M.
 (3) 4.83 B.M.
 (4) 5.91 B.M.
 (KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$)



True statement is

- (1) X can't be MnO_2
 (2) Y can be KMnO_4
 (3) Z can be K_2MnO_4
 (4) Step (1) is oxidation while step (2) is disproportionation



X and Y are oxy anion of Cr than oxidation state of Cr in X, Y, Z will be respectively.

- (1) +3, +6, +6
 (2) +6, +3, +6
 (3) +6, +6, +6
 (4) +6, +6, +3

40. $\text{K}_2\text{Cr}_2\text{O}_7$ in acidic medium converts into:-

- (1) Cr^{2+} (2) Cr^{3+}
 (3) Cr^{4+} (4) Cr^{+5}

41. Which compound is used in manufacturing of $\text{K}_2\text{Cr}_2\text{O}_7$?

- (1) Na_2CrO_4 (2) CrO_5
 (3) FeCr_2O_4 (4) CrO_3

42. KMnO_4 acts as an oxidising agent in alkaline medium. When alkaline KMnO_4 is treated with KI, iodide ion is oxidised to.....

- (1) I_2 (2) IO^-
 (3) IO_3^- (4) IO_4^-

43. When acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution is added to Sn^{2+} salts, then Sn^{2+} changes to

- (1) Sn (2) Sn^{3+}
 (3) Sn^{4+} (4) Sn^+

44. Which of the following ions show highest spin only magnetic moment value?

- (1) Ti^{3+} (2) Mn^{2+}
 (3) Fe^{2+} (4) Co^{3+}

45. Which of the following metal ion form unstable complex with CN^- :-

- (1) Ag (I)
 (2) Zn (II)
 (3) Cu (II)
 (4) Cr (II)

f- Block Elements (Inner Transition Elements)

Position in the Periodic Table:

The lanthanides resemble with Yttrium in most of their properties. So it became necessary to accommodate all the fifteen elements together at one place. This has been done by placing the first element, lanthanum below yttrium and placing the remaining fourteen elements separately in the lower part of the periodic table.

Lanthanide series (Z = 58 – 71) (Ce – Lu)

Actinide series (Z = 90 – 103) (Th – Lr)

Lanthanoids

- (i.) Lanthanides are reactive elements so do not found in free state in nature.
- (ii.) Most important minerals for lighter Lanthanides are – Monazite, cerites and orthite and for heavier lanthanides – Gadolinite and Xenotime

Electronic Configuration

- (i) The general configuration of lanthanides may be given as $4f^{1-14}5s^25p^65d^{0-1}6s^2$.
- (ii) It is to be noted that filling of 4f orbitals in the atoms is not regular. A 5d electron appears in gadolinium (Z = 64) with an outer electronic configuration of $4f^75d^16s^2$ (and not $4f^86s^2$). This is because the 4f and 5d electrons are at about the same potential energy and that the atoms have a tendency to retain stable half-filled configuration.
- (iii) On the other hand, the filling of f-orbitals is regular in tripositive ions.
- (iv) After losing outer electrons, the f-orbitals shrink in size and became more stable.
- (v) Pm is the only synthetic radioactive lanthanide.

Atomic Number	Name	Symbol	Electronic configurations*		
			Ln	Ln ²⁺	Ln ³⁺
57	Lanthanum	La	$5d^16s^2$	$5d^1$	$4f^0$
58	Cerium	Ce	$4f^15d^16s^2$	$4f^2$	$4f^1$
59	Praseodymium	Pr	$4f^36s^2$	$4f^3$	$4f^2$
60	Neodymium	Nd	$4f^46s^2$	$4f^4$	$4f^3$
61	Promethium	Pm	$4f^56s^2$	$4f^5$	$4f^4$
62	Samarium	Sm	$4f^66s^2$	$4f^6$	$4f^5$
63	Europium	Eu	$4f^76s^2$	$4f^7$	$4f^6$
64	Gadolinium	Gd	$4f^75d^16s^2$	$4f^75d^1$	$4f^7$
65	Terbium	Tb	$4f^96s^2$	$4f^9$	$4f^8$
66	Dysprosium	Dy	$4f^{10}6s^2$	$4f^{10}$	$4f^9$
67	Holmium	Ho	$4f^{11}6s^2$	$4f^{11}$	$4f^{10}$
68	Erbium	Er	$4f^{12}6s^2$	$4f^{12}$	$4f^{11}$
69	Thulium	Tm	$4f^{13}6s^2$	$4f^{13}$	$4f^{12}$
70	Ytterbium	Yb	$4f^{14}6s^2$	$4f^{14}$	$4f^{13}$
71	Lutetium	Lu	$4f^{14}5d^16s^2$	$4f^{14}5d^1$	$4f^{14}$

Atomic and Ionic Size:

The overall decrease in atomic and ionic radii from lanthanum to lutetium (the lanthanoid contraction) is a unique feature in the chemistry of the lanthanoids. It has far reaching consequences in the chemistry of the third transition series of the elements. The decrease in atomic radii (derived from the structures of metals) is not quite regular as it is regular in M^{3+} ions. This contraction is, of course, similar to that observed in an ordinary transition series and is attributed to the same cause, the imperfect shielding of one electron by another in the same sub-shell. However, the shielding of one 4f electron by another is less than one d electron by another with the increase in nuclear charge along the series. There is fairly regular decrease in the sizes with increasing atomic number.

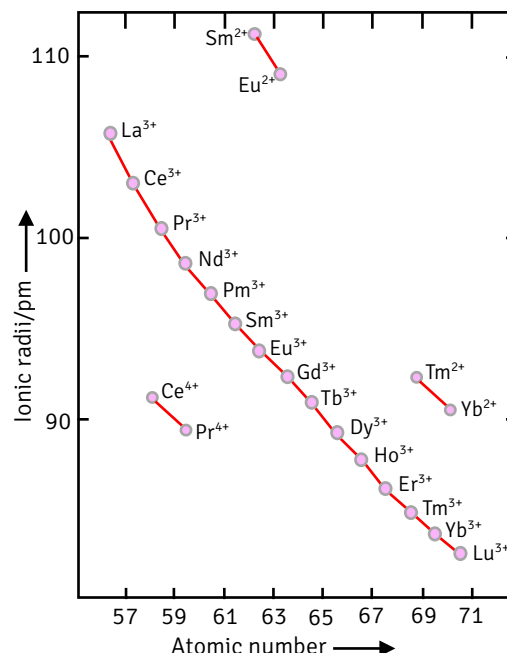


Fig.8.6 : Trends in ionic radii of lanthanoids

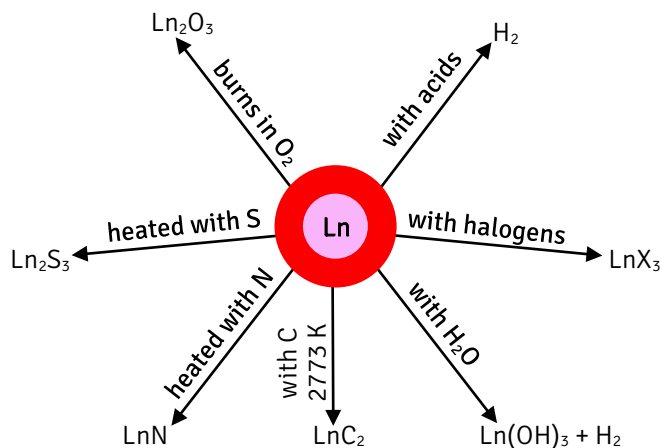
Oxidation states:

In the lanthanoids, La(II) and Ln(III) compounds are predominant species. However, occasionally +2 and +4 ions in solution or in solid compounds are also obtained. This irregularity (as in ionisation enthalpies) arises mainly from the extra stability of empty, half-filled or filled f subshell. Thus, the formation of Ce^{IV} is favoured by its noble gas configuration, but it is a strong oxidant reverting to the common +3 state. The E° value for Ce^{4+}/Ce^{3+} is +1.74 V which suggests that it can oxidise water. However, the reaction rate is very slow and hence $Ce(IV)$ is a good analytical reagent. Pr, Nd, Tb and Dy also exhibit +4 state but only in oxides, MO_2 . Eu^{2+} is formed by losing the two s electrons and its f^7 configuration accounts for the formation of this ion. However, Eu^{2+} is a strong reducing agent changing to the common +3 state. Similarly Yb^{2+} which has f^{14} configuration is a reductant. Tb^{IV} has half-filled f-orbitals and is an oxidant. The behaviour of samarium is very much like europium, exhibiting both +2 and +3 oxidation states.

General Characteristics:

All the lanthanoids are silvery white soft metals and tarnish rapidly in air. The hardness increases with increasing atomic number, samarium being steel hard. Their melting points range between 1000 to 1200 K but samarium melts at 1623 K. They have typical metallic structure and are good conductors of heat and electricity. Density and other properties change smoothly except for Eu and Yb and occasionally for Sm and Tm. Many trivalent lanthanoid ions are coloured both in the solid state and in aqueous solutions. Colour of these ions may be attributed to the presence of f electrons. Neither La^{3+} nor Lu^{3+} ion shows any colour but the rest do so. However, absorption bands are narrow, probably because of the excitation within f level. The lanthanoid ions other than the f^0 type (La^{3+} and Ce^{4+}) and the f^{14} type (Yb^{2+} and Lu^{3+}) are all paramagnetic. The first ionisation enthalpies of the lanthanoids are around 600 kJ mol^{-1} , the second about 1200 kJ mol^{-1} comparable with those of calcium. A detailed discussion of the variation of the third ionisation enthalpies indicates that the exchange enthalpy considerations (as in 3d orbitals of the first transition series), appear to impart a certain degree of stability to empty, half-filled and completely filled orbitals f level. This is indicated from the abnormally low value of the third ionisation enthalpy of lanthanum, gadolinium and lutetium. In their chemical behaviour, in general, the earlier members of the series are quite reactive similar to calcium but, with increasing atomic number, they behave more like aluminium.

Chemical Properties of Lanthanoids



Chemical reactions of the lanthanoids

Uses:

The best single use of the lanthanoids is for the production of alloy steels for plates and pipes. A well known alloy is mischmetall which consists of a lanthanoid metal (~ 95%) and iron (~ 5%) and traces of S, C, Ca and Al. A good deal of mischmetall is used in Mg-based alloy to produce bullets, shell and lighter flint. Mixed oxides of lanthanoids are employed as catalysts in petroleum cracking. Some individual Ln oxides are used as phosphors in television screens and similar fluorescing surfaces.

Actinoids

NCERT Table of Actinoids :

			Electronic configurations*			Radii/pm	
Atomic Number	Name	Symbol	M	M ³⁺	M ⁴⁺	M ³⁺	M ⁴⁺
89	Actinium	Ac	6d ¹ 7s ²	5f ⁰		111	
90	Thorium	Th	6d ² 7s ²	5f ¹	5f ⁰		99
91	Protactinium	Pa	5f ² 6d ¹ 7s ²	5f ²	5f ¹		96
92	Uranium	U	5f ³ 6d ¹ 7s ²	5f ³	5f ²	103	93
93	Neptunium	Np	5f ⁴ 6d ¹ 7s ²	5f ⁴	5f ³	101	92
94	Plutonium	Pu	5f ⁶ 7s ²	5f ⁵	5f ⁴	100	90
95	Americium	Am	5f ⁷ 7s ²	5f ⁶	5f ⁵	99	89
96	Curium	Cm	5f ⁷ 6d ¹ 7s ²	5f ⁷	5f ⁶	99	88
97	Berkelium	Bk	5f ⁹ 7s ²	5f ⁸	5f ⁷	98	87
98	Californium	Cf	5f ¹⁰ 7s ²	5f ⁹	5f ⁸	98	86
99	Einsteinium	Es	5f ¹¹ 7s ²	5f ¹⁰	5f ⁹	–	–
100	Fermium	Fm	5f ¹² 7s ²	5f ¹¹	5f ¹⁰	–	–
101	Mendelevium	Md	5f ¹³ 7s ²	5f ¹²	5f ¹¹	–	–
102	Nobelium	No	5f ¹⁴ 7s ²	5f ¹³	5f ¹²	–	–
103	Lawrencium	Lr	5f ¹⁴ 6d ¹ 7s ²	5f ¹⁴	5f ¹³	–	–

Electronic Configuration:

All the actinoids are believed to have the electronic configuration of $7s^2$ and variable occupancy of the 5f and 6d subshells. The fourteen electrons are formally added to 5f, though not in thorium ($Z = 90$) but from Pa onwards the 5f orbitals are complete at element 103. The irregularities in the electronic configurations of the actinoids, like those in the lanthanoids are related to the stabilities of the f^0 , f^7 and f^{14} occupancies of the 5f orbitals. Thus, the configurations of Am and Cm are $[Rn] 5f^7 7s^2$ and $[Rn] 5f^7 6d^1 7s^2$. Although the 5f orbitals resemble the 4f orbitals in their angular part of the wave-function, they are not as buried as 4f orbitals and hence 5f electrons can participate in bonding to a far greater extent.

Ionic Sizes

The general trend in lanthanoids is observable in the actinoids as well. There is a gradual decrease in the size of atoms or M^{3+} ions across the series. This may be referred to as the actinoid contraction (like lanthanoid contraction). The contraction is, however, greater from element to element in this series resulting from poor shielding by 5f electrons.

Oxidation States

There is a greater range of oxidation states, which is in part attributed to the fact that the 5f, 6d and 7s levels are of comparable energies. The known oxidation states of actinoids are listed in Table.

The actinoids show in general +3 oxidation state. The elements, in the first half of the series frequently exhibit higher oxidation states. For example, the maximum oxidation state increases from +4 in Th to +5, +6 and +7 respectively in Pa, U and Np but decreases in succeeding elements. The actinoids resemble the lanthanoids in having more compounds in +3 state than in the +4 state. However, +3 and +4 ions tend to hydrolyse. Because the distribution of oxidation states among the actinoids is so uneven and so different for the former and later elements, it is unsatisfactory to review their chemistry in terms of oxidation states.

Oxidation states of actinium and Actinoids

Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr
3		3	3	3	3	3	3	3	3	3	3	3	3	3
	4	4	4	4	4	4	4	4						
		5	5	5	5	5								
			6	6	6	6								
				7	7									

Some Important Uses of Actinides are as Follows:

1. Thorium is used in atomic reactors as fuel rods and in the treatment of cancer.
2. Uranium is used as nuclear fuel. Its salts are used in glass industry (for imparting green colour). textile industry and also in medicines.
3. Plutonium is used as fuel for atomic reactors as well as in atomic bombs.

Comparison of Lanthanides and Actinides:**Points of Resemblance:**

- (i) Both lanthanides and actinides show a dominant oxidation state of +3.
- (ii) Both are electropositive and act as strong reducing agents.
- (iii) Cations with unpaired electrons in both of them are paramagnetic.
- (iv) Most of the cations of lanthanides and actinides are coloured.
- (v) Both of them show a steady decrease in their ionic radii along the series. Thus, lanthanides show

General Characteristics and Comparison with Lanthanoids

	Lanthanides	Actinides
1.	Besides the most common oxidation state of +3 lanthanides show +2 and +4 oxidation states in case of certain elements.	Besides the most common oxidation state of +3, actinides show +4, +5 and +6 oxidation states in case of certain elements.
2.	Lanthanides have less tendency towards complex formation.	Actinides have a stronger tendency towards complex formation.
3.	Except promethium, they are non radioactive.	All the actinides are radioactive.
4.	Oxides and hydroxide of lanthanides are less basic.	Oxides and hydroxides of actinides are more basic
5.	Do not form oxo cations	Form oxocations such UO_2^{2+} , PuO_2^{2+} , UO^+
6.	Most of their ions are colourless	Most of the iron are coloured
7.	Magnetic properties can easily explained	Magnetic properties cannot be easily explained

NCERT QUESTIONS (Solve Yourself)

1.	Name a member of the lanthanoid series which is well known to exhibit +4 oxidation state.
2.	Actinoid contraction is greater from element to element than lanthanoid contraction. Why?

Exercise 1.2

- Which of the following lanthanoids has smallest atomic radius?
(1) Pm (2) Ce
(3) Gd (4) Er
- All the lanthanides are non-radioactive except
(1) Pr (2) Sm
(3) Lu (4) Pm
- The correct order of ionic radii of Ce, La, Pm and Yb in +3 oxidation state is :
(1) $\text{Yb}^{3+} > \text{Ce}^{3+} > \text{Pm}^{3+} > \text{La}^{3+}$
(2) $\text{Yb}^{3+} > \text{Pm}^{3+} > \text{Ce}^{3+} > \text{La}^{3+}$
(3) $\text{Yb}^{3+} < \text{Pm}^{3+} < \text{Ce}^{3+} < \text{La}^{3+}$
(4) $\text{La}^{3+} < \text{Pm}^{3+} < \text{Yb}^{3+} < \text{Ce}^{3+}$
- ${}_{63}\text{Eu}^{2+}$ can act as:-
(1) Oxidising agent
(2) Reducing agent
(3) Both oxidising agent and reducing agent
(4) Neither oxidising agent nor reducing agent
- Select from the following that is correct about basic strength of hydroxide:-
(1) $\text{Sc}(\text{OH})_3 > \text{Y}(\text{OH})_3 > \text{La}(\text{OH})_3$
(2) $\text{La}(\text{OH})_3 > \text{Y}(\text{OH})_3 > \text{Sc}(\text{OH})_3$
(3) $\text{La}(\text{OH})_3 = \text{Y}(\text{OH})_3 > \text{Sc}(\text{OH})_3$
(4) $\text{Y}(\text{OH})_3 > \text{La}(\text{OH})_3 > \text{Sc}(\text{OH})_3$
- Correct match of melting point:-
(1) Zr < Mo (2) Re < W
(3) Tc < Ru (4) All of these
- Lanthanide for which (+2) and (+3) oxidation states are common in
(1) La (2) Nd
(3) Ce (4) Eu
- Choose the correct order of basic strength
(1) $\text{Ce}(\text{OH})_3 > \text{Lu}(\text{OH})_3 > \text{Gd}(\text{OH})_3$
(2) $\text{Gd}(\text{OH})_3 > \text{Ce}(\text{OH})_3 > \text{Lu}(\text{OH})_3$
(3) $\text{Lu}(\text{OH})_3 > \text{Gd}(\text{OH})_3 > \text{Ce}(\text{OH})_3$
(4) $\text{Ce}(\text{OH})_3 > \text{Gd}(\text{OH})_3 > \text{Lu}(\text{OH})_3$
- Which of the following oxidation state is common for all lanthanoids?
(1) +2 (2) +3 (3) +4 (4) +5

10. There are 14 elements in actinoid series. Which of the following elements does not belong to this series?
(1) U (2) Np (3) Tm (4) Fm
11. Which of the following Lanthanoid show yellow colour:-
(1) La^{+3} (2) Yb^{+2} (3) Lu^{+3} (4) Ce^{+4}
12. Which of the following is a lanthanide:
(1) Ta (2) Rh (3) Th (4) Gd
13. The element with the electronic configuration $[\text{Xe}]^{54} 4f^{14} 5d^1 6s^2$ is a
(1) representative element
(2) transition element
(3) lanthanide
(4) actinide
14. Maximum no. of unpaired electrons are present in:-
(1) Gd^{+3} (2) Yb^{+2} (3) Tb^{+2} (4) Pm^{+3}
15. Which of the following ion is diamagnetic and colourless?
(1) La^{+3}
(2) Lu^{+3}
(3) Zn^{+2}
(4) All of the above
16. Which oxidation state is not shown lanthanoids?
(1) +2 (2) +3 (3) +4 (4) +6
17. Formula of halide of lanthanoid may be
(1) LnX_3 (2) LnX_5
(3) LnX_2 (4) LnX_6

NCERT QUESTIONS (Solve Yourself)

- What is lanthanoid contraction? What are the consequences of lanthanoid contraction?
- In what way is the electronic configuration of the transition elements different from that of the non transition elements?
- What are the different oxidation states exhibited by the lanthanoids?
- Compare the chemistry of actinoids with that of the lanthanoids with special reference to:
(i) electronic configuration (iii) oxidation state
(ii) atomic and ionic sizes and (iv) chemical reactivity.
- What are alloys? Name an important alloy which contains some of the lanthanoid metals. Mention its uses.
- What are inner transition elements? Decide which of the following atomic numbers are the atomic numbers of the inner transition elements : 29, 59, 74, 95, 102, 104.
- The chemistry of the actinoid elements is not so smooth as that of the lanthanoids. Justify this statement by giving some examples from the oxidation state of these elements.
- Which is the last element in the series of the actinoids? Write the electronic configuration of this element. Comment on the possible oxidation state of this element.
- Name the members of the lanthanoid series which exhibit +4 oxidation states and those which exhibit +2 oxidation states. Try to correlate this type of behaviour with the electronic configurations of these elements.
- Compare the chemistry of the actinoids with that of lanthanoids with reference to:
(i) electronic configuration (ii) oxidation states and (iii) chemical reactivity.
- Write the electronic configurations of the elements with the atomic numbers 61, 91, 101, and 109.

Exercise 2

- Generally transition elements form coloured salts due to the presence of unpaired electrons. Which of the following compounds will be coloured?
 (1) Ag_2SO_4 (2) CuF_2
 (3) ZnF_2 (4) Cu_2Cl_2
- On addition of small amount of KMnO_4 to concentrated H_2SO_4 , a green oily compound is obtained which is highly explosive in nature. Identify the compound from the following:
 (1) Mn_2O_7 (2) MnO_2
 (3) MnSO_4 (4) Mn_2O_3
- The magnetic nature of elements depends on the presence of unpaired electrons. Identify the configuration of transition element, which shows highest magnetic moment.
 (1) $3d^7$ (2) $3d^5$ (3) $3d^8$ (4) $3d^2$
- Gadolinium belongs to 4f series. Its atomic number is 64. Which of the following is the correct electronic configuration of gadolinium?
 (1) $[\text{Xe}] 4f^7 5d^1 6s^2$ (2) $[\text{Xe}] 4f^6 5d^2 6s^2$
 (3) $[\text{Xe}] 4f^8 6s^2$ (4) $[\text{Xe}] 4f^9 5s^1$
- In the form of dichromate, Cr (VI) is a strong oxidising agent in acidic medium but Mo (VI) in MoO_3 and W (VI) in WO_3 are not because...
 (a) Cr (VI) is more stable than Mo (VI) and W (VI).
 (b) Mo (VI) and W (VI) are more stable than Cr (VI).
 (c) Higher oxidation states of heavier members of group-6 of transition series are more stable
 (d) Lower oxidation states of heavier members of group-6 of transition series are more stable
 (1) a,b (2) b,c (3) c,d (4) a,d
- General electronic configuration of actinoids is $(n-2)f^{1-14} (n-1)d^{0-2} ns^2$. Which of the following actinoids have one electron in 6d orbital?
 (a) U (Atomic no. 92) (b) Np (Atomic no. 93)
 (c) Pu (Atomic no. 94) (d) Am (Atomic no. 95)
 (1) a,b (2) b,c (3) c,d (4) a,d
- Which of the following lanthanoids show +2 oxidation state besides the characteristic oxidation state +3 of lanthanoids?
 (a) Ce (b) Eu
 (c) Yb (d) Ho
 (1) a,b (2) b,c
 (3) c,d (4) a,d
- Transition elements form binary compounds with halogens. Which of the following elements will form MF_3 type compounds?
 (a) Cr (b) Co
 (c) Cu (d) Ni
 (1) a,b (2) b,c
 (3) c,d (4) a,d
- Although +3 is the characteristic oxidation state for lanthanoids but cerium also shows +4 oxidation state because
 (a) it has variable ionisation enthalpy
 (b) it has a tendency to attain noble gas configuration
 (c) it has a tendency to attain f^0 configuration
 (d) it resembles Pb^{4+}
 (1) a,b (2) b,c
 (3) c,d (4) a,d
- Which of the following statements is not correct?
 (1) Copper liberates hydrogen from acids
 (2) In its higher oxidation states, manganese forms stable compounds with oxygen and fluorine
 (3) Mn^{3+} and Co^{3+} are oxidising agents in aqueous solution
 (4) Ti^{2+} and Cr^{2+} are reducing agents in aqueous solution
- Highest oxidation state of manganese in fluoride is +4 (MnF_4) but highest oxidation state in oxides is +7 (Mn_2O_7) because.....
 (1) fluorine is more electronegative than oxygen
 (2) fluorine does not possess d-orbitals
 (3) fluorine stabilises lower oxidation state
 (4) in covalent compounds, fluorine can form single bond only while oxygen forms double bond

- 12.** Why is HCl not used to make the medium acidic in oxidation reactions of KMnO_4 in acidic medium?
- (1) Both HCl and KMnO_4 act as oxidising agents
 - (2) KMnO_4 oxidises HCl into Cl_2 which is also an oxidising agent
 - (3) KMnO_4 is a weaker oxidising agent than HCl
 - (4) KMnO_4 acts as a reducing agent in the presence of HCl
- 13.** Generally transition elements and their salts are coloured due to the presence of unpaired electrons in metal ions. Which of the following compounds are coloured?
- (a) KMnO_4
 - (b) CeO_2
 - (c) TiCl_4
 - (d) Cu_2Cl_2
- (1) a,b
 - (2) b,c
 - (3) c,d
 - (4) a,d
- 14.** Magnetic moment of X^{n+} is $\sqrt{24}$ B.M. Hence No. of unpaired electron and value of 'n' respectively. (Atomic number = 26)
- (1) 4, 3
 - (2) 3, 5
 - (3) 4, 2
 - (4) 4, 1
- 15.** Identify the incorrect statement among the following
- (1) Lanthanoid contraction is the accumulation of successive shrinkages.
 - (2) As a result of lanthanoid contraction, the properties of 4d series of the transition elements have no similarities with the 5d series of elements.
 - (3) Shielding power of 4f electrons is quite weak.
 - (4) There is a decrease in the radii of the atoms or ions as one proceeds from La to Lu.
- 16.** Which of the following actinoids show oxidation states upto + 7?
- (a) Am
 - (b) Pu
 - (c) U
 - (d) Np
- (1) a,b
 - (2) b,c
 - (3) c,d
 - (4) b,d
- 17.** When KMnO_4 solution is added to oxalic acid, solution, the decolourisation is slow in beginning but become instantaneous after some time because
- (1) CO_2 is a by product
 - (2) Mn^{2+} acts as a catalyst
 - (3) reaction is exothermic
 - (4) MnO_4^- catalyses the reaction
- 18.** Find incorrect statement among the following
- (1) $\text{Na}_2\text{Cr}_2\text{O}_7$ is more soluble than $\text{K}_2\text{Cr}_2\text{O}_7$ in water
 - (2) $\text{K}_2\text{Cr}_2\text{O}_7$ on heating gives O_2 gas
 - (3) $\text{Cr}_2\text{O}_7^{2-}$ in acidic medium stronger oxidising agent than MnO_4^-
 - (4) In $\text{Cr}_2\text{O}_7^{2-}$ anion, there are only two different type of Cr-O bonds
- 19.** Which of the following salt on heating with solid $\text{K}_2\text{Cr}_2\text{O}_7$ and conc. H_2SO_4 evolved orange red vapour which turn NaOH solution yellow?
- (1) NaI
 - (2) AgCl
 - (3) NaBr
 - (4) NaCl
- 20.** Select from the following that is correct about basic strength of hydroxide
- (1) $\text{Sc}(\text{OH})_3 > \text{Y}(\text{OH})_3 > \text{La}(\text{OH})_3$
 - (2) $\text{La}(\text{OH})_3 > \text{Y}(\text{OH})_3 > \text{Sc}(\text{OH})_3$
 - (3) $\text{La}(\text{OH})_3 = \text{Y}(\text{OH})_3 > \text{Sc}(\text{OH})_3$
 - (4) $\text{Y}(\text{OH})_3 > \text{La}(\text{OH})_3 > \text{Sc}(\text{OH})_3$
- 21.** In 3d-series which ion has positive value of standard reduction potential?
- (1) Ti^{2+}
 - (2) Zn^{2+}
 - (3) Cu^{2+}
 - (4) Fe^{2+}
- 22.** $\text{KMnO}_4 + \text{KI} \xrightarrow{\quad\quad\quad} \begin{cases} \text{Acidic Medium (A)} \\ \text{Alkaline Medium (B)} \end{cases}$
- The product (A) and (B) are respectively
- (1) I_2, I_2
 - (2) KI, I_2
 - (3) $\text{KIO}_4, \text{KIO}_3$
 - (4) I_2, KIO_3
- 23.** Select the statement that is correct
- (1) $\text{K}_2\text{Cr}_2\text{O}_7$ is good oxidising Agent
 - (2) $\text{K}_2\text{Cr}_2\text{O}_7$ is used as primary standard in volumetric Analysis
 - (3) On heating on $\text{K}_2\text{Cr}_2\text{O}_7$ forms a green colour compound
 - (4) All of these
- 24.** Ce and Yb exhibit +4 and +2 oxidation state respectively. This is because
- (1) Both Ce^{+4} and Yb^{+2} acquire f^7 configuration
 - (2) Both Ce^{+4} and Yb^{+2} acquire f^0 configuration
 - (3) Ce^{+4} and Yb^{+2} acquire f^0 and f^{14} configuration respectively
 - (4) Ce^{+4} and Yb^{+2} acquire f^7 and f^{14} configuration

25. $A + Na_2CO_3 + O_2 \rightarrow Na_2CrO_4 + B + C$
In the above reaction (A), (B) and (C) respectively are
(1) $FeCr_2O_4$, Fe_3O_4 , CO_2
(2) $FeCr_2O_4$, Fe_2O_3 , CO_2
(3) $FeCrO_4$, Fe_2O_3 , CO
(4) $FeCr_2O_4$, Fe_3O_4 , CO
26. $E^\circ(M^{2+}/M)$ value for copper is positive because
(1) of high enthalpy of atomization
(2) of high second ionization
(3) of high first ionization enthalpy
(4) both (1) and (2)
27. Find the incorrect statement.
(1) in alkaline medium, $KMnO_4$ oxidises I^- to IO_3^-
(2) in acidic medium, $KMnO_4$ oxidises I^- to I_2
(3) in acidic medium, $KMnO_4$ oxidises I^- to IO_3^-
(4) $KMnO_4$ in acidic medium stronger oxidising agent than $K_2Cr_2O_7$
28. Find the correct statement
(1) $Al(OH)_3$ is more basic than $Ce(OH)_3$
(2) All lanthanides are radioactive
(3) Actinides are radioactive
(4) Maximum lanthanides are man made/artificial
29. $KMnO_4$ acts as an oxidising agent in acidic medium. The number of moles of $KMnO_4$ that will be needed to react with one mole of sulphide ions in acidic solution is
(1) $\frac{2}{5}$ (2) $\frac{3}{5}$
(3) $\frac{4}{5}$ (4) $\frac{1}{5}$
30. Transition elements show magnetic moment due to spin and orbital motion of electrons. Which of the following metallic ions have almost same spin only magnetic moment?
(A) Co^{2+} (B) Cr^{2+}
(C) Mn^{2+} (D) Cr^{3+}
(1) A and B (2) B and C
(3) C and D (4) A and D
31. Which of the following ions show higher spin only magnetic moment value?
(A) Ti^{3+} (B) Mn^{2+}
(C) Fe^{2+} (D) Co^{3+}
(1) A and B (2) B and C
(3) C and D (4) All
32. Which of the following will **not** act as oxidising agents?
(A) CrO_3 (B) MoO_3
(C) WO_3 (D) CrO_4^{2-}
(1) A and B (2) B and C
(3) C and D (4) All
33. The correct order of decreasing second ionisation enthalpy of $Ti(22)$, $V(23)$, $Cr(24)$ and $Mn(25)$ is:-
(1) $Mn > Cr > Ti > V$
(2) $Ti > V > Cr > Mn$
(3) $Cr > Mn > V > Ti$
(4) $V > Mn > Cr > Ti$
34. Which one of the elements with the following outer orbital configurations may exhibit the largest number of oxidation states?
(1) $3d^2 4s^2$ (2) $3d^3 4s^2$
(3) $3d^5 4s^1$ (4) $3d^5 4s^2$
35. Which of the following pairs has the same size?
(1) Zn^{2+} , Hf^{4+} (2) Fe^{2+} , Ni^{2+}
(3) Zr^{4+} , Ti^{4+} (4) Zr^{4+} , Hf^{4+}
36. Which one of the following ions has electronic configuration $[Ar] 3d^6$?
(At. nos. $Mn = 25$, $Fe = 26$, $Co = 27$, $Ni = 28$)
(1) Co^{3+} (2) Ni^{3+}
(3) Mn^{3+} (4) Fe^{3+}
37. Which of the following ions will exhibit colour in aqueous solutions?
($Cr = 24$, $Mn = 25$, $Fe = 26$, $Co = 27$)
(1) $Sc^{3+}(z = 21)$ (2) $La^{3+}(z = 57)$
(3) $Ti^{3+}(z = 22)$ (4) $Lu^{3+}(z = 71)$
38. For the four successive transition elements (Cr , Mn , Fe and Co) the stability of +2 oxidation state will be there in which of the following order?
(1) $Mn > Fe > Cr > Co$ (2) $Fe > Mn > Co > Cr$
(3) $Co > Mn > Fe > Cr$ (4) $Cr > Mn > Co > Cr$

39. Which one of the following does not correctly represent the correct order of the property indicated against it?
 (1) $Ti < V < Cr < Mn$: increasing melting points
 (2) $Ti < V < Mn < Cr$: increasing 2nd ionization enthalpy
 (3) $Ti < V < Cr < Mn$: increasing number of oxidation states
 (4) $Ti^{3+} < V^{3+} < Cr^{3+} < Mn^{3+}$: increasing magnetic moment
40. The catalytic activity of transition metals and their compounds is ascribed mainly to:
 (1) Their ability to adopt variable oxidation states
 (2) Their chemical reactivity
 (3) Their magnetic behaviour
 (4) Their unfilled d-orbitals
41. Electronic configuration of a transition element X in +3 oxidation state is $[Ar] 3d^5$. What is its atomic number?
 (1) 25 (2) 26 (3) 27 (4) 24
42. The oxoanion which contains all equivalent M-O bond is:
 (I) CrO_4^{2-} (II) MnO_4^- (III) $Cr_2O_7^{2-}$
 (1) III only (2) I, II, III
 (3) I, II (4) I only
43. When $KMnO_4$ reacts with Br_2 in alkaline medium gives bromate ion. Then oxidation state of Mn changes from +7 to:
 (1) +6 (2) +4 (3) +3 (4) +2
44. Which of the following does not give chromyl chloride test?
 (1) NaCl (2) KCl
 (3) NH_4Cl (4) $HgCl_2$
45. Which of the statements is not true?
 (1) $K_2Cr_2O_7$ solution in acidic medium is orange
 (2) $K_2Cr_2O_7$ solution becomes yellow on increasing the pH beyond 7
 (3) On passing H_2S through acidified $K_2Cr_2O_7$ solution, a milky colour is observed
 (4) $Na_2Cr_2O_7$ is preferred over $K_2Cr_2O_7$ in volumetric analysis
46. Zn and Cd do not show variable valency, because:-
 (1) They have only two electrons in outermost shell
 (2) Their d-subshells are complete
 (3) Their d-subshells are incomplete
 (4) They are relatively soft metals
47. $(NH_4)_2Cr_2O_7$ (Ammonium dichromate) is used in fire works. The green coloured powder blown in air is:
 (1) Cr_2O_3 (2) CrO_2
 (3) Cr_2O_4 (4) CrO_3
48. Which of the following ion does not exist?
 (1) $[CuI_4]^{2-}$ (2) VO_4^{3-}
 (3) WO_4^{2-} (4) CrO_4^{2-}
49. Blue colour of glass produced by:
 (1) Cr_2O_3 (2) Fe_2O_3
 (3) MnO (4) CoO
50. Highly unstable species is:
 (1) Sc(II) (2) Ti(II)
 (3) Co(II) (4) V(II)
51. Which of the following diamagnetic:
 (1) Aurum (2) Tungsten
 (3) Palladium (4) Niobium
52. Which of the following is ferromagnetic:-
 (1) U (2) Co
 (3) Cr (4) Cu
53. More number of oxidation states are exhibited by the actinoids than by the lanthanoids. The main reason for this is
 (1) More active nature of the actinoids
 (2) More energy difference between 4f and 5d orbitals
 (3) Lesser energy difference between 5f and 6d orbitals than that between 4f and 5d orbitals
 (4) Greater metallic character of the lanthanoids than that of the corresponding actinoids
54. Uranium can be obtained by the electrolysis of:-
 (1) UF_6 (2) UF_4 (3) UCl_6 (4) UI_6

Exercise 3

1. Match column I & Column II and mark correct answer:

Column I		Column II	
(I)	Element with highest IE_2	(a)	V
(II)	Element with highest IE_3	(b)	Cr
(III)	M in $M(CO)_6$	(c)	Cu
(IV)	Element with highest heat of atomization	(d)	Zn
		(e)	Ni

- (1) (I)-(a), (II)-(b), (III)-(c), (IV)-(d)
 (2) (I)-(c), (II)-(d), (III)-(b), (IV)-(a)
 (3) (I)-(d), (II)-(b), (III)-(c), (IV)-(a)
 (4) (I)-(a), (II)-(d), (III)-(b), (IV)-(c)

2. Match the following Process in column A with their Transition metal as catalyst in column B

Column A	Column B
I. Wacker's Process	(a) Rh
II. Haber's Process	(b) Mo
III. Polymerisation of alkenes	(c) Ti
IV. Wilkinson's Catalyst	(d) Pd

- (1) (I)-(a), (II)-(b), (III)-(c), (IV)-(d)
 (2) (I)-(c), (II)-(d), (III)-(b), (IV)-(a)
 (3) (I)-(d), (II)-(b), (III)-(c), (IV)-(a)
 (4) (I)-(a), (II)-(d), (III)-(b), (IV)-(c)

3. Match the catalysts given in Column-I with the processes given in Column-II.

	Column-I (Catalyst)		Column-II (Process)
(i)	Ni in the presence of hydrogen	(a)	Ziegler Natta catalyst
(ii)	Cu_2Cl_2	(b)	Contact process
(iii)	V_2O_5	(c)	Vegetable oil to ghee
(iv)	Finely divided iron	(d)	Sandmeyer reaction
(v)	$TiCl_4 + Al(CH_3)_3$	(e)	Haber's Process
		(f)	Decomposition of $KClO_3$

- (1) (i)→(c) (ii)→(d) (iii)→(b) (iv)→(e) (v)→(a)
 (2) (i)→(d) (ii)→(c) (iii)→(b) (iv)→(e) (v)→(a)
 (3) (i)→(a) (ii)→(d) (iii)→(e) (iv)→(b) (v)→(c)
 (4) (i)→(c) (ii)→(d) (iii)→(b) (iv)→(a) (v)→(e)

4. Match column I & Column II and mark correct answer:

Column-I			Column-II
(A)	Bronze	(I)	Cu, Ni
(B)	Brass	(II)	Fe, Cr, Ni, C
(C)	UK silver coin	(III)	Cu, Zn
(D)	Stainless Steel	(IV)	Cu, Sn

Choose the correct answer from the options given below :

- (1) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)
 (2) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)
 (3) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
 (4) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

5. Match the properties give in Column-I with the metals given in Column-II.

	Column-I (Property)		Column-II (Metal)
(i)	An element which can show +8 oxidation state	(a)	Mn
(ii)	3d block element that can show upto +7 oxidation state	(b)	Cr
(iii)	3d block element with highest melting point	(c)	Os
		(d)	Fe

- (1) (i)→(b) (ii)→(a) (iii)→(c)
 (2) (i)→(d) (ii)→(a) (iii)→(b)
 (3) (i)→(c) (ii)→(a) (iii)→(b)
 (4) (i)→(a) (ii)→(c) (iii)→(b)

6. Match the statements given in Column-I with the oxidation states given in Column-II.

	Column-I		Column-II
(i)	Oxidation state of Mn in MnO_2 is	(a)	+2
(ii)	Most stable oxidation state of Mn is	(b)	+3
(iii)	Most stable oxidation state of Mn in oxides is	(c)	+4
(iv)	Characteristic oxidation state of lanthanoids is	(d)	+5
(v)		(e)	+7

- (1) (i)→(c) (ii)→(a) (iii)→(e) (iv)→(b)
 (2) (i)→(a) (ii)→(c) (iii)→(e) (iv)→(b)
 (3) (i)→(c) (ii)→(a) (iii)→(b) (iv)→(e)
 (4) (i)→(b) (ii)→(e) (iii)→(a) (iv)→(c)

7. Match the solutions given in Column-I and the colour given in Column-II.

	Column-I (Aqueous solution of salt)		Column-II (Colour)
(i)	$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	(a)	Green
(ii)	$\text{NiCl}_2 \cdot 4\text{H}_2\text{O}$	(b)	Light pink
(iii)	$\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$	(c)	Blue
(iv)	$\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$	(d)	Pale green
(v)	Cu_2Cl_2	(e)	Pink
		(f)	Colourless

- (1) (i)→(d) (ii)→(a) (iii)→(b) (iv)→(f) (v)→(e)
 (2) (i)→(a) (ii)→(d) (iii)→(b) (iv)→(f) (v)→(e)
 (3) (i)→(d) (ii)→(a) (iii)→(f) (iv)→(b) (v)→(e)
 (4) (i)→(d) (ii)→(a) (iii)→(b) (iv)→(e) (v)→(f)

8. Match the property given in Column-I with the element given in Column-II.

	Column-I (Property)		Column-II (Element)
(i)	Lanthanoid which shows +4 oxidation state	(a)	Pm
(ii)	Lanthanoid which can show +2 oxidation state	(b)	Ce
(iii)	Radioactive lanthanoid	(c)	Lu
(iv)	Lanthanoid which has $4f^7$ electronic configuration in +3 oxidation state	(d)	Eu
(v)	Lanthanoid which has $4f^{14}$ electronic configuration in +3 oxidation state	(e)	Gd
		(f)	Dy

- (1) (i)→(b) (ii)→(d) (iii)→(a) (iv)→(e) (v)→(c)
 (2) (i)→(b) (ii)→(a) (iii)→(d) (iv)→(e) (v)→(c)
 (3) (i)→(b) (ii)→(d) (iii)→(a) (iv)→(c) (v)→(e)
 (4) (i)→(d) (ii)→(b) (iii)→(e) (iv)→(a) (v)→(c)

9. **Statement I:** Iron (III) catalyst, acidified $\text{K}_2\text{Cr}_2\text{O}_7$ and neutral KMnO_4 have the ability to oxidise I^- to I_2 independently.

Statement II: Manganate ion is paramagnetic nature and involves $p\pi-p\pi$ bonding.

- (1) Both statement I and Statement II are true
 (2) Both statement I and Statement II are false
 (3) Statement I is true but Statement II is false
 (4) Statement I is false but Statement II is true

10. **Statement I:** The E° value of $\text{Ce}^{4+}/\text{Ce}^{3+}$ is +1.74 V
Statement II: Ce is more stable in Ce^{4+} state than Ce^{3+} state.

- (1) Both statement I and statement II are correct
 (2) Statement I is incorrect but statement II is correct
 (3) Both statement I and statement II are incorrect
 (4) Statement I is correct but statement II is incorrect.

11. Assertion: Zn, Cd, Hg are non-transition elements while Cu, Ag, Au are transition element

Reason: In Zn, Cd, Hg (n-1) d orbitals are completely filled in their atomic state where as in Cu, Au they are incomplete.

- (1) Both assertion and reason are true, and reason is the correct explanation of the assertion.
- (2) Both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) Assertion is true but reason is false.
- (4) Both assertion and reason are false.

12. Assertion: Cu^{2+} iodide is not known

Reason: Cu^{2+} oxidises I^- to iodine

- (1) Both assertion and reason are true, and reason is the correct explanation of the assertion.
- (2) Both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) Assertion is not true but reason is true.
- (4) Both assertion and reason are false.

13. Assertion: KMnO_4 is dark pink coloured compound

Reason: In the KMnO_4 charge transfer spectrum occurs.

- (1) Both assertion and reason are true, and reason is the correct explanation of the assertion.
- (2) Both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) Assertion is not true but reason is true.
- (4) Both assertion and reason are false.

14. Assertion: Valency of transition elements is variable

Reason: Low energy gap between ns and (n-1)d orbitals

- (1) Both assertion and reason are true, and reason is the correct explanation of the assertion.
- (2) Both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) Assertion is not true but reason is true.
- (4) Both assertion and reason are false.

15. Assertion: Melting point of Mn is less than that of Fe

Reason: Mn has less number of unpaired e⁻ than Fe in atomic state

- (1) Both assertion and reason are true, and reason is the correct explanation of the assertion.
- (2) Both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) Assertion is true but reason is false.
- (4) Both assertion and reason are false.

16. Assertion: Solution of Na_2CrO_4 in water is intensely colored.

Reason: Oxidation State of Cr in Na_2CrO_4 is +6.

- (1) Both assertion and reason are true, and reason is the correct explanation of the assertion.
- (2) Both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) Assertion is true but reason is false.
- (4) Both assertion and reason are false.

17. Assertion: Ce^{+4} acts as oxidizing agent in aqueous medium

Reason: +4 is common oxidation state of lanthanides

- (1) Both assertion and reason are true, and reason is the correct explanation of the assertion.
- (2) Both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) Assertion is true but reason is false.
- (4) Both assertion and reason are false.

18. Assertion: Change in colour of acidic solution of potassium dichromate by breath is used to test drunk drivers.

Reason: Change in colour is due to the complexation of alcohol with potassium dichromate.

- (1) Both assertion and reason are true, and reason is the correct explanation of the assertion.
- (2) Both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) Assertion is true but reason is false.
- (4) Both assertion and reason are false.

- 19. Assertion:** Eu^{2+} & Yb^{2+} are reducing agents for their ions.
Reason: Both ions have stable half filled configuration.
- (1) Both assertion and reason are true, and reason is the correct explanation of the assertion.
 - (2) Both assertion and reason are true but reason is not the correct explanation of assertion.
 - (3) Assertion is true but reason is false.
 - (4) Both assertion and reason are false.
- 20. Assertion:** La_2O_3 is basic nature.
Reason: La in aqueous solution gives $\text{La}(\text{OH})_3$.
- (1) Both assertion and reason are true, and reason is the correct explanation of the assertion.
 - (2) Both assertion and reason are true but reason is not the correct explanation of assertion.
 - (3) Assertion is true but reason is false.
 - (4) Both assertion and reason are false.
- 21. Assertion:** Actinoids can possess +4 O.S. more easily than lanthanoid.
Reason: All actinoids are radioactive.
- (1) Both assertion and reason are true, and reason is the correct explanation of the assertion.
 - (2) Both assertion and reason are true but reason is not the correct explanation of assertion.
 - (3) Assertion is true but reason is false.
 - (4) Both assertion and reason are false.
- 22. Assertion :** Separation of Zr and Hf is difficult.
Reason : Because Zr and Hf lie in the same group of the periodic table.
- (1) Both assertion and reason are true, and reason is the correct explanation of the assertion.
 - (2) Both assertion and reason are true but reason is not the correct explanation of assertion.
 - (3) Assertion is not true but reason is true.
 - (4) Both assertion and reason are false.
- 23. Assertion :** Actinoids form relatively less stable complexes as compared to lanthanoids.
Reason : Actinoids can utilise their 5f orbitals along with 6d orbitals in bonding but lanthanoids do not use their 4f orbital for bonding.
- (1) Both assertion and reason are true, and reason is the correct explanation of the assertion.
 - (2) Both assertion and reason are true but reason is not the correct explanation of assertion.
 - (3) Assertion is not true but reason is true.
 - (4) Both assertion and reason are false.
- 24. Assertion :** Cu cannot liberate hydrogen from acids.
Reason : Because it has positive electrode potential.
- (1) Both assertion and reason are true, and reason is the correct explanation of the assertion.
 - (2) Both assertion and reason are true but reason is not the correct explanation of assertion.
 - (3) Assertion is not true but reason is true.
 - (4) Both assertion and reason are false.
- 25. Assertion :** The highest oxidation state of osmium is +8.
Reason : Osmium is a 5d-block element.
- (1) Both assertion and reason are true, and reason is the correct explanation of the assertion.
 - (2) Both assertion and reason are true but reason is not the correct explanation of assertion.
 - (3) Assertion is not true but reason is true.
 - (4) Both assertion and reason are false.

Exercise 4 (Previous Year's Questions)

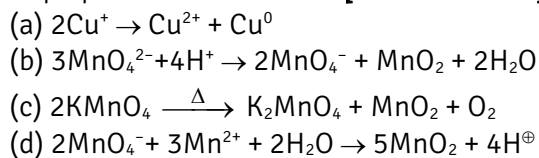
- Which of the following statements about the interstitial compounds is incorrect?
[NEET-UG 2013]
(1) They have higher melting points than the pure metal
(2) They retain metallic conductivity
(3) They are chemically reactive
(4) They are much harder than pure metal
- KMnO_4 can be prepared from K_2MnO_4 as per the reaction:-
$$3\text{MnO}_4^{2-} + 2\text{H}_2\text{O} \rightleftharpoons 2\text{MnO}_4^- + \text{MnO}_2 + 4\text{OH}^-$$

The reaction can go to completion by removing OH^- ions by adding:
[NEET-UG 2013]
(1) SO_2 (2) HCl (3) KOH (4) CO_2
- Magnetic moment 2.84 B.M. is given by:-
(At. no.), $\text{Ni} = 28$, $\text{Ti} = 22$, $\text{Cr} = 24$, $\text{Co} = 27$
[AIPMT 2015]
(1) Ti^{3+} (2) Cr^{2+} (3) Co^{2+} (4) Ni^{2+}
- Which of the following processes does not involve oxidation of iron? [AIPMT 2015]
(1) Decolourization of blue CuSO_4 solution by iron
(2) Formation of $\text{Fe}(\text{CO})_5$ from Fe
(3) Liberation of H_2 from steam by iron at high temperature
(4) Rusting of iron sheets
- Because of lanthanoid contraction, which of the following pairs of elements have nearly same atomic radii? (Numbers in the parenthesis are atomic numbers)
[AIPMT 2015]
(1) Zr (40) and Nb (41) (2) Zr (40) and Hf (72)
(3) Zr (40) and Ta (73) (4) Ti (22) and Zr (40)
- Which one of the following statements related to lanthanons is incorrect?
[NEET-II 2016]
(1) All the lanthanons are much more reactive than aluminum
(2) $\text{Ce}(+4)$ solutions are widely used as oxidizing agent in volumetric analysis
(3) Europium shows +2 oxidation state.
(4) The basicity decreases as the ionic radius decreases from Pr to Lu.
- Name the gas that can readily decolourise acidified KMnO_4 solution:[NEET(UG) 2017]
(1) SO_2 (2) NO_2 (3) P_2O_5 (4) CO_2
- HgCl_2 and I_2 both when dissolved in water containing I^- ions the pair of species formed is:
[NEET(UG) 2017]
(1) HgI_2, I^- (2) $\text{HgI}_4^{2-}, \text{I}_3^-$
(3) $\text{Hg}_2\text{I}_2, \text{I}^-$ (4) $\text{HgI}_2, \text{I}_3^-$
- The reason for greater range of oxidation states in actinoids is attributed to:-
[NEET(UG) 2017]
(1) Actinoid contraction
(2) 5f, 6d and 7s levels having comparable energies
(3) 4f and 5d levels being close in energies
(4) the radioactive nature of actinoids
- Which one of the following ions exhibits d-d transition and paramagnetism as well?
[NEET(UG) 2018]
(1) CrO_4^{2-} (2) $\text{Cr}_2\text{O}_7^{2-}$
(3) MnO_4^- (4) MnO_4^{2-}
- Match the metal ions given in Column I with the spin magnetic moments of the ions given in Column II and assign the correct code:
[NEET UG 2018]

Column I		Column II	
a.	Co^{3+}	i.	$\sqrt{8}$ B.M.
b.	Cr^{3+}	ii.	$\sqrt{35}$ B.M.
c.	Fe^{3+}	iii.	$\sqrt{3}$ B.M.
d.	Ni^{2+}	iv.	$\sqrt{24}$ B.M.
		v.	$\sqrt{15}$ B.M.

	a	b	c	d
(1)	iv	v	ii	i
(2)	i	ii	iii	iv
(3)	iv	i	ii	iii
(4)	iii	v	i	ii

12. Which of the following reactions are disproportionation reaction? [NEET UG 2019]



Select the correct option from the following:

- (1) (a) and (b) only (2) (a), (b) and (c)
 (3) (a), (c) and (d) (4) (a), and (d) only

13. The manganate and permanganate ions are tetrahedral, due to [NEET UG 2019]

- (1) The π -bonding involves overlap of p-orbitals of oxygen with d-orbitals of manganese
 (2) There is no π -bonding
 (3) The π -bonding involves overlap of p-orbitals of oxygen with p-orbitals of manganese
 (4) The π -bonding involves overlap of d-orbitals of oxygen with d-orbitals of manganese.

14. Match the catalyst with the process:

NEET UG 2019 UDISHA]

	Catalyst		Process
(i)	V_2O_5	(a)	The oxidation of ethyne to ethanal
(ii)	$\text{TiCl}_4 + \text{Al}(\text{CH}_3)_3$	(b)	Polymerisation of alkynes
(iii)	PdCl_2	(c)	Oxidation of SO_2 in the manufacture of H_2SO_4
(iv)	Nickel	(d)	Polymerisation of complexes ethylene

Which of the following is the correct option?

- (1) i-c, ii-d, iii-a, iv-b
 (2) i-a, ii-b, iii-c, iv-d
 (3) i-a, ii-c, iii-b, iv-d
 (4) i-c, ii-a, iii-d, iv-b

15. When neutral or faintly alkaline KMnO_4 is treated with potassium iodide, iodide ion is converted into 'X'. 'X' is-

[NEET UG 2019 UDISHA]

- (1) I_2 (2) IO_4^- (3) IO_3^- (4) IO^-

16. The oxidation state of Cr in CrO_5 is:

[NEET UG 2019 UDISHA]

- (1) -6 (2) +12 (3) +6 (4) +4

17. Match the following aspects with the respective metal.

[NEET (UG) 2020(COVID-19)]

Aspects		Metals	
a	The metal which reveals a maximum number of oxidation states	i	Scandium
b	The metal although placed in 3d block is considered not as a transition element	ii	Copper
c	The metal which does not exhibit variable oxidation states	iii	Manganese
d	The metal which in +1 oxidation state in aqueous solution undergoes disproportionation	iv	Zinc

Select the correct option:

- (1) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)
 (2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
 (3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
 (4) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

18. Identify the incorrect statement:

[NEET-2020]

- (1) Interstitial compounds are those that are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals.
 (2) The oxidation states of chromium in CrO_4^{2-} and CrO_7^{2-} are not the same.
 (3) $\text{Cr}^{2+}(\text{d}^4)$ is a stronger reducing agent than $\text{Fe}^{2+}(\text{d}^6)$ in water.
 (4) The transition metals and their compounds are known for their catalytic activity due to their ability to adopt multiple oxidation states and to form complexes.

19. The calculated spin only magnetic moment of Cr^{2+} ion is :

[NEET-2020]

- (1) 5.92 BM (2) 2.84 BM
 (3) 3.87 BM (4) 4.90 BM

20. Match the following aspects with the respective metal.

[NEET 2020 Exam (Covid Re-exam)]

	Aspects		Metal
(a)	The metal which reveals a maximum number of oxidation states	(i)	Scandium
(b)	The metal although placed in 3d block is considered not as a transition element	(ii)	Copper
(c)	The metal which does not exhibit variable oxidation states	(iii)	Manganese
(d)	The metal which in +1 oxidation state in aqueous solution undergoes disproportionation	(iv)	Zinc

Select the correct option :

- (1) (a)-(i) (b)-(iv) (c)-(ii) (d)-(iii)
 (2) (a)-(iii) (b)-(iv) (c)-(i) (d)-(ii)
 (3) (a)-(iii) (b)-(i) (c)-(iv) (d)-(ii)
 (4) (a)-(ii) (b)-(iv) (c)-(i) (d)-(iii)

21. Identify the incorrect statement from the following:

[NEET 2020 Exam (Covid Re-exam)]

- (1) Zirconium and Hafnium have identical radii of 160 pm and 159 pm, respectively as a consequence of lanthanoid contraction.
 (2) Lanthanoids reveal only +3 oxidation state.
 (3) The lanthanoid ions other than the f^0 type and the f^{14} type are all paramagnetic.
 (4) The overall decrease in atomic and ionic radii from lanthanum to lutetium is called lanthanoid contraction.

22. The incorrect statement among the following is: [NEET-2021]

- (1) Actinoids are highly reactive metals, especially when finely divided.
 (2) Actinoid contraction is greater for element to element than Lanthanoid contraction.
 (3) Most of the trivalent Lanthanoid ions are colorless in the solid state.
 (4) Lanthanoids are good conductors of heat and electricity

23. The stability of Cu^{2+} is more than Cu^+ salts in aqueous solution due to – [NEET-2023]

- (1) enthalpy of atomization.
 (2) hydration energy.
 (3) second ionisation enthalpy
 (4) first ionisation enthalpy

24. Which of the following statements are INCORRECT? [NEET-2023]

- A. All the transition metals except scandium form MO oxides which are ionic.
 B. The highest oxidation number corresponding to the group number in transition metal oxides is attained in Sc_2O_3 to Mn_2O_7
 C. Basis character increases from V_2O_3 to V_2O_4 to V_2O_5
 D. V_2O_4 dissolves in acids to give VO_4^{3-} salts.
 E. CrO is basic but Cr_2O_3 is amphoteric.

Choose the correct answer from the options given below:

- (1) B and D only (2) C and D only
 (3) B and C only (4) A and E only

25. The E° value for the $\text{Mn}^{3+}/\text{Mn}^{2+}$ couple is more positive than that of $\text{Cr}^{3+}/\text{Cr}^{2+}$ or $\text{Fe}^{3+}/\text{Fe}^{2+}$ due to change of: [NEET-2024]

- (1) d^4 to d^5 configuration
 (2) d^3 to d^5 configuration
 (3) d^5 to d^4 configuration
 (4) d^5 to d^2 configuration

26. 'Spin only' magnetic moment same for which of the following ions? [NEET-2024]

- (A) Ti^{3+} (B) Cr^{2+}
 (C) Mn^{2+} (D) Fe^{2+}
 (E) Sc^{3+}

Choose the most appropriate answer from the options given below:

- (1) B and C only (2) A and D only
 (3) B and D only (4) A and E only

27. The pair of lanthanoid ions which are diamagnetic is [NEET-2024]

- (1) Gd^{3+} and Eu^{3+} (2) Pm^{3+} and Sm^{3+}
 (3) Ce^{4+} and Yb^{2+} (4) Ce^{3+} and Eu^{2+}

28. Which of the following pairs of ions will have same spin only magnetic moment values within the pair? [RE-NEET-2024]

A. Zn^{2+} , Ti^{2+} B. Cr^{2+} , Fe^{2+}
 C. Ti^{3+} , Cu^{2+} D. V^{2+} , Cu^{+}

Choose the correct answer from the options given below:

- (1) C and D only (2) A and D only
 (3) A and B only (4) B and C only

29. Which of the following set of ions act as oxidising agents? [RE-NEET-2024]

(1) Ce^{4+} and Tb^{4+} (2) La^{3+} and Lu^{3+}
 (3) Eu^{2+} and Yb^{2+} (4) Eu^{2+} and Tb^{4+}

30. The UV-visible absorption bands in the spectra of lanthanoid ions are 'X', probably because of the excitation of electrons involving 'Y'. The 'X' and 'Y', respectively, are: [RE-NEET-2024]

- (1) Broad and f orbitals
 (2) Narrow and f orbitals
 (3) Broad and d and f orbitals
 (4) Narrow and d and f orbitals

31. Match List-I with List-II [NEET 2025]

List-I

List-II

A. Haber process I. Fe catalyst
 B. Wacker oxidation II. PdCl_2
 C. Wilkinson catalyst III. $[(\text{PPh}_3)_3\text{RhCl}]$
 D. Ziegler catalyst IV. TiCl_4 with $\text{Al}(\text{CH}_3)_3$

Chose the correct answer from the options given below :

- (1) A-I, B-II, C-III, D-IV
 (2) A-I, B-IV, C-III, D-II
 (3) A-I, B-II, C-IV, D-III
 (4) A-II, B-III, C-I, D-IV

32. Given below are two statements :

[NEET 2025]

Statement I : Ferromagnetism is considered as an extreme form of paramagnetism.

Statement II : The number of unpaired electrons in a Cr^{2+} ion ($Z=24$) is the same as that of a Nd^{3+} ion ($Z=60$).

- (1) Statement I is true but Statement II is false
 (2) Statement I is false but Statement II is true
 (3) Both Statement I and Statement II are true
 (4) Both Statement I and Statement II are false

ANSWER KEYS

Exercise 1.1

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	2	2	4	4	4	4	1	4	3	3	4	4	4	4	3	3	3	3	3
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	4	1	1	4	3	1	3	4	4	3	4	2	2	4	1	4	2	4	4	2
Que.	41	42	43	44	45															
Ans.	3	3	3	2	3															

Exercise 1.2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17			
Ans.	4	4	3	2	2	4	4	4	2	3	4	4	3	1	4	4	1			

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	1	2	1	2	1	2	1	2	1	4	2	1	3	2	4	2	3	4	2
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	3	4	4	3	2	4	3	3	1	4	2	2	3	4	4	1	3	1	1	1
Que.	41	42	43	44	45	46	47	48	49	50	51	52	53	54						
Ans.	2	3	2	4	4	2	1	1	4	1	3	2	3	1						

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	3	1	2	3	1	4	1	2	4	3	1	1	1	3	2	3	3	3	1
Que.	21	22	23	24	25															
Ans.	2	2	3	1	2															

Exercise 4 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	4	1	2	2	1	1	2	2	4	1	1	1	1	3	3	2	2	4	2
Que.	21	22	23	24	25	26	27	28	29	30	31	32								
Ans.	2	3	2	2	1	3	3	4	1	2	1	1								

Chapter 2

CO-ORDINATION COMPOUNDS

Chapter Summary

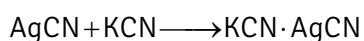
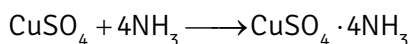
- Introduction
- Various terms Involved in Coordination Compounds
- IUPAC Nomenclature of Coordination Compounds
- Isomerism in Complexes
- Bonding in Coordination Compounds/ Werner's Theory
 - Experimental Evidence of Werner's Theory
- Valence Bond Theory
- Crystal Field Theory
- Organometallic Compounds or C-Metal bond compounds
 - Classification of Organometallic Compounds
 - Synthesis of Organometallic Compounds
 - Application of Organometallic Compounds

Introduction

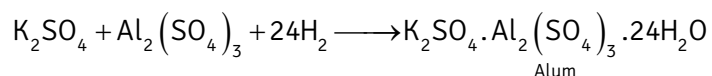
- (a) The concept of coordination compounds arises from the complex formation tendency of transition elements.
- (b) These compounds play a vital role in our lives. Haemoglobin of animal blood and chlorophyll of plants are also coordination compounds of Fe, Mg respectively.

Molecular or Addition Compounds :

- (a) When two or more simple salt solutions are mixed and then subjected for crystallisation, the crystals which are obtained are of addition compounds.
- (b) Some common examples are as follows.



Simple stable compounds Addition or molecular compounds



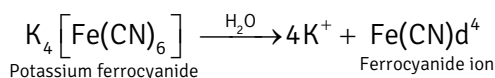
- (c) The molecular or addition compounds are of two types i.e. Double salt or lattice compound and coordination or complex compounds.

Double Salt or Lattice Compounds :

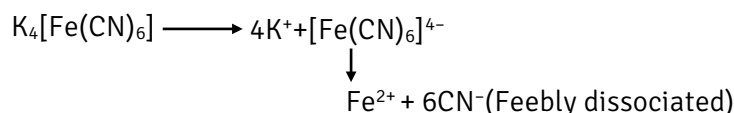
- (a) The addition compound which undergoes complete ionisation to form three types of ions i.e., two types of cations and one type of anions or vice versa, is called as double salt.
- (b) These double salts lose their identity in aqueous solution. Ex. when carnallite ($\text{KCl}_1 \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$) is dissolved in water, it exhibits the properties of KCl and MgCl_2 .

Coordination or Complex Compounds :

- (a) Addition compound of this type undergoes partial ionisation to form a complex ion in aqueous solution is called as complex compound or coordination compound.
- (b) Complexes retain their identity in aqueous solution. i.e., they do not dissociate into separate ion.

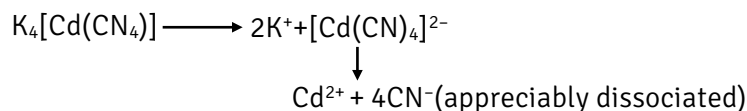


- (c) On the basis of stability of complex ion, complex ions are further divided as follows-
- (i) **Perfect complexes :** The compounds in which complex ion is fairly stable and further dissociation or feeble dissociation is not possible in solution state.

Example:

The ferrocyanide ion $[\text{Fe}(\text{CN})_6]^{4-}$ is so insignificantly dissociated that it can be considered as practically undissociated and does not give the test of Fe^{2+} or CN^- ions.

- (ii) **Imperfect complexes :** Those in which complex ion is less stable and is reversibly dissociated to give enough simple ions and thus imparts their tests.

Example:**Note:**

An imperfect complex may be too unstable to exist and may be completely dissociated in solution, now it behaves as a double salt.

Various terms Involved in Coordination Compounds**(a) Complex ion :**

An aggregate of metal ion with anions cation or neutral molecules is called as complex ion.

(b) Central metal ion :

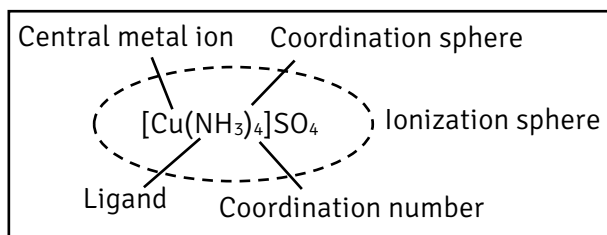
- (i) The metal ion which forms complex ion in combination with anions or neutral molecules is called as central metal ion.
- (ii) Central metal ion acts as an electron pair acceptor and forms coordinate covalent bond.

(c) Coordination number :

- (i) The number of ligands that combine with the central metal ion to form the complex ion is called as coordination number.
- (ii) From every ligand central metal ion accepts lone pair of electrons. Thus the number of lone pair of electrons accepted by the central metal ion in the formation of a complex is called as coordination number.
- (iii) The total number of coordinate covalent bonds formed by central metal ion with ligands is called as coordination number.

(d) Coordination sphere - (entity) :

- (i) The central metal atom and the ligands directly attached to it are collectively termed as the coordination sphere.
- (ii) Coordination sphere is written inside square bracket, for example $[\text{Co}(\text{NH}_3)_6]^{3+}$
- (iii) The part outside the bracket is called ionisation sphere.
- (iv) The species present in the coordination sphere are nonionizable.
- (v) The species present in the ionization sphere are ionisable.

**(e) Oxidation state :**

- (i) It is a number which represents the electric charge on the central metal atom of a complex ion.

Example:

The oxidation number of Fe, Co and Ni in $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $\text{Ni}(\text{CO})_4$ is +2, +3 and zero respectively.

(f) Charge On the complex ion :

It is the algebraic sum of the total charge of the ligands and central metal ion.

Question :

What will be the charge on the given complexes-

- (a) $[\text{Co}_{(\text{III})}\text{F}_6]$

Solution:

Charge on Co is +3

Total charge on F^- is = - 6

So, algebraic sum of charge = - 3

∴ Charge on $[\text{CoF}_6]$ is - 3.

- (b) $[\text{Fe}_{(\text{II})}(\text{NH}_3)_2\text{Cl}_4]$

Solution:

Charge on Fe is given = +2

Total Charge of ligands -4

So algebraic sum of Charge = -2

∴ Charge on $[\text{Fe}(\text{NH}_3)_2\text{Cl}_4]$ is -2.

(g) Ligands :

- (i) The anions or neutral molecules which combine with central metal ion to form complex ion are called as ligands.
- (ii) They act as electron pair donor Lewis bases, but strong crystal field ligands like CO, CN^- etc can accept electron pair from the metal ion, because these ligands are π -acids.

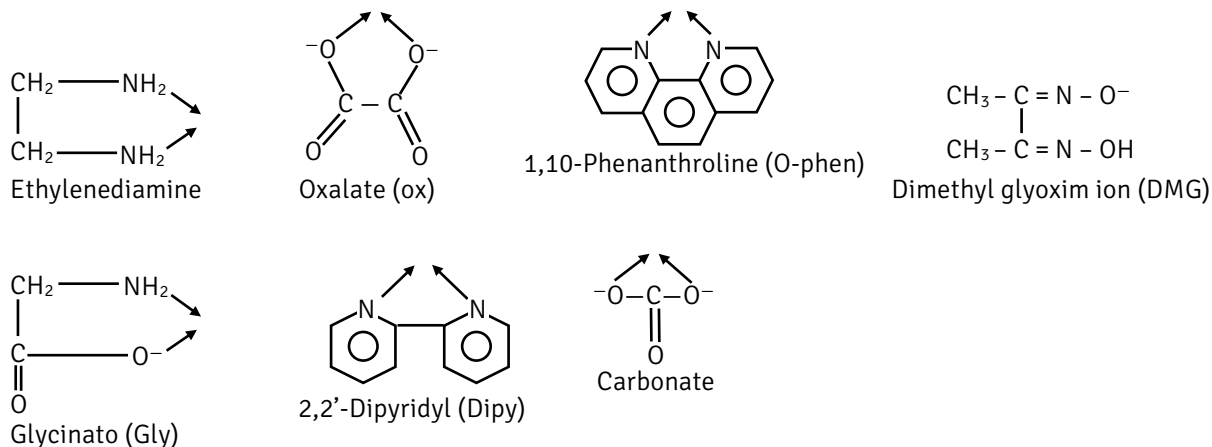
Types of Ligands :**(a) Unidentate Ligands/Monodentate Ligands :**

Ligands which have only one e⁻ donor atom.

X^- , CN^- , NO_2^- , NH_3 , Pyridine, OH^- , NO_3^- , H_2O , SO_3^{2-} , CO, NO, OH^- , O^{2-} , $(\text{C}_6\text{H}_5)_3\text{P}$ etc.

(b) Bidentate Ligands/Didentate Ligands :

Ligands which have two donor atoms and have the ability to link with central metal ion at two positions are called bidentate ligands. Some Ex.

**(c) Tridentate ligands :**

The ligands having three donor atoms are called tridentate ligands.

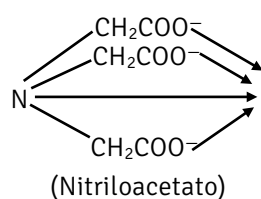
Example:

**(d) Tetradentate ligands :**

Those ligands possess four donor atoms,

Example:

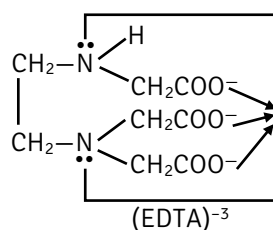
Nitriloacetato

**(e) Pentadentate ligands**

They have five donor atoms.

Example:

Ethylenediamine triacetate ion.

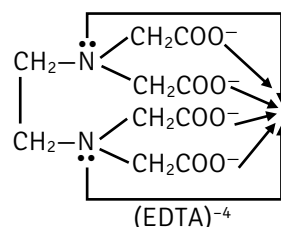


Ethylenediaminetriacetate ion

(f) Hexadentate ligands :

Example:

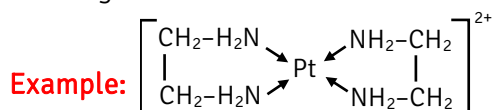
They have six donor atoms.



Ethylenediaminetriacetate ion

(g) Chelating ligands :

- Polydentate ligands whose structures permit the attachment of two or more donor site to metal ion simultaneously, thus closing one or more rings are called chelating ligands and formed are known as chelate compounds.
- A chelate may be defined as a ring structure formed by the combination of a polydentate two or more donor atoms with a metal ion forming part of the ring.
- The process of formation of chelates is called chelation.
- Chelate complexes are more stable than ordinary complexes because chelation is an exothermic process.
- This increased stability of the compound is due to chelation, is called the chelate effect.
- In the complex ion given below, 5 membered rings are formed. So all of these are called chelating ligands.

**(h) Ambident ligands :**

Ligands which can ligate through two different atoms but at a time attach by only one donor site present in it are called ambident ligands.



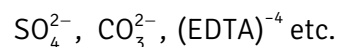
Example: CN^- can coordinate through either the nitrogen or the carbon atom to a central metal ion.

Note: Such possibilities give rise to linkage isomerism in coordination compounds

(i) Flexidentate ligands :

Ligands which sometimes do not use all the linking atoms to get coordinated with central metal ion.

Example:



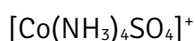
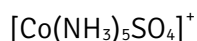
Example:

EDTA is normally hexadentate ligand but in the following complexes it is quadri or penta dentate



Hexadentate

Quadridentate



Sulphate-monodentate

Sulphate-bidentate

Factors Affecting the Stability of Complexes :

The stability of complexes depends on the following factors :

(A) Nature of the central metal ion :

- Greater the charge density on the metal ion, Δ_0 is more greater is the tendency to form the stable complex.

Thus, the complexes formed by metal ion with high charge density are more stable than complexes formed by metal ion with less charge density.

Example:

Fe^{+3} ion forms a more stable complex than Fe^{+2} , Similarly Pt^{4+} complex is more stable than Pt^{2+} complex.

- The stability of the complexes, of some of the cations having same charge but different ionic radii decrease with the increase in ionic radii. (Irving- william's order)

Ion stability	Cu^{+2}	>	Ni^{+2}	>	Co^{+2}	>	Fe^{+2}	>	Mn^{2+}
Ionic radii	0.69		0.78		0.82		0.83		0.91

(B) Nature of ligand :

- A strongly basic ligand forms a more stable complex with the metal ion than a ligand which is weakly basic in nature. Thus a cyano complex will be more stable than a chloro complex.
- When a bidentate ligand combines with the metal ion then a chelated complex is formed. This type of chelated complex will be more stable.

IUPAC Nomenclature of Coordination Compounds

The present system of nomenclature derived from the suggestions of Alfred Werner and recommended by the Inorganic Nomenclature Committee of the I.U.P.A.C. The main rules of naming of complexes are:

- Like simple salts, the positive part of the coordination compound is named first.

Example:

$K_4[Fe(CN)_6]$ the naming of this complex starts with potassium.

- Then after ligands of the coordination sphere are to be named.

- The ligands can be neutral, anionic or cationic.

- The neutral ligands are named as the molecule

Example:

C_5H_5N pyridine, $(C_6H_5)_3P$ Triphenyl phosphine,

$H_2N-CH_2-CH_2-NH_2$ ethylene diamine.

The neutral ligands which are not named as the molecule are –

CO carbonyl, NO nitrosyl, H_2O Aqua, NH_3 ammine –

- Anionic ligands ending with 'ide' are named by replacing the 'ide' with suffix 'O'.

Symbol	Name as ligand	Symbol	Name as ligand
Cl^-	Chloro/Chlorido	N^{3-}	Nitrido
Br^-	Bromo/Bromido	O_2^{2-}	Peroxo, O_2^- superoxol
CN^-	Cyano	O_2H^-	Perhydroxo
O^{2-}	Oxo	S^{2-}	Sulphido
OH^-	Hydroxo	NH_2^-	Amido
H^-	Hydrido	NH^{2-}	Imido

Ligands whose names end in 'ite' or 'ate' become 'ito' i.e., by replacing the ending 'e' with 'o' as follows.

CO_3^{2-}	Carbonato	SO_3^{2-}	Sulphito
$C_2O_4^{2-}$	Oxalato (Ox) $^{2-}$	CH_3COO^-	Acetato
SO_4^{2-}	Sulphato	NO_2^-	(bonded through oxygen) nitrite/nitrito-O
NO_3^-	Nitrato		(bonded through nitrogen) nitro/nitrito-N
$S_2O_3^{2-}$	Thiosulphato	Or	nitrito-N
			nitro-O

- Positive ligands naming ends in 'ium' $NH_2-NH_3^+$ Hydrazinium, NO_2^+ nitronium, NO^+ nitrosonium

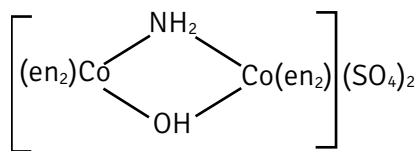
- NH_4^+ (ammonium) is not a ligand because it has no lone pair to donate.

- If ligands are present more than once, then their repetition is indicated by prefixes like di, tri, tetra etc. However, when the name of the ligand includes a number,

Example:

dipyridyl, ethylene diamine, then bis, tris, tetrakis are used in place of di, tri, tetra, etc.

- (e) In case of chelating ligands or ligands having di, tri, tetra, etc., in their name the prefixes bis, tris, tetrakis are used before ligands placed in paranthesis.



Bis(ethylenediamine) cobalt(III)-μ-amido-μ- hydroxobis(ethylenediamine) cobalt (III) sulphate

- (f) When more than one type of ligand is present in the complex, then the ligands are named in the alphabetical order.

(Prior to this naming of ligands was followed in the order - negative, neutral and positive ligands)

- (g) After naming of ligands the central metal ion is to be named followed by its oxidation state in Roman numbers in brackets. (as per IUPAC)

If the complex is neutral or provides a cationic complex ion, then the central metal ion is to be named as it is.

If the complex provides anionic complex ion then the name of central metal ion ends in 'ate'

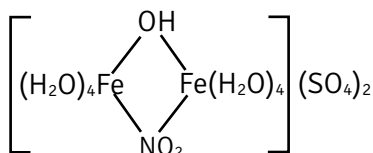
- (h) After the naming of central metal ion, anion which is in the outer sphere is to be named.

The naming of some of the complexes is done as follows – (as per IUPAC)

Complex Compounds		IUPAC Name
(i)	$\text{K}_4[\text{Fe}(\text{CN})_6]$ (anionic complex) so suffix ate is added with metal's name	Potassium hexacyanoferrate (II)
(ii)	$\text{K}_2[\text{Pt Cl}_6]$	Potassium hexachloroplatinate (IV)
(iii)	$[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (Cationic complex) so metal is without any suffix	Hexammine cobalt (III) chloride
(iv)	$[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl}$	Tetra aqua di chloro chromium (III) chloride
(v)	$[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$	Diammine tetra chloroplatinum (IV)
(vi)	$[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ (Neutral complex) So no suffix is used with metal ion	Triammine trichloro cobalt (III)
(vii)	$\text{K}_3[\text{Co}(\text{NO}_2)_6]$	Potassium hexanitro cobaltate (III)
(viii)	$\text{Na}_3[\text{Fe}(\text{CN})_5\text{NO}]$	Sodium pentacyano nitrosyl ferrate (II)
(ix)	$[\text{NiCl}_4]^{2-}$	Tetrachloro nickelate (II) ion
(x)	$[\text{Ru}(\text{NH}_3)_5\text{Cl}]^{+2}$	Pentammine chlororuthenium (III) ion
(xi)	$[\text{Fe}(\text{en})_3]\text{Cl}_3$	Tris (ethylenediamine) iron (III) chloride
(xii)	$[\text{Ni}(\text{Gly})_2]$	Bis (glycinato) nickel (II)

- (i) If a complex ion has two metal atoms then it is termed polynuclear. The ligand which connects the two metal ions is called as Bridging ligand or Bridge group.

A prefix of Greek letter μ, is repeated before the name of each different kind of bridging group.



Octo aqua -μ-hydroxo-μ-nitrodiiron (III) sulphate

NCERT QUESTIONS (Solve Yourself) :

- Write the formulas for the following coordination compounds:
 - tetraamminediaquacobalt(III) chloride
 - potassium tetracyanonickelate(II)
 - tris(ethane-1,2-diamine) chromium(III) chloride
 - amminebromidochloridonitrito-N-platinate(II)
 - dichloridobis(ethane-1,2-diamine)platinum(IV) nitrate
 - iron(III) hexacyanidoferrate(II)
- Write the IUPAC names of the following coordination compounds:
 - $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
 - $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
 - $\text{K}_3[\text{Fe}(\text{CN})_6]$
 - $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$
 - $\text{K}_2[\text{PdCl}_4]$
 - $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NH}_2\text{CH}_3)]\text{Cl}$
- Write down the IUPAC name for each of the following complexes and indicate the oxidation state, electronic configuration and coordination number. Also give stereochemistry and magnetic moment of the complex:
 - $\text{K}[\text{Cr}(\text{H}_2\text{O})_2(\text{C}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O}$
 - $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
 - $[\text{CrCl}_3(\text{py})_3]$
 - $\text{Cs}[\text{FeCl}_4]$
 - $\text{K}_4[\text{Mn}(\text{CN})_6]$
- FeSO_4 solution mixed with $(\text{NH}_4)_2\text{SO}_4$ solution in 1:1 molar ratio gives the test of Fe^{2+} ion but CuSO_4 solution mixed with aqueous ammonia in 1:4 molar ratio does not give the test of Cu^{2+} ion. Explain why?
- Explain with two examples each of the following: coordination entity, ligand, coordination number, coordination polyhedron, homoleptic and heteroleptic.
- What is meant by unidentate, didentate and ambidentate ligands? Give two examples for each.
- What is meant by the chelate effect? Give an example.
- Using IUPAC norms write the formulas for the following:
 - Tetrahydroxidozincate(II)
 - Potassium tetrachloridopalladate(II)
 - Diamminedichloridoplatinum(II)
 - Potassium tetracyanonickelate(II)
 - Pentaamminenitrito-O-cobalt(III)
 - Hexaamminecobalt(III) sulphate
 - Potassium tri(oxalato)chromate(III)
 - Hexaammineplatinum(IV)
 - Tetrabromidocuprate(II)
 - Pentaamminenitrito-N-cobalt(III)
- Using IUPAC norms write the systematic names of the following:
 - $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
 - $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NH}_2\text{CH}_3)]\text{Cl}$
 - $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$
 - $[\text{Co}(\text{NH}_3)_4\text{Cl}(\text{NO}_2)]\text{Cl}$
 - $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$
 - $[\text{NiCl}_4]^{2-}$
 - $[\text{Ni}(\text{NH}_3)_6]\text{Cl}_2$
 - $[\text{Co}(\text{en})_3]^{3+}$
 - $[\text{Ni}(\text{CO})_4]$

Exercise 1.1

- In the complex ion $[\text{Fe}(\text{EDTA})]^-$ the coordination number and oxidation state of central metal ion is:-
 - C.N. = 6 O. N. = + 3
 - C.N. = 1 O. N. = - 1
 - C.N. = 4 O. N. = + 2
 - C.N. = 3 O. N. = + 3
- The coordination number and oxidation number of the central metal ion in the complex $[\text{Pt}(\text{en})_2]^{+2}$ is:-
 - C. N. = 2, O. N. = +2
 - C. N. = 6, O. N. = +4
 - C. N. = 4, O. N. = +4
 - C. N. = 4, O. N. = +2

3. Select bidentate or didentate ligand from the following.
 (1) CO (2) SCN^-
 (3) CH_3COO^- (4) $\text{C}_2\text{O}_4^{2-}$
4. The oxidation and coordination number of Pt in $[\text{Pt}(\text{C}_2\text{H}_4)\text{Cl}_3]^-$ is respectively:-
 (1) + 1, 3 (2) + 2, 4
 (3) + 3, 6 (4) + 2, 5
5. The Co-ordination number and oxidation number of X in the compound $[\text{X}(\text{SO}_4)(\text{NH}_3)_5]$ will be:-
 (1) 10 and 3 (2) 1 and 6
 (3) 6 and 4 (4) 6 and 2
6. What is the oxidation number of Fe in $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{2+}$ ion?
 (1) +2 (2) +3 (3) +1 (4) 0
7. The oxidation state of iron in $\text{Na}_4[\text{Fe}(\text{CN})_5(\text{NOS})]$ is
 (1) +1 (2) +2 (3) +3 (4) Zero
8. Incorrect statement about DMG:-
 (1) It is tetradentate ligand
 (2) Chelating ligand
 (3) Dioxime of diacetyl
 (4) In gravimetric determination of Ni is used
9. Which of the following species is not expected to be a ligand?
 (1) NO (2) NH_4^+
 (3) $\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$ (4) CO
10. Which of the following complexes are homoleptic?
 (a) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (b) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$
 (c) $[\text{Ni}(\text{CN})_4]^{2-}$ (d) $[\text{Ni}(\text{NH}_3)_4\text{Cl}_2]$
 (1) a, c, d (2) b, c, d
 (3) b, c (4) a, c
11. Which of the following complexes are heteroleptic?
 (a) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ (b) $[\text{Fe}(\text{NH}_3)_4\text{Cl}_2]^+$
 (c) $[\text{Mn}(\text{CN})_6]^{4-}$ (d) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$
 (1) a, c, d (2) b, d
 (3) b, c (4) a, c
12. Identify the correct statements for the behaviour of ethane-1,2-diamine as a ligand.
 (a) It is a neutral ligand
 (b) It is a didentate ligand
 (c) It is a chelating ligand
 (d) It is unidentate ligand
 (1) a, b, c (2) a, c
 (3) b, c, d (4) a, b
13. Oxidation state of cobalt in the complex $[\text{Co}(\text{NO}_2)_6]^{3-}$ is:
 (1) + 2 (2) + 3
 (3) + 1 (4) +2 and + 3
14. What is the charge on the complex $[\text{Cr}(\text{C}_2\text{O}_4)_2(\text{H}_2\text{O})_2]$ formed by Cr(III):
 (1) +3 (2) +1 (3) +2 (4) -1
15. Co-ordination number of platinum in $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]^{++}$ ion is:
 (1) 4 (2) 2 (3) 8 (4) 6
16. $\text{K}_3[\text{Fe}(\text{CN})_6]$ is:-
 (a) Potassiumhexacyanoferrate(II)
 (b) Potassiumhexacyanoferrate(III)
 (c) Potassiumferri-cyanide
 (d) Hexacyanoferrate(III) potassium
 Correct answer is:-
 (1) Only (a) and (b) (2) Only (b) and (c)
 (3) Only (a) and (c) (4) Only (b) and (d)
17. Give the IUPAC name of the complex compound $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Br}](\text{NO}_3)_2$.
 (1) Bromoaquatetraamminecobalt (III) nitrate
 (2) Bromoaquatetraaminocobalt (III) nitrate
 (3) Bromoaquatetraamminecobalt (III) nitrate
 (4) Tetraammineaquabromocobalt (III) nitrate
18. Which of the following complex is anion:-
 (1) Fluoropentaamminecobalt (III)
 (2) Trioxalato ferrate (III)
 (3) PentaCarbonyliron(0)
 (4) Dichlorodiammineplatinum
19. The chloro-bis (ethylenediamine) nitro cobalt (III) ion is:-
 (1) $[\text{Co}(\text{NO}_2)_2(\text{en})_2\text{Cl}_2]^+$
 (2) $[\text{CoCl}(\text{NO}_2)_2(\text{en})_2]^+$
 (3) $[\text{Co}(\text{NO}_2)\text{Cl}(\text{en})_2]^+$
 (4) $[\text{Co}(\text{en})\text{Cl}_2(\text{NO}_2)_2]^-$

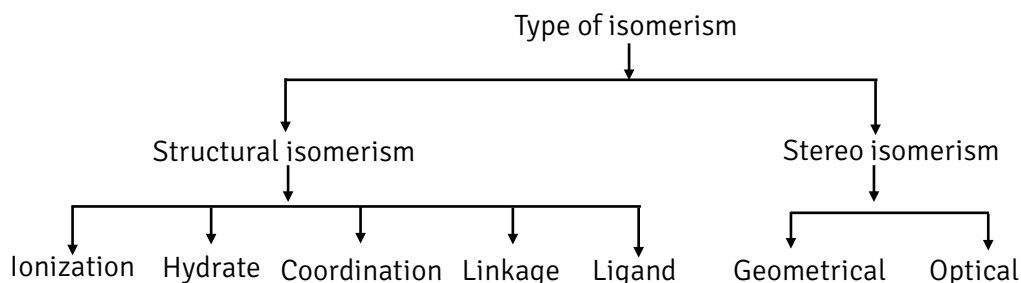
20. Which of the following complexes is not a chelate
 (1) bis(dimethylglyoximato)nickel (II)
 (2) Potassium ethylenediaminetetrathiocyanato chromate (III)
 (3) Tetrammine dichlorocobalt (III) nitrate
 (4) Trans-diglycinatoplatinum (II)
21. The correct IUPAC name of the complex $[\text{Fe}(\text{C}_5\text{H}_5)_2]$ is
 (1) Cyclopentadienyliron (II)
 (2) Bis(cyclopentadienyl)iron (II)
 (3) Dicyclopentadienylferrate (II)
 (4) Ferrocene
22. The correct name of $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]$ $[\text{PtCl}_4]$ is:
 (1) Tetraamminedichloroplatinum(IV) tetrachloroplatinate (II)
 (2) Dichlorotetraammineplatinum(IV) tetrachloroplatinate (II)
 (3) Tetrachloroplatinum (II) tetraammineplatinate(IV)
 (4) Tetrachloroplatinum (II) dichlorotetraamine platinate (IV)
23. The IUPAC name of $\text{K}_2[\text{Cr}(\text{CN})_2\text{O}_2(\text{O})_2(\text{NH}_3)]$ is:-
 (1) Potassiumamminedicyano dioxoperoxochromate(VI)
 (2) Potassiumamminecyanoperoxodioxo chromium (VI)
 (3) Potassiumamminecyanoperoxodioxo chromium (VI)
 (4) Potassiumamminecyanoperoxodioxo chromate (IV)
24. The IUPAC name for $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ is:
 (1) Hexaammine cobalt (III) hexacyanochromate (III)
 (2) Hexacyanochromium cobalt hexaammine (VI)
 (3) Hexaammine cobalt (III) hexacyanochromium (VI)
 (4) Hexacyanochromium (III) hexaammine cobalt (III)
25. The IUPAC name for $[\text{Co}(\text{NCS})(\text{NH}_3)_5]\text{Cl}_2$ is:
 (1) Pentaammine(thiocyanato-N)cobalt(III) chloride
 (2) Pentaammine(thiocyanato-S)cobalt(III) chloride
 (3) Pentaammine(isothiocyanato-N,S)cobalt (III)chloride
 (4) Pentaammine(mercapto-N)cobalt(III) chloride
26. IUPAC name of $\text{K}_2[\text{OsCl}_5\text{N}]$ will be
 (1) Potassium pentachloroazidoosmate (VIII)
 (2) Potassium pentachloroazidoosmate (VI)
 (3) Potassium pentachloronitridoosmate (VI)
 (4) Potassium nitro osmate (III)
27. IUPAC name of $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NO}_2)]$ is
 (1) Platinum diaminechloronitrite
 (2) Chloronitrito-N-ammineplatinum (II)
 (3) Diamminechloridonitrito-N-platinum (II)
 (4) Diamminechloronitrito-N platinate (II)
28. The correct IUPAC name of $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ is
 (1) Diamminedichloridoplatinum (II)
 (2) Diamminedichloridoplatinium (IV)
 (3) Diamminedichloridoplatinum (0)
 (4) Dichloridodiammineplatinum (IV)
29. The correct IUPAC name of $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ is
 (1) Diamminedichloridoplatinum (II)
 (2) Diamminedichloridoplatinum (IV)
 (3) Diamminedichloridoplatinum (0)
 (4) Diamminedichloridoplatinum (V)
30. Which of the following species is not expected to act as a ligand:
 (1) NO^+ (2) NH_4^+
 (3) $\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{NH}_2$ (4) CO
31. How many EDTA molecules are required to make an octahedral complex with a Ca^{+2} ion?
 (1) 1 (2) 3 (3) 4 (4) 2
32. Which of the following is π -acid ligand ?
 (1) H_2O (2) CN^- (3) F^- (4) (en)

33. Which of the following statement is Correct about ethane 1,2 diamine?
 (1) It is a neutral ligand
 (2) It is a bidentate ligand
 (3) It is a chelating ligand
 (4) All of the above
34. IUPAC name of $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NO}_2)]$ is
 (1) platinum diamminechloronitrite
 (2) chloronitrito-N-ammineplatinum(II)
 (3) Diammine chloridonitrito-N-Platinum(II)
 (4) diamminechloronitrito-N-Platinum(I)
35. The IUPAC name for $[\text{Pt}(\text{NH}_3)_3(\text{Br})(\text{NO}_2)\text{Cl}]\text{Cl}$ is
 (1) Triamminechlorobromonitroplatinum (IV)chloride
 (2) Triamminebromochloronitroplatinum (IV)chloride
 (3) Trimminechlorobromoplatinum (IV)chloride
 (4) Triamminechloronitrobromoplatinum (IV)chloride
36. In brown ring complex, the oxidation state of iron will be
 (1) +2 (2) +3 (3) +1 (4) 0
37. The IUPAC name for ionisation isomers of $[\text{Pt}(\text{NH}_3)_3(\text{Br})(\text{NO}_2)(\text{I})]\text{Cl}$ is
 (1) Triamminebromidochloridonitroplatinum (IV) iodide
 (2) Triamminebromidochloridoiodidoplatinum (IV)nitrite
 (3) Triamminechloridoiodidonitroplatinum (IV)bromide
 (4) All are possible
38. The correct formula of diammine dichlorodicyanochromate(III) is:
 (1) $[\text{CrCl}_2(\text{CN})_2(\text{NH}_3)_2]^{3+}$
 (2) $[\text{CrCl}_2(\text{CN})_2(\text{NH}_3)_2]^{3-}$
 (3) $[\text{CrCl}_2(\text{CN})_2(\text{NH}_3)_2]$
 (4) $[\text{CrCl}_2(\text{CN})_2(\text{NH}_3)_2]^{-}$

Isomerism in Complexes

- (a) Compounds which have the same molecular formula, but differ in their properties are called as Isomers.
- (b) Isomerism is commonly considered, to be the characteristic of only organic compounds, it is also found although less frequently among inorganic substances.

Classification of isomerism :



Structural Isomerism:

It arises due to the difference in the type of chemical linkages and distribution of ligands within and outside the coordination sphere.

Ionisation Isomers:

The type of isomerism which is due to the exchange of groups or ion between the coordinating sphere and the ionisation sphere. Ex.

(a) $\text{Co}(\text{NH}_3)_4\text{Br}_2\text{SO}_4$ can represent

(i) $[\text{Co}(\text{NH}_3)_4\text{Br}_2]\text{SO}_4$ (red violet)

(ii) $[\text{Co}(\text{NH}_3)_4\text{SO}_4]\text{Br}_2$ (red).

These complexes give sulphate ion and bromide ion respectively

(b) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Br}_2$ and $[\text{Pt}(\text{NH}_3)_4\text{Br}_2]\text{Cl}_2$

(c) $[\text{Co}(\text{NH}_3)_4(\text{NO}_3)_2]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_4\text{SO}_4](\text{NO}_3)_2$

Hydrate isomers :

This type of isomerism is due to presence of different number of water molecules inside a coordination sphere.

Example:

(a) $\text{Cr}(\text{H}_2\text{O})_6\text{Cl}_3$ has three possible structures

(i) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ violet (ii) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ green (iii) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$ dark green.

These complexes differ from one another with respect to the number of water molecules acting as ligands. Other hydrate isomers are

(i) $[\text{Co}(\text{NH}_3)_4\text{H}_2\text{OCl}]\text{Cl}_2$ (ii) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \cdot \text{H}_2\text{O}$

Linkage or salt isomers :

- (a) This type of isomerism arises due to presence of ambidentate ligands like NO_2^- , CN^- and SCN^-
- (b) These ligands have two donor atoms but at a time only one atom is directly linked to the central metal atom of the complex.
- (c) Such type of isomers are distinguished by infra red (I.R.) spectroscopy. Ex.
- (i) $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$
- (ii) In NO_2^- ligand, The coordinating sites are nitrogen (i.e., NO_2^- Nitro ligand) or through oxygen (i.e. ONO Nitrito ligand)
- (iii) The nitro isomer is yellow and is stable to acids whereas nitrito isomer is red and is decomposed by acids.

Coordination isomers :

- (a) This type of isomerism is exhibited when the complex has two complex ions in it - 'Cationic and anionic'.
- (b) This type of isomerism is caused by the interchange of ligands and metal ions between the two complex ions of the same complex. Ex.
- (i) $[\text{Co}(\text{NH}_3)_6] [\text{Cr}(\text{CN})_6]$ (ii) $[\text{Co}(\text{NH}_3)_6] [\text{Cr}(\text{C}_2\text{O}_4)_3]$
 $[\text{Cr}(\text{NH}_3)_6] [\text{Co}(\text{CN})_6]$ $[\text{Cr}(\text{NH}_3)_6] [\text{Co}(\text{C}_2\text{O}_4)_3]$

Ligand isomers :

- (a) Ligands with $\text{C}_3\text{H}_6(\text{NH}_2)_2$ have two different structures i.e. 1, 3-diamino propane and 1, 2-diaminopropane(propylene diamine).
- (b) Those complexes which have same molecular formula, but differ with respect to their ligands are called as Ligand isomers.

Example:

$[\text{Fe}(\text{H}_2\text{O})_2 \text{C}_3\text{H}_6(\text{NH}_2)_2\text{Cl}_2]$ has two different structures

**Stereo Isomerism:**

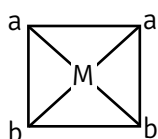
- (a) They have same molecular formula, same constitution, they differ only with respect to the spatial orientation of ligands in space around the metal ion.
- (b) The two stereo isomers which are possible are - Geometrical and optical.

Geometrical or cis - Trans isomers :

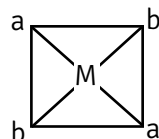
- The ligands occupy different positions around the central metal ion.
- When two identical ligands are coordinated to the metal ion from same side then it is cis isomer. (Latin, cismeans same).
- If the two identical ligands are coordinated to the metal ion from opposite side then it is Trans isomer (in Latin, Trans means across).
- These geometrical isomers differ in physical as well as in chemical properties.
- 4-coordinated complexes with tetrahedral geometry do not exhibit cis- Trans isomerism, because relative position of ligands remain same for each other. It is exhibited by 4-coordinated complexes with square planar geometry.

Geometrical isomers with coordination number = 4 (Square planar complexes) :

- Complexes with general formula, Ma_2b_2 (where both a and b are monodentate) can have Cis-and trans isomers.

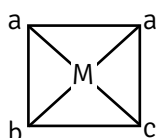


Cis-isomer

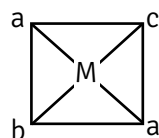


Trans-isomer

- Complexes with general formula Ma_2bc can have Cis - and trans-isomers.

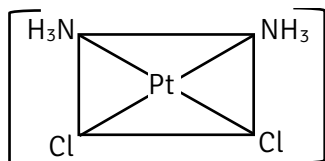


Cis

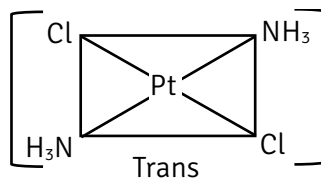


Trans

- $[Pt(NH_3)_2Cl_2]$

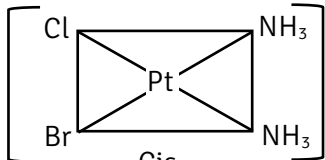


Cis(Cis-platin) anti cancer

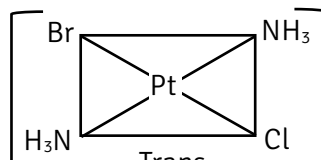


Trans

- $[Pt(NH_3)_2ClBr]$

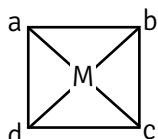


Cis

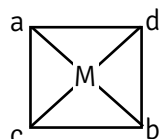


Trans

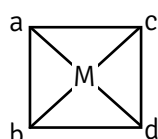
- Complexes with general formula, $Mabcd$ can have three isomers.



(i)



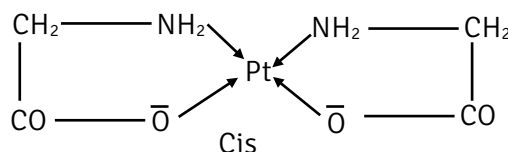
(ii)



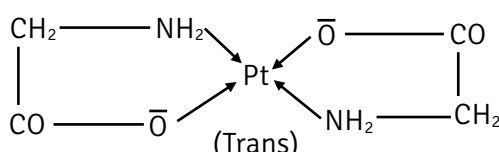
(iii)

- Diglycinato platinum

- complexes

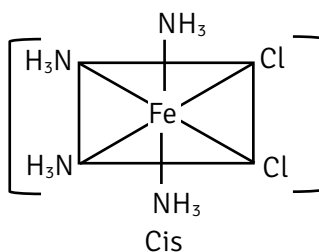
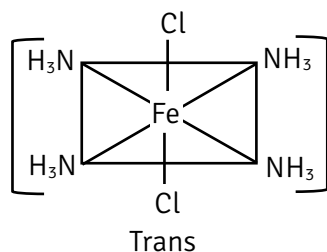
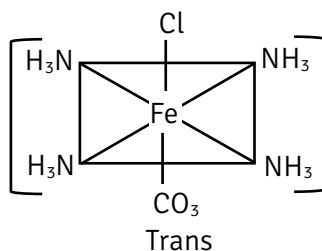
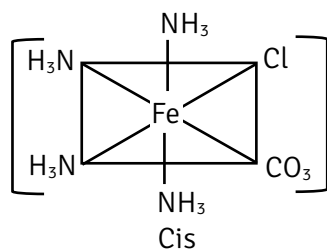
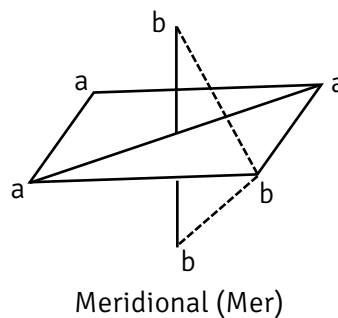
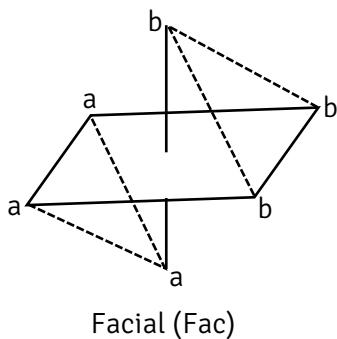


Cis



(Trans)

Geometrical isomers with Coordination number = 6

(i) $[\text{Fe}(\text{NH}_3)_4\text{Cl}_2]$ (ii) $[\text{Fe}(\text{NH}_3)_4\text{ClCO}_3]$ (iii) Facial and Meridional isomerism (Ma_3b_3)**Note :**

Other 6-Coordinated geometrical isomers are –

General formula	Total No. of geometrical isomers
Mabcdef	15
Ma_2bcde	9
$\text{Ma}_2\text{b}_2\text{cd}$	6
$\text{Ma}_2\text{b}_2\text{c}_2$	5
Ma_3bcd	4
$\text{Ma}_3\text{b}_2\text{c}$	3
Ma_3b_3	2
Ma_4bc	2
Ma_4b_2	2
Ma_5b	Nil
Ma_6	Nil
Here M = central atom	
a, b, c, d, e, f = Monodentate ligands	

Optical isomers :

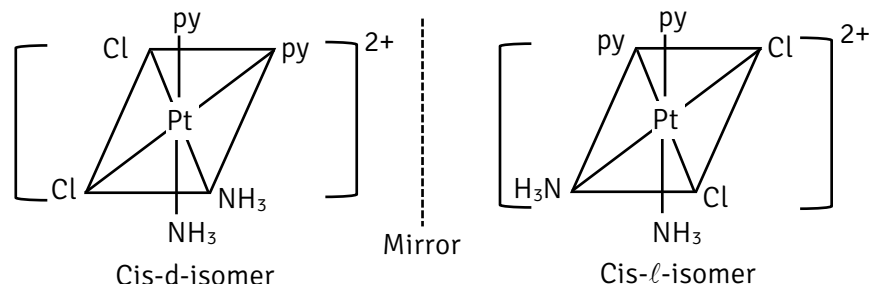
- Optically active complexes are those which are nonsuperimposable over the mirror image structure.
- An optically active complex is one which is asymmetric in nature i.e., not divisible into two identical halves.
- The complex which rotates plane polarised light to left hand side is **laevo rotatory** i.e. ' λ ' or ' $-$ ' and if the complex rotates the plane polarised light to right hand side then it is **dextro rotatory** 'd' or '+'.
 - Thus complexes which have same physical and chemical properties but differ in their action towards plane polarised light are called as **optical isomers**.
- The 'd' and ' λ ' isomers of a compound are called as **Enantiomers or Enantiomorphs**.
- Only those 6-coordinated complexes in which there are chelating agents i.e. bidentate ligands, exhibit optical isomerism.
- This is due to the absence of elements of symmetry in the complex.
- Optical isomerism is expected in tetrahedral complexes of the type $Mabcd$.
- These isomers are not isolable due to their labile nature.

Note :

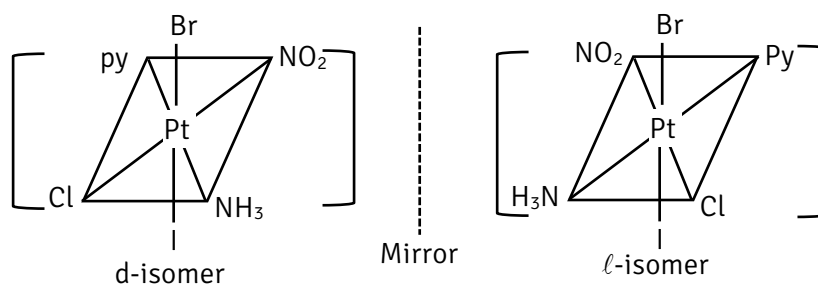
- Optical isomerism is not in square planar complexes on account of the presence of axis of symmetry.
- Optical isomerism is very common in octahedral complexes

Optical isomers with Coordination number = 6

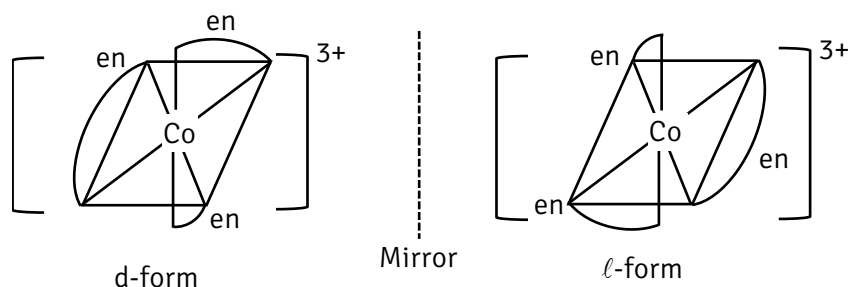
- $[Ma_2b_2c_2]^{n+} \rightarrow [Pt(py)_2(NH_3)_2Cl_2]^{2+}$

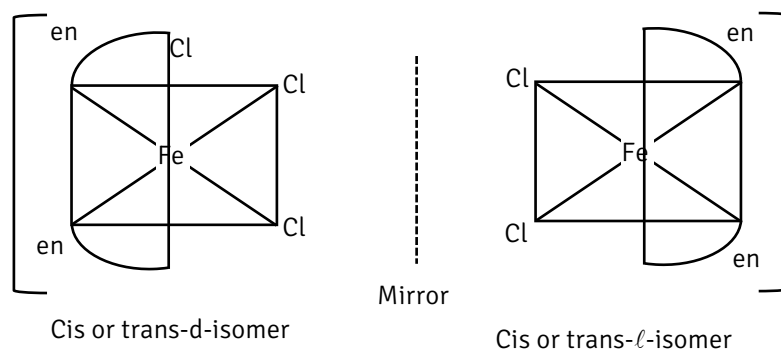
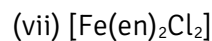
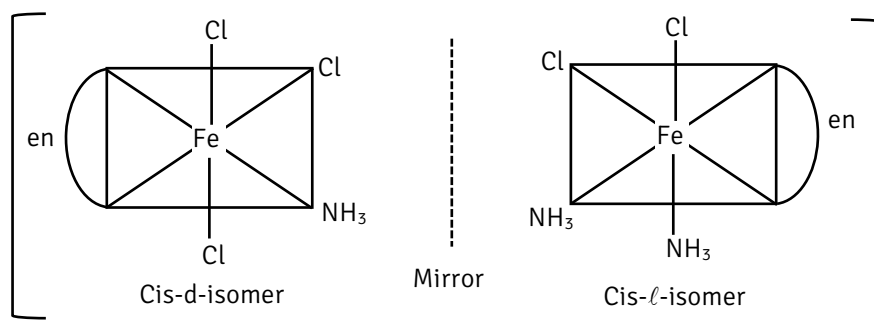
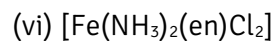
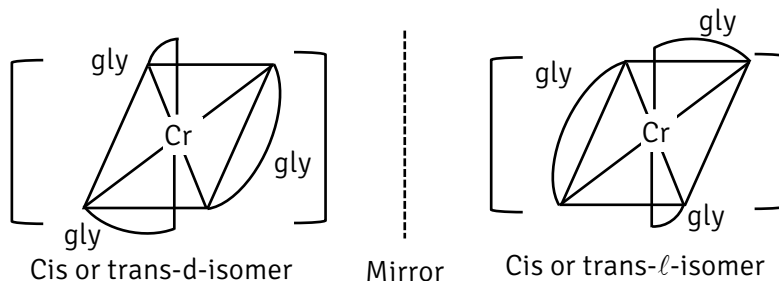
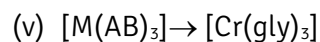
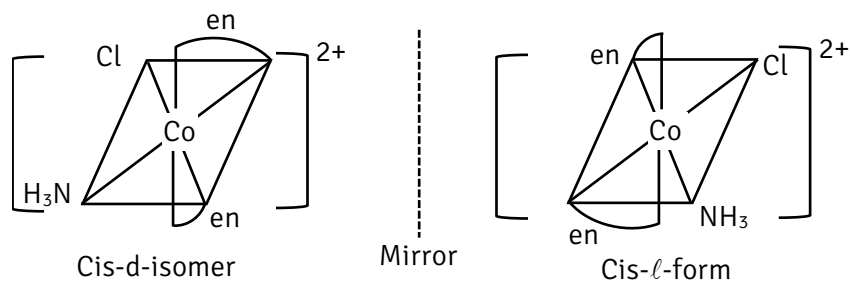
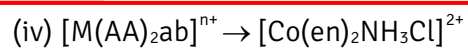


- $[Mabcedf] \rightarrow [Pt(py)NH_3NO_2ClBrI]$



- $[M(AA)_3]^{n+} \rightarrow [Co(en)_3]^{3+}$

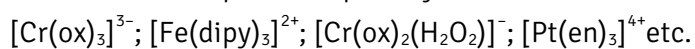




Note :

(i) Trans - $[Fe(en)_2Cl_2]$ do not show optical activity due to superimposition of their mirror image.

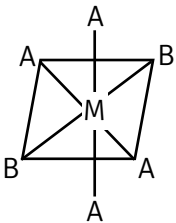
(ii) Some more examples are optically active :



NCERT QUESTIONS (Solve Yourself)

- Indicate the types of isomerism exhibited by the following complexes and draw the structures for these isomers:
 (i) $K[Cr(H_2O)_2(C_2O_4)_2]$ (ii) $[Co(en)_3]Cl_3$
 (iii) $[Co(NH_3)_5(NO_2)](NO_3)_2$ (iv) $[Pt(NH_3)(H_2O)Cl_2]$
- Give evidence that $[Co(NH_3)_5Cl]SO_4$ and $[Co(NH_3)_5(SO_4)]Cl$ are ionisation isomers.
- List various types of isomerism possible for coordination compounds, giving an example of each.
- How many geometrical isomers are possible in the following coordination entities?
 (i) $[Cr(C_2O_4)_3]^{3-}$ (ii) $[Co(NH_3)_3Cl_3]$
- Draw the structures of optical isomers of:
 (i) $[Cr(C_2O_4)_3]^{3-}$ (ii) $[PtCl_2(en)_2]^{2+}$ (iii) $[Cr(NH_3)_2Cl_2(en)]^+$
- Draw all the isomers (geometrical and optical) of:
 (i) $[CoCl_2(en)_2]^+$ (ii) $[Co(NH_3)Cl(en)_2]^{2+}$ (iii) $[Co(NH_3)_2Cl_2(en)]^+$
- Write all the geometrical isomers of $[Pt(NH_3)(Br)(Cl)(py)]$ and how many of these will exhibit optical isomers?
- Give the oxidation state, d orbital occupation and coordination number of the central metal ion in the following complexes:
 (i) $K_3[Co(C_2O_4)_3]$ (ii) $cis-[CrCl_2(en)_2]Cl$ (iii) $(NH_4)_2[CoF_4]$ (iv) $[Mn(H_2O)_6]SO_4$

Exercise 1.2

- A square planar complex is cis platin $[Pt(NH_3)_2Cl_2]$, shows :
 (1) Geometrical isomerism
 (2) Optical isomerism
 (3) Linkage isomerism
 (4) None
- The two compounds sulphatopentaammine cobalt(III)bromide and sulphatopentaammine cobalt(III)chloride represent:
 (1) Linkage isomerism
 (2) Ionisation isomerism
 (3) Co-ordination isomerism
 (4) No isomerism
- Which of the following complex can not exhibit geometrical isomerism:-
 (1) $[Pt(NH_3)_2Cl NO_2]$
 (2) $[Pt(gly)_2]$
 (3) $[Cu(en)_2]^{+2}$
 (4) $[Pt(H_2O)(NH_3)BrCl]$
- Which one of the following compounds will exhibit linkage isomerism:-
 (1) $[Pt(NH_3)_2Cl_2]$
 (2) $[Co(NH_3)_3NO_2]Cl_2$
 (3) $[Co(NH_3)_4Cl_2]Cl$
 (4) $[Co(en)_2Cl_2]Cl$
- Out of the following which complex will show geometrical isomerism ?
 (1) $[Pt(NH_3)_2Cl_2]$ (2) $Ni(CO)_4$
 (3) $Na_3[Ni(CN)_4]$ (4) $K[Ag(CN)_2]$
- Which of the following complexes will show optical isomerism ?
 (1) $[Cr(NH_3)_6]^{2+}$ (2) $[Ni(H_2O)_6]^{2+}$
 (3) $[Pt(NH_3)_3Br]NO_3$ (4) $[Cr(en)_3]Cl_3$
- The compound $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_4Cl_2] Cl \cdot 2H_2O$ represent
 (1) Linkage isomerism
 (2) Hydrate isomerism
 (3) Ligand isomerism
 (4) None of these
- The isomer -


 can be marked as -
 (1) Cis isomer (2) Leavo isomer
 (3) Dextro isomer (4) Trans isomer

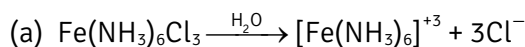
9. A Planar Complex ($Mabcd$) gives:-
 (1) Two Optical isomer
 (2) Two geometrical isomer
 (3) Three optical isomer
 (4) Three geometrical isomers
10. Theoretically the number of geometrical isomers expected for octahedral complex $[Mabcdef]$ is:-
 (1) Zero (2) 30 (3) 15 (4) 9
11. Which of the following complex produce precipitate with $AgNO_3$ and exist in two geometrical isomeric form
 (1) $PtCl_2 \cdot 4NH_3$ (2) $PtCl_2 \cdot 3NH_3$
 (3) $PtCl_4 \cdot 4NH_3$ (4) $PtCl_4 \cdot 2NH_3$
12. $Cis[Pt(en)_2 Cl_2] Br_2 \cdot 2H_2O$ and $trans [Pd(en)_2 Br_2] Cl_2 \cdot 2H_2O$ are:-
 (1) Ionisation isomer
 (2) Geometrical isomer
 (3) Hydrate isomer
 (4) None of these
13. Which of the following octahedral complex have maximum stereoisomers:-
 (1) $[M(CN)_6]^+$ (2) $[M(en)(CN)_4]^+$
 (3) $[M(en)_2(CN)_2]^+$ (4) $[M(en)_3]^+$
14. Which of the following complex does not show geometrical isomerism?
 (1) $[Co(NH_3)_4 Cl_2]^+$ (2) $[Co(NH_3)_3(NO_2)_3]$
 (3) $[Cr(en)_3]^{3+}$ (4) $[Pt(gly)_2]$
15. Both geometrical and optical isomerism are shown by:
 (1) $[Co(en)_2 Cl_2]^+$ (2) $[Co(NH_3)_5 Cl]^{2+}$
 (3) $[Co(NH_3)_4 Cl_2]^+$ (4) $[Cr(ox)_3]^{3-}$
16. Which of the following complex compounds does not exhibits geometrical isomerism?
 (1) $[PtCl_2(NH_3)_2]$
 (2) $[PdCl_2 Br]$
 (3) $[Pt(NH_3)(py)(Cl)(Br)]$
 (4) $[Pt(NH_3)_3(Br)]$
17. The pair of molecules that exhibit geometrical isomerism are
 (1) $[Cr(NH_3)_4 Cl_2]^+$ and $[Co(NH_3)_4 Cl_2]^+$
 (2) $[Cr(NH_3)_5 Cl]^{2+}$ and $[Co(NH_3)_5 Cl]^{2+}$
 (3) $[Cr(NH_3)_6]^{3+}$ and $[Co(NH_3)_6]^{3+}$
 (4) All of the above show geometric isomerism
18. The compounds $[Co(SO_4)(NH_3)_5]Br$ and $[Co(NH_3)_5 Br]SO_4$ represent.
 (1) linkage isomerism
 (2) ionisation isomerism
 (3) Coordination isomerism
 (4) no isomerism
19. What kind of isomerism exists between $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5 Cl]Cl_2 \cdot H_2O$?
 (1) linkage isomerism
 (2) Hydrate isomerism
 (3) ionisation isomerism
 (4) coordination isomerism
20. Indicate the complex ion which shows geometrical isomerism.
 (1) $[Cr(H_2O)_4 Cl_2]^+$ (2) $[Pt(NH_3)_3 Cl]$
 (3) $[Co(NH_3)_6]^{3+}$ (4) $[Co(CN)_5(NC)]^{3-}$
21. Indicate the complex ion which shows geometrical isomerism.
 (1) $[Cr(H_2O)_4 Cl_2]^+$ (2) $[Pt(NH_3)_3 Cl]$
 (3) $[Co(NH_3)_6]^{3+}$ (4) $[Co(CN)_5(NC)]^{-3}$
22. The compounds $[Co(SO_4)(NH_3)_5]Br$ and $[Co(SO_4)(NH_3)_5]Cl$ represent:
 (1) Linkage isomerism
 (2) Ionisation isomerism
 (3) no isomerism
 (4) Coordination isomerism
23. Which complex is optically active?
 (1) $[Co(en)_3]^{+3}$
 (2) $trans - [Co(en)_2 Cl_2]^+$
 (3) $Cis - [Co(en)_2 Cl_2]^+$
 (4) (1) and (3) both
24. For the Given complex $[Co(en)_2(NH_3)_2]^{+3}$, the number of Geometrical isomers, optical isomers and total number of isomers of all type possible respectively are:
 (1) 2, 2, 4 (2) 2, 2, 3
 (3) 2, 0, 2 (4) 0, 2, 2
25. Which is correctly matched
- | Compound | Total stereoisomer |
|-------------------------|--------------------|
| (1) $[Co(en)_3]Cl_3$ | 3 |
| (2) $[Co(en)_2 Cl_2]Cl$ | 2 |
| (3) $[Co(NH_3)_3 Cl_3]$ | 2 |
| (4) $[Pt(NH_3)_2 Cl_2]$ | 3 |
26. Which of the following complex species is not expected to exhibit optical isomerism
 (1) $[Co(en)(NH_3)_2 Cl_2]^+$ (2) $[Co(en)_3]^{+3}$
 (3) $[Co(en)_2 Cl_2]^+$ (4) $[Co(NH_3)_3 Cl_3]$

27. Which of the following have maximum number of isomers
 (1) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ (2) $[\text{Ni}(\text{en})(\text{NH}_3)_4]$
 (3) $[\text{Ni}(\text{C}_2\text{O}_4)(\text{en})_2]^0$ (4) $[\text{Cr}(\text{SCN})_2(\text{NH}_3)_4]^+$
28. Which among the following can exhibit (cis-trans) isomerism
 (1) $\text{CoCl}_3 \cdot 4\text{NH}_3$ (2) $\text{CoCl}_3 \cdot 6\text{NH}_3$
 (3) $\text{CoCl}_3 \cdot 5\text{NH}_3$ (4) $\text{PtCl}_2 \cdot 4\text{H}_2\text{O}$
29. Out of the following coordination entities which is chiral (optically active)
 (1) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (2) $\text{trans}-[\text{CrCl}_2(\text{OX})_2]^{3-}$
 (3) $\text{cis}-[\text{CrCl}_2(\text{NH}_3)_4]^+$ (4) $\text{cis}-[\text{CrCl}_2(\text{OX})_2]^{3-}$
30. Which of the following does not have optical isomer
 (1) $[\text{Co}(\text{en})_3]\text{Cl}_3$
 (2) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
 (3) $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$
 (4) $[\text{Co}(\text{NH}_3)_3(\text{Cl})(\text{Br})(\text{I})]$
31. Geometrical isomerism can be shown by
 (1) $\text{Na}[\text{Ag}(\text{CN})_2]$
 (2) $\text{Na}_2[\text{Cd}(\text{NO}_2)_4]$
 (3) $[\text{PtCl}_4\text{I}_2]^{-2}$
 (4) $[\text{PtCl}(\text{NH}_3)_3][\text{Au}(\text{CN})_2]$
32. Which of the following will give a pair of enantiomorphs
 (1) $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$
 (2) $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$
 (3) $[\text{Pt}(\text{NH}_3)_6][\text{PtCl}_6]$
 (4) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{NO}_2$
33. Total number of possible isomers of complex $[\text{Pd}(\text{NH}_3)_2(\text{SCN})_2]$
 (1) 2 (2) 4 (3) 3 (4) 6
34. Indicate the complex ion which shows geometrical isomerism
 (1) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]^+$
 (2) $[\text{Pt}(\text{NH}_3)_3\text{Cl}]^+$
 (3) $[\text{Co}(\text{NH}_3)_6]^{3+}$
 (4) $[\text{Co}(\text{CN})_5(\text{NC})]^{3-}$

Bonding in Coordination Compounds/ Werner's Theory

Werner's Theory of complexes :

This was postulated by Alfred Werner in 1893. The postulates of Werner's Theory of complexes are –



Every metal forming a complex exhibits two types of valencies - Primary valency and Secondary Valency. In the above complex chlorines act as primary valency and NH_3 acts as secondary valency.

- (b) Primary valency is equal to oxidation state of metal ion so, primary valency is satisfied by anions whereas secondary valency of the metal ion may be satisfied by anions, neutral molecules or rarely by cations.
- (c) Primary valency is ionisable in nature whereas secondary valency is nonionisable.
- (d) The number of secondary valencies of a metal ion in the complex will remain constant and this is called as coordination number.

Example:

- Fe^{+2} , Fe^{+3} , Co^{+3} , Pt^{+4} , Cr^{+3} , Ru^{+2} , Os^{+3} have a coordination number of six
- Pt^{+2} , Ni^{+2} , Cu^{+2} , Zn^{+2} , Au^{+3} , Hg^{+2} have a coordination number of four.
- Ag^{+1} which is an exceptional ion with coordination number of two.
- Mo^{+3} has the highest coordination number of eight.

(e) Shapes and isomerism in complexes :

- The more possible shape of 6-coordinated complexes can be hexagonal planar or octahedral. It can exhibit geometrical or optical isomerism both.
- The shape of 4-coordinated complexes can be square planar (exhibiting geometrical isomerism) or Tetrahedral (exhibiting optical isomerism).

Experimental Evidence of Werner's Theory

(a) Precipitation of primary valencies on the addition of a suitable reagent.

Example:

$\text{Fe}(\text{NH}_3)_6\text{Cl}_3$ forms 3 moles of AgCl in the form of precipitate on addition of AgNO_3 solution.

This indicates that the complex ionises as $\text{Fe}(\text{NH}_3)_6\text{Cl}_3 \xrightarrow{\text{H}_2\text{O}} \text{Fe}(\text{NH}_3)_6^{+3} + 3\text{Cl}^-$

(b) Electrical conductance of complexes : More the number of ions provided greater is the electrical conductance of the complex in aqueous medium. Electrical conductivity $\propto$ no. of ions.

Example:

The electrical conductance of aqueous $[\text{Fe}(\text{NH}_3)_6]\text{Cl}_3$ is greater than that of aqueous solution of $[\text{Fe}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$.

$[\text{Fe}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2 \longrightarrow [\text{Fe}(\text{NH}_3)_5\text{Cl}]^{+2} + 2\text{Cl}^-$ Total 3 ions

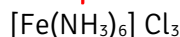
Complex	Modern formula	No. of Cl^- ions precipitated	Total number of ions
$\text{PtCl}_4 \cdot 6\text{NH}_3$	$[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$	4	5
$\text{PtCl}_4 \cdot 5\text{NH}_3$	$[\text{Pt}(\text{NH}_3)_5\text{Cl}]\text{Cl}_3$	3	4
$\text{PtCl}_4 \cdot 4\text{NH}_3$	$[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}_2$	2	3
$\text{PtCl}_4 \cdot 3\text{NH}_3$	$[\text{Pt}(\text{NH}_3)_3\text{Cl}_3]\text{Cl}$	1	2
$\text{PtCl}_4 \cdot 2\text{NH}_3$	$[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$		0 (non-electrolyte)

Werner's representation of complexes :

(i)	$\text{Fe}(\text{NH}_3)_6\text{Cl}_3 \longrightarrow [\text{Fe}(\text{NH}_3)_6]\text{Cl}_3$ Dotted lines indicate primary valency and continuous lines indicate secondary valency of metal ion.	
(ii)	$\text{Fe}(\text{NH}_3)_5\text{Cl}_3 \longrightarrow [\text{Fe}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ In this complex two 'Cl' groups act as primary valencies and one of the 'Cl' acts as secondary valency also. Thus werner's representation of this complex -	
(iii)	$\text{Fe}(\text{NH}_3)_4\text{Cl}_3 \longrightarrow [\text{Fe}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ In this complex one 'Cl' groups act as primary valency and two of the 'Cl' group act as secondary valencies also. Thus werner's representation of this complex -	

Modern Theory Of Complexes Central metal ion along with secondary valencies or ligands are written in square brackets and primary valencies outside the square bracket.

Example:



- (a) **Coordination sphere or inner sphere:** This is written in square bracket. This constitutes the central metal ion along with ligands.
- (b) **Outer sphere or Ionisation sphere:** This is written outside the square bracket. In aqueous solution the groups written outside the square bracket dissociate and are provided in the form of ions.
- (c) **Modern theory of complexes or Electronic Theory of complexes :** Sidgwick and Lowry in 1923 developed this theory, and modified Werner's Theory.
 - (i) According to Sidgwick the primary valency is regarded to be formed by transfer of electron and secondary valency is formed by the sharing of electron pair provided by the donor.
 - (ii) During formation of primary valency the metal loses electrons. Thus the number of electrons decrease during primary valency formation.
 - (iii) In the formation of secondary valency the metal ion gains electrons pairs for the formation of coordinate bond.
 - (iv) Thus during the formation of secondary valency, total number of electrons associated with the metal ion increase.
 - (v) The total number of electrons associated with the metal ion in its complex may be equal to or nearly equal to the atomic number of inert gas of the same period. This total number of electrons is called 'Effective atomic number' (E.A.N.) of the metal.

Example:

Effective atomic number of cobalt in $[\text{Co}(\text{NH}_3)_6]^{3+}$ can be calculated as follows :

Atomic number of cobalt = 27

Oxidation state of cobalt in complex = +3

Number of electrons in Co^{+3} ion are $(27-3 = 24)$

During coordinate covalent bonding, Co^{+3} ion gains 6 pairs of electrons.

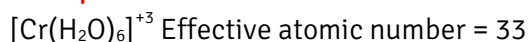
Thus Effective atomic number of cobalt is $[\text{Co}(\text{NH}_3)_6]^{+3} \quad 24+12 = 36$.

(vi) E.A.N. = (No. of electron in the metal ion) + (No. of electrons gained from the ligands)

(vii) As the Effective atomic number of cobalt is 36 in this complex, according to Sidgwick, this complex will be stable.

(viii) Effective atomic number rule is applicable in many cases, there are several examples in which effective atomic number rule is not obeyed.

Example:



Complex	Metal (Oxidation State)	Atomic Number of Metal	Coordination Number	Effective atomic number (E.A.N.)
$\text{K}_4[\text{Fe}(\text{CN})_6]$	+2	26	6	$(26 - 2) + (6 \times 2) = 36 [\text{Kr}]$
$[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$	+2	29	4	$(29 - 2) + (4 \times 2) = 35$
$[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$	+3	27	6	$(27 - 3) + (6 \times 2) = 36 [\text{Kr}]$
$\text{Ni}(\text{CO})_4$	0	28	4	$(28 - 0) + (4 \times 2) = 36 [\text{Kr}]$
$\text{K}_2[\text{Ni}(\text{CN})_4]$	+2	28	4	$(28 - 2) + (4 \times 2) = 34$
$\text{K}_2[\text{PtCl}_6]$	+4	78	6	$(78 - 4) + (6 \times 2) = 86 [\text{Rn}]$
$\text{K}_3[\text{Cr}(\text{C}_2\text{O}_4)_3]$	+3	24	6	$(24 - 3) + (6 \times 2) = 33$
$\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$	+3	26	6	$(26 - 3) + (6 \times 2) = 35$
$\text{K}_2[\text{HgI}_4]$	+2	80	4	$(80 - 2) + (4 \times 2) = 86 [\text{Rn}]$
$[\text{Ag}(\text{NH}_3)_2]\text{Cl}$	+1	47	2	$(47 - 1) + (2 \times 2) = 50$
$\text{K}_2[\text{PdCl}_4]$	+2	46	4	$(46 - 2) + (4 \times 2) = 52$

Valence Bond Theory

This theory was mainly developed by Linus Pauling. The main features of this theory are -

- Every metal ion when it forms a complex compound undergoes formation of coordinate covalent bond.
- During this bond formation, the metal ion acts as electron pair acceptor. For this the metal ion provides vacant orbitals.
- The number of vacant orbitals provided is equal to the coordination number of metal ion.

Example:

In the formation of $[\text{Fe}(\text{NH}_3)_6]^{3+}$, Fe^{3+} ion provides six vacant orbitals.

In $[\text{Cu}(\text{NH}_3)_4]^{2+}$, Cu^{+2} ion provides four vacant orbitals.

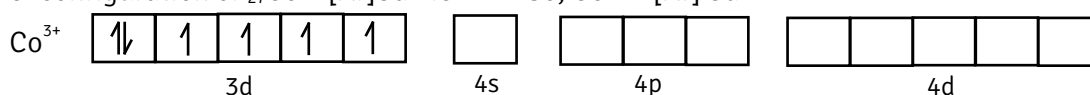
- The metal provides vacant orbitals only after the process of hybridisation, thus vacant hybrid orbitals are provided by the metal ion.
- The vacant hybrid orbitals of metal ion get overlapped by orbitals of ligands containing lone pair of electrons.
- The number of such overlappings is equal to the coordination number of metal ion.
- The empty 'd' orbitals involved in hybridisation may be inner $(n-1)d$ or outer "nd" orbitals.
- These complexes are called as Inner orbital complexes and outer orbital complexes respectively.
- If inner 'd' orbitals are involved in hybridisation, then it is through only the pairing of unpaired electrons in the 'd' orbitals of metal ion.
- Then such type of complexes will be diamagnetic or less paramagnetic and will be called as Low spin complexes.
- All outer orbital complexes have paramagnetic nature and they are called as High spin complexes.

Applications of valence Bond theory :

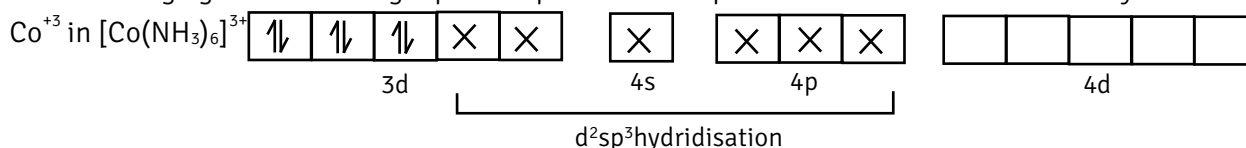
(a) To 6-coordinated complexes :



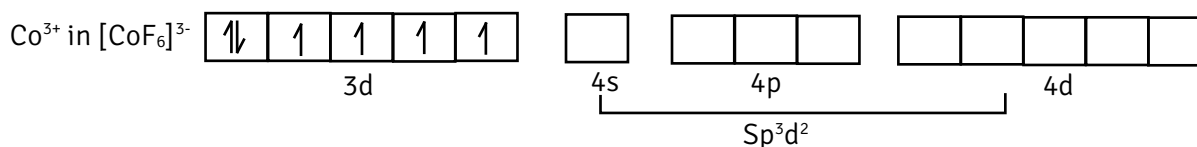
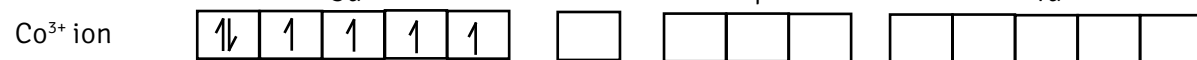
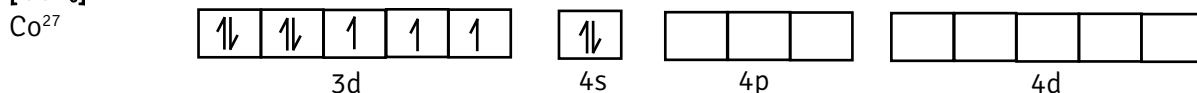
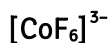
e^- configuration of $_{27}\text{Co} = [\text{Ar}]3d^7 4s^2$ So, $\text{Co}^{+3} = [\text{Ar}] 3d^6$



Due to strong ligand electrons get paired up so metal ion provides vacant '3d' orbitals for hybridisation.

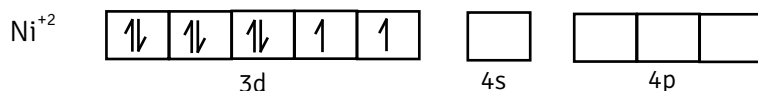


- This is a **diamagnetic complex**.
- As inner 'd' orbitals are involved in hybridisation, hence it is an **inner orbital complex**.
- It is a **low spin complex**.

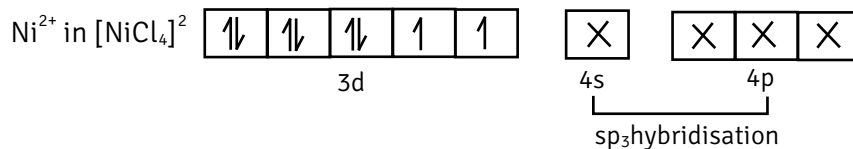


- This is a paramagnetic complex and high spin complex.
- The outer 'd' orbitals are involved in hybridisation, hence it is an outer orbital complex.

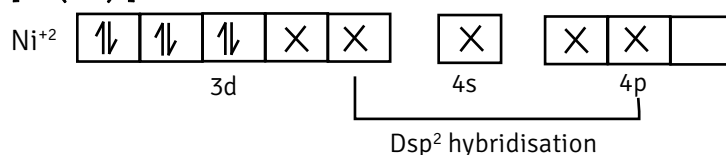
(b) 4-coordinated complexes :



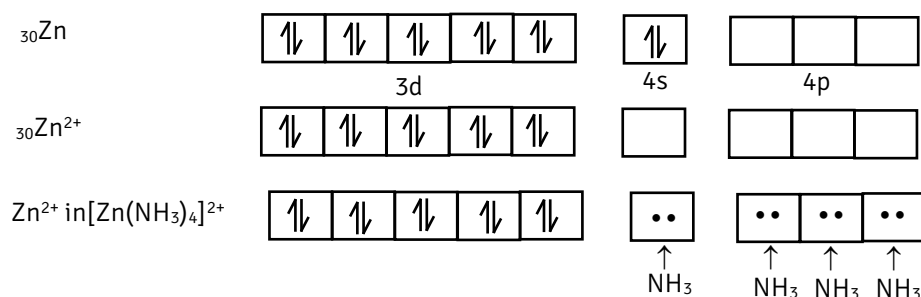
- The unpaired electrons remain as such because Cl^- is a weak ligand as it is a paramagnetic complex.



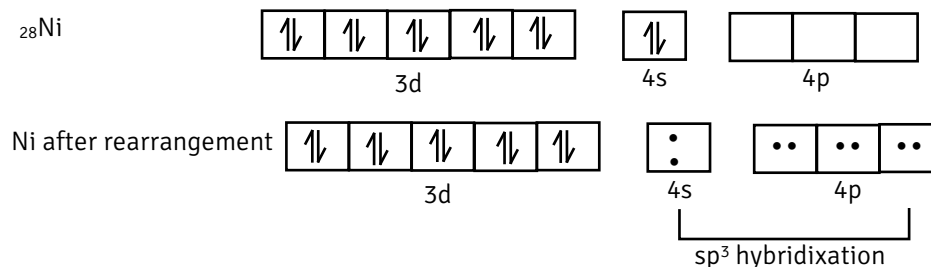
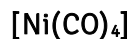
- $[\text{NiCl}_4]^{2-}$ has tetrahedral geometry.
- It is a high spin complex.



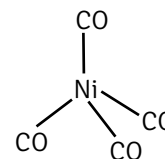
- CN^- is **strong field ligand**. Thus the complex has **square planar geometry**.
- The unpaired electrons in '3d' orbitals of Ni^{2+} get paired up, So, $[\text{Ni}(\text{CN})_4]^{2-}$ is **diamagnetic** in nature.
- This Ni^{2+} complex is an **inner orbital complex** and a **low spin complex**.



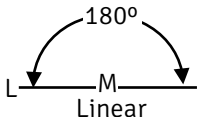
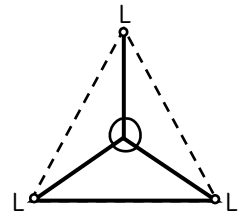
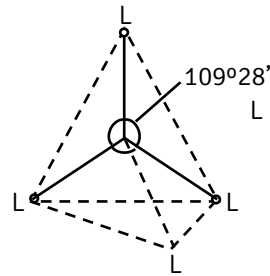
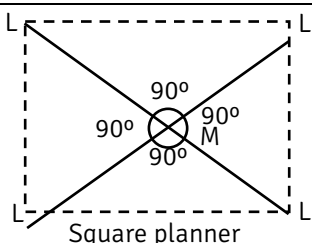
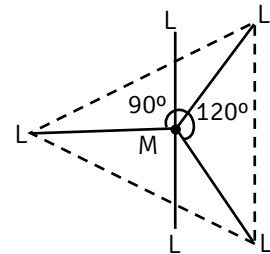
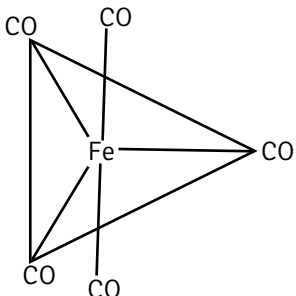
- Because the complex is formed by sp^3 hybridisation, hence it is tetrahedral
- Since all electrons are paired, hence it is diamagnetic



- CO is strong ligand. So it is **high spin complex**.
- The empty one 4s and three 4p orbitals mix to form four new sp^3 hybridised orbitals.
- Each orbital accepts one electron pair from carbon monoxide molecule forming $[\text{Ni}(\text{CO})_4]$



- It is diamagnetic in nature, the shape of nickel tetra carbonyl is tetrahedral as shown below

Coordination Number	Hybridised orbital	Geometrical shape of the Complex	Examples of Complex
2	sp	 Linear	$[Ag(NH_3)_2]^+$ $[Ag(CN)_2]^-$
3	sp^2		$[HgI_3]^-$
4	sp^3	 $109^\circ 28'$	$[CuCl_4]^{-2}$ $[ZnCl_4]^{-2}$ $[FeCl_4]^-$; $[Ni(CO)_4]$ $[Zn(NH_3)_4]^{+2}$
4	dsp^2 The d-orbital Involved is $d_{x^2-y^2}$ orbital	 Square planar	$[PdCl_4]^{2-}$ $[Ni(CN)_4]^{2-}$ $[Pt(NH_3)_4]^{+2}$; $[Cu(H_2O)_4]^{+2}$; $[Cu(NH_3)_4]^{+2}$; $[PtCl_4]^{2-}$
5	sp^3d	 Trigonal bipyramidal	$[CuCl_5]^{-3}$ $[MoCl_5]$
5	dsp^3	 Trigonal bipyramidal	$[Ni(CN)_5]^{-3}$ $[Fe(CO)_5]$

6	$d^2 sp^3$ When d-orbitals are (n-1) d-orbitals Inner orbital complexes) or $sp^3 d^2$ When d-orbitals are nd orbitals (Outer orbital complexes) in both cases d-orbitals are d_{z^2} and $d_{x^2-y^2}$ orbitals	Octahedral	$[\text{Cr}(\text{NH}_3)_6]^{+3}$ $[\text{Ti}(\text{H}_2\text{O})_6]^{+3}$ $[\text{Fe}(\text{CN})_6]^{-3}$ $[\text{Co}(\text{NH}_3)_6]^{+3}$ $[\text{PtCl}_6]^{-2}, [\text{CoF}_6]^{-3}$
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NCERT QUESTIONS (Solve yourself)

- Explain on the basis of valence bond theory that $[\text{Ni}(\text{CN})_4]^{2-}$ ion with square planar structure is diamagnetic and the $[\text{NiCl}_4]^{2-}$ ion with tetrahedral geometry is paramagnetic.
- $[\text{NiCl}_4]^{2-}$ is paramagnetic while $[\text{Ni}(\text{CO})_4]$ is diamagnetic though both are tetrahedral. Why?
- $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ is strongly paramagnetic whereas $[\text{Fe}(\text{CN})_6]^{3-}$ is weakly paramagnetic. Explain.
- Explain $[\text{Co}(\text{NH}_3)_6]^{3+}$ is an inner orbital complex whereas $[\text{Ni}(\text{NH}_3)_6]^{2+}$ is an outer orbital complex.
- Predict the number of unpaired electrons in the square planar $[\text{Pt}(\text{CN})_4]^{2-}$ ion.
- The hexaquo manganese(II) ion contains five unpaired electrons, while the hexacyanoion contains only one unpaired electron. Explain using Crystal Field Theory.
- Explain the bonding in coordination compounds in terms of Werner's postulates.
- Specify the oxidation numbers of the metals in the following coordination entities:
 (i) $[\text{Co}(\text{H}_2\text{O})(\text{CN})(\text{en})_2]^{2+}$ (ii) $[\text{CoBr}_2(\text{en})_2]^+$ (iii) $[\text{PtCl}_4]^{2-}$
 (iv) $\text{K}_3[\text{Fe}(\text{CN})_6]$ (v) $[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$
- Discuss the nature of bonding in the following coordination entities on the basis of valence bond theory: (i) $[\text{Fe}(\text{CN})_6]^{4-}$ (ii) $[\text{FeF}_6]^{3-}$ (iii) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ (iv) $[\text{CoF}_6]^{3-}$
 (i) $[\text{Fe}(\text{CN})_6]^{4-}$ (ii) $[\text{FeF}_6]^{3-}$ (iii) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ (iv) $[\text{CoF}_6]^{3-}$
- Draw figure to show the splitting of d orbitals in an octahedral crystal field.
- What is spectrochemical series? Explain the difference between a weak field ligand and a strong field ligand.
- $[\text{Cr}(\text{NH}_3)_6]^{3+}$ is paramagnetic while $[\text{Ni}(\text{CN})_4]^{2-}$ is diamagnetic. Explain why?

Exercise 1.3

- Which of the following has least conductivity in aqueous solution.
 (1) $\text{CoCl}_3.4\text{NH}_3$ (2) $\text{CoCl}_3.3\text{NH}_3$
 (3) $\text{CoCl}_3.5\text{NH}_3$ (4) $\text{CoCl}_3.6\text{NH}_3$
- If EAN of a central metal ion X^{+2} in a complex is 34. and atomic number of X is 28. The number of monodentate ligands present in complex are:-
 (1) 3 (2) 4 (3) 6 (4) 2
- The EAN of cobalt in the complex ion $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ is:-
 (1) 27 (2) 36
 (3) 33 (4) 35
- The effective atomic number of Cr (atomic no. 24) in $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ is
 (1) 35 (2) 27
 (3) 33 (4) 36

5. In which of the given complex 25% of chloride ion gets precipitated on reaction with excess AgNO_3 ?
 (1) $\text{PtCl}_2 \cdot 4\text{NH}_3$ (2) $\text{PtCl}_4 \cdot 5\text{NH}_3$
 (3) $\text{PtCl}_4 \cdot 4\text{NH}_3$ (4) $\text{PtCl}_4 \cdot 3\text{NH}_3$
6. In the metal carbonyls of general formula $\text{M}(\text{CO})_x$ (Which follows EAN rule) if M is Ni, Fe and Cr the value of x will be respectively:
 (1) 6, 5, 6 (2) 4, 5, 6
 (3) 4, 4, 5 (4) 4, 6, 6
7. When 1mol $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ is treated with excess of AgNO_3 , 3 mol of AgCl are obtained. The formula of the complex is
 (1) $[\text{CrCl}_3(\text{H}_2\text{O})_3] \cdot 3\text{H}_2\text{O}$
 (2) $[\text{CrCl}_2(\text{H}_2\text{O})_4] \cdot \text{Cl} \cdot 2\text{H}_2\text{O}$
 (3) $[\text{CrCl}(\text{H}_2\text{O})_5] \cdot \text{Cl}_2 \cdot \text{H}_2\text{O}$
 (4) $[\text{Cr}(\text{H}_2\text{O})_6] \cdot \text{Cl}_3$
8. In which of the following pair EAN of central metal atom is not same?
 (1) $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{Fe}(\text{NH}_3)_6]^{3+}$
 (2) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ and $[\text{Cr}(\text{CN})_6]^{3-}$
 (3) $[\text{FeF}_6]^{3-}$ and $[\text{Fe}(\text{CN})_6]^{3-}$
 (4) $[\text{Ni}(\text{CO})_4]$ and $[\text{Ni}(\text{CN})_4]^{2-}$
9. Which of the following complexes produces three moles of silver chloride when its one mole is treated with excess of silver nitrate:
 (1) $[\text{Co}(\text{H}_2\text{O})_3\text{Cl}_3]$ (2) $[\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2] \cdot \text{Cl}$
 (3) $[\text{Co}(\text{H}_2\text{O})_5\text{Cl}] \cdot \text{Cl}_2$ (4) $[\text{Co}(\text{H}_2\text{O})_6] \cdot \text{Cl}_3$
10. The number of chloride ions which would be precipitated when the complex $\text{PtCl}_4 \cdot 4\text{NH}_3$ is treated with silver nitrate is:
 (1) Four (2) One
 (3) Three (4) Two
11. Which of the following compound is paramagnetic
 (1) Tetracyanonickelate (II) ion
 (2) Tetraamminezinc (II) ion
 (3) Hexaamine chromium (III) ion
 (4) Diammine silver (I) ion
12. The shape of the complex $[\text{Ag}(\text{NH}_3)_2]^+$ is:
 (1) Octahedral (2) Square planar
 (3) Tetrahedral (4) Linear
13. Hexafluoroferrate (III) ion is an outer orbital complex. The number of unpaired electrons are
 (1) 1 (2) 5
 (3) 4 (4) Unpredictable
14. The shape of $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$ is:
 (1) Square planar (2) Pyramidal
 (3) Octahedral (4) Tetrahedral
15. Among the following ions, which one has the highest paramagnetism?
 (1) $[\text{FeF}_6]^{3-}$ (2) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
 (3) $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ (4) $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$
16. In the complex ion ML_6^{n+} , M^{n+} has five d-electrons and L is weak field ligand. According to crystal field theory, the magnetic properties of the complex ion correspond to how many unpaired electrons.
 (1) 0 (2) 5 (3) 2 (4) 3
17. A magnetic moment of 1.73 BM will be shown by one among of the following compounds.
 (1) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ (2) $[\text{Ni}(\text{CN})_4]^{2-}$
 (3) TiCl_4 (4) $[\text{CoCl}_6]^{3-}$
18. The magnetic property and the shape of $[\text{Cr}(\text{NH}_3)_6]^{3+}$ complex ions are:
 (1) Paramagnetic, Octahedral
 (2) Diamagnetic, square planar
 (3) Paramagnetic, tetrahedral
 (4) None of the above
19. Amongst the following ions which one has the highest paramagnetism
 (1) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ (2) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$
 (3) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ (4) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
20. Which of the following complexes is an inner orbital complex?
 (1) $[\text{CoF}_6]^{3+}$ (2) $[\text{FeF}_6]^{3-}$
 (3) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ (4) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
21. The wrong statement is:-
 (1) Halide ligands forms high spin complex
 (2) Strong ligands form low spin complex
 (3) $[\text{FeF}_6]^{3-}$ is inner orbital complex
 (4) $\text{Na}_3[\text{CoF}_6]$ is outer orbital complex

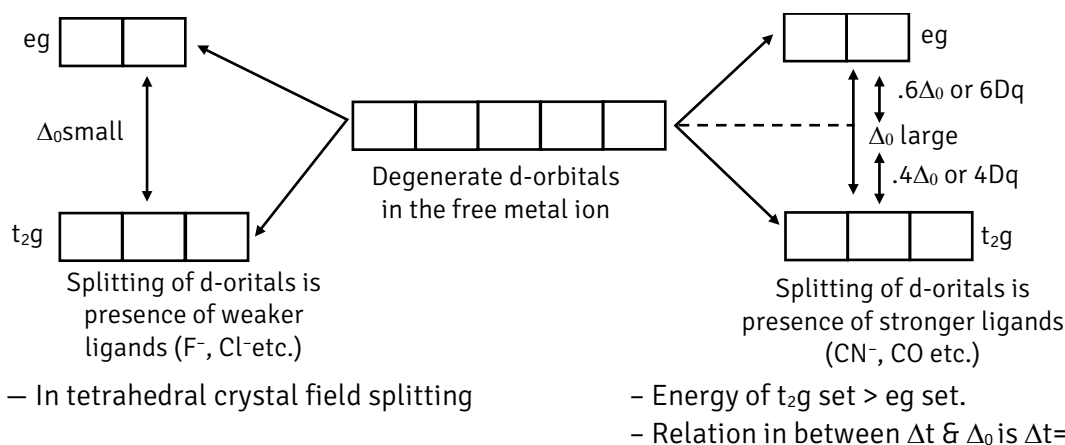
22. What is incorrect for $K_4[Fe(CN)_6]$
 (1) Oxidation state of iron is +2
 (2) It exhibit diamagnetic character
 (3) It exhibit paramagnetic character
 (4) It involved d^2sp^3 hybridisation
23. In which of the following molecules, central atom used orbitals of different quantum number in the hybridisation:-
 (1) $[Fe(CO)_5]$ (2) IF_7
 (3) $Ni(CO)_4$ (4) XeO_4
24. What are the geometric shape and the oxidation number of the copper atom, respectively, for the complex ion, $[Cu(NH_3)_4(OH_2)_2]^{2+}$?
 (1) Tetrahedral; +2 (2) Square planar; -2
 (3) Linear; +3 (4) Octahedral; +2
25. Which of the following contains one unpaired electron in the 4p orbitals:-
 (1) $[Cu(NH_3)_2]^+$ (2) $[Cu(NH_3)_4]^{2+}$
 (3) $[Cu(CN)_4]^{3-}$ (4) $[Ni(CN)_4]^{2-}$
26. Which of the following options are correct for $[Fe(CN)_6]^{3-}$ complex?
 (a) d^2sp^3 hybridisation
 (b) sp^3d^2 hybridisation
 (c) paramagnetic
 (d) diamagnetic
 (1) c,d (2) a, b
 (3) a,c (4) a, d
27. Which of the following complexes are low spin and diamagnetic?
 (a) $K_4[Os(CN)_6]$ (b) $[Mo(CO)_6]$
 (c) $[Mn(CN)_6]^{4-}$
 Select the correct answer using the codes given below:
 (1) a, b and c (2) a and b only
 (3) a and c only (4) b and c only
28. How many unpaired electrons are present in the Brown ring complex $[Fe(H_2O)_5(NO)]SO_4$
 (1) 4 (2) 3 (3) 0 (4) 5
29. Which of the following statements about $Fe(CO)_5$ is correct?
 (1) It is paramagnetic and high spin complex
 (2) It is diamagnetic and high spin complex
 (3) It is diamagnetic and low spin complex
 (4) It is paramagnetic and low spin complex
30. Which is true for $[Ni(en)_2]^{2+}$, $Z(Ni) = 28$?
 (1) paramagnetism, dsp^2 , square planar, C.N. of Ni = 2
 (2) diamagnetism, dsp^2 , square planar, C.N. of Ni = 4.
 (3) diamagnetism, sp^3 , tetrahedral, C.N. of Ni=4
 (4) paramagnetism, sp^3 , tetrahedral, C.N. of Ni=4.
31. Arrange the following in order of decreasing number of unpaired electrons:
 I. $[Fe(H_2O)_6]^{2+}$ II. $[Fe(CN)_6]^{3-}$
 III. $[Fe(CN)_6]^{4-}$ IV. $[Fe(H_2O)_6]^{3+}$
 (1) IV, I, II, III (2) I, II, III, IV
 (3) III, II, I, IV (4) II, III, I, IV
32. The number of unpaired electrons present in complex ion $[FeF_6]^{3-}$ is:
 (1) 5 (2) 4 (3) 6 (4) 0
33. In which of the following co-ordination compounds, the magnetic moment would be maximum:
 (1) $[Co(NH_3)_6]Cl_3$ (2) $K_4[Fe(CN)_6]$
 (3) $K_3[Fe(CN)_6]$ (4) $[Fe(H_2O)_6]SO_4$
34. Which has maximum paramagnetic nature:
 (1) $[Cu(H_2O)_4]^{2+}$ (2) $[Cu(NH_3)_4]^{2+}$
 (3) $[Mn(H_2O)_6]^{2+}$ (4) $[Fe(CN)_6]^{4-}$
35. When $CrCl_3 \cdot 6NH_3$ is treated with excess of $AgNO_3$, 3 moles of $AgCl$ are obtained. The formula of the complex is:
 (1) $[CrCl_3(NH_3)_3]3NH_3$
 (2) $[CrCl_2(NH_3)_4]Cl \cdot 2NH_3$
 (3) $[CrCl(NH_3)_5]Cl_2 \cdot NH_3$
 (4) $[Cr(NH_3)_6]Cl_3$

36. The value of effective atomic no. of Cr in $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ is
 (1) 35 (2) 27
 (3) 33 (4) 36
37. How many ions are produced from the complex, $[\text{Co}(\text{NH}_3)_6]\text{Br}_3$?
 (1) 6 (2) 4
 (3) 3 (4) 2
38. The value of n in the Carbonyl is $(\text{CO})_n-\text{Co}-\text{Co}-(\text{CO})_n$
 (1) 4 (2) 5
 (3) 6 (4) 8
39. Which of the following doesn't follow effective Atomic Number Rule?
 (1) $[\text{Cu}(\text{NH}_3)_4]^{+2}$ (2) $[\text{ZnCl}_4]^{2-}$
 (3) $[\text{Ni}(\text{CO})_4]$ (4) $[\text{Zn}(\text{H}_2\text{O})_4]^{2+}$
40. Which of the following cannot act as an electrolyte
 (1) $\text{CoCl}_3.6\text{NH}_3$ (2) $\text{CoCl}_3.5\text{NH}_3$
 (3) $\text{CoCl}_3.4\text{NH}_3$ (4) $\text{CoCl}_3.3\text{NH}_3$
41. When 1 mole of $\text{CrCl}_3.6\text{H}_2\text{O}$ is treated with excess of AgNO_3 , 3 moles of AgCl are obtained. The formula of the complex is
 (1) $[\text{CrCl}_3(\text{H}_2\text{O})_3].3\text{H}_2\text{O}$
 (2) $[\text{CrCl}_2(\text{H}_2\text{O})_4]\text{Cl}.2\text{H}_2\text{O}$
 (3) $[\text{CrCl}(\text{H}_2\text{O})_5]\text{Cl}_2.\text{H}_2\text{O}$
 (4) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$
42. Which of the following options is correct for $\text{K}_4[\text{Fe}(\text{CN})_6]$ complex?
 (1) d^2sp^3 Hybridisation, diamagnetic
 (2) sp^3d^2 Hybridisation
 (3) Paramagnetic
 (4) None of these
43. Geometrical shapes of the complexes formed by the reaction of Ni with Cl^- , CN^- and H_2O respectively are.
 (1) octahedral, tetrahedral and sq. planar
 (2) Tetrahedral. square planar and octahedral
 (3) Square planar, tetrahedral and octahedral
 (4) octahedral, Square planar and tetrahedral
44. In Cu-ammonia complex the state of Hybridization of Cu^{+2} is
 (1) sp^3 (2) sp^3d (3) sp^3d^2 (4) dsp^2
45. The geometry of $[\text{Ni}(\text{CO})_4]$ and $[\text{Ni}(\text{PPh}_3)_2\text{Cl}_2]$ are
 (1) both are square planar
 (2) tetrahedral and square planar
 (3) both tetrahedral
 (4) square planar and tetrahedral
46. $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ resemble in
 (1) geometry (2) magnetic nature
 (3) hybridation (4) oxidation state

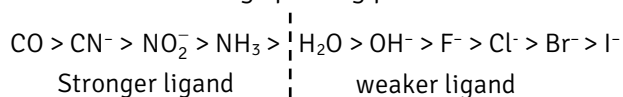
Crystal Field Theory

- (a) This theory was proposed by **Bethe** and **Vleck**.
- (b) According to this theory when a ligand approaches to the metal ion, the 'd' orbitals of the metal ion undergo splitting forming two sets i.e., higher energy set, eg. of $d_{x^2-y^2}$ and d_{z^2} and a lower energy set, t_{2g} of d_{xy} , d_{yz} and d_{xz} .
- (c) If the ligand approaching is strongly basic then more splitting of 'd' orbitals is brought about i.e., the energy difference between 't_{2g}' and 'eg' will be very high.
- (d) Under these conditions the electrons of the 'd' orbitals of metal ion get paired up in the t_{2g} set of orbitals. This results in the formation of a diamagnetic or less paramagnetic complex ($\Delta_0 > \text{PE}$)

- (e) If the ligand approaching is weakly basic, then under its influence the 'd' orbitals of metal ion undergo less splitting i.e., the energy difference between t_{2g} and e_g orbitals will be less. Due to this the electrons of 'd' orbitals of metal ion get arranged in t_{2g} and e_g orbitals forming a paramagnetic complex ion. ($\Delta_0 < PE$) Where Δ_0 is energy difference in octahedral splitting and PE is pairing energy.



- (f) Order of decreasing splitting power is –



- (g) It is also called spectro chemical series.

- (h) Strongly basic ligands are CN^- , NH_3 , NO_2^- Ethylene diamine, CO , etc. The effect of these ligands is that they bring about pairing of unpaired electrons in the 'd' orbitals of metal ion during complex formation.
- (i) According to crystal field Theory $[Ni(CN)_4]^{2-}$ will be diamagnetic. As CN^- is a strongly basic ligand which brings about pairing of unpaired electrons in the $3d^8$ configuration of Ni^{2+} ion. Whereas $2 [NiCl_4]^{2-}$ will be paramagnetic, as Cl^- is a weakly basic ligand and under its influence, no pairing of unpaired electrons can take place. Thus the magnetic nature of complexes can be easily predicted based on the crystal field theory.
- (j) When a bidentate ligand like ethylene diamine, or Glycinate ion or oxalate ion combines with the metal ion in the formation of complexes, then the final complex resulted will be a cyclic complex. This type of cyclic complex is called as chelate.
- (k) The process involving formation of a cyclic complex is called as chelation and the bidentate ligand is called as chelating agent.

NCERT QUESTIONS (Solve Yourself)

- What is crystal field splitting energy? How does the magnitude of Δ_0 decide the actual configuration of d orbitals in a coordination entity?
- Explain the violet colour of the complex $[Ti(H_2O)_6]^{3+}$ on the basis of crystal field theory.
- Amongst the following, the most stable complex is
(i) $[Fe(H_2O)_6]^{3+}$ (ii) $[Fe(NH_3)_6]^{3+}$ (iii) $[Fe(C_2O_4)_3]^{3-}$ (iv) $[FeCl_6]^{3-}$
- What will be the correct order for the wavelengths of absorption in the visible region for the following: $[Ni(NO_2)_6]^{4-}$, $[Ni(NH_3)_6]^{2+}$, $[Ni(H_2O)_6]^{2+}$

Exercise 1.4

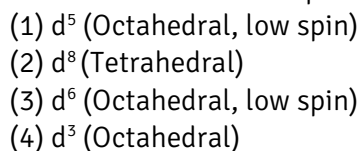
1. Which one of these ions absorbs energy from visible spectrum:-



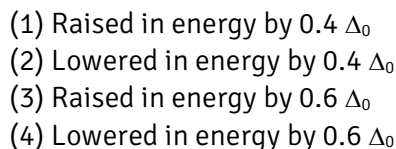
2. In the complex $[\text{Ni}(\text{H}_2\text{O})_2(\text{NH}_3)_4]^{+2}$ the magnetic moment (μ) of Ni is:-



3. Which of the following system has maximum number of unpaired electrons:



4. In an octahedral crystal field, the t_{2g} orbitals are



5. If $\Delta_0 < P$, the correct electronic configuration for d^4 system will be:-



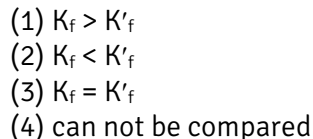
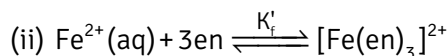
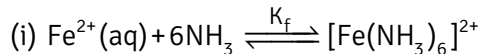
6. Match List-I with List-II and select the correct answer using the codes given below the lists:-

List -I (Complex ions)		List II (Number of Unpaired Electrons)	
A.	$[\text{CrF}_6]^{4-}$	i.	One
B.	$[\text{MnF}_6]^{4-}$	ii.	Two
C.	$[\text{Cr}(\text{CN})_6]^{4-}$	iii.	Three
D.	$[\text{Mn}(\text{CN})_6]^{4-}$	iv.	Four
			Five

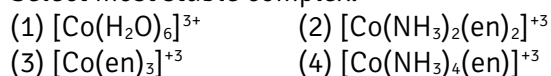
Code:

	A	B	C	D
(1)	iv	i	ii	v
(2)	ii	v	iii	i
(3)	iv	v	ii	i
(4)	ii	i	iii	v

7. Consider the following complex formation reactions and comment on their formation constant value



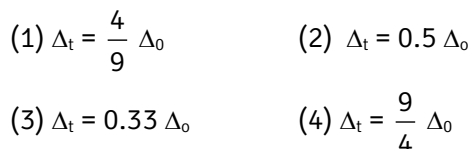
8. Select most stable complex:-



9. What will be the theoretical value of 'spin only' magnetic moment when $\text{Fe}(\text{SCN})_3$ reacts with a solution containing F^- ions to yield a complex?



10. The crystal field splitting energy for octahedral complex (Δ_0) and that for tetrahedral complex (Δ_t) are related as



11. Which one of the following high-spin complexes has the largest CFSE (Crystal field stabilization energy)?



12. Match List-I (complex ions) with List-II (CFSE) and select the correct answer using the codes given below the lists:

Column-I		Column-II		
(P) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$		1. $-0.6 \Delta_0$		
(Q) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$		2. $-0.4 + 1\text{P.E.}$		
(R) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$		3. 0		
(S) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$		4. $-1.2 \Delta_0$		
	P	Q	R	S
(1)	3	1	2	4
(2)	1	2	3	4
(3)	4	3	2	1
(4)	None of these			

13. Crystal field stabilization energy for high spin d^4 octahedral complex is:



14. Among the following ions, which one has the highest paramagnetism?
 (1) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ (2) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
 (3) $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ (4) $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$
15. Which of the following complex formed by Cu^{+2} ions is most stable?
 (1) $\text{Cu}^{+2} + 4\text{NH}_3 \longrightarrow [\text{Cu}(\text{NH}_3)_4]^{+2}$
 (2) $\text{Cu}^{+2} + 4\text{CN}^- \longrightarrow [\text{Cu}(\text{CN})_4]^{-2}$
 (3) $\text{Cu}^{+2} + 2\text{en} \longrightarrow [\text{Cu}(\text{en})_2]^{+2}$
 (4) $\text{Cu}^{+2} + 4\text{H}_2\text{O} \longrightarrow [\text{Cu}(\text{H}_2\text{O})_4]^{+2}$
16. Which complex has highest paramagnetism?
 (1) $[\text{Cr}(\text{H}_2\text{O})_6]^{+3}$ (2) $[\text{Fe}(\text{H}_2\text{O})_6]^{+2}$
 (3) $[\text{Cu}(\text{H}_2\text{O})_6]^{+2}$ (4) $[\text{Zn}(\text{H}_2\text{O})_6]^{+2}$
17. Same number of unpaired electron is observed in which of the following complexes
 (a) $[\text{MnCl}_6]^{3-}$ (b) $[\text{Fe}(\text{CN})_6]^{3-}$
 (c) $[\text{CoF}_6]^{3-}$ (d) $[\text{Ni}(\text{NH}_3)_4]^{2+}$
 (1) a and b (2) a and c
 (3) b and d (4) c and d
18. Amongst the following the most stable compound is
 (1) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ (2) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$
 (3) $[\text{Fe}(\text{NH}_3)_6]^{3+}$ (4) $[\text{FeCl}_6]^{3-}$
19. Which of the following system has maximum value of μ (only spin magnetic moment)?
 (1) d^5 ($\Delta_0 > P$) (2) d^8 (tetrahedral)
 (3) d^6 (high spin) (4) d^9 (octahedral)
20. Hexafluoroferrate(III) ion is an outer orbital Complex, the number of unpaired electrons present in it is
 (1) 1 (2) 5
 (3) 4 (4) unpredictable
21. Among the following ions, which one has highest paramagnetism
 (1) $[\text{Cr}(\text{NH}_3)_6]^{+3}$ (2) $[\text{Fe}(\text{H}_2\text{O})_6]^{+2}$
 (3) $[\text{Zn}(\text{H}_2\text{O})_6]^{+2}$ (4) $[\text{Cu}(\text{H}_2\text{O})_6]^{+2}$
22. Spin only magnetic moments of a d^8 ion in octahedral, square planar and tetrahedral complex, respectively are
 (1) 2.8 BM, 0.4 B.M., 2.8 BM
 (2) 2.8 BM, 0 BM, 2.8 BM
 (3) 0, 0 & 0 B.M.
 (4) None of these
23. Among the following the colored compound is
 (1) CuCl (2) $\text{K}_3[\text{Cu}(\text{CN})_4]$
 (3) CuF_2 (4) $[\text{Cu}(\text{H}_2\text{O})_4]\text{BF}_4$
24. CFSE of high spin d^4 complex is
 (1) $-0.6 \Delta_0$ (2) $-1.8 \Delta_0$
 (3) $-1.6 \Delta_0 + P$ (4) $-1.2 \Delta_0$
25. Which ion would you expect to have the maximum splitting of d-orbitals
 (1) $[\text{Fe}(\text{CN})_6]^{-4}$ (2) $[\text{Fe}(\text{CN})_6]^{-3}$
 (3) $[\text{Fe}(\text{H}_2\text{O})_6]^{+2}$ (4) $[\text{Fe}(\text{H}_2\text{O})_6]^{+3}$

Organometallic Compounds or C-Metal bond compounds

Definition : Compounds containing one or more metal carbon bonds.

(a) The compounds of metalloids (Ge, Sb) and non-metals (B, Si, P, As etc.) which are less electro - ve than carbon are also categorised in this title.

Example: $\left\{ \begin{array}{l} \text{R-Zn-R} \text{ dialkyl zinc (Frankland reagent)} \\ \text{R-Mg-X} \text{ Alkyl magnesium halide (Grignards reagent)} \end{array} \right.$

(b) $\left. \begin{array}{l} \text{Sodium acetate} \quad \text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{ONa} \\ \text{Sodium ethoxide} \quad \text{C}_2\text{H}_5-\text{O}-\text{Na} \\ \text{Sodium Mercaptide} \quad \text{H}_3\text{C}-\text{C}-\text{SNa} \end{array} \right\}$ These are not organometallic compounds because in these Compounds metal is not directly attached with carbon atom

(c) $\text{B}(\text{OCH}_3)_3$ is not OMC $\rightarrow \begin{array}{c} \text{H}_3\text{CO} \\ \diagup \\ \text{B}-\text{OCH} \\ \diagdown \\ \text{H}_3\text{CO} \end{array}$ and $\text{CH}_3\text{B}(\text{OCH}_3)_2$ is OMC $\begin{array}{c} \text{OCH}_3 \\ \diagup \\ \text{CH}_3-\text{B} \\ \diagdown \\ \text{OCH}_3 \end{array}$

(d) Metal cyanides, carbides and carbonates are not considered OMC because in these compounds metal carbon bond has ionic nature.

Classification of Organometallic Compounds

OMC can be classified into three main categories–

σ - bonded OMC

These compounds are formed mostly by non transition and metalloid elements.

Example:

R – Mg – X,	(Grignard's reagent)
(CH ₃) ₂ Zn	(Dimethyl zinc or Frankland's reagent)
(R) ₂ Cd,	(dialkyl cadmium)
(C ₂ H ₅) ₄ Pb	(Tetra ethyl lead)
(CH ₃) ₃ Al, TiCl ₄	(Ziegler natta catalyst)

π - bonded OMC

These are usually formed by transition metals.

Example:

Ferrocene	[Fe(η^5 - C ₅ H ₅) ₂] where η = No. of carbon atoms combined with metal atom.
Zeise's salt	K[PtCl ₃ (η^2 - C ₂ H ₄)]
Dibenzene chromium	[Cr(η^6 - C ₆ H ₆) ₂]

σ and π - bonded OMC

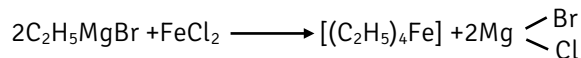
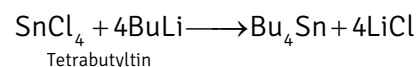
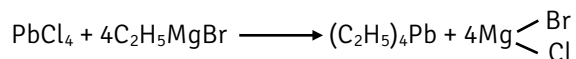
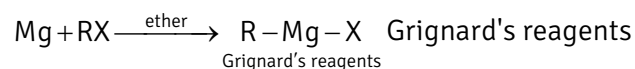
Transition metals of gp. 6, 7, 8, 9 and 10 in their zero-oxidation state form such type of OMC. The carbonyl compounds of these metals have both σ and π bonds.

Example:

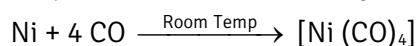
[Ni(CO)₄], [Fe(CO)₅], [Mn₂(CO)₁₀]
[M(CO)₆] where M is Cr, Mo, W.

Synthesis of Organometallic Compounds

(1) By the reaction of metal or metal halide with the desired alkyl halide in presence of dry ether.



(2) Preparation of metal carbonyl

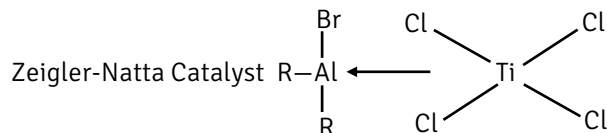


Application of Organometallic Compounds

(a) As homogeneous Catalyst

Wilkinson's catalyst for hydrogenation of olefins is [P(Ph₃)₃RhCl]

(b) As heterogeneous catalyst



Tri alkyl Al ← Titanium tetrachloride used for polymerisation of alkene.

- (c) Organic synthesis – OMC like $R - Mg - X$, $R_2 - Cd$, $CH_3 - Li$ etc. are used for preparation of almost all type of organic compounds .
- (d) As petrol additive - TEL(Tetraethyl lead) is used as antiknock compd. in petrol.
- (e) In medicine -Organo arsenic compounds are used as medicine for syphilis disease.
- (f) In agriculture - Seeds are treated with ethyl Hg Chloride to protect the plants against infection.

NCERT QUESTIONS (Solve Yourself)

- Aqueous copper sulphate solution (blue in colour) gives:
 - a green precipitate with aqueous potassium fluoride and
 - a bright green solution with aqueous potassium chloride. Explain these experimental results.
- What is the coordination entity formed when excess of aqueous KCN is added to an aqueous solution of copper sulphate? Why is it that no precipitate of copper sulphide is obtained when $H_2S(g)$ is passed through this solution?
- A solution of $[Ni(H_2O)_6]^{2+}$ is green but a solution of $[Ni(CN)_4]^{2-}$ is colourless. Explain.
- $[Fe(CN)_6]^{4-}$ and $[Fe(H_2O)_6]^{2+}$ are of different colours in dilute solutions. Why?
- Discuss the nature of bonding in metal carbonyls.
- Discuss briefly giving an example in each case the role of coordination compounds in:
 - biological systems
 - medicinal chemistry and
 - analytical chemistry
 - extraction/metallurgy of metals.
- How many ions are produced from the complex $Co(NH_3)_6Cl_2$ in solution?
 - 6
 - 4
 - 3
 - 2
- Amongst the following ions which one has the highest magnetic moment value?
 - $[Cr(H_2O)_6]^{3+}$
 - $[Fe(H_2O)_6]^{2+}$
 - $[Zn(H_2O)_6]^{2+}$

Exercise 1.5

- Which of the following is π -acid ligand
 - NH_3
 - CO
 - gly.
 - ethylene diamine
- Which of the following statement is/are **wrong**:
 - Al_4C_3 is an organometallic compound
 - Metal carbonyls are organometallic compounds
 - TEL is π bonded organometallic compound
 - Frankland reagent is σ -bonded organometallic compound

The answer is:-

 - c and d
 - a and c
 - a and b
 - All are correct
- Compounds which contain one or more metal-carbon bonds are called:
 - Organic compounds
 - Complex compounds
 - Metal carbides
 - OMC compounds.
- Which one of the following is used as a heterogeneous catalyst?
 - Wilkinson's catalyst
 - Tetraethyl lead
 - Zeigler Natta catalyst
 - Grignard's reagent
- Which of the following statement is **true**
 - $FeCO_3$ and Fe_3C are organometallic compounds.
 - In ferrocene, ligand is cyclopentadienyl.
 - $Pb(C_2H_5)_4$ is π -bonded OMC
 - In zeise salt central metal is sp^3 hybridised.
- Synergic bonding involves:-
 - The transference of electrons from ligands to metal
 - The transference of electrons from filled metal orbitals to anti-bonding orbitals of ligands
 - Both the above
 - None of these

7. OMC form during purification of a metal in Mond's process:-
 (1) Ni(CO)_4 (2) $\text{Pb(C}_2\text{H}_5)_4$
 (3) $\text{Li-C}_4\text{H}_9$ (4) $\text{Na}_2[\text{Ni(CN)}_4]$
8. Which of the following is **not** an organometallic compound?
 (1) $(\text{C}_2\text{H}_5)_2\text{Zn}$ (2) $\text{CH}_3\text{B(OCH}_3)_2$
 (3) $\text{B(OCH}_3)_3$ (4) Ni(CO)_4
9. What is **wrong** about the compound $\text{K}[\text{Pt}(\eta^2\text{-C}_2\text{H}_4)\text{Cl}_3]$
 (1) It is called Zeise's salt
 (2) It is π bonded complex
 (3) Oxidation number of Pt is +4
 (4) Four ligands surrounds the Platinum ion.
10. Which is **not** a π -bonded complex?
 (1) Zeise salt
 (2) Ferrocene
 (3) bis(benzene) chromium
 (4) Tetraethyl lead
11. Oxidation state of central metal in Zeise's salt?
 (1) +2 (2) +1
 (3) +3 (4) 0
12. Which of the following is an organometallic compound
 (1) cis-platin
 (2) Zeise salt
 (3) Tollen's reagent
 (4) Sodium nitroprusside
13. Formula of sodium Nitroprusside is?
 (1) $\text{Na}_4[\text{Fe(CN)}_5\text{NOS}]$
 (2) $\text{Na}_2[\text{Fe(CN)}_5\text{NO}]$
 (3) $\text{NaFe[Fe(CN)}_6]$
 (4) $\text{Na}_2[\text{Fe(CN)}_6\text{NO}_2]$
14. Ziegler - Natta catalyst is an organometallic compound of Metal?
 (1) Fe (2) Zr
 (3) Rh (4) Ti
15. When excess of ammonia is added to CuSO_4 solution. The deep blue colour complex is formed. The complex is
 (1) Tetrahedral & paramagnetic
 (2) tetrahedral & diamagnetic
 (3) square planar & diamagnetic
 (4) square planar & paramagnetic
16. When CuSO_4 , reacts with excess of KCN it forms a soluble complex which is
 (1) $\text{K}_2[\text{Cu(CN)}_4]$
 (2) $\text{K}_3[\text{Cu(CN)}_4]$
 (3) Cu(CN)_2
 (4) CuCN
17. A brown ring is formed in the ring test of NO_3^- ion. It is due to the formation of-
 (1) $[\text{Fe(H}_2\text{O)}_5(\text{NO})]^{2+}$
 (2) $\text{FeSO}_4 \cdot \text{NO}$
 (3) $[\text{Fe(H}_2\text{O)}_4(\text{NO})_2]^{2+}$
 (4) $\text{FeSO}_4 \cdot \text{HNO}_3$

Importance and Applications of Coordination Compounds:

- Coordination compounds are of great importance in biological systems. The pigment responsible for photosynthesis, chlorophyll, is a coordination compound of magnesium. Haemoglobin, the red pigment of blood which acts as oxygen carrier is a coordination compound of iron. Vitamin B_{12} , cyanocobalamin, the anti-pernicious anaemia factor, is a coordination compound of cobalt. Among the other compounds of biological importance with coordinated metal ions are the enzymes like, carboxypeptidase A and carbonic anhydrase (catalysts of biological systems).
- Coordination compounds are used as catalysts for many industrial processes. Examples include rhodium complex, $[(\text{Ph}_3\text{P})_3\text{RhCl}]$, a Wilkinson catalyst, is used for the hydrogenation of alkenes.
- Articles can be electroplated with silver and gold much more smoothly and evenly from solutions of the complexes, $[\text{Ag(CN)}_2]^-$ and $[\text{Au(CN)}_2]^-$ than from a solution of simple metal ions.
- In black and white photography, the developed film is fixed by washing with hypo solution which dissolves the undecomposed AgBr to form a complex ion, $[\text{Ag(S}_2\text{O}_3)_2]^{3-}$.

Exercise 1.6

1. The brown ring test for nitrites and nitrates is due to the formation of a complex ion with formula:-
 (1) $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}^+]^{2+}$ (2) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
 (3) $[\text{Fe}(\text{H}_2\text{O})(\text{NO})_5]^{2-}$ (4) $[\text{Fe}(\text{NO})(\text{CN})_5]^{2+}$
2. In a ferric salt on adding $\text{K}_4[\text{Fe}(\text{CN})_6]$ a prussian blue coloured compound is obtained which is:-
 (1) $\text{K}_3[\text{Fe}(\text{CN})_6]$ (2) $\text{Fe}_3[\text{Fe}(\text{CN})_6]_4$
 (3) $\text{FeSO}_4[\text{Fe}(\text{CN})_6]$ (4) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
3. Hypo is used in photography because it is:-
 (1) A strong reducing agent
 (2) A strong oxidising agent
 (3) A strong Complexing agent
 (4) Photo sensitive Compound
4. The solubility of AgBr in hypo solution is due to the formation of:-
 (1) Ag_2SO_3 (2) $\text{Ag}_2\text{S}_2\text{O}_3$
 (3) $[\text{Ag}(\text{S}_2\text{O}_3)]^-$ (4) $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$
5. Which of the following is related to Nessler's reagent?
 (1) $\text{PtCl}_4 + \text{KCl} \rightarrow$
 (2) $\text{AgCl} + \text{NH}_3 \rightarrow$
 (3) $\text{AgBr} + \text{Na}_2\text{S}_2\text{O}_3 \rightarrow$
 (4) $\text{HgI}_2 + \text{KI} \rightarrow$
6. $\text{K}_4[\text{Fe}(\text{CN})_6]$ (excess) reacts with FeCl_3 to form:-
 (1) $\text{K}_3\text{Fe}(\text{CN})_6$ (2) $\text{K}_4[\text{Fe}(\text{CN})_3\text{Cl}_3]$
 (3) $\text{K}_3[\text{Fe}(\text{CN})_5\text{Cl}]$ (4) $\text{KFe}[\text{Fe}(\text{CN})_6]$
7. A reagent used for identifying nickel ion is:
 (1) Potassium ferrocyanide
 (2) Phenolphthalein
 (3) Dimethyl glyoxime
 (4) EDTA
8. When CuCl reacts with excess KCl then the product obtained will be?
 (1) $\text{K}_3[\text{CuCl}_4]$ (2) $\text{K}_2[\text{CuCl}_4]$
 (3) $\text{K}[\text{CuCl}_2]$ (4) $\text{K}_2[\text{CuCl}_2]$
9. During developing of a photographic plate, one of the step involves washing of plate with hypo solution. The purpose of this is to
 (1) Reduce AgBr to Ag
 (2) Convert AgBr to soluble thiosulphate complex
 (3) Convert AgBr to Ag_2SO_4
 (4) Convert AgBr to $\text{Ag}_2\text{S}_2\text{O}_3$
10. Dimethyl glyoxime is used for the test of in alkaline medium :
 (1) Cobalt (2) Zinc
 (3) Manganese (4) Nickel
11. When dimethyl glyoxime solution is added to an aqueous solution of nickel (II) chloride followed by ammonium hydroxide :
 (1) No precipitate is obtained
 (2) A blue coloured ppt. is obtained
 (3) A rosy red coloured ppt. is obtained
 (4) A black coloured ppt. is obtained
12. During the qualitative analysis of salt with cation y^{2+} , addition of a reagent (X) to alkaline solution of the salt gives a bright red precipitate. The reagent (X) and the cation (y^{2+}) present respectively are :
 (1) Dimethylglyoxime and Ni^{2+}
 (2) Dimethylglyoxime and Cu^{2+}
 (3) Nessler's reagent and Pb^{2+}
 (4) Nessler's reagent and Ni^{2+}

Exercise 2

- Which of the following react with AgCl ?
 (1) KCN (2) NH_4OH
 (3) $\text{Na}_2\text{S}_2\text{O}_3$ (4) All
- The most stable complex among the following is
 (1) $\text{K}_3[\text{Al}(\text{C}_2\text{O}_4)_3]$ (2) $[\text{Pt}(\text{en})_2]\text{Cl}_2$
 (3) $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$ (4) $\text{K}_2[\text{Ni}(\text{EDTA})]$
- The coordination number of Ni^{+2} is = 4
 $\text{NiCl}_2 + \text{KCN}(\text{Complex}) \rightarrow \text{A}(\text{Complex}) + \text{A} + \text{Conc. HCl}(\text{Maximum}) \rightarrow \text{B}(\text{Chloro complex})$
 The IUPAC name of A and B are
 (1) Potassium tetracyanonickelate (II)
 Potassium tetrachloronickelate (II)
 (2) Tetracyanopotassiumnickelate (II),
 Tetrachloro potassiumnickelate (II),
 (3) Tetracyanonickel (II),
 Tetrachloronickel (II)
 (4) Potassiumtetracyanonickel (II),
 Potassiumtetrachloronickel (II).
- The octahedral complex of a metal ion M^{3+} with four monodentate ligands $\text{L}_1, \text{L}_2, \text{L}_3$ and L_4 absorb wavelength in the region of red, green, yellow and blue, respectively. The increasing order of ligand strength of the four ligands is:
 (1) $\text{L}_3 < \text{L}_2 < \text{L}_4 < \text{L}_1$ (2) $\text{L}_1 < \text{L}_2 < \text{L}_4 < \text{L}_3$
 (3) $\text{L}_4 < \text{L}_3 < \text{L}_2 < \text{L}_1$ (4) $\text{L}_1 < \text{L}_3 < \text{L}_2 < \text{L}_4$
- Read the following statements:
 (I) Geometrical isomerism is not observed in square planar complexes whereas optical isomerism may be observed.
 (II) Square planar complexes of the type Ma_2b_2 exhibit optical isomerism and two geometrical isomers.
 (III) Square planar complex of the type Mabcd , shows three isomers—two cis and one trans.
 (IV) In $\text{fac-Co}[(\text{NH}_3)_3(\text{NO}_2)_3]$ the three identical ligands are present on the triangular face of the octahedral.
 The option having **incorrect** statements is:
 (1) I, III & IV (2) I & II
 (3) II only (4) I, II, III & IV
- Choose the correct set representing the given statements as true (T) of false (F).
 (I) Coordination isomerism arises from the interchange of ligands between cationic and anionic entities present in a complex.
 (II) Thiocyanate ligand is an ambidentate ligand.
 (III) Ionisation isomerism arises when the counter ion in a complex salt is itself a potential ligand and can displace a ligand which can then become the counter ion.
 (IV) $[\text{Co}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$ do not exhibit coordination isomerism
 (1) FTTT (2) TTTT
 (3) TTTF (4) FTFT
- Identify the **incorrect** one
 (1) In octahedral complexes, the magnetic behaviour of metal ions having upto three electrons in the d orbitals in their coordination entities is independent on the nature of ligand.
 (2) Complex ions having d^4, d^5, d^6 & d^7 configuration with presence of SFL are low spin complexes.
 (3) Magnetic moment of coordination compounds is independent of nature of ligands.
 (4) $[\text{Mn}(\text{CN})_6]^{-3}$ has magnetic moment of two unpaired electrons whereas $[\text{MnCl}_6]^{-3}$ has four unpaired electrons.
- Among the following statements the **incorrect** one
 (1) The crystal field theory is an electrostatic model which considers the metal ligand bond to be ionic.
 (2) The five d-orbitals of an isolated gaseous metal atom/ion have same energy.
 (3) The degeneracy of d-orbitals of metal is lost if a spherically symmetrical field of negative charges surrounds the metal atom/ion
 (4) The pattern of splitting of d-orbitals depends upon the nature of the crystal field.

9. **Incorrect** statement among the following is:

- (1) If $\Delta_0 < P$ the fourth electron enters one of e_g orbitals giving the configuration $t_{2g}^3 e_g^1$.
- (2) If $\Delta_0 > P$ the fourth electron enters one of t_{2g} orbitals giving the configuration $t_{2g}^4 e_g^0$.
- (3) If $\Delta_0 > P$ then e^- configuration of metal ion d^6 will be represented as $t_{2g}^4 e_g^2$.
- (4) All of the above

10. If the complex ion $[\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{+3}$ absorbs blue-green radiation; then the colour of the radiation absorption by $[\text{Co}(\text{NH}_3)_5(\text{Cl})]^{+2}$ will be:-

- (1) Blue
- (2) Yellow
- (3) Violet
- (4) Indigo

11. Which is **correct**:-

	Compound	Number of Stereoisomers
(1)	$[\text{Cr}(\text{en})_2\text{Cl}_2]\text{Cl}$	4
(2)	$[\text{Cr}(\text{en})_3]\text{Cl}_3$	3
(3)	$[\text{Pt}(\text{gly})_2]$	2
(4)	$[\text{Ni}(\text{dmg})_2]$	2

12. The solubility of AgCN increases by the addition of KCN because of:-

- (1) Complex formation
- (2) Redox change
- (3) Salt formation
- (4) None of these

13. The number of ions gives by $\text{K}_2[\text{PtCl}_6]$ in aqueous solution is:-

- (1) 2
- (2) 3
- (3) 4
- (4) zero

14. Carbon donor ligands are strong ligands and usually forms low spin complexes. Among the complexes given below, select the high spin complex:

- (1) $[\text{Fe}(\text{CN})_6]^{3-}$
- (2) $[\text{Fe}(\text{CN})_6]^{4-}$
- (3) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$
- (4) None of these

15. Which is **correct**:-

Complex	No. of Rings
(1) $[\text{Co}(\text{ox})_2\text{Cl}_2]^{-3}$	4
(2) $[\text{Ni}(\text{dmg})_2]$	4
(3) $[\text{Ca}(\text{EDTA})]^{-4}$	6
(4) Brown ring complex	1

16. Select the **correct** statement:

- (1) Potassium ferrocyanide and potassium ferricyanide can be differentiated by measuring the solid state magnetic moment
- (2) The complex $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ can be differentiated by adding aqueous solution of barium chloride
- (3) The complex $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{Cl}$ can be differentiated by adding aqueous solution of silver nitrate
- (4) All of these

17. Which of the following is **not** blue coloured compound?

- (1) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
- (2) $[\text{Cu}(\text{H}_2\text{O})_4]\text{SO}_4 \cdot \text{H}_2\text{O}$
- (3) $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$
- (4) $\text{Cu}_2[\text{Fe}(\text{CN})_6]$

18. It is an experiment fact that:
 $\text{DMG} + \text{Ni(II)salt} + \text{NH}_4\text{OH} \longrightarrow \text{Red ppt.}$
 Which of the following is wrong about this red ppt:

- (1) It is a non-ionic complex
- (2) It involves intra molecular H-bonding
- (3) Ni(II) is sp^3 hybridised
- (4) It is a diamagnetic complex

19. Zinc (II) ion on reaction with NaOH first give a white precipitate which dissolves in excess of NaOH due to the formation of:

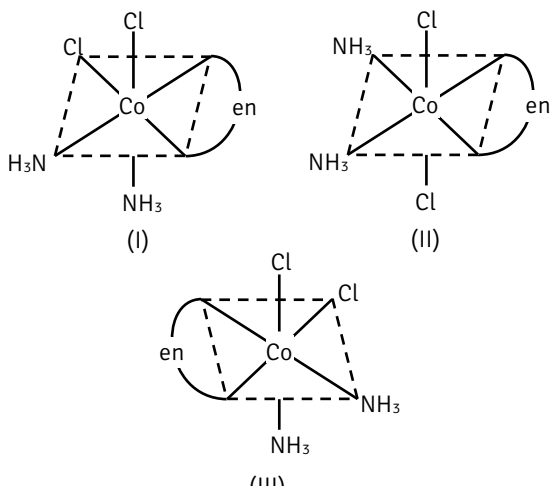
- (1) ZnO
- (2) $\text{Zn}(\text{OH})_2$
- (3) $[\text{Zn}(\text{OH})_4]^{2-}$
- (4) $[\text{Zn}(\text{H}_2\text{O})_4]^{2+}$

20. Which of the following **cannot** show linkage isomerism?

- (1) NO_2^-
- (2) SCN^-
- (3) CN^-
- (4) NH_3

21. The complexes $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_6]$ and $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2][\text{PtCl}_4]$ are

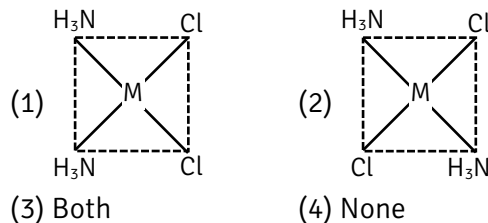
- (1) linkage isomers
- (2) optical isomers
- (3) co-ordination isomers
- (4) ionisation isomers

22. EDTA is used for the estimation of
 (1) Na^+ and K^+ ions
 (2) Halide ions
 (3) Cu^{2+} and Ag^+ ions
 (4) Ca^{2+} and Mg^{2+} ions
23. A compound is made by mixing cobalt (III) nitrite and potassium nitrite solutions in the ratio of 1: 3. The aqueous solution of the compound showed 4 particles per molecule whereas molar conductivity reveals the presence of six electrical charges. The formula of the compound is
 (1) $\text{Co}(\text{NO}_2)_3 \cdot 2\text{KNO}_2$ (2) $\text{Co}(\text{NO}_2)_3 \cdot 4\text{KNO}_2$
 (3) $\text{K}_3[\text{Co}(\text{NO}_2)_6]$ (4) $\text{K}[\text{Co}(\text{NO}_2)_4]$
24. Which of the following will exhibit maximum ionic conductivity?
 (1) $\text{K}_4[\text{Fe}(\text{CN})_6]$ (2) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
 (3) $[\text{Cu}(\text{NH}_3)_4]\text{Cl}_2$ (4) $[\text{Ni}(\text{CO})_4]$
25. Three arrangements are shown for the complex $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$. Pick up the wrong statement.

 (1) I and II are geometrical isomers
 (2) II and III are optical isomers
 (3) I and III are optical isomers
 (4) II and III are geometrical isomers
26. The compound which does not show paramagnetism:
 (1) $[\text{Cu}(\text{NH}_3)_4]\text{Cl}_2$ (2) $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$
 (3) NO (4) NO_2

27. The number of d-electrons of $\text{Cr}(Z = 24)$ in $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ ion:
 (1) 2 (2) 3 (3) 4 (4) 5
28. In sodium nitroprusside, the oxidation number, co-ordination number and EAN of iron are respectively
 (1) +3, 6, 36 (2) +2, 6, 36
 (3) +3, 3, 36 (4) 6, +3, 35
29. Which of the following is copper (I) compound:
 (1) $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ (2) $[\text{Cu}(\text{CN})_4]^{3-}$
 (3) $[\text{Cu}(\text{NH}_3)_4]^{++}$ (4) All of these
30. Low spin complex is formed by:
 (1) sp^3d^2 hybridization
 (2) sp^3d hybridization
 (3) d^2sp^3 hybridization
 (4) sp^3 hybridization
31. Which of the following is a high spin complex:
 (1) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (2) $[\text{Fe}(\text{CN})_6]^{4-}$
 (3) $[\text{Ni}(\text{CN})_4]^{2-}$ (4) $[\text{FeF}_6]^{3-}$
32. The number of ions per mole of the complex $\text{CoCl}_3 \cdot 5\text{NH}_3$ in aqueous solution will be:
 (1) Four (2) Nine (3) Three (4) Two
33. In the complex $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Br}$, the co-ordination number and oxidation state of cobalt are:
 (1) 6 and +3 (2) 3 and +3
 (3) 4 and +2 (4) 6 and +1
34. Pick out from the following complex compounds, a poor electrolytic conductor in solution:
 (1) $\text{K}_2[\text{PtCl}_6]$
 (2) $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$
 (3) $\text{K}_4[\text{Fe}(\text{CN})_6]$
 (4) $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$
35. Co-ordination number of Cr in $\text{CrCl}_3 \cdot 5\text{H}_2\text{O}$ is six. The volume of 0.1 N AgNO_3 needed to ppt. the chlorine in outer sphere in 200 ml of 0.01 M solution of the complex is:
 (1) 140 ml (2) 40 ml
 (3) 80 ml (4) 20 ml

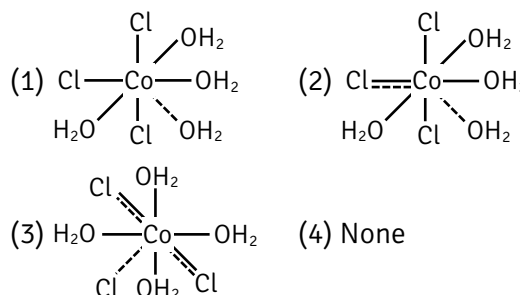
36. Which of the following complexes has a geometry different from others:
 (1) $[\text{NiCl}_4]^{2-}$ (2) $\text{Ni}(\text{CO})_4$
 (3) $[\text{Ni}(\text{CN})_4]^{2-}$ (4) $\text{Zn}(\text{NH}_3)_4^{++}$
37. Which of the following shows maximum molar conductance:
 (1) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (2) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
 (3) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ (4) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
38. Which of the following pairs of name and formula of complexes, is not correct:
 (1) Tetramminecopper(II)sulphate $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$
 (2) Diamminesilver (I) chloride $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$
 (3) Potassium hexacyanoferrate (III) $\text{K}_4[\text{Fe}(\text{CN})_6]$
 (4) Potassium amminepentachloroplatinate (IV) $\text{K}[\text{PtCl}_5(\text{NH}_3)]$
39. Conc. H_2SO_4 will not dehydrate the following complex:
 (1) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$
 (2) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$
 (3) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$
 (4) All of these
40. On adding AgNO_3 solution to a solution of $[\text{Pt}(\text{NH}_3)_3\text{Cl}_3]\text{Cl}$, the percentage of total chloride ion precipitated is:
 (1) 100 (2) 75 (3) 50 (4) 25
41. $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ ion is:
 (1) Coloured and paramagnetic
 (2) Colourless and paramagnetic
 (3) Colourless and diamagnetic
 (4) Coloured and octahedral
42. A co-ordination complex has the formula $\text{PtCl}_4 \cdot 2\text{KCl}$. Electrical conductance measurements indicate the presence of three ion in one formula unit. Treatment with AgNO_3 produces no ppt. of AgCl . What is the co-ordination number of Pt in this complex:
 (1) 5 (2) 6 (3) 4 (4) 3
43. EAN of a metal carbonyl $\text{M}(\text{CO})_x$ is 36. If atomic number of metal M is 26, what is the value of x:
 (1) 4 (2) 8 (3) 5 (4) 6

44. Which of the following isomers of $[\text{M}(\text{NH}_3)_2\text{Cl}_2]$ would react with silver oxalate ($\text{Ag}_2\text{C}_2\text{O}_4$):



45. Among the following ions which has the highest degree of paramagnetism:
 (1) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ (2) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
 (3) $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ (4) $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$

46. $\text{CoCl}_3 \cdot 4\text{H}_2\text{O}$ is an anhydrous binary solute hence its Werner's representation is:



47. The effective atomic number of $\text{Co}(\text{CO})_4$ is 35 and hence is less stable. It attains stability by
 (1) oxidation of Co (2) reduction of Co
 (3) dimerization (4) Both (2) & (3)
48. The value of magnetic moment for a complex ion is 1.73 BM. The complex ion is
 (1) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (2) $[\text{MnF}_6]^{3-}$
 (3) $[\text{Fe}(\text{CN})_5(\text{H}_2\text{O})]^{3-}$ (4) $[\text{Mn}(\text{CN})_6]^{-4}$
49. One mole of the complex compound $\text{CoCl}_3 \cdot 5\text{NH}_3$ gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with two moles of AgNO_3 solution to yield two moles of $\text{AgCl}(s)$. The structure of the complex is
 (1) $[\text{Co}(\text{NH}_3)_4\text{Cl}]\text{Cl}_2 \cdot \text{NH}_3$
 (2) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
 (3) $[\text{Co}(\text{NH}_3)_3\text{Cl}] \cdot 2\text{NH}_3$
 (4) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \cdot \text{NH}_3$

50. For $[\text{Fe}(\text{F})_6]^{3-}$ and $[\text{Fe}(\text{CN})_6]^{3-}$ magnetic moment of the fluoride complex is expected to be:

- (1) The same as the magnetic moment of the cyanide complex
- (2) Larger than the magnetic moment of the cyanide complex because there are more unpaired electrons in the fluoride complex
- (3) Smaller than the magnetic moment of the cyanide complex because there are more unpaired electrons in the fluoride complex
- (4) Larger than the magnetic moment of the cyanide complex because there are fewer unpaired electrons in the fluoride complex

51. $\text{CuCl}_2 + \text{K}_4[\text{Fe}(\text{CN})_6] \rightarrow \text{Chocolate brown ppt(X)}$
Select the **correct** statement for (X):

- (1) Its IUPAC name is copper (I) hexacyanoferrate (II)
- (2) It reacts with excess potassium cyanide forming another soluble complex which has tetrahedral geometry
- (3) It has 'spin only' magnetic moment equal to $\sqrt{36}$ B.M.
- (4) 2 and 3 both

52. S_1 : A solution of $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ is green but a solution of $[\text{Ni}(\text{CN})_4]^{2-}$ is colourless.

S_2 : Energy difference between d levels (i.e. Δ) for H_2O complex is in the visible region and that for the cyano complex is in the UV region.

- (1) Both statements are correct and S_2 is the correct answer of S_1
- (2) Both statements are correct and S_2 is not the correct answer of S_1
- (3) S_1 is correct and S_2 is incorrect
- (4) S_2 is false and S_1 is correct

53. Wilkinson's catalyst react with H_2 to form an octahedral complex in which $\text{Rh}(\text{Z}=45)$ has the following electronic configuration in the ligand field $t_{2g}^{2,2,2}, e_g^{0,0}$. Then which of the following is **correct** about this new complex.

- (1) It is paramagnetic
- (2) Its IUPAC name is chlorodihydridotris(triphenylphosphine) rhodium (III)
- (3) Hybridisation of $\text{Rh}(\text{I})$ is d^2sp^3
- (4) None of these

54. Extraction of metals of other processes is through the complex formation:-

I: cyanide process

II: Mond's process

III: Photographic fixing process

Complexes formed in these methods are

	I	II	III
(1)	$[\text{Ag}(\text{NH}_3)_2]\text{Cl}$	$\text{Ni}(\text{CO})_4$	$[\text{Ag}(\text{CN})_2]^-$
(2)	$[\text{Cd}(\text{CN})_4]^{2-}$	$\text{Ni}(\text{CO})_4$	$[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$
(3)	$[\text{Ag}(\text{CN})_2]^-$	$\text{Ni}(\text{CO})_4$	$[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$
(4)	$[\text{Ag}(\text{CN})_2]^-$	$[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$	$\text{Ni}(\text{CO})_4$

55. Identify the optically active compounds from the following:

(a) $[\text{Co}(\text{en})_3]^{3+}$

(b) $\text{trans-}[\text{Co}(\text{en})_2\text{Cl}_2]^+$

(c) $\text{cis-}[\text{Co}(\text{en})_2\text{Cl}_2]^+$

(d) $[\text{Cr}(\text{NH}_3)_5\text{Cl}]$

(1) a, c

(2) c, d

(3) b, c, d

(4) a, b, c

56. Which of the following complexes formed by Cu^{2+} ions is most stable?

(1) $\text{Cu}^{2+} + 4\text{NH}_3 \rightleftharpoons [\text{Cu}(\text{NH}_3)_4]^{2+}$,
 $\log K=11.6$

(2) $\text{Cu}^{2+} + 4\text{CN}^- \rightleftharpoons [\text{Cu}(\text{CN})_4]^-$,
 $\log K=27.3$

(3) $\text{Cu}^{2+} + 2\text{en} \rightleftharpoons [\text{Cu}(\text{en})_2]^{2+}$,
 $\log K=15.4$

(4) $\text{Cu}^{2+} + 4\text{H}_2\text{O} \rightleftharpoons [\text{Cu}(\text{H}_2\text{O})_4]^{2+}$,
 $\log K = 8.9$

57. When 0.1 mole $\text{CoCl}_3(\text{NH}_3)_5$ is treated with excess of AgNO_3 , 0.2 mole of AgCl are obtained. The conductivity of solution will correspond to

(1) 1: 3 electrolyte (2) 1: 2 electrolyte

(3) 1: 1 electrolyte (4) 3: 1 electrolyte

58. A chelating agent has two or more than two donor atoms to bind to a single metal ion. Which of the following is not a chelating agent?

(1) thiosulphato

(2) oxalato

(3) glycinate

(4) ethane-1, 2-diamine

59. The pair in which both species have same magnetic moment (spin only value) is:
 (1) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$, $[\text{CoCl}_4]^{2-}$
 (2) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
 (3) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$
 (4) $[\text{CoCl}_4]^{2-}$, $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
60. The number of possible isomers of an octahedral complex $[\text{Co}(\text{C}_2\text{O}_4)_2(\text{NH}_3)_2]^-$ is:
 (1) 1 (2) 2 (3) 3 (4) 4
61. The ligands in anti cancer drug cisplatin are:
 (1) NH_3 , Cl (2) NH_3 , H_2O
 (3) Cl , H_2O (4) NO , Cl
62. Given below, catalyst and corresponding process/reaction are matched. The mismatch is:
 (1) $[\text{RhCl}(\text{PPh}_3)_2]$: Hydrogenation
 (2) $\text{TiCl}_4 + \text{Al}(\text{C}_2\text{H}_5)_3$: Polymerization
 (3) V_2O_5 : Haber-Bosch process
 (4) Nickel: Hydrogenation
63. Among the following the species having square planar geometry for central atom are:
 (i) XeF_4 (ii) SF_4
 (iii) $[\text{NiCl}_4]^{2-}$ (iv) $[\text{PdCl}_4]^{2-}$
 (1) i and iv (2) i and ii
 (3) ii and iii (4) iii and iv
64. In which of the following compound C-O bond length is maximum:
 (1) $[\text{Fe}(\text{CO})_5]^-$ (2) $[\text{Ni}(\text{CO})_4]$
 (3) $[\text{Cr}(\text{CO})_6]$ (4) $[\text{Mn}(\text{CO})_5]^+$
65. The wave length of light absorbed is highest in:
 (1) $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ (2) $[\text{Co}(\text{NH}_3)_5\text{H}_2\text{O}]^{3+}$
 (3) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (4) $[\text{Co}(\text{en})_3]^{3+}$
66. Which of the following is diamagnetic:
 (1) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ (2) $[\text{NiCl}_4]^{2-}$
 (3) $[\text{PtCl}_4]^{2-}$ (4) $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$
67. Which of the following compound has minimum coordination number:
 (1) $[\text{Co}_2(\text{CO})_8]$ (2) $[\text{Mn}_2(\text{CO})_{10}]$
 (3) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ (4) $[\text{Co}(\text{NH}_3)_4\text{Cl}]$
68. Which of the following compound show both linkage & ionisation isomerism:
 (1) $[\text{Co}(\text{NH}_3)_5\text{NO}_2]$ ((NO_2)
 (2) $[\text{Co}(\text{NO}_2)(\text{Py})_2(\text{NH}_3)_2]\text{NO}_2$
 (3) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_2$
 (4) $[\text{Co}(\text{NO}_2)\text{Br}(\text{NH}_3)_4]\text{Cl}$
69. Complex in which central atom contain minimum electron density in valence shell
 (1) $[\text{Co}(\text{CO})_4]^+$ (2) $[\text{Mn}(\text{CO})_5]^-$
 (3) $[\text{V}(\text{CO})_6]$ (4) $[\text{Cr}(\text{CO})_6]$
70. Colour of ruby is due to presence of Cr_2O_3 in Al_2O_3 the reason of colour property
 (1) charge transfer of O^{2-} to Al^{+3}
 (2) charge transfer of O_2^- to Cr^{3+}
 (3) charge transfer of O^- to Cr^{3+}
 (4) d-d transition in Cr^{+3}
71. Electronic configuration of Co in a octahedral complex is $t_{2g}^6 e_g^0$ with a ligand L, than L is
 (1) SCN^- (2) oxalate
 (3) F^- (4) Cl^-
72. Which of the following complex is most stable among the following:-
 (1) $\text{Na}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ (2) $[\text{Co}(\text{NH}_3)_6]^{+3}$
 (3) $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ (4) $\text{Na}_3[\text{CoF}_6]$
73. Select the incorrect statement for same type of ligands:-
 (1) Neutral ligands can be strong than anionic ligands
 (2) Chelating ligands are stronger than non chelating
 (3) π -bond containing ligands are classical
 (4) electron pair donating ability depends on electronegativity of donar atom.
74. Which of the following homoleptic complex has square planar geometry:-
 (1) $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{-3}$ (2) $[\text{Co}(\text{EDTA})]^{-2}$
 (3) $[\text{Pt}(\text{trien})]^{+2}$ (4) $[\text{Ni}(\text{H}_2\text{O})_4]^{+2}$

Exercise 3

1. Match the column-I to column-II.

Column-I (Molar conductance at infinite dilution)		Column-II (Complex compound)	
(A)	229	(P)	$[\text{Pt}(\text{NH}_3)_5\text{Cl}]\text{Cl}_3$
(B)	0	(Q)	$[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$
(C)	404	(R)	$[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}_2$
(D)	523	(S)	$[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$

- (1) A-R B-Q C-P D-S
 (2) A-S B-Q C-P D-R
 (3) A-Q B-P C-R D-S
 (4) A-S B-R C-P D-Q

2. Match the column-I to column-II.

Column-I (Complex ions)		Column-II (Number of unpaired electrons)	
(A)	$[\text{CrF}_6]^{4-}$	(P)	One
(B)	$[\text{MnF}_6]^{4-}$	(Q)	Two
(C)	$[\text{Cr}(\text{CN})_6]^{4-}$	(R)	Three
(D)	$[\text{Mn}(\text{CN})_6]^{4-}$	(S)	Four
		(T)	Five

- (1) A-T B-Q C-S D-P
 (2) A-P B-Q C-R D-S
 (3) A-Q B-S C-R D-P
 (4) A-S B-T C-Q D-P

3. Match the column-I to column-II.

Column-I (Complexes)		Column-II (Hybridization of central atom)	
(A)	$\text{Ni}(\text{CO})_4$	(P)	sp^3
(B)	$[\text{Ni}(\text{CN})_4]^{2-}$	(Q)	dsp^2
(C)	$[\text{Fe}(\text{CN})_6]^{4-}$	(R)	sp^3d^2
(D)	$[\text{MnF}_6]^{4-}$	(S)	d^2sp^3

- (1) A-P B-Q C-S D-R
 (2) A-R B-Q C-S D-P
 (3) A-P B-Q C-R D-S
 (4) A-Q B-S C-P D-R

4. Match the column-I to column-II.

Column-I (Complexes)		Column-II (Types)	
(A)	$[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ & $[\text{Co}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$	(P)	Ionisation isomerism
(B)	$[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$ & $[\text{Cr}(\text{CN})_6][\text{Co}(\text{NH}_3)_6]$	(Q)	Coordination isomerism
(C)	$[\text{Co}(\text{NH}_3)_5(\text{SO}_4)]\text{Br}$ & $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$	(R)	Linkage isomerism
(D)	$[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ & $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$	(S)	Solvate isomerism

- (1) A-R, B-P, C-Q, D-S
 (2) A-Q, B-R, C-S, D-P
 (3) A-R, B-Q, C-P, D-S
 (4) A-S, B-R, C-Q, D-P

5. Match the column-I to column-II.

Column-I		Column-II	
(A)	$\text{K}_n[\text{Fe}(\text{CN})_6]$	(P)	(n = two)
(B)	$[\text{Ag}(\text{CN})_n]^-$	(Q)	(n = three)
(C)	Facial- $[\text{PtCl}_3\text{Br}_n]^{2-}$	(R)	(n = four)
(D)	$[\text{Pt}(\text{NH}_3)_n]\text{Cl}_4$ Octahedral	(S)	(n = six)

- (1) A-S B-P C-Q D-R
 (2) A-R B-P C-Q D-S
 (3) A-Q B-P C-R D-S
 (4) A-P B-R C-Q D-S

6. Match the column-I to column-II.

Column-I (Complexes)		Column-II (Property)	
(A)	$[\text{ZnCl}_4]^{2-}$	(P)	Tetrahedral
(B)	$[\text{CuCl}_4]^{2-}$	(Q)	Sq. Planar
(C)	$[\text{MnCl}_4]^{2-}$	(R)	One unpaired electron
(D)	$[\text{PtCl}_4]^{2-}$	(S)	Five unpaired electrons

- (1) A-Q B-P C-S D-Q
 (2) A-P B-Q C-S D-R
 (3) A-R B-P C-P D-Q
 (4) A-P B-R C-S D-Q

7. Match the column-I to column-II.

Column-I (C.F.S.E.)		Column-II (Configuration)	
(A)	$-0.8 \Delta_0$	(P)	t_{2g}^3, e_g^2
(B)	Zero	(Q)	t_{2g}^5, e_g^0
(C)	$-1.2 \Delta_0$	(R)	t_{2g}^2, e_g^0
(D)	$-2.0 \Delta_0$	(S)	t_{2g}^3, e_g^0

- (1) A-R B- P C-S D-Q
 (2) A-S B- R C-P D-Q
 (3) A-R B- Q C-S D-R
 (4) A-Q B- P C-S D-R

8. Match the column-I to column-II.

Column-I (Complex)		Column-II (Orbitals)	
(A)	$[\text{NiCl}_4]^{2-}$	(P)	$3d_{z^2}, 3d_{x^2-y^2}$
(B)	$[\text{Ni}(\text{NH}_3)_6]^{2+}$	(Q)	$4d_{z^2}, 4d_{x^2-y^2}$
(C)	$[\text{Ni}(\text{CN})_4]^{2-}$	(R)	Two 4p-orbitals
(D)	$[\text{Fe}(\text{CN})_6]^{3-}$	(S)	No d-orbital

- (1) A-P B- Q C-R D-S
 (2) A-P B- R C-S D-Q
 (3) A-S B- R C-Q D-P
 (4) A-S B- Q C-R D-P

9. Match the column-I to column-II.

Column-I (Ions)		Column-II (Spin-magnetic moment)	
(A)	Cr^+	(P)	$\sqrt{24}$ B.M
(B)	Cu^{+1}	(Q)	$\sqrt{8}$ B.M
(C)	V^{+3}	(R)	$\sqrt{35}$ B.M
(D)	Fe^{+2}	(S)	0 B.M.

- (1) A-R B- P C-Q D-S
 (2) A-R B-S C-Q D-P
 (3) A-Q B-S C-R D-P
 (4) A-S B-Q C-R D-P

10. Match the column-I to column-II.

Column-I (Hybridisation)		Column-II (Complex)	
(A)	dsp^2	(P)	$[\text{FeF}_6]^{4-}$
(B)	sp^3	(Q)	$[\text{Ti}(\text{H}_2\text{O})_3\text{Cl}_3]$
(C)	sp^3d^2	(R)	$[\text{Cr}(\text{NH}_3)_6]^{3+}$
(D)	d^2sp^3	(S)	$[\text{FeCl}_4]^{2-}$
		(T)	$\text{Ni}(\text{CO})_4$
		(U)	$[\text{Ni}(\text{CN})_4]^{2-}$

- (1) A-T B-S, U C-Q, R D-P
 (2) A-T, U B-S, C-R, D-P, Q
 (3) A-U B-S, T C-P, D-Q, R
 (4) A-S, T B-T, U C-P, Q D-R

11. **Assertion (A)** : In N_2H_4 , any one N-atom can coordinate with central metal cation in a coordination compound.**Reason (R)**: N_2H_4 , can also act as ambidentate ligand.

- (1) If A is true but the R is false.
 (2) If A is false but R is true.
 (3) If both A and R are true and the R is the correct explanation of A.
 (4) If both A and R are true but R is not the correct explanation of A.

12. **Assertion (A)** : $[\text{Ti}(\text{H}_2\text{O})_6]^{4+}$ is coloured while $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ is colourless.**Reason (R)**: d-d transition is not possible in $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$.

- (1) If A is true but the R is false.
 (2) If A is false but R is true.
 (3) If both A and R are true and the R is the correct explanation of A.
 (4) If both A and R are true but R is not the correct explanation of A.

13. Assertion (A) : $[\text{Fe}(\text{EDTA})]^-$ complex is octahedral in shape.

Reason (R): EDTA is a hexadentate ligand which undergoes sp^3d^2 hybridisation.

- (1) If A is true but the R is false.
- (2) If A is false but R is true.
- (3) If both A and R are true and the R is the correct explanation of A.
- (4) If both A and R are true but R is not the correct explanation of A.

14. Assertion (A) : Tetrahedral complexes with chiral structure exhibit optical isomerism.

Reason (R): They lack plane of symmetry.

- (1) If A is true but the R is false.
- (2) If A is false but R is true.
- (3) If both A and R are true and the R is the correct explanation of A.
- (4) If both A and R are true but R is not the correct explanation of A.

15. Assertion (A) : Zeise's salt is a π -bonded organometallic compound.

Reason (R): Zeise's salt contain C_2H_4 molecule as one of the ligand.

- (1) If A is true but the R is false.
- (2) If A is false but R is true.
- (3) If both A and R are true and the R is the correct explanation of A.
- (4) If both A and R are true but R is not the correct explanation of A.

16. Assertion (A) : $[\text{CoCl}_3(\text{NH}_3)_3]$ does not give white precipitate with AgNO_3 solution.

Reason (R): $[\text{CoCl}_3(\text{NH}_3)_3]$ complex is optically inactive.

- (1) If A is true but the R is false.
- (2) If A is false but R is true.
- (3) If both A and R are true and the R is the correct explanation of A.
- (4) If both A and R are true but R is not the correct explanation of A.

17. Assertion (A) : Transition metal ion forming octahedral complexes undergo sp^3d^2 or d^2sp^3 hybridisation.

Reason (R): Strong field ligands force the unpaired electrons of central metal ion to pair up causing d^2sp^3 hybridisation whereas weak field ligands do not affect electronic configuration of the metal ion undergoes in sp^3d^2 hybridisation.

- (1) If A is true but the R is false.
- (2) If A is false but R is true.
- (3) If both A and R are true and the R is the correct explanation of A.
- (4) If both A and R are true but R is not the correct explanation of A.

18. Assertion (A) : Complex ion $[\text{Co}(\text{NH}_3)_6]^{2+}$ is readily oxidized to $[\text{Co}(\text{NH}_3)_6]^{3+}$.

Reason (R): Unpaired electron in complex ion $[\text{Co}(\text{NH}_3)_6]^{2+}$ is present in 4p orbital.

- (1) If A is true but the R is false.
- (2) If A is false but R is true.
- (3) If both A and R are true and the R is the correct explanation of A.
- (4) If both A and R are true but R is not the correct explanation of A.

19. Assertion (A) : Hydrazine is a neutral ligand.

Reason (R): It has two N as donor atoms and behaves as a chelating ligand

- (1) If A is true but the R is false.
- (2) If A is false but R is true.
- (3) If both A and R are true and the R is the correct explanation of A.
- (4) If both A and R are true but R is not the correct explanation of A.

20. Assertion (A) : Complex anion $[\text{Re}_2\text{Cl}_8]^{2-}$ has one δ -bond, one sigma and one π -bond

Reason (R): d_{z^2} orbital can never form δ -bond.

- (1) If A is true but the R is false.
- (2) If A is false but R is true.
- (3) If both A and R are true and the R is the correct explanation of A.
- (4) If both A and R are true but R is not the correct explanation of A.

Exercise 4 (Previous Year's Questions)

- A magnetic moment of 1.73 BM will be shown by one among the following:
[NEET-UG 2013]
(1) $[\text{CoCl}_6]^{4-}$ (2) $[\text{Cu}(\text{NH}_3)_4]^{2+}$
(3) $[\text{Ni}(\text{CN})_4]^{2-}$ (4) TiCl_4
- Which of the following complexes is used to be as an anticancer agent? [AIPMT 2014]
(1) $\text{mer-}[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ (2) $\text{cis-}[\text{PtCl}_2(\text{NH}_3)_2]$
(3) $\text{cis-K}_2[\text{PtCl}_2\text{Br}_2]$ (4) $\text{Na}_2[\text{CoCl}_4]$
- Cobalt (III) chloride forms several octahedral complexes with ammonia. Which of the following will not give test of chloride ions with silver nitrate at 25°C ? [AIPMT 2015]
(1) $\text{CoCl}_3 \cdot 4\text{NH}_3$ (2) $\text{CoCl}_3 \cdot 5\text{NH}_3$
(3) $\text{CoCl}_3 \cdot 6\text{NH}_3$ (4) $\text{CoCl}_3 \cdot 3\text{NH}_3$
- Which of these statements about $[\text{Co}(\text{CN})_6]^{3-}$ is true: [AIPMT 2015]
(1) $[\text{Co}(\text{CN})_6]^{3-}$ has four unpaired electrons and will be in a low-spin configuration
(2) $[\text{Co}(\text{CN})_6]^{3-}$ has four unpaired electrons and will be in a high spin configuration
(3) $[\text{Co}(\text{CN})_6]^{3-}$ has no unpaired electrons and will be in a high-spin configuration
(4) $[\text{Co}(\text{CN})_6]^{3-}$ has no unpaired electrons and will be in a low-spin configuration
- The name of complex ion, $[\text{Fe}(\text{CN})_6]^{3-}$ is: [Re-AIPMT 2015]
(1) Tricyanoferrate (III) ion
(2) Hexacyanidoferrate (III) ion
(3) Hexacyanoiron (III) ion
(4) Hexacyanito ferrate (III) ion
- The hybridization involved in complex $[\text{Ni}(\text{CN})_4]^{2-}$ is (At. No. Ni = 28) [Re-AIPMT 2015]
(1) d^2sp^2 (2) d^2sp^3 (3) dsp^2 (4) sp^3
- The sum of coordination number and oxidation number of the metal M in the complex $[\text{M}(\text{en})_2(\text{C}_2\text{O}_4)]\text{Cl}$ (where en is ethylenediamine) is- [Re-AIPMT 2015]
(1) 7 (2) 8 (3) 9 (4) 6
- Number of possible isomers for the complex $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ will be: (en - ethylenediamine) [Re-AIPMT 2015]
(1) 3 (2) 4 (3) 2 (4) 1
- Which of the following has longest C-O bond length? (Free C-O bond length in CO is $1.128 \text{ \AA}$). [NEET-2016]
(1) $\text{Ni}(\text{CO})_4$ (2) $[\text{Co}(\text{CO})_4]^-$
(3) $[\text{Fe}(\text{CO})_6]^{2-}$ (4) $[\text{Mn}(\text{CO})_6]^+$
- The correct increasing order of trans effect of the following species is: [NEET-2016]
(1) $\text{Br}^- > \text{CN}^- > \text{NH}_3 > \text{C}_6\text{H}_5^-$
(2) $\text{CN}^- > \text{Br}^- > \text{C}_6\text{H}_5^- > \text{NH}_3$
(3) $\text{NH}_3 > \text{CN}^- > \text{Br}^- > \text{C}_6\text{H}_5^-$
(4) $\text{CN}^- > \text{C}_6\text{H}_5^- > \text{Br}^- > \text{NH}_3$
- Jahn-Teller effect **not** observed in high spin complexes of: [NEET-2016]
(1) d^4 (2) d^9 (3) d^7 (4) d^8
- An example of a sigma bonded organometallic compound is: [NEET 2017]
(1) Ruthenocene (2) Ferrocene
(3) Cobaltocene (4) Grignard's reagent
- The correct order of the stoichiometries of AgCl formed when AgNO_3 in excess is treated with the complexes: $\text{CoCl}_3 \cdot 6\text{NH}_3$, $\text{CoCl}_3 \cdot 5\text{NH}_3$, $\text{CoCl}_3 \cdot 4\text{NH}_3$ respectively is : [NEET 2017]
(1) $1\text{AgCl}, 3\text{AgCl}, 2\text{AgCl}$
(2) $3\text{AgCl}, 1\text{AgCl}, 2\text{AgCl}$
(3) $3\text{AgCl}, 2\text{AgCl}, 1\text{AgCl}$
(4) $2\text{AgCl}, 3\text{AgCl}, 1\text{AgCl}$
- Pick out the correct statement with respect to $[\text{Mn}(\text{CN})_6]^{3-}$: [NEET-2017]
(1) It is sp^3d^2 hybridised and tetrahedral
(2) It is d^2sp^3 hybridised and octahedral
(3) It is dsp^2 hybridised and square planar
(4) It is sp^3d^2 hybridised and octahedral

15. Correct increasing order for the wavelengths of absorption in the visible region the complexes of Co^{3+} is:-

[NEET-2017]

- (1) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Co}(\text{en})_3]^{3+}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$
 (2) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$; $[\text{Co}(\text{NH}_3)_6]^{3+}$, $(\text{Co}(\text{en})_3]^{3+}$
 (3) $[\text{Co}(\text{NH}_3)_6]^{3+}$; $[\text{Co}(\text{en})_3]^{3+}$, $(\text{Co}(\text{H}_2\text{O})_6]^{3+}$
 (4) $[\text{Co}(\text{en})_3]^{3+}$; $[\text{Co}(\text{NH}_3)_6]^{3+}$; $(\text{Co}(\text{H}_2\text{O})_6]^{3+}$

16. The type of isomerism shown by the complex $[\text{CoCl}_2(\text{en})_2]$ is: [NEET(UG) 2018]

- (1) Geometrical isomerism
 (2) Coordination isomerism
 (3) Ionization isomerism
 (4) Linkage isomerism

17. The geometry and magnetic behaviour of the complex $(\text{Ni}(\text{CO})_4)$ are

[NEET(UG) 2018]

- (1) Square plane geometry and diamagnetic
 (2) Tetrahedral geometry and diamagnetic
 (3) Square planar geometry and paramagnetic
 (4) Tetrahedral geometry and paramagnetic

18. Iron carbonyl, $\text{Fe}(\text{CO})_5$ is [NEET(UG) 2018]

- (1) tetranuclear (2) mononuclear
 (3) trinuclear (4) dinuclear

19. What is the correct electronic configuration of the central atom in $\text{K}_4[\text{Fe}(\text{CN})_6]$ based on crystal field theory? [NEET(UG) 2019]

- (1) $t_{2g}^4 e_g^2$ (2) $t_{2g}^6 e_g^0$
 (3) $e^3 t_2^3$ (4) $e^4 t_2^2$

20. The Crystal Field Stabilisation Energy (CFSE) for $[\text{CoCl}_6]^{4-}$ is 18000 cm^{-1} . The CFSE for $[\text{CoCl}_4]^{2-}$ will be-

[NEET(UG) 2019 (ODISHA)]

- (1) 6000 cm^{-1} (2) 16000 cm^{-1}
 (3) 18000 cm^{-1} (4) 8000 cm^{-1}

21. Urea reacts with water to form A which will decompose to form B. B when passed through $\text{Cu}^{2+}(\text{aq})$, deep blue colour solution C is formed.

What is the formula of C from the following?

[NEET(UG) 2020]

- (1) $\text{Cu}(\text{OH})_2$ (2) $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
 (3) CuSO_4 (4) $[\text{Cu}(\text{NH}_3)_4]^{2+}$

22. Which of the following is the correct order of increasing field strength of ligands to form coordination compounds?

[NEET(UG) 2020]

- (1) $\text{F}^- < \text{SCN}^- < \text{C}_2\text{O}_4^{2-} < \text{CN}^-$
 (2) $\text{CN}^- < \text{C}_2\text{O}_4^{2-} < \text{SCN}^- < \text{F}^-$
 (3) $\text{SCN}^- < \text{F}^- < \text{C}_2\text{O}_4^{2-} < \text{CN}^-$
 (4) $\text{SCN}^- < \text{F}^- < \text{CN}^- < \text{C}_2\text{O}_4^{2-}$

23. Ethylene diaminetetraacetate (EDTA) ion is [NEET(UG) 2021]

- (1) Tridentate ligand with three "N" donor atoms
 (2) Hexadentate ligand with four "O" and two "N" donor atoms
 (3) Unidentate ligand
 (4) Bidentate ligand with two "N" donor atoms

24. Match list-I with List-II [NEET(UG) 2021]

List-I		List-II	
(a)	$[\text{Fe}(\text{CN})_6]^{3-}$	(i)	5.92 BM
(b)	$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	(ii)	0 BM
(c)	$[\text{Fe}(\text{CN})_6]^{4-}$	(iii)	4.90 BM
(d)	$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	(iv)	1.73 BM

Choose the **correct** answer from the option given below

- (1) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
 (2) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
 (3) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
 (4) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

25. The order of energy absorbed which is responsible for the color of complexes [NEET(UG) 2022]

(a) $[\text{Ni}(\text{H}_2\text{O})_2(\text{en})_2]^{2+}$ (b) $[\text{Ni}(\text{H}_2\text{O})_4(\text{en})]^{2+}$ (c) $[\text{Ni}(\text{en})_3]^{2+}$ is:

- (1) (c) > (b) > (a) (2) (c) > (a) > (b)
 (3) (b) > (a) > (c) (4) (a) > (b) > (c)

26. The IUPAC name of the complex - $[\text{Ag}(\text{H}_2\text{O})_2][\text{Ag}(\text{CN})_2]$ is: [NEET(UG) 2022]

- (1) diaquasilver(II) dicyanidoargentate(II)
 (2) dicyanidosilver(I) diaquaargentate(I)
 (3) diaquasilver(I) dicyanidoargentate(I)
 (4) dicyanidosilver(II) diaquaargentate(II)

27. Which complex compound is most stable?
[NEET-2023]

- (1) $[\text{Co}(\text{NH}_3)_3(\text{NO}_3)_3]$
- (2) $[\text{CoCl}_2(\text{en})_2]\text{NO}_3$
- (3) $[\text{Co}(\text{NH}_3)_6]_2(\text{SO}_4)_3$
- (4) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Br}](\text{NO}_3)_2$

28. Homoleptic complex from the following complexes is :
[NEET-2023]

- (1) Diamminechloridonitrito -N-platinum (II)
- (2) Pentaamminecarbonatocobalt (III) chloride
- (3) Triamminetriaquachromium (III) chloride
- (4) Potassium trioxalatoaluminate (III)

29. Match List I with List II.
[NEET-2024]

	List I (Complex)		List II (Type of isomerism)
A.	$[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$	i.	Solvate isomerism
B.	$[\text{Co}(\text{NH}_3)_5(\text{SO}_4)]\text{Br}$	ii.	Linkage isomerism
C.	$[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$	iii.	Ionization isomerism
D.	$[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$	iv.	Coordination isomerism

Choose the correct answer from the options given below:

- (1) A-I, B-IV, C-III, D-II
- (2) A-II, B-IV, C-III, D-I
- (3) A-II, B-III, C-IV, D-I
- (4) A-I, B-III, C-IV, D-II

30. Given below are two statements:

Statement I : Both $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{CoF}_6]^{3-}$ complexes are octahedral but differ in their magnetic behavior.

Statement II : $[\text{Co}(\text{NH}_3)_6]^{3+}$ is diamagnetic whereas $[\text{CoF}_6]^{3-}$ is paramagnetic.

In the light of the above statements, Choose the correct answer from the options given below :
[NEET-2024]

- (1) Statement I is true but Statement II is false.
- (2) Statement I is false but Statement II is true
- (3) Both Statement I and Statement II are true
- (4) Both Statement I and Statement II are false.

31. **Statement I :** $[\text{Co}(\text{NH}_3)_6]^{3+}$ is a homoleptic complex whereas $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ is a heteroleptic complex.

Statement II : Complex $[\text{Co}(\text{NH}_3)_6]^{3+}$ has only one kind of ligands but $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ has more than one kind of ligands.

In the light of the above statements, Choose the correct answer from the options given below :
[NEET-2024]

- (1) Statement I is true but Statement II is false.
- (2) Statement I is false but Statement II is true
- (3) Both Statement I and Statement II are true
- (4) Both Statement I and Statement II are false

32. Ethylene diaminetetraacetate ion is a/an:
[RE-NEET-2024]

- (1) hexadentate ligand
- (2) ambidentate ligand
- (3) monodentate ligand
- (4) bidentate ligand

33. Which of the following is not an ambidentate ligand?
[RE-NEET-2024]

- (1) $\text{C}_2\text{O}_4^{2-}$
- (2) SCN^-
- (3) NO_2^-
- (4) CN^-

34. $[\text{Mn}_2(\text{CO})_{10}]$ and $[\text{Co}_2(\text{CO})_8]$ structures have:
[RE-NEET-2024]

- A. Metal-Metal linkage
- B. Terminal CO groups
- C. Bridging CO groups
- D. Metal in zero oxidation state

Choose the correct answer from the options given below :

- (1) Only A, B, C
- (2) Only B, C, D
- (3) Only A, C, D
- (4) Only A, B, D

35. Out of the following complex compounds, which of the compound will be having the minimum conductance in solution?
[NEET 2025]

- (1) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
- (2) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}$
- (3) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
- (4) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$

36. Which of the following are paramagnetic?
[NEET 2025]
- A. $[\text{NiCl}_4]^{2-}$ B. $\text{Ni}(\text{CO})_4$
C. $[\text{Ni}(\text{CN})_4]^{2-}$ D. $[\text{Ni}(\text{H}_2\text{O}_6)]^{2+}$
E. $\text{Ni}(\text{PPh}_3)_4$
- Choose the **correct** answer from the options given below:
- (1) A and D only (2) A, D and E only
(3) A and C only (4) B and E only

37. The correct order of the wavelength of light absorbed by the following complexes is
[NEET 2025]
- A. $[\text{Co}(\text{NH}_3)_6]$ B. $[\text{Co}(\text{CN})_6]^{3-}$
C. $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ D. $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$
- Choose the correct answer from the options given below :
- (1) $C < D < A < B$ (2) $C < A < D < B$
(3) $B < D < A < C$ (4) $B < A < D < C$

ANSWER KEYS

Exercise 1.1

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	4	4	2	4	3	2	1	2	4	2	1	2	4	4	2	4	2	3	3
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38		
Ans.	2	1	1	1	1	3	3	1	1	2	1	2	4	3	2	2	4	4		

Exercise 1.2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	4	3	2	1	4	2	4	4	3	3	4	3	3	1	4	1	2	2	1
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34						
Ans.	1	3	4	2	3	4	4	1	4	2	3	2	4	1						

Exercise 1.3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	2	2	3	4	2	4	4	4	4	3	4	2	1	1	2	1	1	2	3
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	3	3	1	4	2	3	2	2	3	2	1	1	4	3	4	3	2	1	1	4
Que.	41	42	43	44	45	46														
Ans.	4	1	2	4	3	4														

Exercise 1.4

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	2	4	2	2	3	2	3	3	1	4	1	4	2	3	2	2	2	3	2
Que.	21	22	23	24	25															
Ans.	2	2	3	1	2															

Exercise 1.5

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17			
Ans.	2	2	4	3	2	3	1	3	3	4	1	2	2	4	4	2	1			

Exercise 1.6

Que.	1	2	3	4	5	6	7	8	9	10	11	12								
Ans.	1	4	3	4	4	4	3	4	2	4	3	1								

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	4	4	1	4	2	3	3	3	3	2	3	1	2	3	2	4	4	3	3	4
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	3	4	3	1	2	2	2	2	2	3	4	3	1	2	2	3	1	3	3	4
Que.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	3	2	3	1	2	3	4	4	2	2	2	1	2	3	1	2	2	1	2	3
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74						
Ans.	1	3	1	1	1	3	4	4	1	4	2	1	3	3						

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	4	1	3	2	4	1	4	2	3	1	2	1	3	3	4	3	1	1	2

Exercise 4 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	2	4	4	2	3	3	1	3	4	4	1	3	2	4	1	2	2	2	4
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37			
Ans.	4	3	2	1	2	3	2	4	3	3	3	1	1	4	3,4	2	4			

Chapter 3

THE p-BLOCK ELEMENTS

Chapter Summary

- Introduction
- Inert Pair Effect
- Group 13 Elements and their Properties
- Group 14 Elements and their Properties
 - Allotropes of Carbon
- Group 15 Elements and their Properties
 - Compound of Nitrogen
 - Allotropic Forms of Phosphorus
 - Compounds of Phosphorus
- Group 16 Elements and their Properties
 - Properties of Oxygen and Ozone
 - Sulphuric Acid
- Group 17 Elements and Properties of Halogens
- Group 18 Elements and their Properties

Introduction

In p-block elements the last electron enters the outermost p orbital. As we know that the number of p orbitals is three and, therefore, the maximum number of electrons that can be accommodated in a set of p orbitals is six. Consequently there are six groups of p-block elements in the periodic table numbering from 13 to 18.

Inert Pair Effect

- | | | |
|------------|------------|-------------------------------------|
| Group-13 | Group-14 | Group-15
(group oxidation state) |
| ns^2np^1 | ns^2np^2 | ns^2np^3 |
| +3 | +4 | +5 |
- While moving down the group the stability of lower oxidation state (2 less than group oxidation state) progressively increases, for the last element of the group the stability of lower oxidation state becomes ever greater than group oxidation state. This is called **inert pair effect**.
 - For the last element group oxidation state is highly oxidising in nature.

Group-13

B	+3
Al	+3

Group-14

C	+4
Si	+4

Group-15

N	+5
P	+5

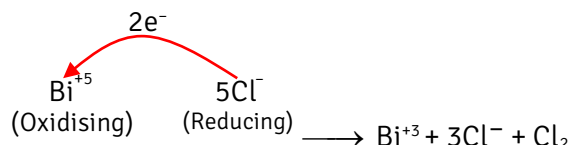
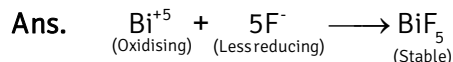
Ga	+3 > +1	Ge	+4 > +2	As	+5 > +3
In	+3 > +1	Sn	+4 > +2	Sb	+5 > +3
Tl	+3 < +1	Pb	+4 < +2	Bi	+5 < +3

Reason:

As we move down the group there is presence of d & f-orbitals in inner shells which have poor shielding effect hence Z_{eff} increase. As a result, the ns^2 electron pair becomes more and more tightly held to the nucleus and becomes reluctant to participate in bonding.

Examples:

- (1.) PbCl_2 is more stable than PbCl_4
- (2.) TlCl is more stable than TlCl_3
- (3.) GaCl_3 is more stable than TlCl_3
- (4.) SnCl_4 is more stable than PbCl_4
- (5.) TlI_3 does not exist
- (6.) PbI_4 does not exist
- (7.) Only BiF_5 exists but BiCl_5 does not exist?

**Back Bonding:**

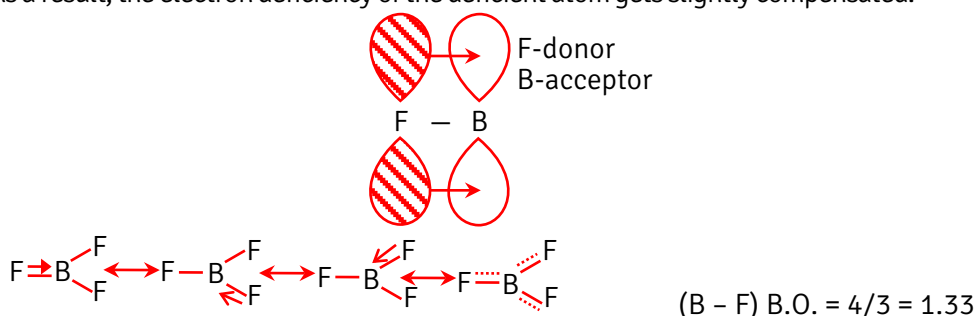
It is formed between two covalently bonded adjacent atoms.

Conditions for Back Bonding:

- (i) Both bonded atoms must belong to 2nd period or one bonded atom must belong to 2nd period and the other must belong to 3rd period.
 - (ii) One bonded atom must possess vacant orbital and the other bonded atom must possess lone pair.
- As a result of back bonding, bond length decreases and bond energy increases.

Back Bonding:

It is a partial sideways overlap in which there is a slight electron transference from electron rich atom to electron deficient atom. As a result, the electron deficiency of the deficient atom gets slightly compensated.

**Types of Back Bonding:**

Based on type of orbital: (2 types)

- (1) $p\pi - p\pi$ back bonding
- (2) $p\pi - d\pi$ back bonding

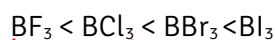
(1) $p\pi - p\pi$ Back Bonding:

Order of strength: $2p - 2p > 2p - 3p > 2p - 4p \dots$

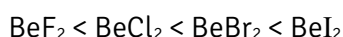
Size $\uparrow$ extent of B.B. $\downarrow$

It is used to explain following observations

- (a) Abnormal bond length and bond energy of B-F bond in BF_3 .
- (b) Lewis acidic order of Boron and Beryllium halides.



BB effective
so Lewis acid
strength $\downarrow$



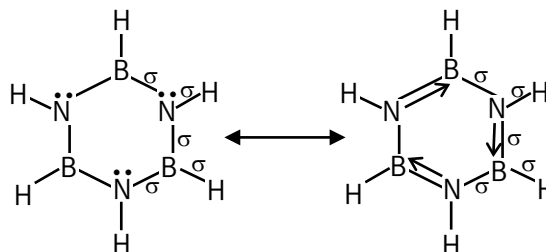
(c) Hybridisation:

If a lone pair participates in back bonding, then it is not considered in hybridisation.

Example:

$B_3N_3H_6$ (Inorganic Benzene or Borazine or Borazole)

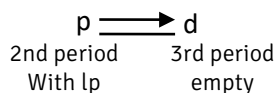
- Hybridisation of B and N = sp^2



- Inorganic benzene is more reactive than organic benzene due to which the bond in it is polar. Although the entire molecule is non-polar.

(d) If back bonding is present then tendency to form dimer or polymer decreases.**Example:**

BF_3 , BeF_2

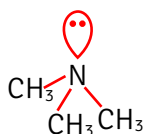
(2) $p\pi - d\pi$ Back Bonding:

It is used to explain following observations:

(a) Hybridisation:**Example:**

Trimethyl amine

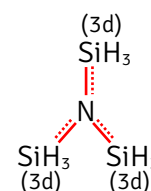
$(CH_3)_3N$



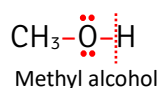
- sp^3 hybrid(N)
- trigonal pyramidal
- Lewis base (due to presence of lp)

Trisilyl amine

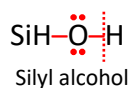
$(SiH_3)_3N$



- sp^2 hybrid(N)
- trigonal planar
- Not Lewis base
- Bond angle increases

(b) Acidic Strength:

- No back bonding
- Less acidic



- Back bonding present in conjugate base
- More acidic

Group 13 Elements and their Properties

Group	13	14	15	16	17	18
General Electronic configuration	ns^2np^1	ns^2np^2	ns^2np^3	ns^2np^4	ns^2np^5	ns^2np^6 ($1s^2$ for He)
First member of the group	B	C	N	O	F	He
Group oxidation state	+3	+4	+5	+6	+7	+8
Other oxidation states	+1	+2, -4	+3, -3	+4, +2, -2	+5, +3, +1, -1	+6, +4, +2

Table 11.2 Atomic and Physical Properties of Group 13 Elements

Property	Element				
	Boron B	Aluminium Al	Gallium Ga	Indium In	Thallium Tl
Atomic number	5	13	31	49	81
Atomic mass(g mol ⁻¹)	10.81	26.98	69.72	114.82	204.38
Electronic Configuration	[He]2S ² 2p ¹	[Ne]3s ² 3p ¹	[Ar]3d ¹⁰ 4s ² 4p ¹	[Kr]4d ¹⁰ 5s ² 5p ¹	[Xe]4f ¹⁴ 5d ¹⁰ 6s ² 6p ¹
Atomic radius/pm ^a	(88)	143	135	167	170
Ionic radius M ³⁺ /pm ^b	(27)	53.5	62.0	80.0	88.5
Ionic radius M ⁺ /pm	-	-	120	140	150
Ionization enthalpy (kJ mol ⁻¹)	$\Delta_i H_1$ $\Delta_i H_2$ $\Delta_i H_3$	801 2427 3659	577 1816 2744	579 1979 2962	558 1820 2704
Electronegativity ^c	2.0	1.5	1.6	1.7	1.8
Density/g cm ⁻³ at 298 K	2.35	2.70	5.90	7.31	11.85
Melting point / K	2453	933	303	430	576
Boiling point / K	3923	2740	2676	2353	1730
E ⁰ /V for (M ³⁺ /M)	-	-1.66	-0.56	-0.34	+1.26
E ⁰ /V for (M ⁺ /M)	-	+0.55	-0.79(acid) -1.39(alkali)	-0.18	-0.34

Boron is a typical non-metal, aluminium is a metal but shows many chemical similarities to boron, and gallium, indium and thallium are almost exclusively metallic in character.

Electronic Configuration: $ns^2 np^1$.

Atomic Radii:

$B < Ga < Al < In < Tl$, Here $Ga < Al$ is due to transition contraction.

Ionization Enthalpy:

$B > Tl > Ga > Al > In$,

Greater IE of Tl is due to lanthanoid contraction

Greater IE of Ga is due to transition contraction

Electronegativity:

$B > Tl > In > Ga > Al$

This is because of the discrepancies in atomic size of the elements.

Physical Properties:

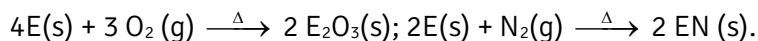
Boron is non-metallic in nature. It is extremely hard and black coloured solid. It exists in many allotropic forms. Due to very strong crystalline lattice, boron has unusually high melting point. **Gallium with low melting point (303 K) (exist in liquid state during summer) and high boiling point (2676 K) makes it a useful material for measuring high temperatures (Thermometer).**

Chemical Properties:**Oxidation State and Trends in Chemical Reactivity:**

- Due to small size of boron, the sum of its first three ionization enthalpies is very high. This prevents it to form + 3 ions and force it to form only covalent compounds. But as we move from B to Al, the sum of the first three ionisation enthalpies of Al considerably decreases, and is therefore able to form Al^{+3} ions.
- The relative stability of + 1 oxidation state progressively increases for heavier elements: $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$. The compound in +1 oxidation state, as expected from energy considerations, are more ionic than those in + 3 oxidation state.
- In trivalent state, the number of electrons around the central atom in a molecule of the compounds of these elements (e.g., boron in BF_3) will be only six. Such electron deficient molecules behave as Lewis acids. The tendency to behave as Lewis acid decreases with the increases in the size down the group. BCl_3 easily accepts a lone pair of electrons from ammonia to form $\text{BCl}_3 \cdot \text{NH}_3$. In trivalent state most of the compounds being covalent are hydrolysed in water. The trichloride on hydrolysis in water form tetrahedral $[\text{M}(\text{OH})_4]^-$ species; Aluminium chloride in acidified aqueous solution form octahedral $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ ion.

(i) Reactivity Towards Air:

Boron is unreactive in crystalline form. Aluminium forms a very thin oxide layer on the surface which protects the metal from further attack. Amorphous boron and aluminium metal on heating in air form B_2O_3 and Al_2O_3 respectively. With dinitrogen at high temperature, they form nitrides.

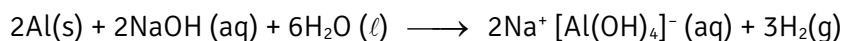


The nature of these oxides varies down the group. Boron trioxide is acidic and reacts with basic (metallic) oxides forming metal borates. Aluminium and gallium oxides are amphoteric and those of indium and thallium are basic in their properties.

(ii) Reactivity Towards Acids and Alkalies:

Boron does not react with acids and alkalies even at moderate temperature; but aluminium dissolves in mineral acids and aqueous alkalies and thus shows amphoteric character.

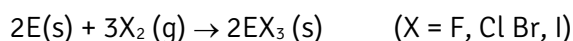
Aluminium dissolved in dilute HCl and liberates dihydrogen. However, concentrated nitric acid renders aluminium passive by forming protective oxide layer on the surface. Aluminium also reacts with aqueous alkali and liberates dihydrogen.



Sodium tetrahydroxoaluminate (III)

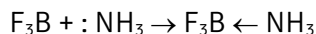
(iii) Reactivity Towards Halogens:

These elements react with halogen to form trihalides (except TlI_3).



Important Trends and Anomalous Properties of Boron:

Certain important trends can be observed in the chemical behaviour of group 13 elements. The tri-chlorides, bromides and iodides of all these elements being covalent in nature are hydrolysed in water. Species like tetrahedral $[M(OH)_4]^-$ and octahedral $[M(H_2O)_6]^{3+}$, except in boron, exist in aqueous medium. The monomeric trihalides, being electron deficient, are strong Lewis acids. Boron trifluoride easily reacts with Lewis bases such as NH_3 to complete octet around boron.



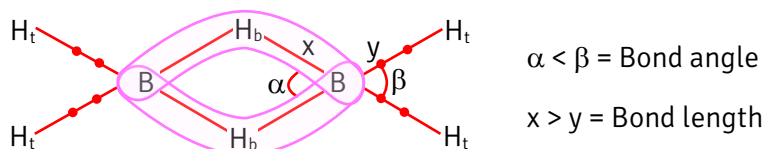
It is due to the absence of d orbitals that the maximum covalence of B is 4. Since the d orbitals are available with Al and other elements, the maximum covalence can be expected beyond 4. Most of the other metal halides (e.g., $AlCl_3$) are dimerised through halogen bridging (e.g., Al_2Cl_6). The metal species completes its octet by accepting electrons from halogen in these halogens bridged molecules.

Some Important Structure:

(1) Banana bond or 3C - 2e bond or e^- low bond

Diborane (B_2H_6):

(a) B_2H_6



$$(\alpha = 97^\circ, \beta = 120^\circ)$$

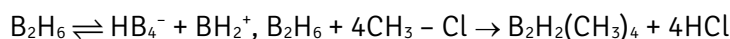
$$2C-2e^- \text{ bonds} = 4$$

$$3C-2e^- \text{ bonds} = 2$$

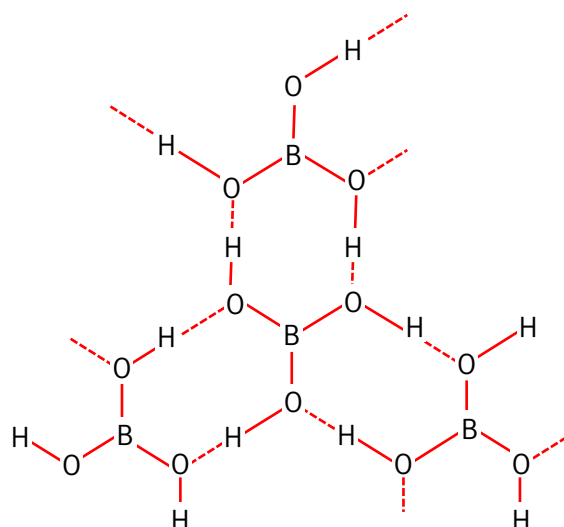
$$\text{Hybridisation state} = sp^3$$

Electron deficient molecule (act as Lewis acid)

- All four terminal hydrogen and two boron atoms are present in same plane and both bridging H are present in perpendicular plane.
- If substitution reaction occurs, only the four terminal hydrogen atoms will be replaced.



(2) Boric acid (H_3BO_3)

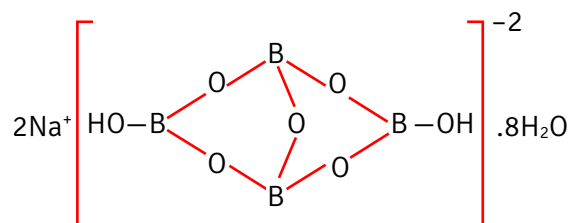


(3) Borax ($\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$)/ $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$:

Important points:

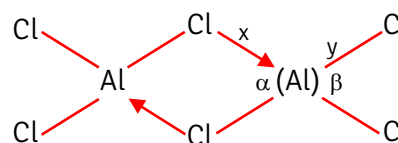
B—O—B linkage = 5

Total B—O bonds = 14

Two Boron— sp^2 hybridisedTwo Boron— sp^3 hybridised(a) Al_2Cl_6 (dimer of AlCl_3 in liquid or solid state)BA = $a < b$ BL = $x > y$ Hybridisation state = sp^3

Non-polar

Octet complete



NCERT QUESTIONS (Solve Yourself)

- Standard electrode potential values, $E^\ominus$ for Al^{3+}/Al is -1.66 V and that of Tl^{3+}/Tl is $+1.26 \text{ V}$. Predict about the formation of M^{3+} ion in solution and compare the electropositive character of the two metals.
- White fumes appear around the bottle of anhydrous aluminium chloride. Give reason.
- Boron is unable to form BF_6^{3-} ion. Explain.
- Why is boric acid considered as a weak acid?
- How can you explain higher stability of BCl_3 as compared to TlCl_3 ?
- Why does boron trifluoride behave as a Lewis acid?
- Consider the compounds, BCl_3 and CCl_4 . How will they behave with water? Justify.
- Is boric acid a protic acid? Explain.
- Explain what happens when boric acid is heated.
- Describe the shapes of BF_3 and BH_4^- . Assign the hybridisation of boron in these species.
- Write reactions to justify amphoteric nature of aluminium.
- What are electron deficient compounds? Are BCl_3 and SiCl_4 electron deficient species? Explain.
- Suggest reasons why the B—F bond lengths in BF_3 (130 pm) and BF_4^- (143 pm) differ.
- If B—Cl bond has a dipole moment, explain why BCl_3 molecule has zero dipole moment.
- Aluminium trifluoride is insoluble in anhydrous HF but dissolves on addition of NaF. Aluminium trifluoride precipitates out of the resulting solution when gaseous BF_3 is bubbled through. Give reasons.
- Explain structures of diborane and boric acid.
- What happens when
 - Borax is heated strongly,
 - Boric acid is added to water,
 - Aluminium is treated with dilute NaOH,
 - BF_3 is reacted with ammonia?
- Give reasons:
 - Conc. HNO_3 can be transported in aluminium container.
 - A mixture of dilute NaOH and aluminium pieces is used to open drain.
 - Graphite is used as lubricant.
 - Diamond is used as an abrasive.
 - Aluminium alloys are used to make aircraft body.
 - Aluminium utensils should not be kept in water overnight.
 - Aluminium wire is used to make transmission cables.
- How would you explain the lower atomic radius of Ga as compared to Al?
- In some of the reactions thallium resembles aluminium, whereas in others it resembles with group I metals. Support this statement by giving some evidences.

21. When metal X is treated with sodium hydroxide, a white precipitate (A) is obtained, which is soluble in excess of NaOH to give soluble complex (B). Compound (A) is soluble in dilute HCl to form compound (C). The compound (A) when heated strongly gives (D), which is used to extract metal. Identify (X), (A), (B), (C) and (D). Write suitable equations to support their identities.
22. What do you understand by (a) inert pair effect (b) allotropy and (c) catenation?
23. A certain salt X, gives the following results.
 (i) Its aqueous solution is alkaline to litmus.
 (ii) It swells up to a glassy material Y on strong heating.
 (iii) When conc. H_2SO_4 is added to a hot solution of X, white crystal of an acid Z separates out. Write equations for all the above reactions and identify X, Y and Z.
24. Write balanced equations for:
 (i) $\text{BF}_3 + \text{LiH} \rightarrow$ (ii) $\text{B}_2\text{H}_6 + \text{H}_2\text{O} \rightarrow$ (iii) $\text{NaH} + \text{B}_2\text{H}_6 \rightarrow$
 (iv) $\text{H}_3\text{BO}_3 \rightarrow$ (v) $\text{Al} + \text{NaOH} \rightarrow$ (vi) $\text{B}_2\text{H}_6 + \text{NH}_3 \rightarrow$
25. An aqueous solution of borax is
 (a) neutral (b) amphoteric (c) basic (d) acidic

Exercise 1.1

1. The decrease in stability of higher oxidation state in p-block with increasing atomic number is due to:
 (1) Increase in bond energy as going down the group
 (2) Energy required to unpair ns^2 -electrons is not compensated by the energy released in forming the two additional bonds
 (3) Both are correct
 (4) None is correct.
2. Al dissolves in molten NaOH with the formation of:
 (1) Sodium aluminate (Na_3AlO_3)
 (2) Sodium metaluminate (NaAlO_2)
 (3) Aluminium hydroxide
 (4) Alumina
3. Stable compounds in +1 oxidation state is formed by:
 (1) B (2) Al (3) Ga (4) Tl
4. Aluminium is more reactive than iron. But aluminium is less easily corroded than iron because:
 (1) Aluminium is a noble metal
 (2) Oxygen forms a protective oxide layer
 (3) Iron undergoes reaction easily with water
 (4) Iron forms both mono and divalent ions
5. The dissolution of $\text{Al}(\text{OH})_3$ by a solution of NaOH results in the formation of:
 (1) $[\text{Al}(\text{H}_2\text{O})_4(\text{OH})]^{2+}$ (2) $[\text{Al}(\text{H}_2\text{O})_2(\text{OH})_4]^-$
 (3) $[\text{Al}(\text{H}_2\text{O})_3(\text{OH})_3]$ (4) $[\text{Al}(\text{H}_2\text{O})_6(\text{OH})_3]$
6. Aluminium does not react with:
 (1) NaOH (2) HCl
 (3) N_2 (4) HNO_3
7. An aqueous solution of BCl_3 is:
 (1) Weak acid (2) Weak base
 (3) Neutral (4) Strong base
8. Boric acid is polymeric because of:
 (1) Its acidic nature
 (2) Presence of hydrogen bonds
 (3) Its monobasic nature
 (4) Its geometry
9. $4\text{B(s)} + 3\text{O}_2(\text{g}) \xrightarrow{\Delta} \text{I}$
 $2\text{Al(s)} + \text{N}_2(\text{g}) \xrightarrow{\Delta} \text{II}$
 Here, I and II are
 (1) $\text{I} \rightarrow \text{B}_2\text{O}_3(\text{s})$, $\text{II} \rightarrow \text{AlN}(\text{g})$
 (2) $\text{I} \rightarrow \text{B}_2\text{O}_3(\text{g})$, $\text{II} \rightarrow \text{AlN}(\text{s})$
 (3) $\text{I} \rightarrow \text{B}_2\text{O}_3(\text{g})$, $\text{II} \rightarrow \text{AlN}(\text{g})$
 (4) $\text{I} \rightarrow \text{B}_2\text{O}_3(\text{s})$, $\text{II} \rightarrow \text{AlN}(\text{s})$

10. $2\text{Al(s)} + 2\text{NaOH(aq)} + 6\text{H}_2\text{O(l)} \rightarrow 2\text{A} + 3\text{H}_2\text{(g)}$

Here, A is

- (1) $\text{Na}^+[\text{Al}(\text{OH})_4]^- \text{(aq)}$
Sodium tetrahydroxoaluminate (III)
- (2) $\text{Na}_2[\text{Al}(\text{OH})_4]^- \text{(aq)}$
Sodium tetrahydroxoaluminate (III)
- (3) $\text{Na}^+[\text{Al}(\text{OH})_6]^- \text{(aq)}$
Sodium hexahydroxoaluminate (II)
- (4) $\text{Na}[\text{Al}(\text{OH})_6]^- \text{(aq)}$
Sodium hexahydroxoaluminate (III)

11. Dry ice is

- (1) Solid NH_3 (2) Solid SO_2
(3) Solid CO_2 (4) Solid N_2

12. Due to inert pair effect +1 oxidation state is most stable for :

- (1) Al (2) Ga
(3) In (4) Tl

13. Which molecule does not exist ?

- (1) BF_4^- (2) AlF_6^{3-}
(3) BF_6^{3-} (4) All

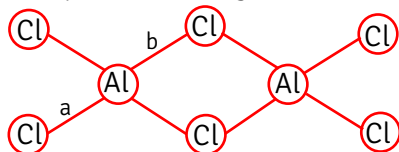
14. H_3BO_3 is an example of :

- (1) Weak monobasic acid
(2) Strong dibasic acid
(3) Weak dibasic acid
(4) Weak tribasic acid

15. AlCl_3 fumes in moist air due to :

- (1) Formation of Al_2O_3
(2) Formation of HCl vapours
(3) Formation of HOCl vapours
(4) None

16. Compare bond length a and b



- (1) $a > b$ (2) $b > a$
(3) $a = b$ (4) can't say

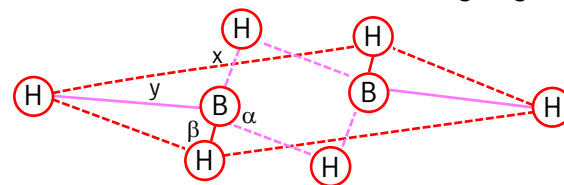
17. Which element is used in measurement of very high temperature ?

- (1) Carbon (2) Aluminium
(3) Gallium (4) Mercury

18. What is the nature of aqueous solution of borax?

- (1) Acidic (2) Basic
(3) Oxidising (4) None

19. Which is true from the following diagram?



- (1) $x > y, \alpha > \beta$ (2) $x > y, \alpha < \beta$
(3) $x < y, \alpha > \beta$ (4) $x < y, \alpha < \beta$

20. True statement about B_2H_6 is:

- (1) Electron deficient hydride
(2) Toxic gas
(3) Give exothermic combustion
(4) All

21. Lake test confirm the presence of :

- (1) Fe^{3+} ion (2) Cr^{3+} ion
(3) Al^{3+} ion (4) Mn^{2+} ion

22. In the lake test of Al^{3+} ion 'lake' is :

- (1) Colourless $\text{Al}(\text{OH})_3$ precipitate
(2) Blue mass of $\text{Al}(\text{OH})_3$ floating in the colourless solution
(3) Blue mass of NaAlO_2
(4) Brown mass of $\text{Al}(\text{OH})_3$

23. The compound formed in the borax bead test of Cu^{2+} ion in oxidising flame is :

- (1) Cu
(2) CuBO_2
(3) $\text{Cu}(\text{BO}_2)_2$
(4) None of these

24. In borax bead test, which of the following compound is formed :

- (1) Meta borate (2) Tetra borate
(3) Tri borate (4) Ortho borate

25. The metal that does not give the borax-bead test is :

- (1) Copper (2) Nickel
(3) Lead (4) Iron

26. Give below are two statements :
Statement-1 : Colourless cupric metaborate is reduced to cuprous metaborate in a luminous flame.
Statement-2 : Cuprous metaborate is obtained by heating boric anhydride and copper sulphate in a non-luminous flame.
 In the light of the above statements, choose the most appropriate answer from the options given below :
 (1) Statement-I is true but Statement-II is false
 (2) Both Statement-I and Statement-II are false
 (3) Statement-I is false but Statement-II is true
 (4) Both Statement-I and Statement-II are true
27. AlCl_3 solution give white gelatinous precipitate with small amount of NaOH which is soluble in excess of NaOH due to the formation of :
 (1) Al(OH)_3 (2) Na[Al(OH)]_4
 (3) NaAlO_2 (4) Both (2) and (3)
28. When blue litmus is added to the solution of AlCl_3 , a red colouration is obtained, this indicates that :
 (1) Aqueous solution of AlCl_3 is acidic
 (2) Aqueous solution of AlCl_3 is basic
 (3) Aqueous solution of AlCl_3 is neutral
 (4) None of these
29. A compound is soluble in water. If ammonium hydroxide is added to aqueous solution of compound a brownish precipitate appears which is soluble in dilute HCl . The basic radical of compound is:
 (1) Aluminium (III) (2) Zinc (II)
 (3) Iron (III) (4) Iron (II)
30. On addition of aqueous NaOH to a salt solution, a white gelatinous precipitate is formed, which dissolves in excess alkali. The salt solution contains :
 (1) Calcium ions
 (2) Aluminium ions
 (3) Barium ions
 (4) Iron ions

Group 14 Elements and their Properties

Property	Element				
	Carbon C	Silicon Si	Germanium Ge	Tin Sn	Lead Pb
Atomic Number	6	14	32	50	82
Atomic mass (g mol^{-1})	12.01	28.09	72.60	118.71	207.2
Electronic configuration	$[\text{He}]2s^2 2p^2$	$[\text{Ne}]3s^2 3p^2$	$[\text{Ar}]3d^{10} 4s^2 4p^2$	$[\text{Kr}]4d^{10} 5s^2 5p^2$	$[\text{Xe}]4f^{14} 5d^{10} 6s^2 6p^2$
Covalent radius/ pm^a	77	118	122	140	146
Ionic radius $\text{M}^{4+}/\text{pm}^b$	–	40	53	69	78
Ionic radius $\text{M}^{2+}/\text{pm}^b$	–	–	73	118	119
Ionization enthalpy/ kJ mol^{-1}	$\Delta_i H_1$	1086	786	761	708
	$\Delta_i H_2$	2352	1577	1537	1411
	$\Delta_i H_3$	4620	3228	3300	2942
	$\Delta_i H_4$	6220	4354	4409	3929
Electronegativity ^c	2.5	1.8	1.8	1.8	1.9
Density ^d / gcm^{-3}	3.51 ^e	2.34	5.32	7.26 ^f	11.34
Melting point/K	4373	1693	1218	505	600
Boiling point/K	–	3550	3123	2896	2024
Electrical resistivity/ ohm cm (293 K)	$10^{14} - 10^{16}$	50	50	10^{-5}	2×10^{-5}

- **Electronic Configuration:** ns^2np^2 .
- **Covalent radius :**
 $C < Si < Ge < Sn < Pb$
From Si to Pb a small increase in radius is due to the presence of completely filled d and f orbitals in heavier members.
- **Ionization Enthalpy:** $C > Si > Ge > Pb > Sn$
Slight increase in $\Delta_i H$ from Sn to Pb is the consequence of poor shielding effects of intervening d and f-orbitals.
- **Electronegativity:**
Due to small size, the elements of this group are slightly more electronegative than group 13 elements. The electronegativity value for elements from Si to Pb are almost the same.

Physical Properties:

Carbon and silicon are non-metals; germanium is metalloid whereas tin and lead are soft metals with low melting points. Melting points and boiling points of group 14 elements are much higher than those of corresponding elements of group 13.

Chemical Properties:**Oxidation States and Trends in Chemical Reactivity:**

The group 14 elements have four electrons in outermost shell. The common oxidation states exhibited by these elements are + 4 and + 2. Carbon also exhibits negative oxidation states. Since the sum of the first four ionization enthalpies is very high, compound in +4 oxidation state are generally covalent in nature. In heavier members the tendency to show +2 oxidation state increases in the sequence $Ge < Sn < Pb$. It is due to the inability of ns^2 electrons of valence shell to participate in bonding.

(i) Reactivity Towards Oxygen:

There are mainly two types of oxides, i.e. monoxide and dioxide of formula MO and MO_2 respectively. SiO only exists at high temperature. Oxides in higher oxidation states of elements are generally more acidic than those in lower oxidation state. The dioxides – CO_2 , SiO_2 and GeO_2 are acidic, whereas SnO_2 and PbO_2 are amphoteric in nature. Among monoxides, CO is neutral, GeO is distinctly acidic whereas SnO and PbO are amphoteric.

(ii) Reactivity Towards Water:

Carbon, silicon and germanium are not affected by water. Tin decomposes steam to form dioxide and dihydrogen gas. Lead is unaffected by water, probably because of a protective oxide film formation.

(iii) Reactivity Towards Halogen:

These elements can form halides of formula MX_2 and MX_4 (where X = F, Cl Br, I). Except carbon all other members react directly with halogen under suitable condition to make halides. Most of the MX_4 are covalent in nature. Exceptions are SnF_4 and PbF_4 , which are ionic in nature. PbI_4 does not exist because Pb–I bond initially formed during the reaction does not release enough energy to unpair $6s^2$ electrons and excite one of them to higher orbital to have four unpaired electrons around lead atom. Heavier members Ge to Pb are able to make halides of formula MX_2 . Stability of dihalides increases down the group. Except CCl_4 other tetrachlorides are easily hydrolysed by water because the central atom can accommodate the lone pair of electrons from oxygen atom of water molecules in d orbital.

Like first member of other groups, carbon also differs from rest of the members of its group. It is due to its smaller size, higher electronegativity, higher ionisation enthalpy and unavailability of d orbitals.

In carbon, only s and p orbitals are available for bonding and, therefore, it can accommodate only four pairs of electrons around it. This would limit the maximum covalence to four whereas other members can expand their covalence due to the presence of d orbitals.

Carbon also has unique ability to form $p\pi-p\pi$ multiple bonds with itself and with other atoms of small size and high electronegativity. Few examples of multiple bonding are: $C=C$, $C\equiv C$, $C=O$, $C=S$, and $C\equiv N$. Heavier elements do not form $p\pi-p\pi$ bonds because their atomic orbitals are too large and diffuse to have effective overlapping.

Carbon atoms have the tendency to link with one another through covalent bonds to form chains and rings. This property is called catenation. This is because $C-C$ bonds are very strong. Down the group the size increases and electronegativity decreases, and, thereby, tendency to show catenation decreases. This can be clearly seen from bond enthalpies values. The order of catenation is $C > Si > Ge \approx Sn$. Lead does not show catenation.

Bond	Bond enthalpy/ kJ mol^{-1}
$C-C$	348
$Si-Si$	297
$Ge-Ge$	260
$Sn-Sn$	240

Allotropes of Carbon

Crystalline form:

Diamond, Graphite, Fullerenes,

Amorphous form:

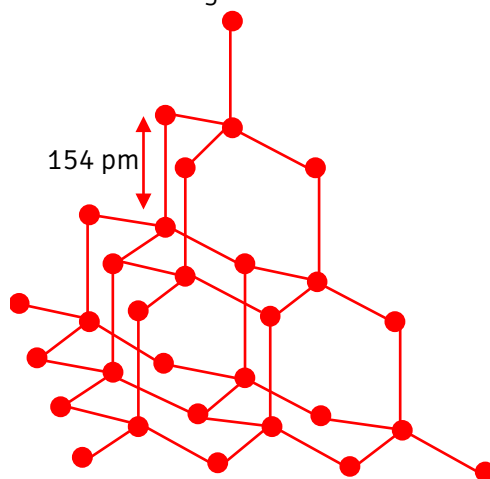
Coke, Charcoal, Coals

- **Diamond:**

sp^3 hybridisation

3-D Covalent Solid (Directional Covalent Bond)

Diamond is the hardest substance on earth. It is used as an abrasive for sharpening hard tools in making dyes and in the manufacture of tungsten filament for electric light bulbs.



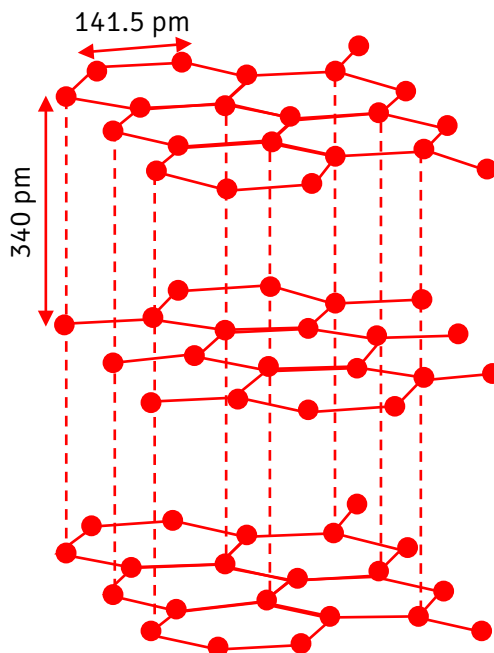
- **Graphite:**

Graphite has a layered structure. Layers are held by van der Waals' forces.

sp^2 hybridisation

The electrons are delocalised over the whole sheet. Graphite conducts electricity along the sheet. Graphite cleaves easily between the layers and therefore, it is very soft and slippery. For this reason, graphite is used as a dry lubricant in machines running at high temperature, where oil cannot be used as a lubricant.

Graphite is also known as plumbago (used as lead pencil)

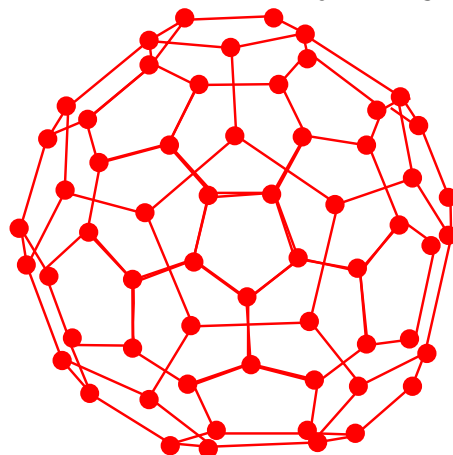
**Thermal Stability:**

Diamond < Graphite

In diamond and graphite some surface carbon atoms have free valency. These carbon atoms form new bonds with impurities. These new bonds are known as dangling bond.

Fullerenes:

Fullerenes are made by the heating of graphite in an electrical arc in the presence of inert gases such as helium or argon. Fullerenes are the only pure form of carbon because they have smooth structure without having 'dangling' bonds. Fullerenes are cage like molecules having even number of carbon atoms like C_{60} , C_{70} and C_{350} . C_{60} molecule has a shape like soccer ball and called Buckminsterfullerene. It contains twenty, six-membered rings and twelve, five membered rings. A six membered ring is fused with six or five membered rings but a five membered ring can only fuse with six membered rings. All the carbon atoms are equal and they undergo sp^2 hybridisation. The remaining electron at each carbon atom is delocalised in molecular orbitals, which in turn give aromatic character to molecule. This ball shaped molecule has 60 vertices and each one is occupied by one carbon atom and it also contains both single and double bonds with C-C distance of 143.5 pm and 138.3 pm respectively. Spherical fullerenes are also called bucky balls in short. It is very important to know that graphite is thermodynamically most stable allotrope of carbon and, therefore, $\Delta_f H^{(-)}$ values of diamond and fullerene, C_{60} are 1.90 and 38.1 kJ mol^{-1} , respectively. Carbon black is obtained by burning hydrocarbons in a limited supply of air.



Use of Carbon and Silicon :

Graphite fibres embedded in plastic material form high strength, lightweight composites. The composites are used in products such as tennis rackets, fishing rods, aircrafts and canoes. Being good conductor, graphite is used for electrodes in batteries and industrial electrolysis. Crucibles made from graphite are inert to dilute acids and alkalies. Being highly porous, activated charcoal is used in adsorbing poisonous gases; also used in water filters to remove organic contaminants and in airconditioning system to control odour. Carbon black is used as black pigment in black ink and as filler in automobile tyres. Coke is used as a fuel and largely as a reducing agent in metallurgy. Diamond is a precious stone and used in jewellery. It is measured in carats (1 carat = 200 mg).

NCERT QUESTIONS (Solve Yourself)

- Select the member(s) of group 14 that
 - forms the most acidic dioxide,
 - is commonly found in +2 oxidation state,
 - used as semiconductor.
- $[\text{SiF}_6]^{2-}$ is known whereas $[\text{SiCl}_6]^{2-}$ not. Give possible reasons.
- Diamond is covalent, yet it has high melting point. Why?
- What are silicones?
- Discuss the pattern of variation in the oxidation states of
 - B to Tl and
 - C to Pb.
- Write the resonance structures of CO_3^{2-} and HCO_3^- .
- What is the state of hybridisation of carbon in (a) CO_3^{2-} (b) diamond (c) graphite?
- Explain the difference in properties of diamond and graphite on the basis of their structures.
- Rationalise the given statements and give chemical reactions :
 - Lead(II) chloride reacts with Cl_2 to give PbCl_4 .
 - Lead(IV) chloride is highly unstable towards heat.
 - Lead is known not to form an iodide, PbI_4 .
- Suggest a reason as to why CO is poisonous.
- How is excessive content of CO_2 responsible for global warming?
- Complete the following reactions:
 - $\text{C}_2\text{H}_4 + \text{O}_2 \rightarrow$
 - $4\text{Al} + 3\text{O}_2 \rightarrow$
- Explain the following reactions
 - Silicon is heated with methyl chloride at high temperature in the presence of copper;
 - Silicon dioxide is treated with hydrogen fluoride;
 - CO is heated with ZnO;
 - Hydrated alumina is treated with aqueous NaOH solution.
- Explain why is there a phenomenal decrease in ionization enthalpy from carbon to silicon?
- What are allotropes? Sketch the structure of two allotropes of carbon namely diamond and graphite. What is the impact of structure on physical properties of two allotropes?
- Classify following oxides as neutral, acidic, basic or amphoteric:
CO, B_2O_3 , SiO_2 , CO_2 , Al_2O_3 , PbO_2 , Tl_2O_3
 - Write suitable chemical equations to show their nature.
- Give one method for industrial preparation and one for laboratory preparation of CO and CO_2 each.
- Boric acid is polymeric due to
 - its acidic nature
 - the presence of hydrogen bonds
 - its monobasic nature
 - its geometry

19.	The type of hybridisation of boron in diborane is			
	(a) sp	(b) sp ²	(c) sp ³	(d) dsp ²
20.	Thermodynamically the most stable form of carbon is			
	(a) diamond	(b) graphite	(c) fullerenes	(d) coal
21.	Elements of group 14			
	(a) exhibit oxidation state of +4 only		(b) exhibit oxidation state of +2 and +4	
	(c) form M ²⁻ and M ⁴⁺ ions		(d) form M ²⁺ and M ⁴⁺ ions	
22.	If the starting material for the manufacture of silicones is RSiCl ₃ , write the structure of the product formed.			

Exercise 1.2

- Which of the following statements are correct?
 (a) Fullerenes have dangling bonds
 (b) Fullerenes are cage-like molecules
 (c) Graphite is thermodynamically most stable allotrope of carbon
 (d) Graphite is slippery and hard and therefore used as dry lubricant in machines
 (1) a, b, c (2) b, c
 (3) a, b, d (4) a, c
- Which of the following is a good conductor of electricity?
 (1) diamond (2) graphite
 (3) coal (4) none
- Glass is soluble in
 (1) HF (2) H_2SO_4
 (3) $HClO_4$ (4) aqua-regia
- Which one of the oxides is neutral?
 (1) CO (2) SnO_2 (3) ZnO (4) SiO_2
- Inert form of carbon is
 (1) Diamond (2) Graphite
 (3) Coal (4) Charcoal
- Elements of group 14
 (1) exhibit oxidation state of +4 only
 (2) exhibit oxidation state of +2 and +4
 (3) form M^{-2} and M^{4+} ions
 (4) form M^{2+} and M^{4+} ions
- The hybridisation of the central atom in SiF_6^{2-} , $[GeCl_6]^{2-}$, $[Sn(OH)_6]^{6-}$ is
 (1) sp^3d (2) sp^3d^2
 (3) sp^3 (4) sp^3d^3
- Consider the following oxides
 GeO , CO_2 , CO , GeO_2 , SiO_2 , SnO_2 , SnO ,
 Which of the following option is correct?
 (1) Acidic – GeO_2 , CO , SiO_2 ; Neutral – GeO ;
 Amphoteric – CO_2 , SnO , SnO_2
 (2) Acidic – SnO_2 , GeO_2 , CO_2 ; Neutral – CO ,
 SnO , SiO_2 ; Amphoteric – GeO
 (3) Acidic – CO_2 , GeO_2 , SiO_2 ; Neutral – CO
 Amphoteric – SnO_2 , SnO
 (4) Acidic – GeO , SiO_2 , SnO ; Neutral – CO ,
 CO_2 ; Amphoteric – SnO_2 , GeO_2
- In laboratory carbon dioxide is prepared by the reaction of A and B. Here A and B are respectively
 (1) A → dil. HCl; B → calcium carbonate
 (2) A → conc. HCl; B → calcium carbide
 (3) A → dil. HCl; B → calcium carbide
 (4) A → conc. HCN; B → magnesium carbonate
- Carbon dioxide can be obtained as a solid in the form of
 (1) Dry ice (2) Wet ice
 (3) Both (1) and (2) (4) None of these
- Configuration of Pb is
 (1) $[Kr]4d^{10}5s^25p^2$
 (2) $[Kr]4d^{10}5s^25p^3$
 (3) $[Xe]4f^{14}5d^{10}6s^26p^2$
 (4) $[Xe]4f^{14}5d^{10}6s^26p^3$
- Nature of CO and CO_2 is :
 (1) Acidic and amphoteric respectively
 (2) Neutral and acidic respectively
 (3) Neutral and amphoteric respectively
 (4) Basic and Acidic respectively

- 13.** CCl_4 does not hydrolyse because
 (1) Carbon has negative charge
 (2) Chlorine has positive charge
 (3) Carbon does not have vacant d-orbitals
 (4) Electronegativity of chlorine is not high
- 14.** Reason for Anomalous behaviour of carbon are
 (1) Absence of vacant d
 (2) Small size
 (3) High IP and EN
 (4) All
- 15.** Diamond is an example of
 (1) molecular solid (2) ionic solid
 (3) covalent solid (4) metallic solid
- 16.** Fullerenes are only pure form of carbon due to
 (1) Cage like molecules
 (2) Soccer ball structure
 (3) Absence of dangling bonds
 (4) Presence of six membered ring
- 17.** $\text{X}=\text{X}$ is stable if X is :
 (1) C (2) Si
 (3) Ge (4) Sn
- 18.** Carbon is sp^2 hybridised in
 (1) diamond and graphite
 (2) graphite and fullerenes
 (3) diamond and fullerenes
 (4) only diamond
- 19.** Hybridisation of silicon in SiO_2 is :
 (1) sp (2) sp^2
 (3) sp^3 (4) sp^3d
- 20.** On addition of aqueous NaOH to a salt solution, a white gelatinous precipitate is formed, which dissolves in excess alkali. The salt solution contains :
 (1) Calcium ions
 (2) Aluminium ions
 (3) Barium ions
 (4) Iron ions
- 21.** A salt contain Pb^{2+} ; its presence can not be identified by :
 (1) H_2S (2) K_2CrO_4
 (3) CH_3COONa (4) KI
- 22.** A gas is obtained by addition of dil. H_2SO_4 to a mixture which turns lead acetate paper black. It is :
 (1) SO_2 (2) CO_2
 (3) H_2S (4) NO_2
- 23.** An aqueous solution of a substance gives a white precipitate on treatment with dil. HCl which dissolves on heating. When H_2S is passed through the hot acidic solution, a black precipitate is obtained. The substance is :
 (1) Hg_2^{2+} salt (2) Cu^{2+} salt
 (3) Ag^+ salt (4) Pb^{2+} salt
- 24.** $\text{Pb}(\text{CH}_3\text{COO})_2$ gives colour with H_2S :
 (1) Black (2) White
 (3) Red (4) Orange
- 25.** Which of the following salt is soluble in hot water but insoluble in cold water and its aqueous solution also gives yellow ppt. with a solution containing CrO_4^{2-} ions :
 (1) CaSO_4 (2) CaCl_2
 (3) PbCl_2 (4) NH_4Cl
- 26.** An aqueous solution of a substance gives a white ppt. on treatment with dil. HCl , which dissolved on heating. When hydrogen sulphite is passed through the hot acidic solution, a black ppt. is obtained. The substance is a :
 (1) Ba^{2+} salt
 (2) Cu^{2+} salt
 (3) Al^{3+} salt
 (4) Pb^{2+} salt

Group 15 Elements and their Properties

Property	N	P	As	Sb	Bi
Atomic number	7	15	33	51	83
Atomic mass/g mol ⁻¹	14.01	30.97	74.92	121.75	208.98
Electronic configuration	[He]2s ² 2p ³	[Ne]3s ² 3p ³	[Ar]3d ¹⁰ 4s ² 4p ³	[Kr]4d ¹⁰ 5s ² 5p ³	[Xe]4f ¹⁴ 5d ¹⁰ 6s ² 6p ³
Ionisation enthalpy	I	1402	1012	947	834
(D ₁ H)/(kJ mol ⁻¹)	II	2856	1903	1798	1595
	III	4577	2910	2736	2443
Electronegativity	3.0	2.1	2.0	1.9	1.9
Covalent radius/pm ^a	70	110	121	141	148
Ionic radius/pm	171 ^b	212 ^b	222 ^b	76 ^c	103 ^c
Melting point/K	63 [*]	317 ^d	1089 ^e	904	544
Boiling point/K	77.2 [*]	554 ^d	888 ^f	1860	1837
Density/[g cm ⁻³ (298 K)]	0.879 ^g	1.823	5.778 ^h	6.697	9.808

^aE^{III} single bond (E = element); ^bE³⁻; ^cE³⁺; ^dWhite phosphorus; ^eGrey α -form at 38.6 atm; ^fSublimation temperature; ^gAt 63 K; ^hGrey α -form; ^{*}Molecular N₂.

Physical Properties:

All the elements of this group are polyatomic. Dinitrogen is a diatomic gas while all others are solids. Metallic character increases down the group. Nitrogen and phosphorus are non – metals, arsenic and antimony metalloids and bismuth are metals. Except nitrogen, all the elements show allotropy.

Chemical Properties:

- In the case of nitrogen, all oxidation states from +1 to +4 tend to disproportionate in acid solution.

Example:

- Similarly, in case of phosphorus nearly all intermediate oxidation states disproportionate into +5 and –3 both in alkali and acid. However, +3 oxidation state in case of arsenic, antimony and bismuth become increasingly stable with respect to disproportionation.

(i) Reactivity Towards Hydrogen:

All the elements of Group 15 form hydrides of the type EH₃ where E = N, P, As, Sb or Bi. The stability of hydrides decreases from NH₃ to BiH₃ which can be observed from their bond dissociation enthalpy. Consequently, the reducing character of the hydrides increases. BiH₃ is the strongest reducing agent amongst all the hydrides. Basicity order NH₃ > PH₃ > AsH₃ > SbH₃

MP Order NH₃ > SbH₃ > AsH₃ > PH₃

BP Order SbH₃ > NH₃ > AsH₃ > PH₃

(ii) Reactivity Towards Oxygen:

All these elements form two types of oxides: E₂O₃ and E₂O₅. The oxides of the type E₂O₃ of nitrogen and phosphorus are purely acidic, that of arsenic and antimony amphoteric and those of bismuth is predominantly basic.

(iii) Reactivity Towards Halogens:

These elements react to form two series of halides: EX_3 and EX_5 . Pentahalides are more covalent than trihalides. All the trihalides of these elements except those of nitrogen are stable. In case of nitrogen, only NF_3 is known to be stable. Trihalides except BiF_3 are predominantly covalent in nature.

(iv) Reactivity Towards Metals:

These elements react with metals to form their binary compounds exhibiting -3 oxidation state, such as, Ca_3N_2 (calcium nitride) Ca_3P_2 (calcium phosphide), Na_3As_2 (sodium arsenide), Zn_3Sb_2 (zinc antimonide) and Mg_3Bi_2 (magnesium bismuthide).

(v) Anomalous Behavior of Nitrogen:

Nitrogen differs from the rest of the members of this group due to its small size, high electronegativity, high ionisation enthalpy and non-availability of d orbitals. Nitrogen has unique ability to form $p\pi-p\pi$ multiple bonds with itself and with other elements having small size and high electronegativity (e.g., C, O). Heavier elements of this group do not form $p\pi-p\pi$ bonds as their atomic orbitals are so large and diffuse that they cannot have effective overlapping. Thus, nitrogen exists as a diatomic molecule with a triple bond (one s and two p) between the two atoms. Consequently, its bond enthalpy ($941.4 \text{ kJ mol}^{-1}$) is very high. On the contrary, phosphorus, arsenic and antimony form single bonds as P–P, As–As and Sb–Sb while bismuth forms metallic bonds in elemental state. However, the single N–N bond is weaker than the single P–P bond because of high interelectronic repulsion of the non-bonding electrons, owing to the small bond length. As a result the catenation tendency is weaker in nitrogen. Another factor which affects the chemistry of nitrogen is the absence of d orbitals in its valence shell. Besides restricting its covalency to four, nitrogen cannot form $d\pi-p\pi$ bond as the heavier elements can e.g., $R_3P=O$ or $R_3P=CH_2$ ($R = \text{alkyl group}$). Phosphorus and arsenic can form $d\pi-d\pi$ bond also with transition metals when their compounds like $P(C_2H_5)_3$ and $As(C_2H_5)_3$ act as ligands.

Nitrogen Gas (N_2):**(a) Properties:**

- (i) N_2 is a colourless, odourless gas insoluble in water.
- (ii) It is non-polar covalent and neutral molecule.
- (iii) It is neither combustible nor a supporter of combustion.
- (iv) It is absorbed by heated Mg and Al. The nitrides formed thus react with water to form NH_3 .

$$3Mg + N_2 \longrightarrow Mg_3N_2 (+ 6H_2O) \longrightarrow 3Mg(OH)_3 + 2NH_3 \uparrow$$

$$2Al + N_2 \longrightarrow 2AlN (+ 6H_2O) \longrightarrow 2Al(OH)_3 + 2NH_3 \uparrow$$
- (v) Reaction with oxygen: When air free from CO_2 and moisture is passed over an electric arc at about 2000 K, nitric oxide is formed. This reaction is endothermic.

$$N_2 + O_2 \longrightarrow 2NO$$
- (vi) Reaction with CaC_2 and BaC_2 : At 1100°C , these carbides react with N_2 forming $CaCN_2$ and $Ba(CN)_2$ respectively.

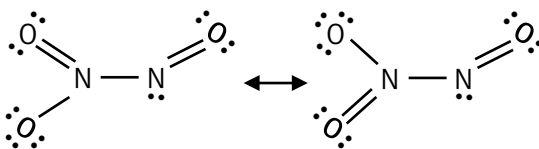
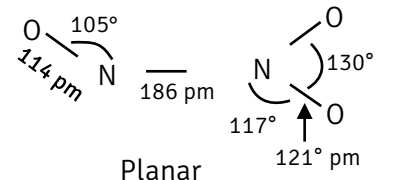
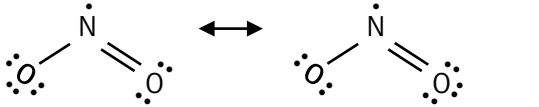
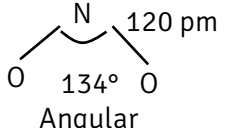
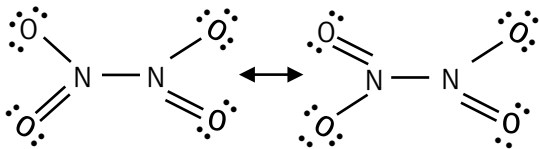
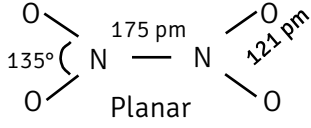
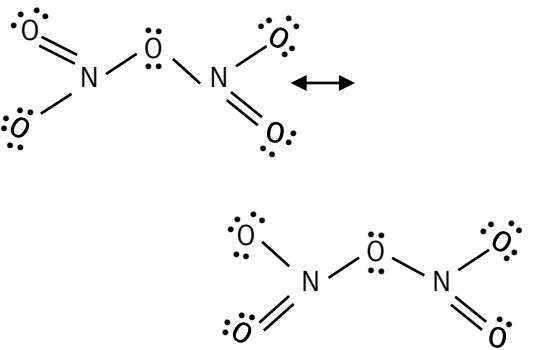
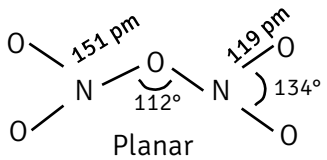
$$CaC_2 + N_2 + \xrightarrow{\Delta} CaCN_2 + C \text{ (nitrolim, a fertilizer)}$$

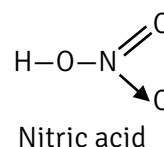
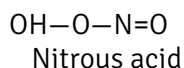
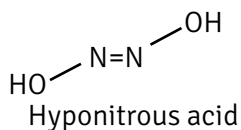
(b) Uses:

- (i) for providing an inert atmosphere during many industrial processes where presence of air or O_2 is to be avoided.
- (ii) for manufacture of NH_3 by the Haber's process.
- (iii) for manufacture of nitrolim.

Compound of Nitrogen**(1) Oxides of Nitrogen:**

Nitrogen forms a number of oxides, N_2O , NO , N_2O_3 , NO_2 or N_2O_4 and N_2O_5 , and also very unstable NO_3 and N_2O_6 . All these oxides of nitrogen exhibit $p\pi-p\pi$ multiple bonding between nitrogen and oxygen.

Name	Formula	Resonance Structures	Bond parameters
Nitrous oxide	N_2O	$\ddot{\text{N}}=\ddot{\text{N}}=\ddot{\text{O}} \longleftrightarrow :\text{N}\equiv\text{N}-\ddot{\text{O}}:$	$\text{N}-\text{N}-\text{O}$ 113 pm 119 pm Linear
Nitric oxide	NO	$:\text{N}=\ddot{\text{O}}: \longleftrightarrow :\dot{\text{N}}=\ddot{\text{O}}:$	$\text{N}-\text{O}$ 115 pm
Dinitrogen trioxide	N_2O_3		 Planar
Nitrogen dioxide	NO_2		 Angular
Dinitrogen tetroxide	N_2O_4		 Planar
Dinitrogen pentoxide	N_2O_5		 Planar

(2) Oxyacids of Nitrogen:**Allotropic Forms of Phosphorus**

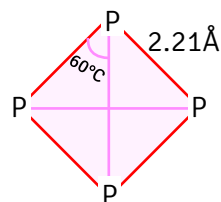
Phosphorus is a very reactive non-metal. It catches fire in air. It occurs in nature in the form of stable phosphates. (Animal bones also contain calcium (58%) phosphate.) Important minerals Fluoropeptides

(a) White Phosphorus**(b) Red Phosphorus****(c) Black phosphorus**

White Phosphorus

Waxy solid

Poisonous

Soluble in CS_2 , Insoluble in waterMonomer of P_4 (discrete tetrahedral)

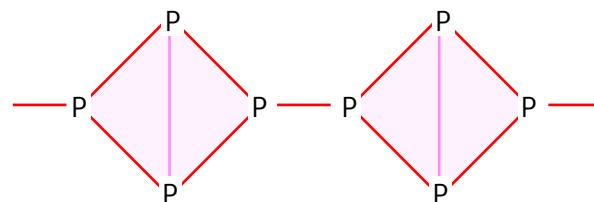
Highly reactive due to bond angle strain

It glows in dark due to slow oxidation (phosphorescence)

It gives phosphene (PH_3) on reaction with NaOH **Red Phosphorus**

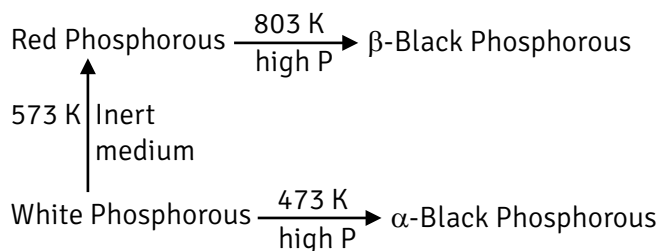
Brittle Powder

Non poisonous

Insoluble in water & CS_2 Polymer of P_4 (tetrahedra linked together)

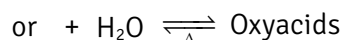
More stable than white phosphorus

It does not glow in dark

It gives hypophosphoric acid when on reaction with NaOH **Order of Stability of MP or Density** $\rightarrow$ White < Red < Black**Compounds of Phosphorus****OXY-Acids:** General Formula of Oxy acid: $\text{H}_x\text{Z}_y\text{O}_z$ (Z = non metal)

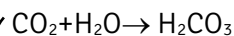
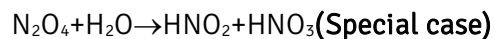
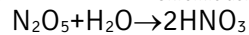
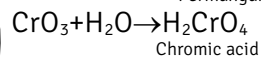
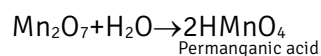
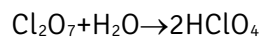
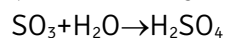
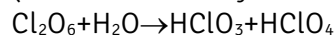
(1) Compound which contain Z-OH bond are known as oxyacids.

(2) Acidic oxide

Non metallic oxide

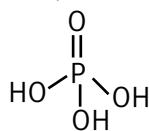
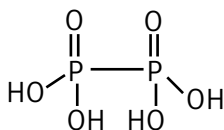
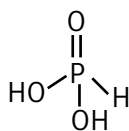
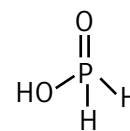
Anhydrides of oxyacids

(3) Oxidation state of central atom does not change.

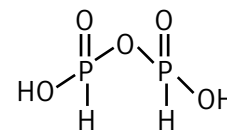
Anhydrides of their
Corresponding oxyacids(So CO_2 is anhydride of H_2CO_3)(N₂O₄ is mixed anhydride of HNO_2 & HNO_3)(Cl₂O₆ is mixed anhydride of HClO_3 and HClO_4)

Pyro	Meta	Per	Hypo	
$\downarrow$ 2 mole oxy acid $\downarrow -H_2O$ Pyro acid	$\downarrow$ 1 mole oxy acid $\downarrow -H_2O$ meta acid	$\downarrow$ oxy acid $\downarrow -H_2O$ $Z-O-O-H$	$\downarrow$ us acid $\downarrow -O$ Hypo us acid	$\downarrow$ ic acid $\downarrow -O$ Hypo ic acid
Example: $H_2S_2O_5$ $H_2S_2O_7$ $H_4P_2O_7$	Example: HPO_3 HBO_2	Example: H_2SO_4 $H_2S_2O_8$ HNO_4 CH_3CO_3H $C_6H_5CO_3H$	Example: HOX H_3PO_2	Example: $H_4P_2O_6$

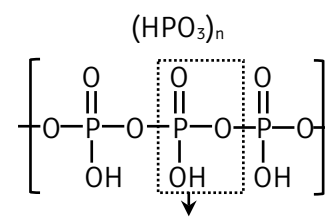
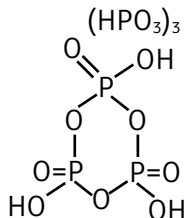
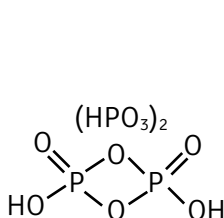
(4) Oxy Acid of Phosphorus:

Orthophosphoric acid
/Phosphoric acidHypophosphoric acid
Pyrophosphoric acidOrthophosphorus acid/
Phosphorus acid/Phosphonic acidHypophosphorus acid/
Phosphinic acid

Pyro phosphorus acid

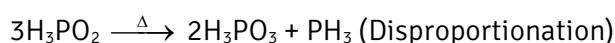
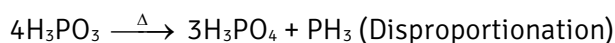
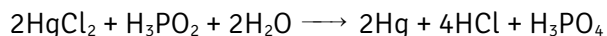
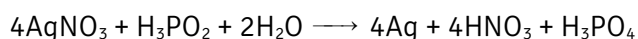


Metaphosphoric acid exists as dimer, cyclic trimer or polymer

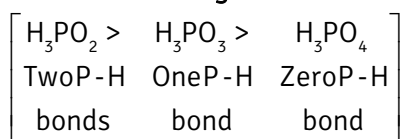


(a) Properties:

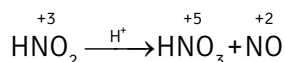
- (i) Oxy acid of phosphorus having +1 & +3 oxidation state (having P-H bond) can act as reducing agent or give disproportionation reaction on heating



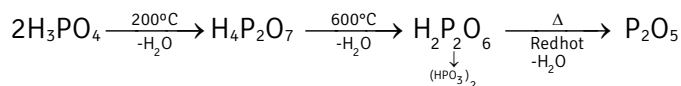
(ii) Order of reducing nature.



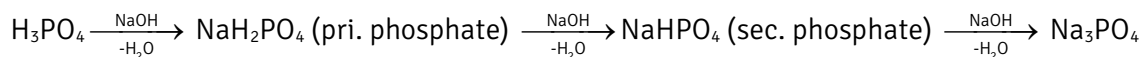
Note: Nitrogen compound having +1 & +3 oxidation state gives disproportionation reaction in acidic medium.



(iii) Heating effect of Phosphoric acid

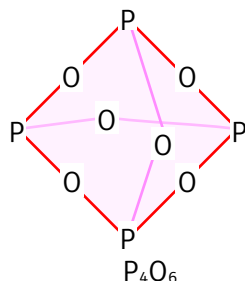


(iv) H_3PO_4 gives three series of salt

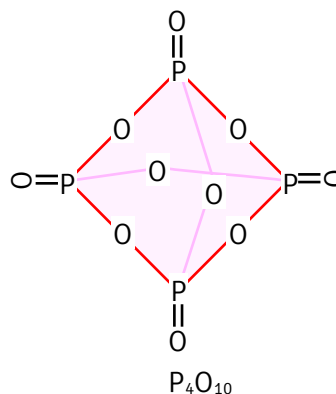


(tert. phosphate)

Oxides of Phosphorus:



- sp^3 hybrid 'P'
- 6 P-O-P bonds
- l.p. on 'P' = 4



- sp^3 hybrid 'P'
- 6 P-O-P bonds
- 4 P=O bonds

NCERT QUESTIONS (Solve Yourself)

- Why are pentahalides more covalent than trihalides?
- Why is BiH_3 the strongest reducing agent amongst all the hydrides of Group 15 elements?
- Why is N_2 less reactive at room temperature?
- Mention the conditions required to maximise the yield of ammonia.
- How does ammonia react with a solution of Cu^{2+} ?
- What is the covalence of nitrogen in N_2O_5 ?
- Bond angle in PH_4^+ is higher than that in PH_3 . Why?
- What happens when white phosphorus is heated with concentrated NaOH solution in an inert atmosphere of CO_2 ?
- What happens when PCl_5 is heated?
- Write a balanced equation for the hydrolytic reaction of PCl_5 in heavy water.
- What is the basicity of H_3PO_4 ?
- What happens when H_3PO_3 is heated?
- Discuss the general characteristics of Group 15 elements with reference to their electronic configuration, oxidation state, atomic size, ionisation enthalpy and electronegativity.
- Why does the reactivity of nitrogen differ from phosphorus?
- Discuss the trends in chemical reactivity of group 15 elements.
- Why does NH_3 form hydrogen bond but PH_3 does not?
- How is nitrogen prepared in the laboratory? Write the chemical equations of the reactions involved.

18. How is ammonia manufactured industrially?
19. Illustrate how copper metal can give different products on reaction with HNO_3 .
20. Give the resonating structures of NO_2 and N_2O_5 .
21. The HNH angle value is higher than HPH, HAsH and HSbH angles. Why?
[Hint: Can be explained on the basis of sp^3 hybridisation in NH_3 and only s-p bonding between hydrogen and other elements of the group].
22. Why does $\text{R}_3\text{P}=\text{O}$ exist but $\text{R}_3\text{N}=\text{O}$ does not (R = alkyl group)?
23. Explain why NH_3 is basic while BiH_3 is only feebly basic.
24. Nitrogen exists as diatomic molecule and phosphorus as P_4 . Why?
25. Write main differences between the properties of white phosphorus and red phosphorus.
26. Why does nitrogen show catenation properties less than phosphorus?
27. Give the disproportionation reaction of H_3PO_3 .
28. Can PCl_5 act as an oxidising as well as a reducing agent? Justify.
29. What are the oxidation states of phosphorus in the following:
(i) H_3PO_3 (ii) PCl_3 (iii) Ca_3P_2 (iv) Na_3PO_4 (v) POF_3 ?

Exercise 1.3

1. Which of the compounds will act as strongest reducing agent?
(1) NH_3 (2) PH_3 (3) AsH_3 (4) SbH_3
2. The thermal stability of the hydrides of group 15 follows the order
(1) $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3 < \text{BiH}_3$
(2) $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3 > \text{BiH}_3$
(3) $\text{PH}_3 > \text{NH}_3 > \text{AsH}_3 > \text{SbH}_3 < \text{BiH}_3$
(4) $\text{AsH}_3 < \text{PH}_3 > \text{SbH}_3 > \text{BiH}_3 > \text{NH}_3$
3. Which of the following oxides is amphoteric in nature?
(1) N_2O_3 (2) P_4O_6 (3) Sb_2O_3 (4) Bi_2O_3
4. Which of the following is (are) paramagnetic in nature?
(1) NO (2) NO_2
(3) N_2O_4 (4) 1 and 2 both
5. Which of the following compounds does give N_2 on heating?
(1) NH_4NO_2 (2) NH_4NO_3
(3) NaN_3 (4) Both (1) and (3)
6. Which of the following acids can form two types of salts?
(1) Hyponitrous acid (2) Nitrous acid
(3) Nitric acid (4) Pernitric acid
7. Nitric acid oxidises P into
(1) PH_3 (2) P_2O_5 (3) HPO_3 (4) H_3PO_4
8. Which of the following metals does not dissolve in conc. HNO_3 ?
(1) Pb (2) Cu (3) Au (4) Hg
9. Red phosphorus can be prepared from white phosphorus by
(1) adding red colour to white phosphorus
(2) heating white phosphorus to red heat
(3) heating white phosphorus with few crystals of iodine
(4) dissolving white phosphorus in NaOH
10. Nitrolim is obtained by passing nitrogen over:
(1) heated mixture of Al_2O_3 and carbon
(2) carborundum
(3) calcium carbide
(4) heated aluminium
11. Which is a set of acid salts and can react with base?
(1) NaH_2PO_2 , Na_2HPO_3 , NaH_2PO_4
(2) Na_2HPO_3 , NaH_2PO_3 , Na_2HPO_4
(3) NaH_2PO_4 , NaH_2PO_3 , Na_2HPO_4
(4) All of these
12. The wrong statement about N_2O is
(1) It is nitrous oxide
(2) It is least reactive oxide of nitrogen
(3) It is not a linear molecule
(4) It is known as laughing gas

- 13.** The true statement for the acids of phosphorus. H_3PO_2 , H_3PO_3 and H_3PO_4 is.
 (1) H_3PO_3 on heating does not disproportionate
 (2) All of them are reducing in nature
 (3) All of them are tribasic acids
 (4) The geometry of phosphorus is tetrahedral in all the three
- 14.** Following are neutral oxides except:
 (1) NO (2) N_2O
 (3) CO (4) NO_2
- 15.** The high reactivity and low volatility of white phosphorus is due to:
 (1) Tetrahedrally arranged P_4 units
 (2) Bond angle of 60° increases steric (strain) factor
 (3) Weak van der Waals forces of attraction
 (4) Both (2) and (3)
- 16.** Nitrogen forms N_2 but phosphorus is converted into P_4 from P_2 . The reason is
 (1) weak $p\pi-p\pi$ bonding
 (2) multiple bond is formed easily
 (3) strong $p\pi-p\pi$ bonding
 (4) triple bond is present in phosphorus atoms
- 17.** Arrange the following hydrides in the decreasing order of their basicities : NH_3 , AsH_3 , SbH_3 , BiH_3 and PH_3
 (1) $\text{NH}_3 > \text{AsH}_3 > \text{SbH}_3 > \text{BiH}_3 > \text{PH}_3$
 (2) $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3 \geq \text{BiH}_3$
 (3) $\text{BiH}_3 > \text{SbH}_3 > \text{AsH}_3 > \text{PH}_3 \geq \text{NH}_3$
 (4) $\text{PH}_3 > \text{BiH}_3 > \text{SbH}_3 > \text{AsH}_3 > \text{NH}_3$
- 18.** Extra pure N_2 can be obtained by heating
 (1) NH_3 with CuO (2) NH_4NO_3
 (3) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ (4) $\text{Ba}(\text{N}_3)_2$
- 19.** The species in which the N-atoms is in a state of sp -hybridisation?
 (1) NO_2^- (2) NO_3^-
 (3) NO_2 (4) NO_2^+
- 20.** Very pure nitrogen is obtained by heating
 (1) $\text{Ba}(\text{N}_3)_2$ (2) NH_4Cl
 (3) NH_4Br (4) All
- 21.** Which are neutral oxides
 (1) N_2O & NO_2 (2) NO & NO_2
 (3) N_2O & NO (4) N_2O_3 & N_2O_5
- 22.** Basic nature of ammonia is due to :
 (1) N-H bond
 (2) Planar structure
 (3) Lone pair at nitrogen
 (4) Pyramidal structure
- 23.** Anhydride of HNO_3 is
 (1) N_2O (2) N_2O_3
 (3) N_2O_4 (4) N_2O_5
- 24.** Which oxide can form dimer
 (1) N_2O (2) NO_2
 (3) Both (4) None
- 25.** Correct statement about NO_2 is :
 (1) Coloured gas (2) Paramagnetic
 (3) Bent structure (4) All
- 26.** Which ammonium salt, on thermal decomposition will give residue ?
 (1) NH_4Cl (2) NH_4NO_2
 (3) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ (4) All
- 27.** Which of the following statement is correct for allotropes of P ?
 (1) White P is soluble in CS_2 but red P is insoluble in CS_2
 (2) All P-P-P bond angle is 90° in white P
 (3) On heating in air, red P change into white P
 (4) black P is least stable allotropes
- 28.** Similarity between white P and red P :
 (1) Both are gaseous at room temperature
 (2) Both are insoluble in water
 (3) Both show chemiluminescence
 (4) Both are highly poisonous

29. Chromyl chloride test is performed for the confirmation of the presence of the following in a mixture :
(1) Sulphate (2) Carbonate
(3) Chloride (4) Nitrate
30. Nitrate salt + conc. H_2SO_4 + Cu turnings $\frac{3}{4}$ ®
Brown fumes and blue solution
The correct statement is :
(1) Brown fumes due to formation of NO_2
(2) Brown fumes due to formation of N_2O
(3) Blue solution due to formation of CuSO_4
(4) Both (1) and (3)
31. The brown ring test is performed for the qualitative detection of :
(1) Bromides (2) Iodides
(3) Nitrates (4) Phosphates
32. The brown ring test for nitrates depends on:
(1) The reduction of nitrate to nitric oxide
(2) Oxidation of nitric oxide to nitrogendioxide
(3) Reduction of ferrous sulphate to Iron
(4) Oxidising action of sulphuric acid
33. The brown ring test for nitrates depends on:
(1) The reduction of nitrate to nitric oxide
(2) Oxidation of nitric oxide to nitrogendioxide
(3) Reduction of ferrous sulphate to Iron
(4) Oxidising action of sulphuric acid
34. Nitrate is confirmed by ring test. The brown colour of the ring is due to formation of :
(1) ferrous nitrite
(2) nitroso ferrous sulphate
(3) ferrous nitrate
(4) ferrous sulphate
35. Nitrate is confirmed by ring test. The brown colour of the ring is due to the formation of:
(1) Ferrous nitrite (2) $\text{FeSO}_4 \cdot \text{NO}$
(3) FeSO_2NO_2 (4) Ferrous nitrate
36. Chromyl chloride test is performed for the confirmation of the presence of the following in a mixture :
(1) Sulphate (2) Carbonate
(3) Chloride (4) Nitrate
37. A solution of a salt in dilute sulphuric acid imparts deep blue colour with starch iodine solution it confirms the presence of which of the following :
(1) NO_2^- (2) I^-
(3) NO_3^- (4) CH_3COO^-
38. Griss-Ilosvary test is used for the confirmation of :
(1) Sulphite ion
(2) Nitrite ion
(3) Nitrate ion
(4) Carbonate ion
39. Nitrate is confirmed by ring test. The brown colour of the ring is due to formation of :
(1) ferrous nitrite
(2) nitroso ferrous sulphate
(3) ferrous nitrate
(4) ferrous sulphate
40. An aqueous solution of the salt (X) and freshly prepared FeSO_4 are mixed in the test tube and then add conc. H_2SO_4 along the sides of the test tube, a dark brown ring is formed at the junction of the two solutions. Which of the following statement is incorrect regarding this experiment :
(1) Salt contains nitrate ion
(2) Brown ring is formed due to formation of $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$
(3) Salt solution (X) will give brown fumes with conc. H_2SO_4
(4) Salt solution (X) will give yellow colour fumes with conc. H_2SO_4 followed by copper turnings

Group 16 Elements and their Properties

Property	O	S	Se	Te	Po
Atomic number	8	16	34	52	84
Atomic mass/g mol ⁻¹	16.00	32.06	78.96	127.60	210.00
Electronic configuration	[He]2s ² 2p ⁴	[Ne]3s ² 3p ⁴	[Ar]3d ¹⁰ 4s ² 4p ⁴	[Kr]4d ¹⁰ 5s ² 5p ⁴	[Xe]4f ¹⁴ 5d ¹⁰ 6s ² 6p ⁴
Covalent radius/(pm) ^a	66	104	117	137	146
Ionic radius, E ²⁻ /pm	140	184	198	221	230 ^b
Electron gain enthalpy, /Δ _{eg} H kJ mol ⁻¹	-141	-200	-195	-190	-174
Ionisation enthalpy (Δ _i H ₁) /kJ mol ⁻¹	1314	1000	941	869	813
Electronegativity	3.50	2.58	2.55	2.01	1.76
Density/g cm ⁻³ (298 K)	1.32 ^c	2.06 ^d	4.19 ^e	6.25	–
Melting point/K	55	393 ^f	490	725	520
Boiling point/K	90	718	958	1260	1235
Oxidation states*	-2, -1, 1, 2	-2, 2, 4, 6	-2, 2, 4, 6	-2, 2, 4, 6	2, 4

Physical Properties:

Oxygen and sulphur are non-metal, selenium and tellurium metalloids, whereas polonium is a metal. Polonium is radioactive and is short lived (Half-life 13.8 days). All these element exhibit allotropy. The larger difference between the melting and boiling points of oxygen and sulphur may be explained on the basis of their atomicity; oxygen exist as diatomic molecules (O₂) whereas sulphur exists as polyatomic molecule (S₈).

Chemical Properties:**(i) Reactivity with Hydrogen:**

Thermal stability and bond energy: H₂O > H₂S > H₂Se > H₂Te
 Reducing power and acidic strength H₂O < H₂S < H₂Se < H₂Te
 Order of bond angle NH₃ > H₂O > PH₃ > H₂S
 M.P. order: H₂O > H₂Te > H₂Se > H₂S
 B.P. order: H₂O > H₂Te > H₂Se > H₂S

(ii) Reactivity with Oxygen:

All these elements form oxides of the EO₂ and EO₃ types where E = S, Se, Te or Po. Ozone (O₃) and sulphur dioxide (SO₂) are gases while selenium dioxide (SeO₂) is solid. Reducing property of dioxide decreases from SO₂ to TeO₂; SO₂ is reducing while TeO₂ is an oxidising agent.

(iii) Reactivity Toward Halogens:

Elements of group 16 form a larger number of halides of the type EX₆, EX₄ and EX₂. The stabilities of the halides decrease in the order F > Cl > Br > I. Amongst hexahalides, hexafluorides are the only stable halides. Sulphur hexafluoride SF₆ is exceptionally stable for steric reasons. Amongst tetrafluorides, SF₄ is a gas, SeF₄ liquid and TeF₄ a solid. All elements except selenium form dichlorides and dibromides. The well known monohalides are dimeric in nature, Examples are S₂F₂, S₂Cl₂, S₂Br₂, Se₂Cl₂ and Se₂Br₂. These dimeric halides undergo disproportionation as given below: 2Se₂Cl₂ → SeCl₄ + 3Se.

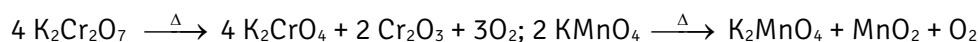
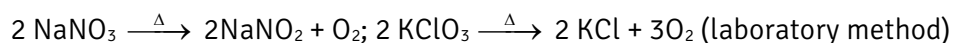
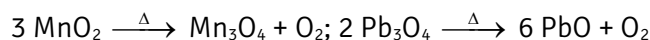
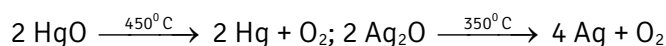
Anomalous Behaviour of Oxygen :

The anomalous behaviour of oxygen, like other members of p-block present in second period is due to its small size and high electronegativity. One typical example of effects of small size and high electronegativity is the presence of strong hydrogen bonding in H_2O which is not found in H_2S .

The absence of d orbitals in oxygen limits its covalency to four and in practice, rarely exceeds two. On the other hand, in case of other elements of the group, the valence shells can be expanded and covalence exceeds four.

Properties of Oxygen and Ozone**Dioxygen (O_2):****(a) Preparation:**

(i) By thermal decomposition of oxides of metals.

**(b) Properties:**

Colourless, odourless and tasteless gas. It is paramagnetic and exhibits allotropy. Three isotopes of oxygen are $^{16}_8\text{O}$, $^{17}_8\text{O}$ and $^{18}_8\text{O}$. Oxygen does not burn but is a strong supporter of combustion.

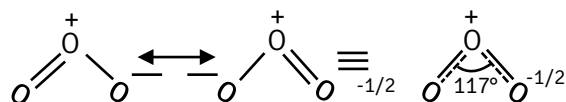
Dioxygen directly reacts with nearly all metals and nonmetals except some metals (e.g. Au, Pt).

(c) Use:

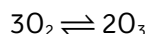
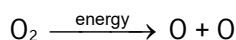
(i) Oxygen mixed with helium or CO_2 is used for artificial respiration.

(ii) Liquid oxygen is used as oxidising agent in rocket fuels.

(iii) Oxygen is used for production of oxy-hydrogen or oxy-acetylene flames employed for cutting and welding.

Oxygen Compound :**Ozone (O_3):****(a) Preparation:**

It is prepared by passing silent electric discharge through pure and dry oxygen



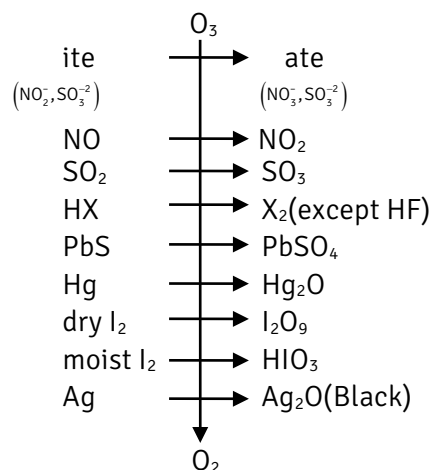
(b) Properties:

- (i) Pale blue gas which forms a blue liquid on cooling and on solidification forms violet black crystals. O_3 molecule is diamagnetic.
- (ii) Ozone is thermodynamically unstable with respect to oxygen since its decomposition into oxygen results in the liberation of heat (ΔH is negative) and an increase in entropy (ΔS is positive). These two effects reinforce each other, resulting in large negative Gibbs energy change (ΔG) for its conversion into oxygen. Therefore, high conservations of ozone can be dangerously explosive.

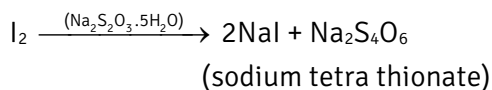
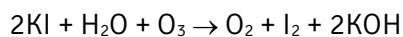
(iii) Oxidising Behaviour of Ozone:

Due to the ease with which it liberates atoms of nascent oxygen ($O_3 \rightarrow O_2 + O$), it acts as a powerful oxidising agent.

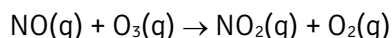
Example: it oxidises lead sulphide to lead sulphate and iodide ions to iodine.

**Estimation of Ozone:**

When ozone reacts with an excess of potassium iodide solution buffered with a borate buffer (pH 9.2), iodine is liberated which can be titrated against a standard solution of sodium thiosulphate. This is quantitative method for estimating O_3 gas.

**Depletion of Ozone Layer:**

Experiments have shown that nitrogen oxides (particularly nitric oxide) combine very rapidly with ozone and there is, thus, the possibility that nitrogen oxides emitted from the exhaust systems of supersonic jet aeroplanes might be slowly depleting the concentration of the ozone layer in the upper atmosphere.



Another threat to this ozone layer is probably posed by the use of freons which are used in aerosol sprays and as refrigerants.

(c) Uses:

- (i) As a germicide and disinfectant for sterilising water and improving the atmosphere of crowded places.
- (ii) For detecting the position of double bond in the unsaturated organic compounds.
- (iii) In mfg. of artificial silk, synthetic camphor, $KMnO_4$ etc.

Allotropes of Sulphur

Crystalline	Amorphous
Rhombic sulphur (α -S) most stable form Monoclinic sulphur (β -S) most stable form $\alpha - S \xrightleftharpoons[<95.6^\circ\text{C}]{>95.6^\circ\text{C}} \beta - S$ 95.6°C = transition Temp. both are soluble in CS_2 but insoluble in water	Milk of sulphur Plastic sulphur (γ -S) Collodial sulphur $\text{H}_2\text{S} + 2\text{HNO}_3 \longrightarrow \text{S} + 2\text{NO}_2 + 2\text{H}_2\text{O}$

- (a) Density of $\alpha\text{S} > \beta\text{S}$, M.P. $\alpha\text{S} < \beta\text{S}$
 (b) Both are puckered crown shape having S_8 units
 (c) S_2 is paramagnetic sulphur which exists in vapour form at high temperature ($\sim 1000\text{K}$).
 (d) Both rhombic and monoclinic sulphur have S_8 molecules. These S_8 molecules are packed to give different crystal structures. The S_8 ring in both the forms is puckered and has a crown shape. In cyclo- S_6 , the ring adopts the chair form

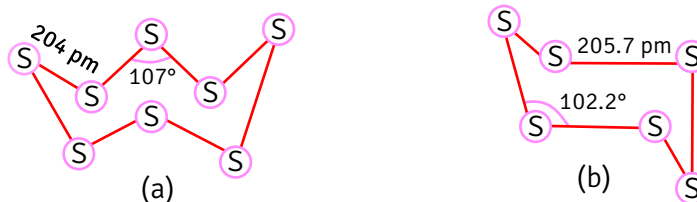
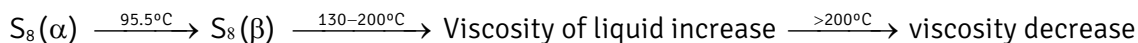


Fig.: The structures of (a) S_8 ring in rhombic sulphur and (b) S_6 form

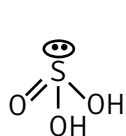
Effect of Temperature:



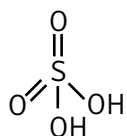
Use of Sulphur:

- (i) Used in manufacture of H_2SO_4 & paper from wood pulp.
 (ii) As a bleaching agent for delicate articles like wool, silk and straw.
 (iii) Used in refining of petroleum and sugar.

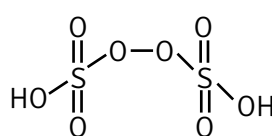
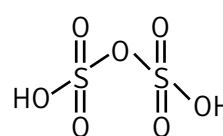
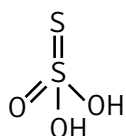
Oxy Acids of Sulphur:



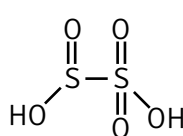
Sulphurous acid



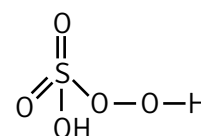
Sulphuric acid

Peroxydisulphuric acid
(Marshall's acid)Pyrosulphuric acid/Fuming
sulphuric acid/Oleum

Thiosulphuric acid

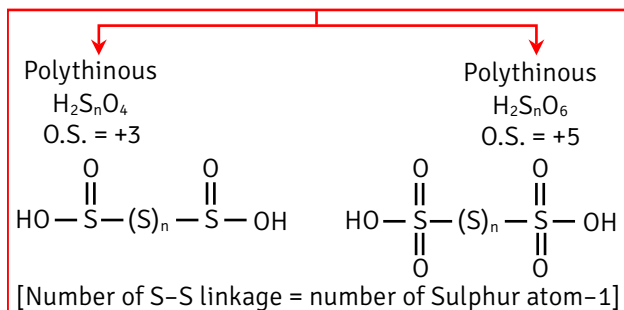


Pyrosulphurous acid

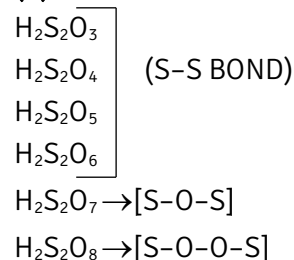


Peroxymonosulphuric acid (Caro's acid)

(a) Oxy Acids Containing S-S Linkage



(b)



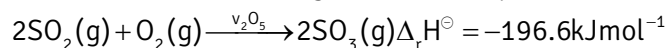
Sulphuric Acid

Manufacture Sulphuric acid is one of the most important industrial chemicals worldwide.

Sulphuric acid is manufactured by the Contact Process which involves three steps:

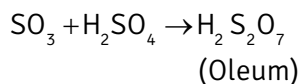
- burning of sulphur or sulphide ores in air to generate SO_2 .
- conversion of SO_2 to SO_3 by the reaction with oxygen in the presence of a catalyst (V_2O_5), and
- absorption of SO_3 in H_2SO_4 to give Oleum ($\text{H}_2\text{S}_2\text{O}_7$).

A flow diagram for the manufacture of sulphuric acid is shown in The SO_2 produced is purified by removing dust and other impurities such as arsenic compounds. The key step in the manufacture of H_2SO_4 is the catalytic oxidation of SO_2 with O_2 to give SO_3 in the presence of V_2O_5 (catalyst).



The reaction is exothermic, reversible and the forward reaction leads to a decrease in volume. Therefore, low temperature and high pressure are the favourable conditions for maximum yield. But the temperature should not be very low otherwise rate of reaction will become slow.

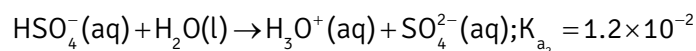
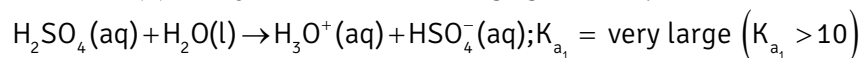
In practice, the plant is operated at a pressure of 2 bar and a temperature of 720 K. The SO_3 gas from the catalytic converter is absorbed in concentrated H_2SO_4 to produce oleum. Dilution of oleum with water gives H_2SO_4 of the desired concentration. In the industry two steps are carried out simultaneously to make the process a continuous one and also to reduce the cost.



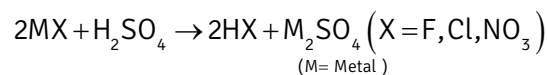
The sulphuric acid obtained by Contact process is 96-98% pure.

Properties:

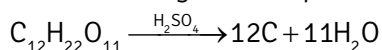
Sulphuric acid is a colourless, dense, oily liquid with a specific gravity of 1.84 at 298 K. The acid freezes at 283 K and boils at 611 K. It dissolves in water with the evolution of a large quantity of heat. Hence, care must be taken while preparing sulphuric acid solution from concentrated sulphuric acid. The concentrated acid must be added slowly into water with constant stirring. The chemical reactions of sulphuric acid are as a result of the following characteristics: (a) low volatility (b) strong acidic character (c) strong affinity for water and (d) ability to act as an oxidising agent. In aqueous solution, sulphuric acid ionises in two steps.



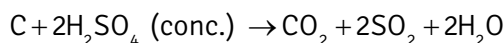
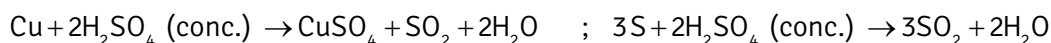
The larger value of K_{a_1} ($K_{a_1} > 10$) means that H_2SO_4 is largely dissociated into H^+ and HSO_4^- . Greater the value of dissociation constant (K_a), the stronger is the acid. The acid forms two series of salts: normal sulphates (such as sodium sulphate and copper sulphate) and acid sulphates (e.g., sodium hydrogen sulphate). Sulphuric acid, because of its low volatility can be used to manufacture more volatile acids from their corresponding salts.



Concentrated sulphuric acid is a strong dehydrating agent. Many wet gases can be dried by passing them through sulphuric acid, provided the gases do not react with the acid. Sulphuric acid removes water from organic compounds; it is evident by its charring action on carbohydrates.



Hot concentrated sulphuric acid is a moderately strong oxidising agent. In this respect, it is intermediate between phosphoric and nitric acids. Both metals and non-metals are oxidised by concentrated sulphuric acid, which is reduced to SO_2 .



Uses:

Sulphuric acid is a very important industrial chemical. A nation's industrial strength can be judged by the quantity of sulphuric acid it produces and consumes. It is needed for the manufacture of hundreds of other compounds and also in many industrial processes. The bulk of sulphuric acid produced is used in the manufacture of fertilisers (e.g., ammonium sulphate, superphosphate). Other uses are in: (a) petroleum refining (b) manufacture of pigments, paints and dyestuff intermediates (c) detergent industry (d) metallurgical applications (e.g., cleansing metals before enameling, electroplating and galvanizing (e) storage batteries (f) in the manufacture of nitrocellulose products and (g) as a laboratory reagent.

NCERT QUESTIONS (Solve Yourself)

1. List the important sources of sulphur.
2. Write the order of thermal stability of the hydrides of Group 16 elements.
3. Why is H_2O a liquid and H_2S a gas?
4. Which of the following does not react with oxygen directly? (Zn, Ti, Pt, Fe)
5. Why does O_3 act as a powerful oxidising agent?
6. How is O_3 estimated quantitatively?
7. What happens when sulphur dioxide is passed through an aqueous solution of Fe(III) salt?
8. Comment on the nature of two S–O bonds formed in SO_2 molecule. Are the two S–O bonds in this molecule equal?
9. How is the presence of SO_2 detected?
10. Mention three areas in which H_2SO_4 plays an important role.
11. Write the conditions to maximise the yield of H_2SO_4 by Contact process.
12. Why is $K_{a_2} \ll K_{a_1}$ for H_2SO_4 in water?
13. Justify the placement of O, S, Se, Te and Po in the same group of the periodic table in terms of electronic configuration, oxidation state and hydride formation.
14. Why is dioxygen a gas but sulphur a solid?
15. Knowing the electron gain enthalpy values for $\text{O} \rightarrow \text{O}^-$ and $\text{O} \rightarrow \text{O}^{2-}$ as -141 and 702 kJ mol^{-1} respectively, how can you account for the formation of a large number of oxides having O^{2-} species and not O^- ? (Hint: Consider lattice energy factor in the formation of compounds).
16. Which aerosols deplete ozone?
17. Describe the manufacture of H_2SO_4 by contact process?
18. How is SO_2 an air pollutant?

Exercise 1.4

- The oxide which on strong heating gives oxygen is :
(1) AgNO_3 (2) BaO_2
(3) Both (1) and (2) (4) Na_2O
- Sulphur on oxidation with hot sulphuric acid gives:
(1) SO_3 (2) SO_2
(3) H_2SO_3 (4) $\text{H}_2\text{S}_2\text{O}_3$
- Estimation of ozone can be made quantitatively by :
(1) decomposition into O_2 and absorption of O_2 into pyrogallol
(2) volumetric method using KI and titration of the liberated iodine using hypo solution
(3) oxidative ozonolysis method
(4) all methods given above
- Oxidation state of sulphur in $\text{H}_2\text{S}_2\text{O}_8$
(1) +2 (2) +4 (3) +6 (4) +8
- Arrange the following elements in the correct order of increasing ionic radii :
Se, Te, Po, O, S
(1) $\text{O} < \text{S} < \text{Se} < \text{Te} < \text{Po}$
(2) $\text{Po} < \text{Te} < \text{Se} < \text{S} < \text{O}$
(3) $\text{S} < \text{O} < \text{Se} < \text{Te} < \text{Po}$
(4) $\text{S} < \text{O} < \text{Po} < \text{Se} < \text{Te}$
- Oxygen shows generally negative oxidation state as -2 except in the case of
(1) OH_2 (2) OF_2 (3) OCl_2 (4) CO_2
- SO_2 and TeO_2 act as respectively are
(1) reducing and reducing
(2) reducing and oxidising
(3) oxidising and oxidising
(4) oxidising and reducing
- Water is much less volatile than H_2S because
(1) H_2O has a bond angle of nearly 150°
(2) hydrogen is loosely bonded with the sulphur
(3) S-atom is less electronegative than O-atom
(4) S-atom is more electronegative than O-atom
- The oxyacids of S having -S-S- bonds are
I. $\text{H}_2\text{S}_2\text{O}_4$ II. $\text{H}_2\text{S}_2\text{O}_7$
III. $\text{H}_2\text{S}_2\text{O}_6$ IV. $\text{H}_2\text{S}_2\text{O}_3$
Choose the correct option.
(1) I and III (2) II and IV
(3) I and II (4) II and III
- Which of the following property of hydrides of group-16 decreases on moving down the group?
(1) Acidic nature (2) Reducing nature
(3) Thermal stability (4) All
- Which is incorrect ?
(1) Polonium is decay product of thorium and uranium minerals
(2) Lv (116) is synthetic radioactive element
(3) oxygen is most abundant element on earth crust (46.6% by mass)
(4) EA of oxygen is maximum in its group
- Which of the following molecule having maximum boiling and melting point :
(1) H_2Te (2) H_2S (3) H_2O (4) H_2Se
- H-X bond strength is minimum if X is :
(1) O (2) S (3) Se (4) Te
- Which of the following accounts for anomalous behaviour of oxygen in the group 16 ?
(1) High electronegativity of oxygen
(2) Small size of oxygen
(3) Absence of vacant d orbitals in oxygen
(4) All
- Which is correct ?
(1) $\text{O} > \text{S} > \text{Se} > \text{Te} > \text{Po}$ (size)
(2) $\text{O} > \text{S} > \text{Se} > \text{Te} > \text{Po}$ (IP)
(3) $\text{S} > \text{O} > \text{Se} > \text{Te} > \text{Po}$ (EA)
(4) $\text{O} > \text{S} > \text{Se} > \text{Te} > \text{Po}$ (MP/BP)
- X = X is stable if X is :
(1) O (2) S (3) Se (4) Te
- Examples of allotropes are :
(1) O_2 and CO_2 (2) CO and CO_2
(3) O_2 and O_3 (4) None
- O_2 can be prepared by heating :
(1) KClO_3 (2) KMnO_4
(3) $\text{K}_2\text{Cr}_2\text{O}_7$ (4) All
- O-O bond distance is shorter in :
(1) O_2 (2) H_2O_2
(3) O_3 (4) same in all

Group 17 Elements and Properties of Halogens

Property	F	Cl	Br	I	At ^a
Atomic number	9	17	35	53	85
Atomic mass/g mol ⁻¹	19.00	35.45	79.90	126.90	210
Electronic configuration	[He]2s ² 2p ⁵	[Ne]3s ² 3p ⁵	[Ar]3d ¹⁰ 4s ² 4p ⁵	[Kr]4d ¹⁰ 5s ² 5p ⁵	[Xe]4f ¹⁴ 5d ¹⁰ 6s ² 6p ⁵
Covalent radius/pm	64	99	114	133	–
Ionic radius X ⁻ /pm	133	184	196	220	–
Ionisation enthalpy/kJ mol ⁻¹	1680	1256	1142	1008	–
Electron gain enthalpy/kJ mol ⁻¹	–333	–349	–325	–296	–
Electronegativity ^b	4	3.2	3.0	2.7	2.2
D _{Hyd} H(X ⁻)/kJmol ⁻¹	515	381	347	305	–
	F ₂	Cl ₂	Br ₂	I ₂	–
Melting point/K	54.4	172.0	265.8	386.6	–
Boiling point/K	84.9	239.0	332.5	458.2	–
Density/g cm ⁻³	1.5 (85) ^c	1.66 (203) ^c	3.19(273) ^c	4.94(293) ^d	–
Distance X – X/pm	143	199	228	266	–
Bond dissociation enthalpy/(kJ mol ⁻¹)	158.8	242.6	192.8	151.1	–
E ^o /V ^e	2.87	1.36	1.09	0.54	–

^aRadioactive; ^bPauling scale; ^cFor the liquid at temperatures (K) given in the parentheses; ^dsolid; ^eThe half-cell reaction is X₂(g) + 2e⁻ → 2X⁻(aq).

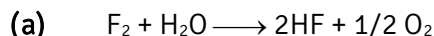
Physical Properties:

Fluorine and chlorine are gases, bromine is a liquid and iodine is a solid. Their melting and boiling points steadily increase with atomic number. F₂, has yellow, Cl₂, greenish yellow, Br₂, red and I₂, violet colour. Fluorine and chlorine react with water. Bromine and iodine are only sparingly soluble in water. But are soluble in organic solvents such as chloroform, carbon tetrachloride, carbon disulphide and hydrocarbons to give coloured solutions. Except the smaller enthalpy of dissociation of F₂ compared to that of Cl₂ whereas X-X bond dissociation enthalpies from chlorine onwards show the expected trend: Cl – Cl > Br – Br > F – F > I – I. A reason for this anomaly is the relatively larger electrons- electron repulsion among the lone pairs in F₂ molecule where they are much closer to each other than in case of Cl₂.

Chemical Properties:

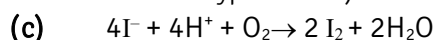
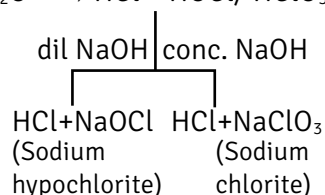
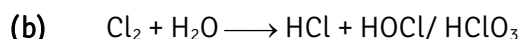
Oxidation States and Trends in Chemical Reactivity

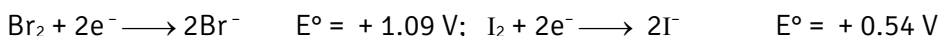
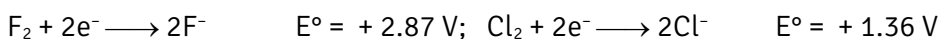
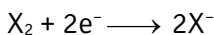
Reaction with Water



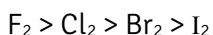
(O₃ is also produced in small amount & mixture of O₂ & O₃ is known as ozonide)

'F₂' is the only non-metal that can oxidise H₂O as it is a powerful oxidising agent. Other halogens & nonmetals undergo disproportionation with water in basic medium. (Redox reaction)



Standard Reduction Potential (SRP):

More the value of the SRP, more powerful is the oxidising agent. Hence the order of oxidising power is:

**(i) Reactivity Towards Hydrogen:**

Thermal stability and bond energy: $H - F > H - Cl > H - Br > H - I$.

Acidic strength and reducing power: $H - F < H - Cl < H - Br < H - I$.

M.P. order: $H - Cl < H - Br < H - F < H - I$.

B.P. order: $H - Cl < H - Br < H - I < H - F$.

(ii) Reactivity Towards Oxygen:

Fluorine forms two oxides OF_2 and O_2F_2 . However, only OF_2 is the thermally stable at 298 K. These oxide are essentially oxygen fluorides because of the higher electronegativity of fluorine than oxygen. Both are strong fluorinating agents. O_2F_2 oxidises plutonium to PuF_6 and the reaction is used in removing plutonium as PuF_6 from spent nuclear fuel.

Chlorine, bromine and iodine form oxides in which the oxidation states of these halogen range from + 1 to + 7. A combination of kinetic and thermodynamic factors lead to the generally decreasing order of stability oxides formed by halogens, $I > Cl > Br$. The higher oxides of halogens tend to be more stable than the lower ones. (Middle row anomaly)

Chlorine oxides, Cl_2O , ClO_2 , Cl_2O_6 and Cl_2O_7 are highly reactive oxidising agents and tend to explode. ClO_2 is used as a bleaching agent for paper pulp and textiles and in water treatment.

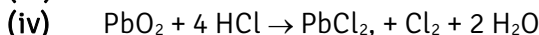
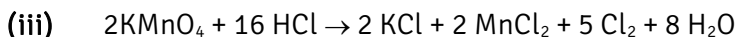
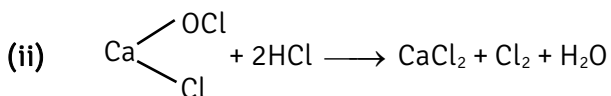
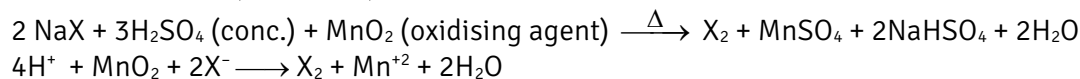
The bromine oxides, Br_2O , BrO_2 , BrO_3 are the least stable halogen oxides and exist only at low temperature. They are very powerful oxidising agents.

The iodine oxides, I_2O_4 , I_2O_5 , I_2O_7 are insoluble solids and decompose on heating. I_2O_5 is very good oxidising agent and is used in the estimation of carbon monoxide.

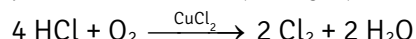
Anomalous behaviour of Fluorine:

Like other elements of p-block present in second period of the periodic table, fluorine is anomalous in many properties. For example, ionisation enthalpy, electronegativity, and electrode potentials are all higher for fluorine than expected from the trends set by other halogens. Also, ionic and covalent radii, m.p. and b.p., enthalpy of bond dissociation and electron gain enthalpy are quite lower than expected. The anomalous behaviour of fluorine is due to its small size, highest electronegativity, low F-F bond dissociation enthalpy, and non availability of d orbitals in valence shell.

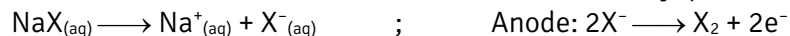
Most of the reactions of fluorine are exothermic (due to the small and strong bond formed by it with other elements). It forms only one oxoacid while other halogens form a number of oxoacids. Hydrogen fluoride is a liquid (b.p. 293 K) due to strong hydrogen bonding. Other hydrogen halides are gases.

Chlorine (Cl₂):**(a) Preparation:****(i) Common method (Cl₂, Br₂, I₂):****(v) Manufacture of Chlorine:**

(vi) Deacon's process: By oxidation of hydrogen chloride gas by atmospheric oxygen in the presence of CuCl₂ (catalyst) at 723 K.



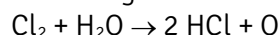
(vii) Electrolytic Process: Chlorine is obtained by the electrolysis of brine (concentrated NaCl solution). Chlorine is liberated at anode. It is also obtained as a by-product in many chemical industries.

**(b) Properties:**

(i) It is a greenish-yellow gas with pungent and suffocating odour. It is about 2–5 times heavier than air. It can be liquefied into greenish-yellow liquid which boils at 239 K. It is soluble in water.

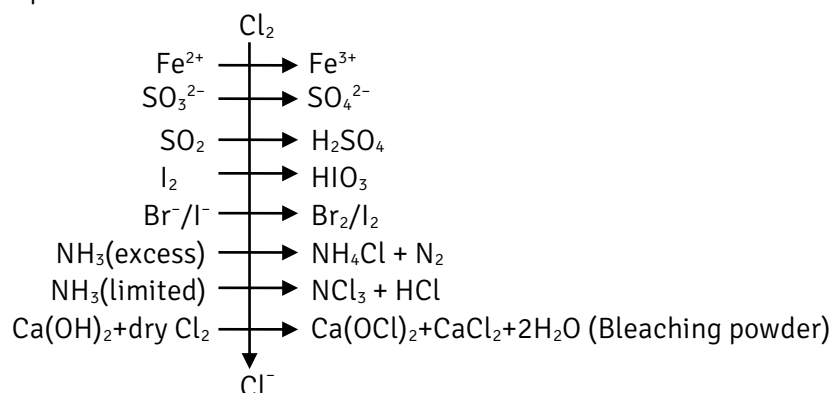
(ii) Oxidising & Bleaching Properties: Chlorine water on standing loses its yellow colour due to the formation of HCl and HOCl. Hypochlorous acid (HOCl) so formed, gives nascent oxygen which is responsible for oxidising and bleaching properties of chlorine.

Bleaching action is due to oxidation.

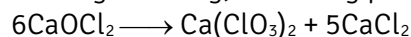


Coloured substance + O → Colourless substance

It bleaches vegetable or organic matter in the presence of moisture. Bleaching effect of chloride is permanent.



On long standing, bleaching powder undergoes auto-oxidation

**(c) Uses:** Cl₂ is used

- for bleaching woodpulp (required for the manufacture of paper and rayon). bleaching cotton and textiles,
- in the manufacture of dyes, drugs and organic compounds such as CCl₄, CHCl₃, DDT, refrigerants, etc.
- in the extraction of gold and platinum.
- in sterilising drinking water
- preparation of poisonous gases such as phosgene (COCl₂), tear gas (CCl₃NO₂), mustard gas (ClCH₂CH₂SCH₂CH₂Cl).

NCERT QUESTIONS (Solve Yourself)

1. Considering the parameters such as bond dissociation enthalpy, electron gain enthalpy and hydration enthalpy, compare the oxidising power of F_2 and Cl_2 .
2. Give two examples to show the anomalous behaviour of fluorine.
3. Sea is the greatest source of some halogens. Comment.
4. Give the reason for bleaching action of Cl_2 .
5. Name two poisonous gases which can be prepared from chlorine gas.
6. Why is ICl more reactive than I_2 ?
7. Why are halogens strong oxidising agents?
8. Explain why fluorine forms only one oxoacid, $HOCl$.
9. Explain why in spite of nearly the same electronegativity, oxygen forms hydrogen bonding while chlorine does not.
10. Write two uses of ClO_2 .
11. Why are halogens coloured?
12. Write the reactions of F_2 and Cl_2 with water.
13. How can you prepare Cl_2 from HCl and HCl from Cl_2 ? Write reactions only.
14. Write balanced equations for the following:
(i) $NaCl$ is heated with sulphuric acid in the presence of MnO_2 .
(ii) Chlorine gas is passed into a solution of NaI in water.
15. With what neutral molecule is ClO^- isoelectronic? Is that molecule a Lewis base?
16. Arrange the following in the order of property indicated for each set:
(i) F_2 , Cl_2 , Br_2 , I_2 - increasing bond dissociation enthalpy.
(ii) HF , HCl , HBr , HI - increasing acid strength.
(iii) NH_3 , PH_3 , AsH_3 , SbH_3 , BiH_3 - increasing base strength.
17. Which one of the following does not exist?
(i) $XeOF_4$ (ii) NeF_2 (iii) XeF_2 (iv) XeF_6
18. Give the formula and describe the structure of a noble gas species which is isostructural with:
(i) ICl_4^- (ii) IBr_2^- (iii) BrO_3^-

Exercise 1.5

- 1.** Arrange them in decreasing order of oxidising power.
Ion ClO_4^- IO_4^- BrO_4^-
(1) $\text{ClO}_4^- > \text{IO}_4^- > \text{BrO}_4^-$
(2) $\text{IO}_4^- > \text{BrO}_4^- > \text{ClO}_4^-$
(3) $\text{BrO}_4^- > \text{IO}_4^- > \text{ClO}_4^-$
(4) $\text{BrO}_4^- > \text{ClO}_4^- > \text{IO}_4^-$

 - 2.** Which one of the hydricid does not form any precipitate with AgNO_3 ?
(1) HF (2) HCl
(3) HBr (4) HI

 - 3.** Chlorine gas is dried over :
(1) CaO (2) NaOH
(3) H_2SO_4 (4) HBr

 - 4.** Which of the following pairs is not correctly matched ?
(1) A halogen which is liquid at room temperature – Bromine
(2) The most electronegative element – Fluorine
(3) The most reactive halogen – Fluorine
(4) The strongest oxidising agent – Iodine

5. Bleaching powder is obtained by the interaction of chlorine and :
 (1) dilute solution of Ca(OH)_2
 (2) concentrated solution of Ca(OH)_2
 (3) dry calcium oxide
 (4) dry slaked lime
6. On heating KClO_3 we get :
 (1) $\text{KClO}_2 + \text{O}_2$
 (2) $\text{KCl} + \text{O}_2$
 (3) $\text{KCl} + \text{O}_3$
 (4) $\text{KCl} + \text{O}_2 + \text{O}_3$
7. The reaction
 $3\text{ClO}^- (\text{aq.}) \longrightarrow \text{ClO}_3^- (\text{aq.}) + 2\text{Cl}^- (\text{aq.})$
 is an example of :
 (1) oxidation reaction
 (2) reduction reaction
 (3) disproportionation
 (4) decomposition reaction
8. Which of the following will displace the halogen from the solution of the halide ?
 (1) Br_2 added to NaI
 (2) Br_2 added to NaCl
 (3) Cl_2 added to KCl
 (4) Cl_2 added to NaF
9. The high oxidising power of fluorine is due to:
 (1) high electron affinity
 (2) high heat of dissociation and low heat of hydration
 (3) low heat of dissociation and high heat of hydration
 (4) high heat of dissociation and high heat of hydration
10. Arrange the following acids in the correct increasing order of their acidic strength :
 HCl , HBr , HI and HF
 (1) $\text{HI} < \text{HBr} < \text{HCl} < \text{HF}$
 (2) $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$
 (3) $\text{HCl} < \text{HBr} < \text{HI} < \text{HF}$
 (4) $\text{HBr} < \text{HI} < \text{HF} < \text{HCl}$
11. $4\text{HCl} + \text{O}_2 \xrightarrow{\text{CuCl}_2} 2\text{Cl}_2 + 2\text{H}_2\text{O}$
 The above reaction of chlorine preparation is of
 (1) Deacon's process
 (2) contact process
 (3) electrolytic process
 (4) None of these
12. Composition of bleaching powder is
 (1) $\text{Ca(OCl)}_2 \cdot \text{CaCl}_2 \cdot \text{Ca(OH)}_2 \cdot 2\text{H}_2\text{O}$
 (2) $\text{CaCl}_2 \cdot \text{CaOCl}_2 \cdot \text{Ca(OH)}_2 \cdot 2\text{H}_2\text{O}$
 (3) $\text{Ca(OCl)}_2 \cdot \text{CaCl}_2 \cdot \text{Ca(OH)}_2 \cdot 4\text{H}_2\text{O}$
 (4) None of the above
13. The non-metal that does not exhibit positive oxidation state is
 (1) fluorine (2) oxygen
 (3) chlorine (4) iodine
14. The correct order of the decreasing acidic strengths of HClO , HClO_2 , HClO_3 and HClO_4 is
 (1) $\text{HClO} > \text{HClO}_2 > \text{HClO}_3 > \text{HClO}_4$
 (2) $\text{HClO}_4 > \text{HClO}_3 > \text{HClO}_2 > \text{HClO}$
 (3) $\text{HClO}_4 > \text{HClO}_2 > \text{HClO} > \text{HClO}_3$
 (4) $\text{HClO}_3 > \text{HClO} > \text{HClO}_4 > \text{HClO}_2$
15. The hybrid state of halogen atom is sp^3 in
 (1) ClO_4^- (2) ClO^-
 (3) ClO_3^- (4) All of these
16. Which is correct order :
 (1) $\text{H}_2\text{SO}_4 > \text{H}_2\text{SO}_3$ (acidic nature)
 (2) $\text{H}_2\text{SO}_4 > \text{H}_2\text{SO}_3$ (oxidising behaviour)
 (3) Both
 (4) None
17. Which is incorrect reaction :
 (1) $\text{NaF} + \text{conc. H}_2\text{SO}_4 \rightarrow \text{HF} + \text{Na}_2\text{SO}_4$
 (2) $\text{NaCl} + \text{conc. H}_2\text{SO}_4 \rightarrow \text{HCl} + \text{Na}_2\text{SO}_4$
 (3) $\text{NaBr} + \text{conc. H}_2\text{SO}_4 \rightarrow \text{HBr} + \text{Na}_2\text{SO}_4$
 (4) None

18. Which has highest negative electron gain enthalpy (maximum electron affinity) ?
(1) F (2) Cl
(3) Br (4) I
19. Which is strongest oxidising agent ?
(1) F₂ (2) Cl₂
(3) Br₂ (4) I₂
20. Which is most reactive halogen ?
(1) F₂ (2) Cl₂
(3) Br₂ (4) I₂
21. Which is strongest reducing agent ?
(1) KF (2) KCl
(3) KBr (4) KI
22. Which is most responsible factor for strongest oxidising behaviour of fluorine ?
(1) High IP
(2) High EA
(3) High hydration energy of F⁻ ion
(4) None
23. Middle row anomaly is related with
(1) Halogens oxides
(2) Halogens acids
(3) Sulphur oxides
(4) Phosphorus oxides
24. Which reaction is not possible ?
(1) F₂ + KCl → KF + Cl₂
(2) Cl₂ + KI → KCl + I₂
(3) Br₂ + KF → KBr + F₂
(4) Cl₂ + KBr → KCl + Br₂
25. Which halogen can oxidised water ?
(1) F₂ (2) Cl₂
(3) Br₂ (4) I₂
26. Which compound is formed in excess ammonia react with Cl₂
(1) NCl₃ and HCl
(2) N₂ and HCl
(3) N₂ and NH₄ Cl
(4) NCl₃ and N₂
27. What is the shape of ClF₃
(1) Trigonal planar
(2) Bent T-shape
(3) Pyramidal
(4) None
28. Which is strongest acid
(1) HOCl (2) HClO₂
(3) HClO₃ (4) HClO₄
29. Violet vapours are given out when is treated with conc. H₂SO₄ :
(1) Bromide
(2) Iodide
(3) Chloride
(4) Nitrate
30. Chromyl chloride test is performed for the confirmation of the presence of the following in a mixture :
(1) Sulphate
(2) Carbonate
(3) Chloride
(4) Nitrate
31. Two test tubes containing a nitrate and a bromide are treated separately with H₂SO₄; brown fumes evolved are passed in water. The water be coloured by vapours evolved from the test tube containing :
(1) Nitrate
(2) Bromide
(3) Both (1) and (2)
(4) None of these

Group 18 Elements and their Properties

Property	He	Ne	Ar	Kr	Xe	Rn*
Atomic number	2	10	18	36	54	86
Atomic mass/ g mol ⁻¹	4.00	20.18	39.95	83.80	131.30	222.00
Electronic configuration	1s ²	[He]2s ² 2p ⁶	[Ne] 3s ² 3p ⁶	[Ar] 3d ¹⁰ 4s ² 4p ⁶	[Kr] 4d ¹⁰ 5s ² 5p ⁶	[Xe] 4f ¹⁴ 5d ¹⁰ 6s ² 6p ⁶
Atomic radius/pm	120	160	190	200	220	–
Ionisation enthalpy /kJmol ⁻¹	2372	2080	1520	1351	1170	1037
Electron gain enthalpy /kJmol ⁻¹	48	116	96	96	77	68
Density (at STP)/gcm ⁻³	1.8×10 ⁻⁴	9.0×10 ⁻⁴	1.8×10 ⁻³	3.7×10 ⁻³	5.9×10 ⁻³	9.7×10 ⁻³
Melting point/K	–	24.6	83.8	115.9	161.3	202
Boiling point/K	4.2	27.1	87.2	119.7	165.0	211
Atmospheric content (% by volume)	5.24×10 ⁻⁴	–	1.82×10 ⁻³	0.934	1.14×10 ⁻⁴	8.7×10 ⁻⁶

Physical Properties:

All the noble gases are monoatomic. They are colourless, and tasteless. They are sparingly soluble in water. They have very low melting and boiling points because the only type of interatomic interaction in these elements is weak dispersion forces, Helium has the lowest boiling point (4.2K) of any known substance. He has a unusual property of diffusing through most commonly used laboratory materials such as rubber, glass or plastics.

Chemical Properties:

In general, noble gases are least reactive. Their inertness to chemical reactivity is attributed to the following reasons:

- The noble gases except helium (1s²) have completely filled ns² np⁶ electronic configuration in their valence shell.
- They have high ionisation enthalpy and more positive electron gain enthalpy.

The reactivity of noble gases has been investigated occasionally ever since their discovery, but all attempt to force them to react to form the compounds were unsuccessful for quite a few years. In March 1962, Neil Bartlett, then at the University of British Columbia, observed the reaction of a noble gas. First, he prepared a red compound which is formulated as O₂⁺ PtF₆⁻. He, then realised that the first ionisation enthalpy of molecular oxygen (1175 kJ mol⁻¹) was almost identical with that xenon (1170 kJ mol⁻¹). He made efforts to prepare same type of compound with Xe⁺ PtF₆⁻ by mixing Pt F₆ and Xenon. After this discovery, a number of xenon compounds mainly with most electronegative elements like fluorine and oxygen, have been synthesized.

The compounds of krypton are fewer. Only the difluoride (KrF₂) has been studied in detail. Compounds of radon have not been isolated but only identified (e.g., RnF₂) by radiotracer technique. No true compounds of Ar, Ne or He are yet known.

Uses:

Helium is a non-inflammable and light gas. Hence, it is used in filling balloons for meteorological observations. It is also used in gas-cooled nuclear reactors. Liquid helium (b.p. 4.2 K) finds use as cryogenic agent for carrying out various experiments at low temperatures. It is used to produce and sustain powerful superconducting magnets which form an essential part of modern NMR spectrometers and Magnetic Resonance Imaging (MRI) systems for clinical diagnosis. It is used as a diluent for oxygen in modern diving apparatus because of its very low solubility in blood.

Neon is used in discharge, tube and incandescent bulbs for advertising display. Uses of Neon Bulbs It is done in botanical gardens and greenhouses.

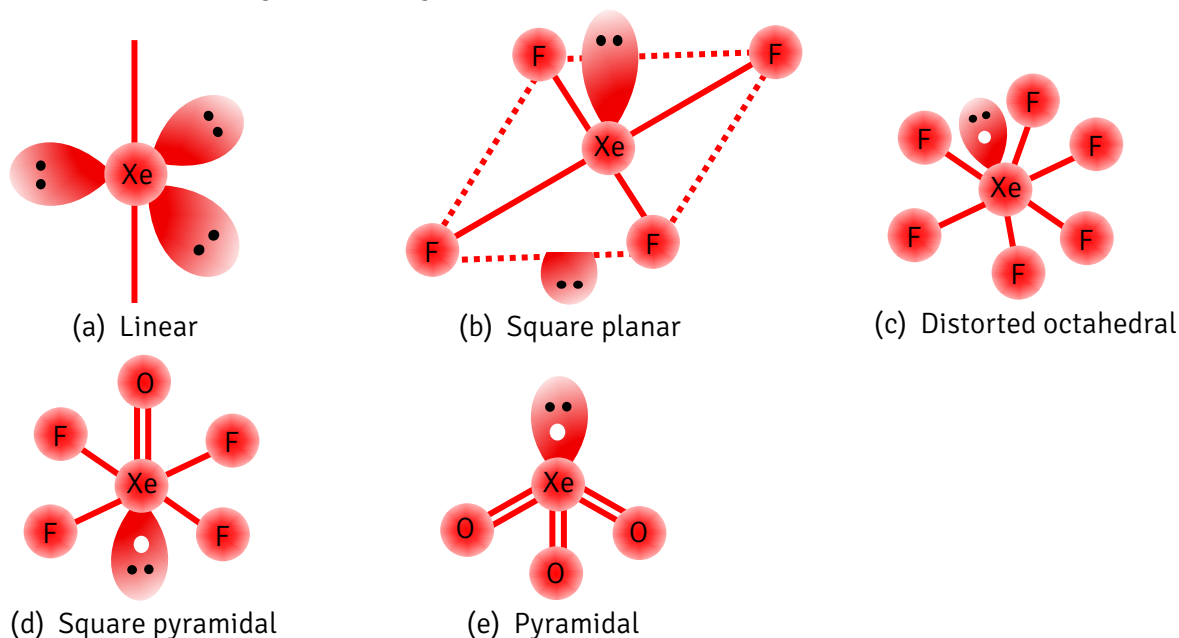


Fig. The structures of

(a) XeF_2

(b) XeF_4

(c) XeF_6

(d) XeOF_4

(e) XeO_3

Neon is used in discharge tubes and fluorescent bulbs for advertisement display purposes. Neon bulbs are used in botanical gardens and in green houses.

Argon is used mainly to provide an inert atmosphere in high temperature metallurgical process (arc welding of metals or alloys) and for filling electric bulbs. It is also used in the laboratory for handling substances that are air-sensitive.

Xenon and Krypton are used in light bulbs designed for special purposes.

NCERT QUESTIONS (Solve Yourself)

1. Why is helium used in diving apparatus?
2. Balance the following equation: $\text{XeF}_6 + \text{H}_2\text{O} \rightarrow \text{XeO}_2\text{F}_2 + \text{HF}$
3. Why has it been difficult to study the chemistry of radon?
4. What inspired N. Bartlett for carrying out reaction between Xe and PtF_6 ?
5. How are xenon fluorides XeF_2 , XeF_4 and XeF_6 obtained?
6. How are XeO_3 and XeOF_4 prepared?
7. Why do noble gases have comparatively large atomic sizes?
8. List the uses of neon and argon gases.

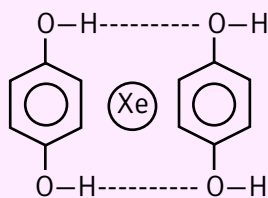
Exercise 1.6

- Of the following species, one which is non-existent :
 (1) XeF_6 (2) XeF_5
 (3) XeF_4 (4) XeF_2
- The coloured discharge tube for advertisement mainly contains:
 (1) xenon (2) helium
 (3) neon (4) argon
- Maximum number of compounds are known in the case of :
 (1) Neon
 (2) Xenon
 (3) Krypton
 (4) Argon
- Which one of the following noble gases is not found in atmosphere ?
 (1) Rn (2) Kr
 (3) Ne (4) Ar
- Helium is added to oxygen used by deep sea divers because :
 (1) It is less soluble in blood than nitrogen under high pressure
 (2) It is lighter than nitrogen
 (3) It is readily miscible with oxygen
 (4) It is less poisonous than nitrogen
- Which one of the following does not exist?
 (1) XeOF_4 (2) NeF_2
 (3) XeF_2 (4) XeF_6
- IP of Xe is nearly equal to
 (1) O (2) O_2
 (3) Cl (4) Br
- Which is the first compound of Xenon prepared in lab ?
 (1) XeF_2 (2) XeF_4
 (3) $\text{Xe} + [\text{PtF}_6]^-$ (4) XeF_6
- Structure of XeF_4 is :
 (1) Tetrahedral (2) Octahedral
 (3) Square Planar (4) Square Pyramidal
- Which is more soluble in water ?
 (1) Xe (2) Ne (3) Ar (4) He
- Which of the following is having linear shape
 (1) XeF_2 (2) XeF_6 (3) Both (4) None
- Which is an explosive solid ?
 (1) XeF_2 (2) XeF_4
 (3) XeO_3 (4) XeOF_2
- Which gas is common as a diluent of oxygen in modern diving apparatus ?
 (1) He (2) Ne (3) Ar (4) Kr
- Which is used in discharge tubes and fluorescent bulbs for advertisement display purposes.
 (1) He (2) Ne (3) Ar (4) Kr

Points to ponder :

Clathrate Compounds: Larger inert gas atoms are trapped into cavity or crystal structure of some organic compounds and form clathrate compounds. For e.g. xenon diquinal.

Force of attraction between noble gas and organic compound is dipole-induced dipole attraction.



He, Ne do not form clathrate compounds due to smaller size.

Exercise 2

- Which of the following statements is incorrect in relation to the structure of diborane?
 - The terminal B-H bond distances are the same as the bond distances measured in compounds which are not electron deficient, and which have no bridge structure.
 - The terminal B-H bond is a 2-centre 3-electron bond.
 - The terminal B-H bond is a 2-centre 2-electron bond.
 - The bridge $\text{B} \begin{array}{c} \text{H} \\ \diagup \quad \diagdown \\ \text{B} \end{array} \text{B}$ is a 3-centre 2-electron bond.
- In diborane, the number of electrons that account for bonding in the bridges is
 - six
 - two
 - eight
 - four
- Consider the following statements for diborane:
 - Boron is approximately sp^3 hybridized
 - B-H-B angle is 180°
 - There are two terminal B-H bonds for each boron atom
 - There are only 12 bonding electrons available.
 - A, C and D are correct
 - A, B and C are correct
 - B, C, and D are correct
 - A, B and D are correct
- The ion(s) that act/s as an oxidising agent in solution is/are
 - Tl and Al^{3+}
 - B^{3+} and Al^{3+}
 - Tl only
 - B^{3+} only
- Aluminium vessels should be washed with materials containing washing soda because:
 - washing soda reacts with aluminium to form soluble aluminate
 - washing soda is expensive
 - washing soda is easily decomposed
 - washing soda reacts with aluminium to form insoluble aluminium oxide
- Carbon-60 contains
 - 20 pentagons and 12 hexagons
 - 12 pentagons and 20 hexagons
 - 30 pentagons and 30 hexagons
 - 24 pentagons and 36 hexagons
- Which of the following structural features of graphite best accounts for its use as a lubricant?
 - Delocalized electrons
 - Strong covalent bonds between carbon atoms
 - Van der Waals' forces between layers
 - Limited three covalency of carbon
- Which of the following allotropic forms of carbon is isomorphous with crystalline silicon?
 - Graphite
 - Coal
 - Coke
 - Diamond
- Silicon dissolves in excess of HF due to formation of:
 - SiF_4
 - SiH_4
 - H_2SiF_6
 - H_2SiF_4
- Which of the following is incorrect for group 14 elements?
 - The stability of dihalides are in the order $\text{CX}_2 < \text{SiX}_2 < \text{GeX}_2 < \text{SnX}_2 < \text{PbX}_2$
 - The ability to form $\text{p}\pi\text{-p}\pi$ multiple bonds among themselves increases down the group
 - The tendency for catenation decreases down the group
 - They all form oxides with the formula MO_2
- Which of the following statement is correct for allotropes of P :-
 - White P is soluble in CS_2 but red P is insoluble in CS_2
 - P-P-P bond angle is 60° in red P
 - On heating in air, white P change into red P
 - White P change in black P at ordinary temp.

12. With excess of Cl_2 , ammonia forms
 (1) Nitrogen
 (2) Nitrosyl chloride
 (3) Ammonium chloride
 (4) Nitrogen trichloride
13. The reaction of white phosphorus with aqueous NaOH gives phosphine along with another compound. The reaction type; the oxidation states of phosphorus in phosphine and the other product are respectively :
 (1) Redox reaction: -3 & -5
 (2) Redox reaction: $+3$ & $+5$
 (3) Disproportionation reaction: -3 & -5
 (4) Disproportionation reaction: -3 & $+1$
14. Which of the following is incorrect?
 (1) O_2 is weaker oxidant than O_3
 (2) O_2 has small bond length than O_3
 (3) Both O_2 and O_3 are paramagnetic
 (4) O_3 is angular is shape
15. Which of the following is a mixed anhydride
 (1) P_4O_{10} (2) SO_3
 (3) Cl_2O_6 (4) SO_2
16. S-S bond is present in :
 (1) $\text{H}_2\text{S}_4\text{O}_6$ (2) $\text{H}_2\text{S}_2\text{O}_6$
 (3) $\text{H}_2\text{S}_2\text{O}_5$ (4) All
17. Which of the following does not have any S-S linkage :-
 (1) $\text{Na}_2\text{S}_2\text{O}_4$
 (2) $\text{H}_2\text{S}_2\text{O}_5$
 (3) $\text{Na}_2\text{S}_4\text{O}_6$
 (4) None
18. What is true about $\text{H}_2\text{S}_n\text{O}_6$ (polythionic acid):
 (1) Total S-S bond = n
 (2) No. of S with '0' state = $n - 1$
 (3) No. of S with +5 state = $n - 2$
 (4) No. of $d\pi p\pi$ bond = 4
19. Which is incorrect order :-
 (1) $\alpha\text{S} > \beta\text{S}$ (Specific gravity)
 (2) White P > Red P (toxic nature)
 (3) $\alpha\text{S} > \beta\text{S}$ (melting point)
 (4) White P > Red P (Reactivity)
20. Which does not contain X-X bond?
 (X = Non metal)
 (1) $\text{H}_2\text{S}_2\text{O}_5$
 (2) $\text{H}_2\text{S}_2\text{O}_6$
 (3) $\text{H}_2\text{S}_2\text{O}_7$
 (4) $\text{H}_4\text{P}_2\text{O}_6$
21. Which of the following statements(s) is incorrect?
 (1) Rhombic sulphur is stable at room temperature
 (2) Both α and β sulphur are soluble in water
 (3) Both rhombic and monoclinic sulphur has the molecular formula S_8
 (4) Both rhombic and monoclinic sulphur are soluble in CS_2
22. Identify the incorrect statement(s) :
 (1) Ozone is a powerful oxidising agent as compared to O_2
 (2) Ozone reacts with KOH and gives an orange coloured solid KO_3
 (3) There is a decrease in volume when ozone decomposed to form O_2
 (4) the decomposition of O_3 to O_2 is exothermic

23. Consider the oxy acids HClO_n series, here value of n is 1 to 4. Then incorrect statement regarding these oxy acids is :
- (1) Acidic character of oxy acids increases with increasing value of n
 - (2) Oxidising power of oxy acids increases with decreasing value of n
 - (3) Thermal stability of oxy acids decreases with increasing value of n
 - (4) 'Cl—O' bond order decreases with decreasing value of n
24. Only iodine forms hepta-fluoride IF_7 , but chlorine and bromine give penta-fluorides. The reason for this is :
- (1) Low electron affinity of iodine
 - (2) Unusual pentagonal bipyramidal structure of IF_7
 - (3) That the larger iodine atom can accommodate more number of smaller fluorine atom around it
 - (4) Low chemical reactivity of IF_7
25. Of the following statements :
- (a) Cl_2 gas is dried using conc. H_2SO_4
 - (b) Fluorine have highest oxidising power
 - (c) Oxidising power of halogens follow the order $\text{I}_2 > \text{Br}_2 > \text{Cl}_2$
 - (d) HI is the strongest acid among HI , HBr , HCl
- (1) a, b and d are correct
 - (2) a, c are correct
 - (3) b, c are correct
 - (4) c, d are correct
26. The first compound of noble gases prepared by Neil-Bartlett was :-
- (1) $\text{Xe}^+ [\text{PtF}_6]^-$
 - (2) XeF_4
 - (3) XeF_6
 - (4) XeOF_4
27. The compound that cannot be formed by xenon is :
- (1) XeO_3
 - (2) XeF_4
 - (3) XeCl_4
 - (4) XeO_2F_2
28. Which of the following xenon compound has the same number of lone pairs as in I_3^-
- (1) XeO_4
 - (2) XeF_4
 - (3) XeF_2
 - (4) XeO_3
29. $\text{Xe} + 3\text{F}_2 \rightarrow \text{A}$ (M^+ = Alkali metal cation)
The state of hybridisation of the central atom in 'A' and shape of the species are :
- (1) sp^3d , TBP
 - (2) sp^3d^3 , distorted octahedral
 - (3) sp^3d^3 , pentagonal bipyramidal
 - (4) No compound formed at all
30. Which of the following is not true about helium?
- (1) It has the lowest boiling point
 - (2) It has the highest first ionization energy
 - (3) It can diffuse through rubber and plastic material
 - (4) It can form clathrate compounds

Exercise 3

1. Match the column-I (Structures) and column-II (Oxoacids of halogens) and choose the correct option from the codes given below.

	Column-I		Column-II
A.		1.	Perchloric acid
B.		2.	Chloric acid
C.		3.	Chlorous acid
D.		4.	Hypochlorous acid

- (1) A-1, B-2, C-3, D-4
 (2) A-3, B-2, C-4, D-1
 (3) A-4, B-3, C-2, D-1
 (4) A-2, B-1, C-4, D-3

2. Match the oxidation state given in Column I with oxides of sulphur given in Column II.

	Column-I		Column-II
A.	+6	p.	SO_4^{2-}
B.	+4	q.	$\text{S}_2\text{O}_7^{2-}$
C.	+2	r.	SO_3
		s.	SO_3^{2-}
		t.	$\text{S}_2\text{O}_3^{2-}$

- (1) A-s; B-t; C-p, q, r
 (2) A-p, q; B-r, s; C-t
 (3) A-p, q; B-r, s; C-t
 (4) A-p; B-r, s; C-q, t

3. Match the following columns.

	Column-I		Column-II
A.	XeF_4	p.	Linear
B.	XeOF_4	q.	Sp^3d^2 -hybridisation
C.	XeF_2	r.	Sp^3d -hybridisation
D.	XeO_2F_2	s.	See-saw

- (1) A-q; B-q; C-p, r; D-r, s
 (2) A-p; B-q; C-r, s; D-p, r
 (3) A-p, r; B-r, s; C-q; D-p
 (4) A-r, s; B-p, r; C-q; D-q

4. Match Column I with Column II and choose the correct combination from the options given.

	Column-I		Column-II
A.	BF_4^-	I.	Oxidation state of central atom is +4
B.	AlCl_3	II.	Strong oxidizing agent
C.	SnO	III.	Lewis acid
D.	PbO_2	IV.	Can be further oxidized
		V.	Tetrahedral shape

- (1) A-I; B-II; C-III; D-V, IV
 (2) A-V; B-III; C-IV; D-I, II
 (3) A-V; B-VI; C-I; D-III, II
 (4) A-III; B-II; C-I; D-IV, V

5. Match the species given in Column I with the properties mentioned in Column II.

	Column-I		Column-II
(i)	BF_4^-	(a)	Oxidation state of central atom is +4
(ii)	AlCl_3	(b)	Strong oxidising agent
(iii)	SnO	(c)	Lewis acid
(iv)	PbO_2	(d)	Can be further oxidised
		(e)	Tetrahedral shape

- (1) (i) → (e) (ii) → (c) (iii) → (d) (iv) → (a), (b)
 (2) (i) → (e) (ii) → (c) (iii) → (d) (iv) → (a), (b)
 (3) (i) → (e) (ii) → (c) (iii) → (d) (iv) → (a), (b)
 (4) (i) → (e) (ii) → (c) (iii) → (d) (iv) → (a), (b)

6. Match the species given in Column I with properties given in Column II.

	Column-I		Column-II
(i)	Diborane	(a)	Used as a flux for soldering metals
(ii)	Gallium	(b)	Crystalline form of silica
(iii)	Borax	(c)	Banana bonds
(iv)	Aluminosilicate	(d)	Low melting, high boiling, useful for measuring high temperatures
(v)	Quartz	(e)	Used as catalyst in petrochemical industries

- (1) (i) → (c), (ii) → (d), (iii) → (a), (iv) → (e), (v) → (b)
 (2) (i) → (c), (ii) → (d), (iii) → (a), (iv) → (e), (v) → (b)
 (3) (i) → (c), (ii) → (d), (iii) → (a), (iv) → (e), (v) → (b)
 (4) (i) → (c), (ii) → (d), (iii) → (a), (iv) → (e), (v) → (b)

7. Match the species given in Column I with the hybridisation given in Column II.

	Column-I		Column-II
(i)	Boron in $[B(OH)_4]^-$	(a)	sp^2
(ii)	Aluminium in $[Al(H_2O)_6]^{3+}$	(b)	sp^3
(iii)	Boron in B_2H_6	(c)	sp^3d^2
(iv)	Carbon in Buckminsterfullerene		
(v)	Silicon in SiO_4^{4-}		
(vi)	Germanium in $[GeCl_6]^{2-}$		

- (1) (i) $\rightarrow$ (b), (ii) $\rightarrow$ (c), (iii) $\rightarrow$ (b), (iv) $\rightarrow$ (a)
 (v) $\rightarrow$ (b), (vi) $\rightarrow$ (c)
 (2) (i) $\rightarrow$ (b), (ii) $\rightarrow$ (c), (iii) $\rightarrow$ (b), (iv) $\rightarrow$ (a)
 (v) $\rightarrow$ (b), (vi) $\rightarrow$ (c)
 (3) (i) $\rightarrow$ (b), (ii) $\rightarrow$ (c), (iii) $\rightarrow$ (b), (iv) $\rightarrow$ (a)
 (v) $\rightarrow$ (b), (vi) $\rightarrow$ (c)
 (4) (i) $\rightarrow$ (b), (ii) $\rightarrow$ (c), (iii) $\rightarrow$ (b), (iv) $\rightarrow$ (a)
 (v) $\rightarrow$ (b), (vi) $\rightarrow$ (c)

8. Match the compounds given in Column I with the hybridisation and shape given in Column II and mark the correct option.

	Column-I		Column-II
(A)	XeF_6	(i)	sp^3d^3 -distorted octahedral
(B)	XeO_3	(ii)	sp^3d^2 -square planar
(C)	$XeOF_4$	(iii)	sp^3 -pyramidal
(D)	XeF_4	(iv)	sp^3d^2 -square pyramidal

Code :

- (1) A(i); B(iii); C(iv); D(ii)
 (2) A(i); B(ii); C(iv); D(iii)
 (3) A(iv); B(iii); C(i); D(ii)
 (4) A(iv); B(i); C(ii); D(iii)

9. Match the formulas of oxides given in Column I with the type of oxide given in Column II and mark the correct option.

	Column-I		Column-II
(A)	Pb_3O_4	(i)	Neutral oxide
(B)	N_2O	(ii)	Acidic oxide
(C)	Mn_2O_7	(iii)	Basic oxide
(D)	Bi_2O_3	(iv)	Mixed oxide

Code :

- (1) A(i); B(ii); C(iii); D(iv)
 (2) A(iv); B(i); C(ii); D(iii)
 (3) A(iii); B(ii); C(iv); D(i)
 (4) A(iv); B(iii); C(i); D(ii)

10. Match the items of Columns I and II and mark the correct option.

	Column-I		Column-II
(A)	H_2SO_4	(i)	Highest electron gain enthalpy
(B)	CCl_3NO_2	(ii)	Chalcogen
(C)	Cl_2	(iii)	Tear gas
(D)	Sulphur	(iv)	Storage batteries

Code :

- (1) A(iv); B(iii); C(i); D(ii)
 (2) A(iii); B(iv); C(i); D(ii)
 (3) A(iv); B(i); C(ii); D(iii)
 (4) A(ii); B(i); C(iii); D(iv)

11. Match the species given in Column I with the shape given in Column II and mark the correct option.

	Column-I		Column-II
(A)	SF_4	(i)	Tetrahedral
(B)	BrF_3	(ii)	Pyramidal
(C)	BrO_3^-	(iii)	Sea-saw shaped
(D)	NH_4^+	(iv)	Bent T-shaped

Code :

- (1) A(iii); B(ii); C(i); D(iv)
 (2) A(iii); B(iv); C(ii); D(i)
 (3) A(i); B(ii); C(iii); D(iv)
 (4) A(i); B(iv); C(iii); D(ii)

12. Match the items of Columns I and II and mark the correct option.

Column I

- (A) Its partial hydrolysis does not change oxidation state of central atom
 (B) It is used in modern diving apparatus
 (C) It is used to provide inert atmosphere for filling electrical bulbs
 (D) Its central atom is in sp^3d^2 hybridisation

Column II

- (i) He (ii) XeF_6
 (iii) XeF_4 (iv) Ar

Code :

- (1) A(i); B(iv); C(ii); D(iii)
 (2) A(i); B(ii); C(iii); D(iv)
 (3) A(ii); B(i); C(iv); D(iii)
 (4) A(i); B(iii); C(ii); D(iv)

- 13. Assertion (A):** If aluminium atoms replace a few silicon atoms in three-dimensional network of silicon dioxide, the overall structure acquires a negative charge.
Reason (R) : Aluminium is trivalent while silicon is tetravalent.
- (1) Both A and R are correct and R is the correct explanation of A.
 - (2) Both A and R are correct but R is not the correct explanation of A.
 - (3) Both A and R are not correct
 - (4) A is not correct but R is correct.
- 14. Assertion (A):** Silicons are water repelling in nature.
Reason (R) : Silicons are organosilicon polymers, which have $(-R_2SiO-)$ as repeating unit.
- (1) A and R both are correct and R is the correct explanation of A.
 - (2) Both A and R are correct but R is not the correct explanation of A.
 - (3) A and R both are not true.
 - (4) A is not true but R is true.
- 15. Assertion :** N_2 is less reactive than P_4 .
Reason : Nitrogen has more electron gain enthalpy than phosphorus.
- (1) Both assertion and reason are correct statements, and reason is the correct explanation of the assertion.
 - (2) Both assertion and reason are correct statements, but reason is not the correct explanation of the assertion.
 - (3) Assertion is correct, but reason is wrong statement.
 - (4) Assertion is wrong but reason is correct statement.
- 16. Assertion :** HNO_3 makes iron passive.
Reason : HNO_3 forms a protective layer of ferric nitrate on the surface of iron.
- (1) Both assertion and reason are correct statements, and reason is the correct explanation of the assertion.
 - (2) Both assertion and reason are correct statements, but reason is not the correct explanation of the assertion.
 - (3) Assertion is correct, but reason is wrong statement.
 - (4) Assertion is wrong but reason is correct statement.
- 17. Assertion :** HI cannot be prepared by the reaction of KI with concentrated H_2SO_4
Reason : HI has lowest H-X bond strength among halogen acids.
- (1) Both assertion and reason are correct statements, and reason is the correct explanation of the assertion.
 - (2) Both assertion and reason are correct statements, but reason is not the correct explanation of the assertion.
 - (3) Assertion is correct, but reason is wrong statement.
 - (4) Assertion is wrong but reason is correct statement.
- 18. Assertion :** Both rhombic and monoclinic sulphur exist as S_8 but oxygen exists as O_2 .
Reason : Oxygen forms $p\pi - p\pi$ multiple bond due to small size and small bond length but $p\pi - p\pi$ bonding is not possible in sulphur.
- (1) Both assertion and reason are correct statements, and reason is the correct explanation of the assertion.
 - (2) Both assertion and reason are correct statements, but reason is not the correct explanation of the assertion.
 - (3) Assertion is correct, but reason is wrong statement.
 - (4) Assertion is wrong but reason is correct statement.
- 19. Assertion :** NaCl reacts with concentrated H_2SO_4 to give colourless fumes with pungent smell. But on adding MnO_2 the fumes become greenish yellow.
Reason : MnO_2 oxidises HCl to chlorine gas which is greenish yellow.
- (1) Both assertion and reason are correct statements, and reason is the correct explanation of the assertion.
 - (2) Both assertion and reason are correct statements, but reason is not the correct explanation of the assertion.
 - (3) Assertion is correct, but reason is wrong statement.
 - (4) Assertion is wrong but reason is correct statement.

20. Assertion : SF_6 cannot be hydrolysed but SF_4 can be.

Reason : Six F atoms in SF_6 prevent the attack of H_2O on sulphur atom of SF_6 .

- (1) Both assertion and reason are correct statements, and reason is the correct explanation of the assertion.
- (2) Both assertion and reason are correct statements, but reason is not the correct explanation of the assertion.
- (3) Assertion is correct, but reason is wrong statement.
- (4) Assertion is wrong but reason is correct statement.

21. Statement I : Dinitrogen is inert due to high bond enthalpy of $\text{N}\equiv\text{N}$ bond.

Statement II : With rapid increase in temperature the reactivity decreases.

- (1) Statement I and Statement II are correct and Statement II is the correct explanation of Statement I.
- (2) Statement I and Statement II are correct but Statement II is not the correct explanation of Statement I.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement II is correct but Statement I is incorrect.

22. I. Atomic and ionic radii of group-16 elements decreases from top to bottom.
 II. Ionisation enthalpy decreases down the group-16.
 III. Oxygen has less negative electron gain enthalpy than sulphur.

The correct statement(s) is/are

- (1) Only I
- (2) Only II
- (3) Both II and III
- (4) All of these

23. Due to stable electronic configurations of noble gases

- I. these gases exhibit very high ionisation enthalpy.
- II. these have no tendency to accept the electron.
- III. these have large positive values of electron gain enthalpy.

The incorrect set of statements is

- (1) I, II and III
- (2) II and III
- (3) I and II
- (4) None of these

24. Statement I : Carbon shows maximum covalency of 4 only.

Statement II : Carbon has any s- and p-orbitals in its valence shell.

- (1) Statement I and Statement II are correct and Statement II is the correct explanation of Statement I.
- (2) Statement I and Statement II are correct but Statement II is not the correct explanation of Statement I.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement II is correct but Statement I is incorrect.

25. The element which exists in liquid state for a wide range of temperature and can be used for measuring high temperature is

- (1) B
- (2) Al
- (3) Ga
- (4) In

26. Which of the following is a Lewis acid?

- (1) AlCl_3
- (2) MgCl_2
- (3) CaCl_2
- (4) BaCl_2

27. The geometry of a complex species can be understood from the knowledge of type of hybridisation of orbitals of central atom. The hybridisation of orbitals of central atom in $[\text{Be}(\text{OH})_4]^-$ and the geometry of the complex are respectively

- (1) sp^3 , tetrahedral
- (2) sp^3 , square planar
- (3) sp^3d^2 , octahedral
- (4) dsp^2 , square planar

28. Which of the following oxides is acidic in nature?
(1) B_2O_3 (2) Al_2O_3
(3) Ga_2O_3 (4) In_2O_3
29. The exhibition of highest co-ordination number depends on the availability of vacant orbitals in the central atom. Which of the following elements is not likely to act as central atom in MF_6^{3-} ?
(1) B (2) Al
(3) Ga (4) In
30. Boric acid is an acid because its molecule
(1) contains replaceable H^+ ion
(2) gives up a proton
(3) accepts OH^- from water releasing proton
(4) combines with proton from water molecule
31. Silicon has a strong tendency to form polymers like silicones. The chain length of silicone polymer can be controlled by adding
(1) $MeSiCl_3$ (2) Me_2SiCl_2
(3) Me_3SiCl (4) Me_4Si
32. A compound X, of boron reacts with NH_3 on heating to give another compound Y which is called inorganic benzene. The compound X can be prepared by treating BF_3 with Lithium aluminium hydride. The compounds X and Y are represented by the formulas.
(1) B_2H_6 , $B_3N_3H_6$ (2) B_2O_3 , $B_3N_3H_6$
(3) BF_3 , $B_3N_3H_6$ (4) $B_3N_3H_6$, B_2H_6
33. Quartz is extensively used as a piezoelectric material, it contains.
(1) Pb (2) Si
(3) Ti (4) Sn
34. The most commonly used reducing agent is
(1) $AlCl_3$ (2) $PbCl_2$
(3) $SnCl_4$ (4) $SnCl_2$
35. Cement, the important building material is a mixture of oxides of several elements. Besides calcium, iron and sulphur, oxides of elements of which of the group (s) are present in the mixture?
(1) group 2 (2) groups 2, 13 and 14
(3) groups 2 and 13 (4) groups 2 and 14
36. The reason for small radius of Ga compared to Al is.
(1) poor screening effect of d and f orbitals
(2) increase in nuclear charge
(3) presence of higher orbitals
(4) higher atomic number
37. The linear shape of CO_2 is due to.
(1) sp^3 hybridisation of carbon
(2) sp hybridisation of carbon
(3) $p\pi-p\pi$ bonding between carbon and oxygen
(4) sp^2 hybridisation of carbon
38. Me_3SiCl is used during polymerisation of organosilicons because
(1) The chain length of organo silicone polymers can be controlled by adding Me_3SiCl
(2) Me_3SiCl blocks the end terminal of silicone polymer
(3) Me_3SiCl improves the quality and yield of the polymer
(4) Me_3SiCl acts as a catalyst during polymerization
39. Which of the following statements are correct?
(1) Fullerenes have dangling bonds
(2) Fullerenes are cage-like molecules
(3) Graphite is thermodynamically most stable allotrope of carbon
(4) Graphite is slippery and hard and therefore used as a dry lubricant in machines

40. Which of the following statements are correct. Answer on the basis of Fig.11.1.

- (1) The two bridged hydrogen atoms and the two boron atoms lie in one plane;
- (2) Out of six B-H bonds two bonds can be described in terms of 3 centre 2-electron bonds.
- (3) Out of six B-H bonds four B-H bonds can be described in terms of 3 centre 2 electron bonds;

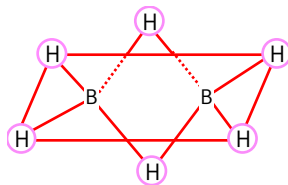


Fig. 11.1

- (4) The four terminal B-H bonds are two centre-two electron regular bonds.

41. Identify the correct resonance structures of carbon dioxide from the ones given below:

- (1) $O \equiv C \equiv O$ (2) $O = C = O$
- (3) ${}^{-}O \equiv C - O^{+}$ (4) ${}^{-}O - C \equiv O^{+}$

42. On addition of conc. H_2SO_4 to a chloride salt, colourless fumes are evolved but in case of iodide salt, violet fumes come out. This is because

- (1) H_2SO_4 reduces HI to I_2
- (2) HI is of violet colour
- (3) HI gets oxidised to I_2
- (4) HI changes to HIO_3

43. In qualitative analysis when H_2S is passed through an aqueous solution of salt acidified with dil. HCl, a black precipitate is obtained. On boiling the precipitate with dil. HNO_3 , it forms a solution of blue colour. Addition of excess of aqueous solution of ammonia to this solution gives .

- (1) deep blue precipitate of $Cu(OH)_2$
- (2) deep blue solution of $[Cu(NH_3)_4]^{2+}$
- (3) deep blue solution of $Cu(NO_3)_2$
- (4) deep blue solution of $Cu(OH)_2.Cu(NO_3)_2$

44. In a cyclotrimetaphosphoric acid molecule, how many single and double bonds are present?

- (1) 3 double bonds; 9 single bonds
- (2) 6 double bonds; 6 single bonds
- (3) 3 double bonds; 12 single bonds
- (4) Zero double bonds; 12 single bonds

45. Which of the following pairs of ions are isoelectronic and isostructural?

- (1) CO_3^{2-}, NO_3^- (2) ClO_3^-, CO_3^{2-}
- (3) SO_3^{2-}, NO_3^- (4) ClO_3^-, SO_3^{2-}

46. Affinity for hydrogen decreases in the group from fluorine to iodine. Which of the halogen acids should have highest bond dissociation enthalpy?

- (1) HF (2) HCl
- (3) HBr (4) HI

47. Bond dissociation enthalpy of E-H (E=element) bonds is given below. Which of the compounds will act as strongest reducing agent?

Compound	NH_3	PH_3	AsH_3	SbH_3
$\Delta_{diss}(E-H)/$ $kJmol^{-1}$	389	322	297	255
(1) NH_3		(2) PH_3		
(3) AsH_3		(4) SbH_3		

48. On heating with concentrated NaOH solution in an inert atmosphere of CO_2 , white phosphorus gives a gas. Which of the following statement is incorrect about the gas?

- (1) It is highly poisonous and has smell like rotten fish.
- (2) It's solution in water decomposes in the presence of light.
- (3) It is more basic than NH_3 .
- (4) It is less basic than NH_3 .

49. Which of the following acids forms three series of salts?
 (1) H_3PO_2 (2) H_3BO_3
 (3) H_3PO_4 (4) H_3PO_3
50. Which of the following statements is wrong?
 (1) Single N – N bond is stronger than the single P – P bond.
 (2) PH_3 can act as a ligand in the formation of coordination compound with transition elements.
 (3) NO_2 is paramagnetic in nature.
 (4) Covalency of nitrogen in N_2O_5 is four.
51. A brown ring is formed in the ring test for NO_3^- ion. It is due to the formation of
 (1) $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{2+}$ (2) $\text{FeSO}_4 \cdot \text{NO}_2$
 (3) $[\text{Fe}(\text{H}_2\text{O})_4(\text{NO})_2]^{2+}$ (4) $\text{FeSO}_4 \cdot \text{HNO}_3$
52. In the preparation of HNO_3 , we get NO gas by catalytic oxidation of ammonia. The moles of NO produced by the oxidation of two moles of NH_3 will be.
 (1) 2 (2) 3 (3) 4 (4) 6
53. Which of the following is not tetrahedral in shape?
 (1) NH_4^+ (2) SiCl_4
 (3) SF_4 (4) SO_4^{2-}
54. Hot conc. H_2SO_4 acts as moderately strong oxidising agent. It oxidises both metals and nonmetals. Which of the following element is oxidised by conc. H_2SO_4 into two gaseous products?
 (1) Cu (2) S
 (3) C (4) Zn
55. In the preparation of compounds of Xe, Bartlett had taken $\text{O}_2^+\text{PtF}_6^-$ as a base compound. This is because
 (1) both O_2 and Xe have same size.
 (2) both O_2 and Xe have same electron gain enthalpy.
 (3) both O_2 and Xe have almost same ionisation enthalpy.
 (4) both Xe and O_2 are gases.
56. In solid state PCl_5 is a.
 (1) covalent solid
 (2) octahedral structure
 (3) ionic solid with $[\text{PCl}_6]^+$ octahedral and $[\text{PCl}_4]^-$ tetrahedra
 (4) ionic solid with $[\text{PCl}_4]^+$ tetrahedral and $[\text{PCl}_6]^-$ octahedra
57. Reduction potentials of some ions are given below. Arrange them in decreasing order of oxidising power.
- | Ion | Reduction Potential $E^\ominus / \text{V}$ |
|------------------|--|
| ClO_4^- | $E^\ominus = 1.19 \text{ V}$ |
| IO_4^- | $E^\ominus = 1.65 \text{ V}$ |
| BrO_4^- | $E^\ominus = 1.74 \text{ V}$ |
- (1) $\text{ClO}_4^- > \text{IO}_4^- > \text{BrO}_4^-$
 (2) $\text{IO}_4^- > \text{BrO}_4^- > \text{ClO}_4^-$
 (3) $\text{BrO}_4^- > \text{IO}_4^- > \text{ClO}_4^-$
 (4) $\text{BrO}_4^- > \text{ClO}_4^- > \text{IO}_4^-$
58. Which of the following is isoelectronic pair?
 (1) ICl_2 , ClO_2 (2) BrO_2^- , BrF_2^+
 (3) ClO_2 , BrF (4) CN^- , O_3
59. Which of the following options are not in accordance with the property mentioned against them?
 (1) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$ Oxidising power.
 (2) $\text{MI} > \text{MBr} > \text{MCl} > \text{MF}$ Ionic character of metal halide.
 (3) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$ Bond dissociation enthalpy.
 (4) $\text{HI} < \text{HBr} < \text{HCl} < \text{HF}$ Hydrogen-halogen bond strength.
60. Which of the following statements are correct?
 (1) Among halogens, radius ratio between iodine and fluorine is maximum.
 (2) Leaving F – F bond, all halogens have weaker X – X bond than X – X' bond in interhalogens.
 (3) Among interhalogen compounds maximum number of atoms are present in iodine fluoride.
 (4) Interhalogen compounds are more reactive than halogen compounds.

- 61.** Which of the following statements are correct for SO_2 gas?
- (1) It acts as bleaching agent in moist conditions.
 - (2) It's molecule has linear geometry.
 - (3) It's dilute solution is used as disinfectant.
 - (4) It can be prepared by the reaction of dilute H_2SO_4 with metal sulphide.
- 62.** Which of the following statements are correct?
- (1) All the three N — O bond lengths in HNO_3 are equal.
 - (2) All P — Cl bond lengths in PCl_5 molecule in gaseous state are equal.
 - (3) P_4 molecule in white phosphorus have angular strain therefore white phosphorus is very reactive.
 - (4) PCl_5 is ionic in solid state in which cation is tetrahedral and anion is octahedral.
- 63.** Which of the following orders are correct as per the properties mentioned against each?
- (1) $\text{As}_2\text{O}_3 < \text{SiO}_2 < \text{P}_2\text{O}_3 < \text{SO}_2$ Acid strength.
 - (2) $\text{AsH}_3 < \text{PH}_3 < \text{NH}_3$ Enthalpy of vapourisation.
 - (3) $\text{S} < \text{O} < \text{Cl} < \text{F}$ More negative electron gain enthalpy.
 - (4) $\text{H}_2\text{O} > \text{H}_2\text{S} > \text{H}_2\text{Se} > \text{H}_2\text{Te}$ Thermal stability.
- 64.** Which of the following statements are correct?
- (1) S — S bond is present in $\text{H}_2\text{S}_2\text{O}_6$.
 - (2) In peroxosulphuric acid (H_2SO_5) sulphur is in +6 oxidation state.
 - (3) Iron powder along with Al_2O_3 and K_2O is used as a catalyst in the preparation of NH_3 by Haber's process.
 - (4) Change in enthalpy is positive for the preparation of SO_3 by catalytic oxidation of SO_2 .
- 65.** In which of the following reactions conc. H_2SO_4 is used as an oxidising reagent?
- (1) $\text{CaF}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + 2\text{HF}$
 - (2) $2\text{HI} + \text{H}_2\text{SO}_4 \rightarrow \text{I}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$
 - (3) $\text{Cu} + 2\text{H}_2\text{SO}_4 \rightarrow \text{CuSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$
 - (4) $\text{NaCl} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HCl}$
- 66.** Ionisation enthalpy for the elements of Group 13 follows the order.
- (1) $\text{B} > \text{Al} > \text{Ga} > \text{In} > \text{Tl}$
 - (2) $\text{B} < \text{Al} < \text{Ga} < \text{In} < \text{Tl}$
 - (3) $\text{B} < \text{Al} > \text{Ga} < \text{In} > \text{Tl}$
 - (4) $\text{B} > \text{Al} < \text{Ga} > \text{In} < \text{Tl}$
- 67.** In the structure of diborane
- (1) All hydrogen atoms lie in one plane and boron atoms lie in a plane perpendicular to this plane.
 - (2) 2 boron atoms and 4 terminal hydrogen atoms lie in the same plane and 2 bridging hydrogen atoms lie in the perpendicular plane.
 - (3) 4 bridging hydrogen atoms and boron atoms lie in one plane and two terminal hydrogen atoms lie in a plane perpendicular to this plane
 - (4) All the atoms are in same plane.
- 68.** The tendency of catenation in Group 14 elements follows the order:
- (1) $\text{C} > \text{Si} > \text{Ge} > \text{Sn}$
 - (2) $\text{C} \gg \text{Si} > \text{Ge} \approx \text{Sn}$
 - (3) $\text{Si} > \text{C} > \text{Sn} > \text{Ge}$
 - (4) $\text{Ge} > \text{Sn} > \text{Si} > \text{C}$
- 69.** Which of the following elements can be involved in $p\pi-d\pi$ bonding?
- (1) Carbon
 - (2) Nitrogen
 - (3) Phosphorus
 - (4) Boron
- 70.** On heating lead nitrate forms oxides of nitrogen and lead. The oxides formed are,
- (1) N_2O , PbO
 - (2) NO_2 , PbO
 - (3) NO , PbO
 - (4) NO , PbO_2

71. which of the following elements does not show allotropy?
(1) Nitrogen
(2) Bismuth
(3) Antimony
(4) Arsenic
72. Maximum covalency of nitrogen is.....,
(1) 3 (2) 5
(3) 4 (4) 6
73. Elements of group-15 form compounds in +5 oxidation state. However, bismuth forms only one well characterised compound in +5 oxidation state. The compound is
(1) Bi_2O_5 (2) BiF_5
(3) BiCl_5 (4) Bi_2S_5
74. On heating ammonium dichromate and barium azide separately we get
(1) N_2 in both cases
(2) N_2 with ammonium dichromate and NO with barium azide
(3) N_2O with ammonium dichromate and N_2 with barium azide
(4) N_2O with ammonium dichromate and NO_2 with barium azide
75. the oxidation state of central atom in the anion of compound NaH_2PO_2 will be
(1) +3 (2) +5
(3) +1 (4) -3
76. A black compound of manganese reacts with a halogen acid to give greenish yellow gas. When excess of this gas reacts with NH_3 an unstable trihalide is formed. In this process the oxidation state of nitrogen changes from
(1) -3 to +3
(2) -3 to 0
(3) -3 to +5
(4) 0 to -3
77. Which of the following is correct for P_4 molecule of white phosphorus?
(a) It has 6 lone pairs of electrons
(b) It has six P-P single bonds
(c) It has three P-P single bonds
(d) It has four lone pairs of electrons
(1) b, d
(2) c, d
(3) a, d
(4) a, b
78. Which of the following are peroxoacids of sulphur?
(1) H_2SO_5 and $\text{H}_2\text{S}_2\text{O}_8$
(2) H_2SO_5 and $\text{H}_2\text{S}_2\text{O}_7$
(3) $\text{H}_2\text{S}_2\text{O}_7$ and $\text{H}_2\text{S}_2\text{O}_8$
(4) $\text{H}_2\text{S}_2\text{O}_6$ and $\text{H}_2\text{S}_2\text{O}_7$
79. if chlorine gas is passed through hot NaOH solution, two changes are observed in the oxidation number of chlorine during the reaction. These are and
(a) 0 to +5
(b) 0 to +3
(c) 0 to -1
(d) 0 to +1
(1) a, b
(2) a, c
(3) a, d
(4) a, b, c
80. Which of the following statements are true?
(a) Only type of interactions between particles of noble gases are due to weak dispersion forces
(b) Ionisation enthalpy of molecular oxygen is very close to that of xenon
(c) Hydrolysis of XeF_6 is a redox reaction
(d) Xenon fluorides are not reactive
(1) a, c
(2) a, b, c
(3) a, d
(4) a, b

Exercise 4 (Previous Year's Questions)

- Roasting of sulphide gives the gas X as a by product. This is colourless gas with coking smell of burnt sulphur and causes great damage to respiratory organs as a result of acid rain. Its aqueous solution is acidic, acts as reducing agent and its acid has never been isolated. The gas X is: [NEET UG 2013]
(1) SO_3 (2) H_2S (3) SO_2 (4) CO_2
- Strong reducing behaviour of H_3PO_2 is due to: [Re AIPMT 2015]
(1) High oxidation state of phosphorus
(2) Presence of two $-\text{OH}$ groups and one $\text{P}-\text{H}$ bond
(3) Presence of one $-\text{OH}$ groups and two $\text{P}-\text{H}$ bonds
(4) High electron gain enthalpy of phosphorus
- Which of the statement given below is incorrect? [Re AIPMT 2015]
(1) ONF is isoelectronic with O_2N^-
(2) OF_2 is an oxide of fluorine
(3) Cl_2O_7 is an anhydride of perchloric acid
(4) O_3 , molecule is bent
- The stability of +1 oxidation state among Al, Ga, In and Tl increase in the sequence: [Re AIPMT 2015]
(1) $\text{Tl} < \text{In} < \text{Ga} < \text{Al}$ (2) $\text{In} < \text{Tl} < \text{Ga} < \text{Al}$
(3) $\text{Ga} < \text{In} < \text{Al} < \text{Tl}$ (4) $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$
- Among the following, the correct order of acidity is [NEET-I 2016]
(1) $\text{HClO}_3 < \text{HClO}_4 < \text{HClO}_2 < \text{HClO}$
(2) $\text{HClO} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$
(3) $\text{HClO}_2 < \text{HClO} < \text{HClO}_3 < \text{HClO}_4$
(4) $\text{HClO}_4 < \text{HClO}_2 < \text{HClO} < \text{HClO}_3$
- Which is the correct statement for the given acids? [NEET-I 2016]
(1) Phosphinic acid is a diprotic acid while phosphonic acid is a monoprotic acid
(2) Phosphinic acid is a monoprotic acid while phosphonic acid is a diprotic acid
(3) Both are triprotic acids
(4) Both are diprotic acids
- In which pair of ions both the species contain S-S bond? [NEET (UG) 2017]
(1) $\text{S}_4\text{O}_6^{2-}$, $\text{S}_2\text{O}_3^{2-}$ (2) $\text{S}_2\text{O}_7^{2-}$, $\text{S}_2\text{O}_8^{2-}$
(3) $\text{S}_4\text{O}_6^{2-}$, $\text{S}_2\text{O}_7^{2-}$ (4) $\text{S}_4\text{O}_7^{2-}$, $\text{S}_2\text{O}_3^{2-}$
- It is because of inability of ns^2 electrons of the valence shell to participate in bonding that :- [NEET (UG) 2017]
(1) Sn^{2+} is oxidising while Pb^{4+} is reducing
(2) Sn^{2+} and Pb^{2+} are both oxidising and reducing
(3) Sn^{4+} is reducing while Pb^{4+} is oxidising
(4) Sn^{2+} is reducing while Pb^{4+} is oxidising
- The liquified gas that is used in dry cleaning along with a suitable detergent is:- [NEET (UG) 2017 (ODISHA)]
(1) Water gas (2) Petroleum gas
(3) NO_2 (4) CO_2
- Which of the following oxoacids of phosphorus has strongest reducing property? [NEET (UG) 2017 (ODISHA)]
(1) $\text{H}_4\text{P}_2\text{O}_7$ (2) H_3PO_3
(3) H_3PO_2 (4) H_3PO_4
- Identify the correct formula of oleum from the following [NEET (UG) 2017 (ODISHA)]
(1) $\text{H}_2\text{S}_2\text{O}_7$ (2) H_2SO_3
(3) H_2SO_4 (4) $\text{H}_2\text{S}_2\text{O}_8$
- Which of the following statements is not true for halogens? [NEET (UG) 2018]
(1) All form monobasic oxyacids.
(2) All are oxidizing agent.
(3) All except fluorine show positive oxidation states.
(4) Chlorine has the highest electron - gain enthalpy.
- The correct order of N-compounds in its decreasing order of oxidation states is - [NEET (UG) 2018]
(1) HNO_3 , NO, N_2 , NH_4Cl
(2) HNO_3 , NO, NH_4Cl , N_2
(3) HNO_3 , NH_4Cl , NO, N_2
(4) NH_4Cl , N_2 , NO, HNO_3

14. Which of the following is incorrect statement? [NEET (UG) 2019]
(1) PbF_4 is covalent in nature
(2) SiCl_4 is easily hydrolysed
(3) GeX_4 ($X = \text{F, Cl, Br, I}$) is more stable than GeX_2
(4) SnF_4 is ionic in nature
15. Which of the following oxoacid of sulphur has $-\text{O}-\text{O}-$ linkage? [NEET-2020]
(1) $\text{H}_2\text{S}_2\text{O}_8$, peroxodisulphuric acid
(2) $\text{H}_2\text{S}_2\text{O}_7$, pyrosulphuric acid
(3) H_2SO_3 , sulphurous acid
(4) H_2SO_4 , sulphuric acid
16. Identify the correct statements from the following: [NEET-2020]
(a) $\text{CO}_2(\text{g})$ is used as refrigerant for ice-cream and frozen food.
(b) The structure of C_{60} contains twelve six carbon rings and twenty-five carbon rings.
(c) ZSM-5, a type of zeolite, is used to convert alcohols into gasoline.
(d) CO is colourless and odourless gas.
(1) (b) and (c) only
(2) (c) and (d) only
(3) (a), (b) and (c) only
(4) (a) and (c) only
17. Which of the following is not correct about carbon monoxide? [NEET (UG)-2020]
(1) It is produced due to incomplete combustion
(2) It forms carboxyhaemoglobin
(3) It reduce oxygen carrying ability of blood
(4) The carboxyhaemoglobin (haemoglobin bound to CO) is less stable than oxyhaemoglobin.
18. Which of the following oxide is amphoteric in nature? [NEET 2020 EXAM (COVID RE-EXAM)]
(1) SnO_2 (2) SiO_2
(3) GeO_2 (4) CO_2
19. **Statement-I:** Acid strength increases in the order given as $\text{HF} \ll \text{HCl} \ll \text{HBr} \ll \text{HI}$.
Statement-II: As the size of the elements F, Cl, Br, I increases down the group, the bond strength of HF, HCl, HBr and HI decreases and so the acid strength increases.
In the light of the above statements, choose the correct answer from the options given below. [NEET-2021]
(1) Statement I is incorrect but Statement II is true.
(2) Both Statement I and Statement II are true.
(3) Both Statement I and Statement II are false
(4) Statement I is correct but Statement II is false.
20. Noble gases are named because of their inertness towards reactivity. Identify an incorrect statement about them. [NEET-2021]
(1) Noble gases have large positive values of electron gain enthalpy
(2) Noble gases are sparingly soluble in water.
(3) Noble gases have very high melting and boiling points
(4) Noble gases have weak dispersion forces.
21. In which one of the following arrangements the given sequence is not strictly according to the properties indicated against it? [NEET-2021]
(1) $\text{CO}_2 < \text{SiO}_2 < \text{SnO}_2 < \text{PbO}_2$: Increasing oxidizing power
(2) $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$: Increasing acidic strength
(3) $\text{H}_2\text{O} < \text{H}_2\text{S} < \text{H}_2\text{Se} < \text{H}_2\text{Te}$: Increasing pK_a values
(4) $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$: Increasing acidic character

22. Which of the following statement is not correct about diborane? [NEET-2022]

- (1) The four terminal B-H bonds are two centre two electron bonds.
- (2) The four terminal Hydrogen atoms and the two Boron atoms lie in one plane.
- (3) Both the Boron atoms are sp^2 hybridised.
- (4) There are two 3-centre-2-electron bonds.

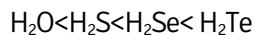
23. Choose the correct statement:[NEET-2022]

- (1) Diamond is covalent and graphite is ionic.
- (2) Diamond is sp^3 hybridised and graphite is sp^2 hybridized.
- (3) Both diamond and graphite are used as dry lubricants.
- (4) Diamond and graphite have two dimensional network.

24. Given below are two statements:

[NEET-2022]

Statement-I: The boiling points of the following hydrides of group 16 elements increases in the order–



Statement-II: The boiling points of these hydrides increase with increase in molar mass. In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement I and Statement II are incorrect.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Both Statement I and Statement II are correct.

25. Match List-I with List-II: [NEET-2023]

	List-I		List-II
A.	Coke	I.	Carbon atoms are sp^3 hybridised
B.	Diamond	II.	Used as a dry lubricant
C.	Fullerene	III.	Used as a reducing agent
D.	Graphite	IV.	Cage like molecules

Choose the correct answer from the options given below:

- (1) A-IV, B-I, C-II, D-III
- (2) A-III, B-I, C-IV, D-II
- (3) A-III, B-IV, C-I, D-II
- (4) A-II, B-IV, C-I, D-III

26. Taking stability as the factor. Which one of the following represents correct relationship? [NEET-2023]

- (1) $InI_3 > InI$
- (2) $AlCl > AlCl_3$
- (3) $TlI > TlI_3$
- (4) $TlCl_3 > TlCl$

27. Match List-I with List-II: [NEET-2023]

	List-I (Oxoacids of Sulphur)		List-II (Bonds)
A.	Peroxodisulphuric Acid	I.	Two S-OH, Four S = O, One S-O-S
B.	Sulphuric acid	II.	Two S-OH, One S = O
C.	Pyrosulphuric acid	III.	Two S-OH, Four S = O, One S-O-O-S
D.	Sulphurous acid	IV.	Two S-OH, Two S = O

Choose the correct answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-I, B-III, C-IV, D-II
- (3) A-III, B-IV, C-II, D-I
- (4) A-I, B-III, C-II, D-IV

28. Among group 16 elements, which one does NOT show –2 oxidation state: [NEET-2024]

- (1) Te
- (2) Po
- (3) O
- (4) Se

29. Identify the **incorrect** statement from the following: [RE-NEET-2024]

- (1) The acidic strength of HX (X = F, Cl, Br and I) follows the order: $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
- (2) Fluorine exhibits - 1 oxidation state where as other halogens exhibit +1, +3, +5 and +7 oxidation states also.
- (3) The enthalpy of dissociation of F_2 is smaller than that of Cl_2 .
- (4) Fluorine is stronger oxidising agent than chlorine.

30. Match List-I with List-II: [RE-NEET-2024]

	List-I (Atom/Molecule)		List-II (Property)
A.	Nitrogen atom	I.	Paramagnetic
B.	Fluorine molecule	II.	Most reactive element in group 18
C.	Oxygen molecule	III.	Element with Highest Ionization enthalpy in group 15
D.	Xenon atom	IV.	Strongest oxidizing agent

Identify the correct answer from the options given below:

- (1) A-III, B-I, C-IV, D-II (2) A-I, B-IV, C-III, D-II
(3) A-II, B-IV, C-I, D-III (4) A-III, B-IV, C-I, D-II

31. Given below are two statements : [NEET-2025]

Statement I : Like nitrogen that can form ammonia, arsenic can form arsine.

Statement II : Antimony cannot form antimony pentoxide.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Statement I is correct but Statement II is incorrect
- (2) Statement I is incorrect but Statement II is correct
- (3) Both Statement I and Statement II are correct
- (4) Both Statement I and Statement II are incorrect

32. Match List-I with List-II [NEET-2025]

List-I	List-II
A. XeO_3	I. Sp^3d ; linear
B. XeF_2	II. Sp^3 ; pyramidal
C. XeOF_4	III. Sp^3d^3 ; distorted octahedral
D. XeF_6	IV. Sp^3d^2 ; square pyramidal

Choose the correct answer from the options given below :

- (1) A-IV, B-II, C-III, D-I
- (2) A-IV, B-II, C-I, D-III
- (3) A-II, B-I, C-IV, D-III
- (4) A-II, B-I, C-III, D-IV

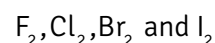
33. Anomalous behavior of oxygen is due to its. [JEE-MAIN-ONLINE 2024]

- (1) small size and low electronegativity
- (2) Large size and high electronegativity
- (3) Small size and high electronegativity
- (4) Large size and low electronegativity

34. Identify the incorrect pair from the following: [JEE-MAIN-ONLINE 2024]

- (1) Cryolite – Na_3AlF_6
- (2) Fluoroapatite – $3\text{Ca}_3(\text{PO}_4)_2 \cdot \text{CaF}_2$
- (3) Fluorspar – BF_3
- (4) Carnallite – $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$

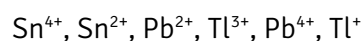
35. Among the following halogens



Which can undergo disproportionation reactions? [JEE-MAIN-ONLINE 2024]

- (1) Cl_2, Br_2 and I_2
- (2) F_2, Cl_2 and Br_2
- (3) F_2 and Cl_2
- (4) Only I_2

36. The number of ions from the following that are expected to behave as oxidising agent is:



[JEE-MAIN-ONLINE 2024]

- (1) 1 (2) 3
- (3) 2 (4) 4

ANSWER KEYS

Exercise 1.1

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	2	4	2	2	4	1	2	4	1	3	4	3	1	2	2	3	2	2	4
Que.	21	22	23	24	25	26	27	28	29	30										
Ans.	3	2	3	1	3	2	4	1	3	2										

Exercise 1.2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	2	1	1	1	2	2	3	1	1	1	2	3	4	3	3	1	2	3	2
Que.	21	22	23	24	25	26														
Ans.	3	3	4	1	3	4														

Exercise 1.3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	4	2	3	4	4	1	4	3	3	3	3	3	4	4	2	1	2	4	4	1
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	3	3	4	2	4	3	1	2	3	4	3	1	1	2	2	3	1	2	2	4

Exercise 1.4

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	
Ans.	3	2	2	3	1	2	2	3	1	3	4	3	4	4	2	1	3	4	1	

Exercise 1.5

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	1	3	4	4	2	3	1	3	2	1	1	1	2	4	3	3	2	1	1
Que.	21	22	23	24	25	26	27	28	29	30	31									
Ans.	4	3	1	3	1	3	2	4	2	3	2									

Exercise 1.6

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14						
Ans.	2	3	2	1	1	2	2	3	3	1	1	3	1	2						

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	4	1	3	1	2	3	4	3	2	1	4	4	3	3	4	4	4	3	3
Que.	21	22	23	24	25	26	27	28	29	30										
Ans.	2	3	3	3	1	1	3	3	3	4										

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	2	1	2	1	1	1	1	2	1	2	3	1	2	3	3	2	1	1	1
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	3	3	4	1	3	1	1	1	1	3	3	1	2	4	2	2	2,3	2	2	1,2,4
Que.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	4	3	2	1	1	1	4	3	3	1	1	1	3	3	3	4	3	2	2	1
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
Ans.	1	3	1	1	2	4	2	2	3	2	1	3	2	1	3	1	1	1	2	4

Exercise 4 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	3	2	4	2	2	1	4	4	3	1	1	1	1	1	4	4	3	2	3
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36				
Ans.	3	3	2	1	2	3	1	2	1	4	1	3	3	3	1	2				

Chapter 4

SALT ANALYSIS

Chapter Summary

- Introduction
- Identification of Anion
- Identification of Cations

Introduction

ANALYSIS always does not mean breaking of substance into its ultimate

constituents. Finding out the nature of substance and identity of its constituents is also analysis and is known as qualitative analysis. Qualitative analysis of inorganic salts means the identification of cations and anions present in the salt or a mixture of salts. Inorganic salts may be obtained by complete or partial neutralisation of acid with base or vice-versa. In the formation of a salt, the part contributed by the acid is called anion and the part contributed by the base is called cation.

Qualitative analysis is carried out through the reactions which are easily perceptible to our senses such as sight and smell.

Such reactions involve:

- (a) Formation of a precipitate
- (b) Change in colour
- (c) Evolution of gas etc.

Systematic analysis of an inorganic salt involves the following steps:

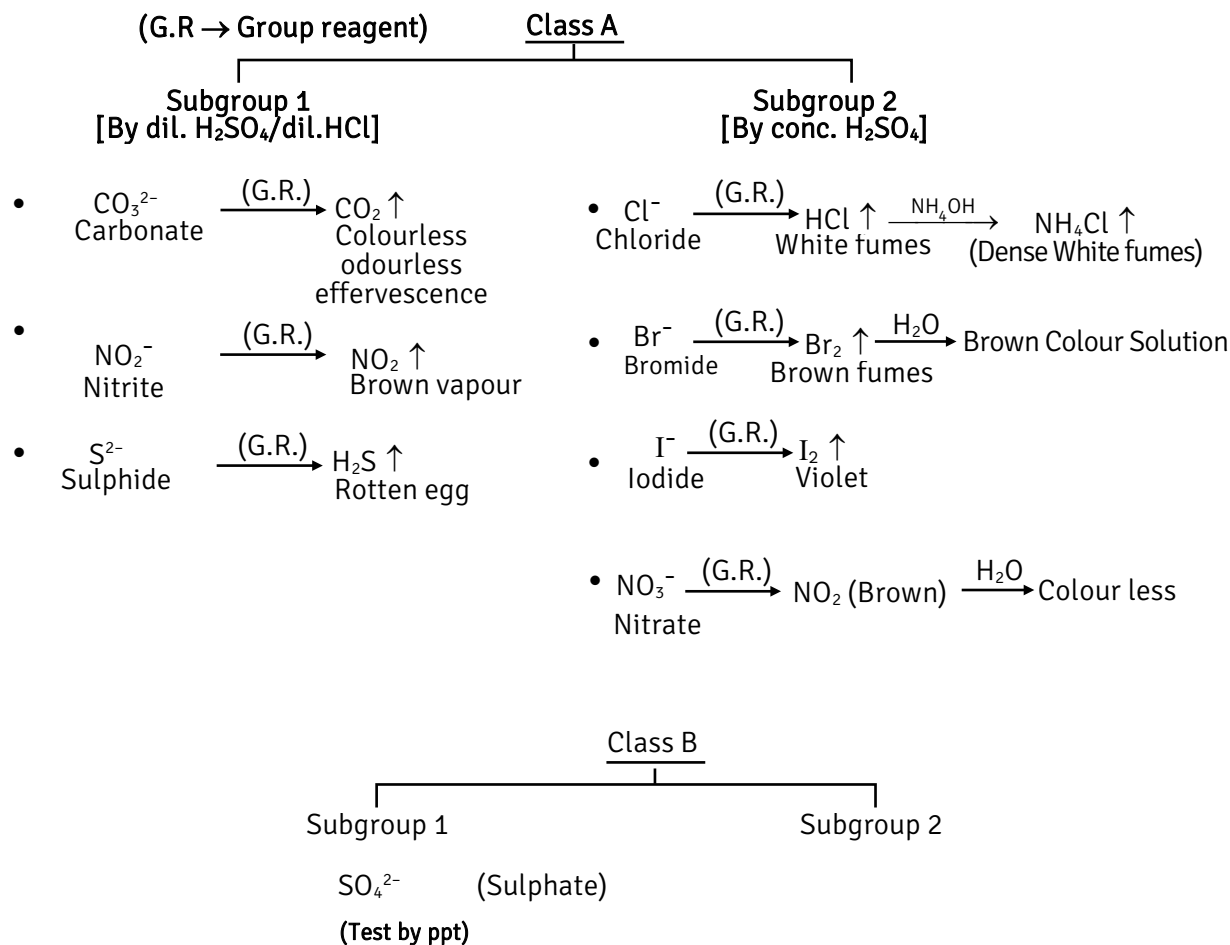
- (i) Preliminary examination of solid salt and its solution.
- (ii) Determination of anions by reactions carried out in solution (wet tests) and confirmatory tests.
- (iii) Determination of cations by reactions carried out in solution (wet tests) and confirmatory tests.

NEET Syllabus:

Cation: Pb^{2+} , Cu^{2+} , Al^{3+} , Fe^{3+} , Zn^{2+} , Ni^{2+} , Ca^{2+} , Ba^{2+} , Mg^{2+} , NH_4^+

Anion: CO_3^{2-} , NO_2^- , Cl^- , Br^- , I^- , NO_3^- , SO_4^{2-}

Identification of Anion



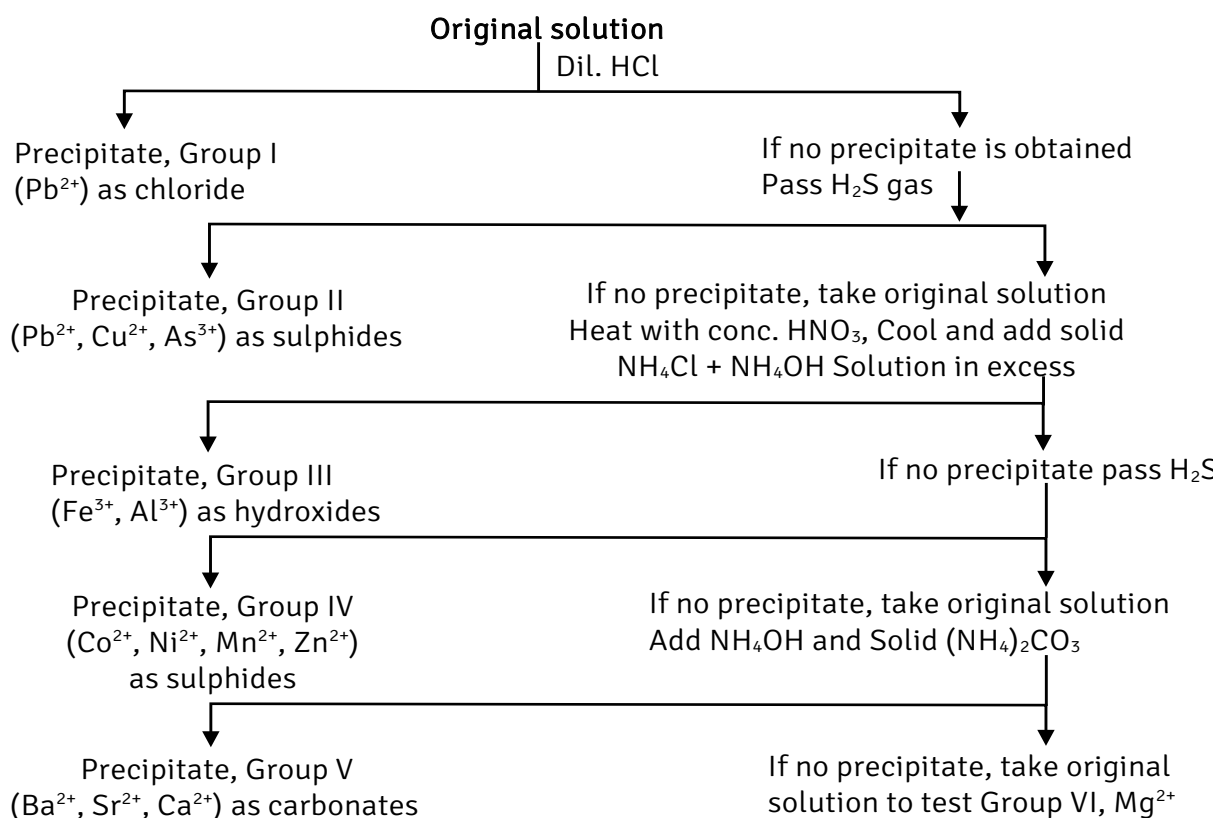
Identification of Cations

In the first step, prepare the clear aqueous solution of the salt. This is called original solution. Cations of different groups with group reagent have been depicted in the table given below.

Group reagents for precipitating ions

Group	Cations*	Group Reagent
Group zero	NH_4^+	None
Group-I	Pb^{2+}	Dilute HCl
Group-II	Pb^{2+} , Cu^{2+} , As^{3+}	H_2S gas in presence of dil. HCl
Group-III	Al^{3+} , Fe^{3+}	NH_4OH in presence of NH_4Cl
Group-IV	Co^{2+} , Ni^{2+} , Mn^{2+} , Zn^{2+}	H_2S in presence of NH_4OH
Group-V	Ba^{2+} , Sr^{2+} , Ca^{2+}	$(\text{NH}_4)_2\text{CO}_3$ in presence of NH_4OH
Group-VI	Mg^{2+}	None

To identify metal cations, these are precipitated from the original solution by using the group reagents according to the flow chart given below:



Exercise 1

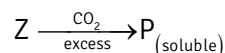
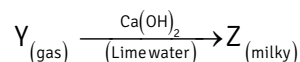
1. Which salt does not give any gas on Reaction with dil. H_2SO_4 .

(1) NaNO_2 (2) Na_2S
(3) K_2CO_3 (4) KCl

2. Salt X, does not give any gaseous product with dil. H_2SO_4 , but produce red brown fumes with conc. H_2SO_4 is:

(1) Na_2CO_3 (2) Na_2SO_4
(3) KNO_3 (4) KNO_2

3. $\text{X}(\text{salt}) \xrightarrow[\text{H}_2\text{SO}_4]{\text{dil.}} \text{Y}_{(\text{gas})} [\text{Colourless}]$



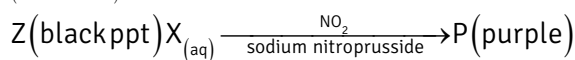
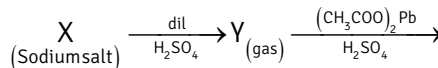
Which is not correct:

(a) Anion of X can be CO_3^{2-}
(b) Gas Y is NO_2
(c) Z is CaCO_3
(d) P is $\text{Ca(HCO}_3)_2$

Options are:

(1) a & b (2) b & c
(3) only b (4) only c

4.



Select the true statement(s) which are given below:

(a) Y is red brown colored gas
(b) Z is PbSO_4
(c) X is Na_2S
(d) P is an Iron complex

Correct option are:

(1) a & b (2) b & c (3) c & d (4) b & d

5.

When salt (NaNO_2) reacts with dil H_2SO_4 , and warming, reddish brown fumes of X (gas) are evolved.

Statement-1: Salt + KI + CH_3COOH + Freshly prepared Starch = No Colouration

Statement-2: Aq. salt + CH_3COOH + Sulphanilic Acid + 1-Naphthylamine = Red dye

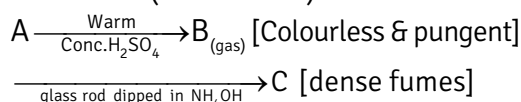
Statement-3: Aq. Salt $\xrightarrow[\text{Solution}]{\text{AgNO}_3}$ white ppt

Pick correct option for **Statement-1**, **Statement-2**, **Statement-3** respectively:

(T = True, F = False)

(1) TTT (2) FTF (3) FTT (4) FFT

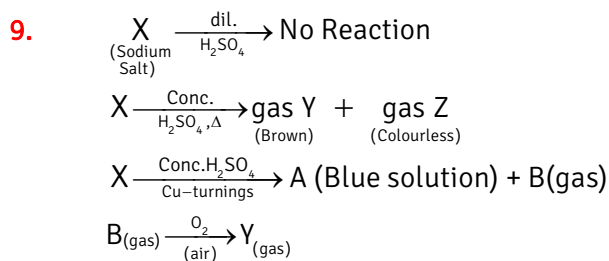
6. Note the observations.
Salt (A) + dil. $\text{H}_2\text{SO}_4 \rightarrow$ No gaseous substance
(Sodium salt)



Find correct possibility.

- (1) Salt(A), on heating with conc. H_2SO_4 & MnO_2 , evolves a greenish yellow pungent gas.
(2) Salt (A), on Reaction with AgNO_3 gives curdy white ppt which is soluble in NH_4OH solution.
(3) Salt can give chromyl chloride test
(4) All
7. True statement for NaBr.
(A) Brown coloured salt.
(B) Gives vigorous fumes with dil. H_2SO_4
(C) Gives brown fumes on Reaction with Conc. H_2SO_4 & fumes intensifies on addition of MnO_2 .
(D) Give orange brown colouration of aqueous layer during layer test.
(E) Gives pale yellow ppt on Reaction with aq. AgNO_3 .
(1) A, C, D, E (2) B, C, E
(3) C, E (4) C, D, E

8. When NaI react with Conc. H_2SO_4 .
(1) I_2 is evolved which can turn starch paper blue
(2) Some amount of HI, SO_2 , H_2S and S are also formed
(3) Violet fumes become dense on addition of MnO_2 .
(4) All are correct.

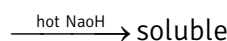
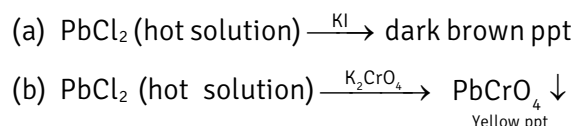


correct statement:

- (1) X will show brown ring test.
(2) Y is NO
(3) A & B are CuSO_4 & NO respectively
(4) Z is NO
10. NO_2^- and NO_3^- can be distinguished by which of the following reagent:
(1) dil. H_2SO_4 (2) conc. H_2SO_4
(3) dil. HCl (4) None of these

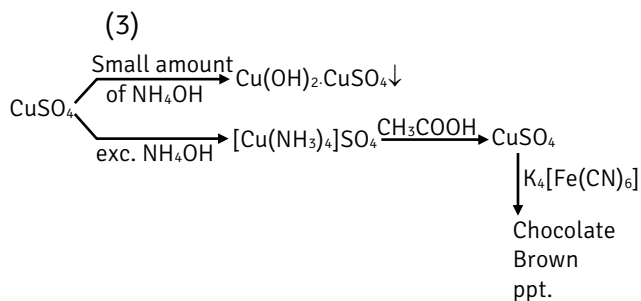
11. $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$ is unstable because:
(1) It liberates NO gas on warming
(2) It liberates NO gas on shaking
(3) The charge of central atom is +1 (relatively low enough)
(4) None of these

12. Which is correct:



- (c) $\text{Pb}(\text{NO}_3)_2 \xrightarrow[\text{(aq)}]{\text{dil. H}_2\text{SO}_4} \text{white ppt}$
(d) Lead sulphide ppt. dissolves in dil. HNO_3
Correct options are:
(1) a, d (2) b, c, d
(3) a, b, d (4) a, c

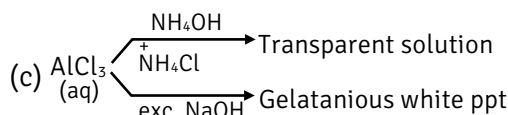
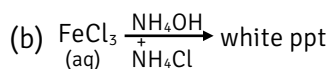
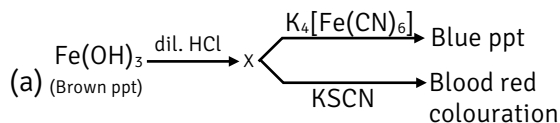
13. Which is correctly matched
(1) CuS (ppt.) is soluble in $\text{HNO}_3 \rightarrow \text{True}$
(2) Reaction mixture of (1), on heating for longtime CuSO_4 is formed & solution turns blue $\rightarrow \text{True}$



- (4) All

14. **Assertion :** If group II is absent we add few drops of HNO_3 & solution is heated
Reason : Reagent of group II is H_2S in presence of dil. HCl
(1) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
(3) Assertion is correct but Reason is not correct.
(4) Assertion is not correct but Reason is correct.

15. Which of the following are true



(d) $\text{AlCl}_3(\text{aq})$ can give a $\oplus$ lake test

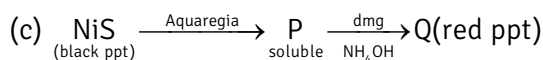
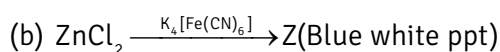
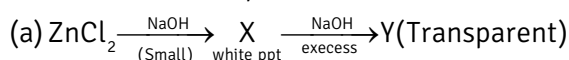
Correct code is:

(1) a & d (2) a, b, d (3) c, d (4) a, b, c

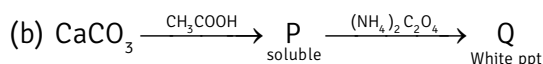
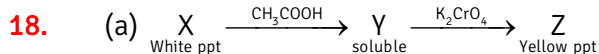
16. **Assertion :** Group reagent for group IV cations is H_2S in presence of $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$
Reason : K_{SP} of group II sulphides $> K_{\text{SP}}$ of group IV sulphides

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (2) Both Assertion and Reason are correct but Reason is not correct explanation of Assertion.
 (3) Assertion is correct but Reason is not correct.
 (4) Assertion is not correct but Reason is correct.

17. Find unknown compounds?



- (1) X: $\text{Zn}(\text{OH})_2$; Y: Na_2ZnO_2 ; Z: FeCl_3 ;
 P: $[\text{Ni}(\text{OH})_2]$; Q: $\text{Ni}(\text{dmg})_2$
 (2) X: $\text{Zn}(\text{OH})_2$; Y: Na_2ZnO_2 ; Z: $\text{Zn}_2[\text{Fe}(\text{CN})_6]$;
 P: NiCl_2 ; Q: $\text{Ni}(\text{dmg})_2$
 (3) X: Na_2ZnO_2 ; Y: $\text{Zn}(\text{OH})_2$; Z: $\text{Zn}_2[\text{Fe}(\text{CN})_6]$;
 P: NiCl_4 ; Q: $[\text{Ni}(\text{dmg})_3]^{2-}$
 (4) X: $\text{Zn}(\text{OH})_2$; Y: $\text{Zn}(\text{OH})_2$; Z: $\text{Zn}(\text{CN})_2$;
 P: $\text{Ni}(\text{OH})_2$; Q: $[\text{Ni}(\text{dmg})_2]$



Incorrect statement

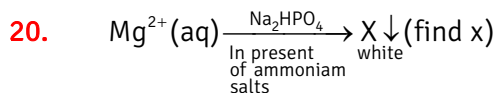
- (1) X can be BaCO_3 (2) Y can be $(\text{CH}_3\text{COO})_2\text{Ba}$
 (3) Q can be CaCrO_4 (4) Z can be BaCrO_4

19. **Assertion :** MgCO_3 is soluble in presence of NH_4^+ salt.

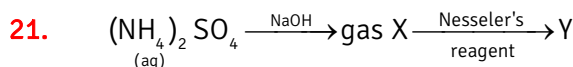
Reason : $\text{NH}_4^+ + \text{CO}_3^{2-} \rightleftharpoons \text{NH}_3 + \text{HCO}_3^-$ shifts towards left hand side.

$\therefore \text{MgCO}_3$ can't be precipitated

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
 (2) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
 (3) Assertion is correct but Reason is not correct.
 (4) Assertion is not correct but Reason is correct.



- (1) $(\text{NH}_4)_3\text{PO}_4$ (2) $\text{Mg}_3(\text{PO}_4)_2$
 (3) $\text{Mg}(\text{NH}_4)\text{PO}_4$ (4) MgCO_3



- (1) X is N_2
 (2) Y is brown ppt
 (3) X turns blue litmus red
 (4) Y contains Hg & Cd

22. Match the List-I with List-II:

List-I Cations		List-II Group reagent	
(P)	Pb^{2+} , Cu^{2+}	(i)	H_2S gas in presence of dilute HCl
(Q)	Al^{3+} , Fe^{3+}	(ii)	$(\text{NH}_4)_2\text{CO}_3$ in presence of NH_4OH
(R)	Co^{2+} , Ni^{2+}	(iii)	NH_4OH in presence of NH_4Cl
(S)	Ba^{2+} , Ca^{2+}	(iv)	H_2S in presence of NH_4OH

Option are :

- (1) P $\rightarrow$ i, Q $\rightarrow$ iii, R $\rightarrow$ ii, S $\rightarrow$ iv
 (2) P $\rightarrow$ iv, Q $\rightarrow$ ii, R $\rightarrow$ iii, S $\rightarrow$ i
 (3) P $\rightarrow$ iii, Q $\rightarrow$ i, R $\rightarrow$ iv, S $\rightarrow$ ii
 (4) P $\rightarrow$ i, Q $\rightarrow$ iii, R $\rightarrow$ iv, S $\rightarrow$ ii

23. Match the following:

Column-I
Metal

- (a) Li
(b) Na
(c) Ba
(d) Ca

Column -II
Colour

- (p) Brick red
(q) Crimson red
(r) Golden yellow
(s) Apple green

Option are :

- (1) $a \rightarrow q, b \rightarrow r, c \rightarrow s, d \rightarrow p$
(2) $a \rightarrow p, b \rightarrow r, c \rightarrow s, d \rightarrow q$
(3) $a \rightarrow s, b \rightarrow r, c \rightarrow q, d \rightarrow p$
(4) $a \rightarrow r, b \rightarrow q, c \rightarrow s, d \rightarrow p$

24. Ammonium salt gives brown colour with alkaline solution of Nessler's reagent to form iodide of Millon's base. The formula of this compound is:

- (1) $\text{NH}_2\text{-Hg-OHg-I}$ (2) $\text{NH}_2\text{-O-Hg-HgI}$
(3) K_2HgI_4 (4) $\text{NH}_2\text{-Hg-I}$

25. Carbonates of Ba and Ca are:

- (1) White (2) Blue (3) Green (4) Black

26. Disodium hydrogen phosphate is used to test:

- (1) Mg^{2+} (2) Na^{2+}
(3) Ca^{2+} (4) All of these

27. Reddish-brown (chocolate) ppt. are formed with:

- (1) Cu^{2+} and $[\text{Fe}(\text{CN})_6]^{4-}$ (2) Ba^{2+} and SO_4^{2-}
(3) Pb^{2+} and I^- (4) None of these

28. Salt X $\xrightarrow[\text{Prusside}]{\text{Sodium Nitro}}$ purple solution

Salt Y $\xrightarrow{\text{dil. H}_2\text{SO}_4}$ gas A

[this can decolourise KMnO_4]

If anions of X and Y are same, than gas A is:

- (1) NO_2 (2) CO_2 (3) H_2S (4) CO

29. Layer test of I^- involves

- (1) Oxidation of I^- by Cl^- (2) Oxidation of Cl^- by I_2
(3) Oxidation of I_2 by Cl^- (4) Oxidation of I^- by Cl_2

30. Chromyl chloride test is not given by

- (1) SnCl_2 (2) NaCl (3) CHCl_3 (4) KCl

31. The brown colour during the test of NO_3^- is due to the formation of complex ion with the formula

- (1) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ (2) $[\text{Fe}(\text{NO}(\text{CN})_5)]^{2+}$
(3) $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$ (4) $[\text{Fe}(\text{H}_2\text{O})(\text{NO})_5]^{2+}$

32. Match List I with List II

List (Anion)		List II (Gas evolved on reaction with dil. H_2SO_4)	
A.	CO_3^{2-}	I	Colourless gas which turns lead acetate paper black
B.	S^{2-}	II	Colourless gas which turns acidified potassium dichromate solution green
C.	SO_3^{2-}	III	Brown fumes which turns acidified KI solution containing starch blue.
D.	NO_2^-	IV	Colourless gas evolved with brisk effervescence, which turns lime water milky

Choose the correct answer from the options given below:

- (1) A-III, B-I, C-II, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-IV, B-I, C-III, D-II
(4) A-IV, B-I, C-II, D-III

33. $\text{Fe}(\text{OH})_3$ can be separated from $\text{Al}(\text{OH})_3$ by addition of

- (1) dil. HCl
(2) NaCl solution
(3) NaOH solution
(4) NH_4Cl and NH_4OH

34. Which of the following is insoluble in excess of NaOH ?

- (1) $\text{Al}(\text{OH})_3$ (2) $\text{Cr}(\text{OH})_3$
(3) $\text{Fe}(\text{OH})_3$ (4) $\text{Zn}(\text{OH})_2$

35. The cations present in slightly acidic solution are Fe^{3+} , Zn^{2+} and Cu^{2+} . The reagent which when added in excess to this solution would identify and separate Fe^{3+} in one step is:

- (1) 2 M HCl (2) 6 M NH_3
(3) 6 M NaOH (4) H_2S gas

36. Which of the following gives blood red colour with KSCN ?

- (1) Cu^{2+} (2) Fe^{3+}
(3) Al^{3+} (4) Zn^{2+}

37. **Statement-I:** $\bar{\text{NO}}_2$ give white precipitate with AgNO_3 solution.

Statement-II: Both AgNO_3 and AgNO_2 are water insoluble

- (1) Both statement-I and Statement-II are correct and statement-II is the correct explanation of statement-I
- (2) Both statement-I and Statement-II are correct and statement-II is not the correct explanation of statement-I
- (3) Statement-I is correct, Statement-II is incorrect
- (4) Statement-I is incorrect, Statement-II is correct

38. **Statement-I:** Aq. ZnCl_2 does not give precipitate if H_2S gas is passed through it in acidic condition

Statement-II: K_{sp} of $\text{ZnS} < K_{sp}$ of CuS

- (1) Both statement-I and Statement-II are correct and statement-II is the correct explanation of statement-I
- (2) Both statement-I and Statement-II are correct and statement-II is not the correct explanation of statement-I
- (3) Statement-I is correct, Statement-II is incorrect
- (4) Statement-I is incorrect, Statement-II is correct

39. **Statement-I:** Cu gives opaque red colour in oxidising flame in borax bead test

Statement-II: Inoxidising flame, $\text{Cu}(\text{BO}_2)_2$ is formed

- (1) Both statement-I and Statement-II are correct and statement-II is the correct explanation of statement-I
- (2) Both statement-I and Statement-II are correct and statement-II is not the correct explanation of statement-I
- (3) Statement-I is correct, Statement-II is incorrect
- (4) Statement-I is incorrect, Statement-II is correct

40. **Assertion :** $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$ can show chromyl chloride test

Reason : Uncomplexed Cl^- are present in solution

- (1) Both assertion and reason are true and reason is the correct explanation of assertion.
- (2) Both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) Assertion is true but reason is false.
- (4) Assertion is false and reason is true

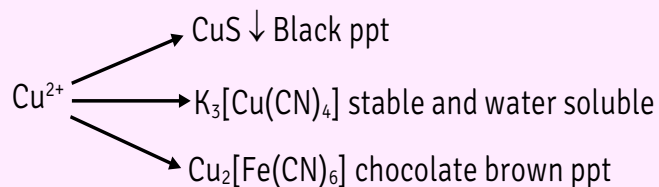
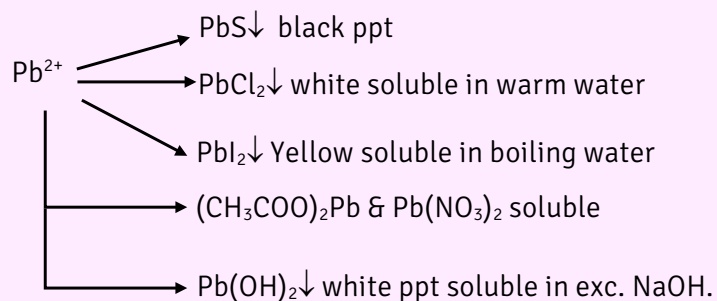
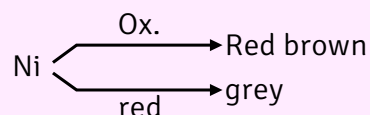
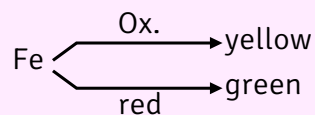
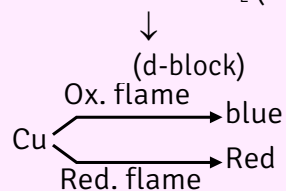
Points to ponder :

Information

Flame test

Li	Crimson red	}X	Be
Na	Golden yellow		Mg
K	Pale violet		Ca brick red
Rb	Red violet		Sr Crimson red
Cs	Blue		Ba Apple green

Borax Bead Test

Borax bead: $\text{NaBO}_2 + \text{B}_2\text{O}_3$ Colour in last: MBO_2 (metaborate)

Exercise 2 (Previous Year's Questions)

- In the precipitation of the iron group (III) in qualitative analysis, ammonium chloride is added before adding ammonium hydroxide to [IIT Main-Online-2024]
 - (1) Decrease concentration of OH^- ions
 - (2) Prevent interference by phosphate ions
 - (3) Increase concentration of NH_4^+ ions
 - (4) Increase concentration of Cl^- ions
- On passing a gas, 'X', through Nessler's reagent, a brown precipitate is obtained. The gas 'X' is : [IIT Main-Online-2024]
 - (1) H_2S
 - (2) NH_3
 - (3) Cl_2
 - (4) CO_2
- An ammoniacal metal salt solution gives a brilliant red precipitate on addition of dimethylglyoxime. The metal ion is : [IIT Main-Online-2023]
 - (1) Ni^{2+}
 - (2) Fe^{2+}
 - (3) Co^{2+}
 - (4) Cu^{2+}
- Element not present in Nessler's reagent is: [IIT Main-Online-2023]
 - (1) I
 - (2) N
 - (3) Hg
 - (4) K
- In chromyl chloride, the number of d-electrons present on chromium is same as in (Given at no. of Ti : 22, V : 23, Cr : 24, Mn : 25, Fe : 26) [IIT Main-Online-2023]
 - (1) Mn (VII)
 - (2) Fe (III)
 - (3) Ti (III)
 - (4) V (IV)
- Passing H_2S gas into a mixture of Mn^{2+} , Ni^{2+} , Cu^{2+} and Hg^{2+} ions in an acidified solution precipitates :
 - (1) CuS and HgS
 - (2) MnS and CuS
 - (3) MnS and NiS
 - (4) NiS and HgS
- Given below are certain cations. Using inorganic qualitative analysis, arrange them in increasing group number from 0 to VI.
 - Al^{3+}
 - Cu^{2+}
 - Ba^{2+}
 - Co^{2+}
 - Mg^{2+}
 Choose the correct answer from the options given below: [NEET-2024]
 - (1) E, C, D, B, A
 - (2) E, A, B, C, D
 - (3) B, A, D, C, E
 - (4) B, C, A, D, E

- During the preparation of Mohr's salt solution (Ferrous ammonium sulphate), which of the following acid is added to prevent hydrolysis of Fe^{2+} ion [NEET-2024]
 - (1) Dilute nitric acid
 - (2) Dilute sulphuric acid
 - (3) dilute hydrochloric acid
 - (4) Concentrated sulphuric acid

- Match List-I with List-II. [RE-NEET-2024]

List-I (Test/reagent)		List-II (Radical identified)	
A.	Lake Test	I.	NO_3^-
B.	Nessler's Reagent	II.	Fe^{3+}
C.	Potassium sulpho cyanide	III.	Al^{3+}
D.	Brown Ring Test	IV.	NH_4^+

- (1) A-IV, B-II, C-III, D-I
- (2) A-II, B-IV, C-III, D-I
- (3) A-II, B-III, C-IV, D-I
- (4) A-III, B-IV, C-II, D-I

- Match List-I with List-II. [RE-NEET 2024]

List-I Solid salt treated with dil. H_2SO_4		List-II Anion detected	
A.	Effervescence of colourless gas	I.	NO_2^-
B.	Gas with smell of rotten egg	II.	CO_3^{2-}
C.	Gas with pungent smell	III.	S^{2-}
D.	Brown fumes	IV.	SO_3^{2-}

Choose the correct answer from the options given below :

- (1) A-II, B-III, C-IV, D-I
- (2) A-IV, B-III, C-II, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-III, C-I, D-IV

- Match List I with List II [NEET-2025]

List I (Ion)		List II (Group Number in Cation Analysis)	
A.	Co^{2+}	I.	Group-I
B.	Mg^{2+}	II.	Group-III
C.	Pb^{2+}	III.	Group-IV
D.	Al^{3+}	IV.	Group-VI

Choose the correct answer from the options given below:

- (1) A-III, B-II, C-IV, D-I
- (2) A-III, B-II, C-I, D-IV
- (3) A-III, B-IV, C-II, D-I
- (4) A-III, B-IV, C-I, D-II

ANSWER KEYS

Exercise 1

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	4	3	3	3	3	4	3	4	3	1	3	2	4	2	1	3	2	3	3	3
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	2	4	1	1	1	1	1	3	4	3	3	4	3	3	2	2	3	3	4	1

Exercise 2 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7	8	9	10	11								
Ans.	1	2	1	2	1	1	3	2	4	1	4								